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(54) **MULTI-SPECIFIC BINDING PROTEINS AND METHODS OF USE THEREOF**

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(52) **U.S. Cl.**

CPC **C07K 16/2809** (2013.01); **A61P 35/00** (2018.01); **C07K 16/18** (2013.01); **C07K 16/2803** (2013.01); **A61K 2039/505** (2013.01); **C07K 2317/31** (2013.01); **C07K 2317/565** (2013.01); **C07K 2317/569** (2013.01); **C07K 2317/622** (2013.01); **C07K 2317/73** (2013.01); **C07K 2317/76** (2013.01); **C07K 2317/92** (2013.01)

(58) **Field of Classification Search**

CPC **C07K 16/2809**; **C07K 2317/31**; **C07K 2317/565**; **C07K 2317/569**; **C07K 2317/622**

See application file for complete search history.

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(57) **ABSTRACT**

The invention relates to multi-specific binding proteins that bind CD19, CD3, and serum albumin. The invention also relates to pharmaceutical compositions comprising these multi-specific binding proteins, expression vectors and host cells for making these multi-specific binding proteins, and methods of use of these multi-specific binding proteins in treating hematologic cancers.

26 Claims, 2 Drawing Sheets

Specification includes a Sequence Listing.

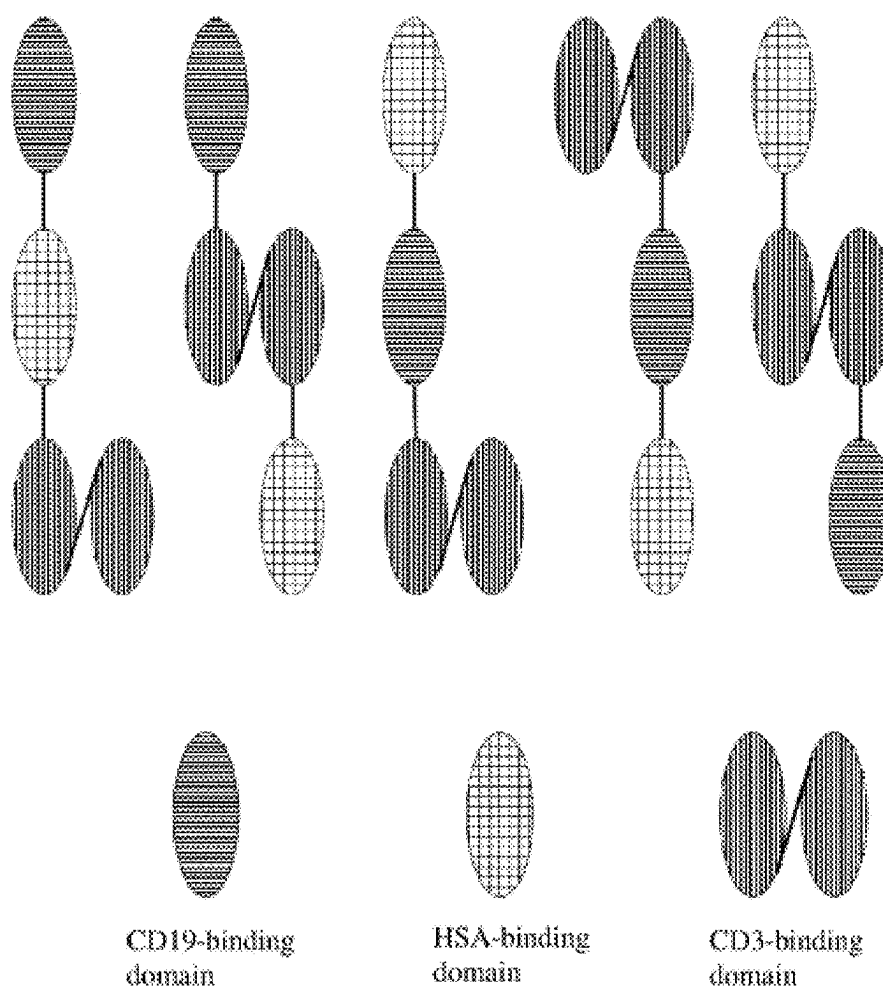


FIG. 1

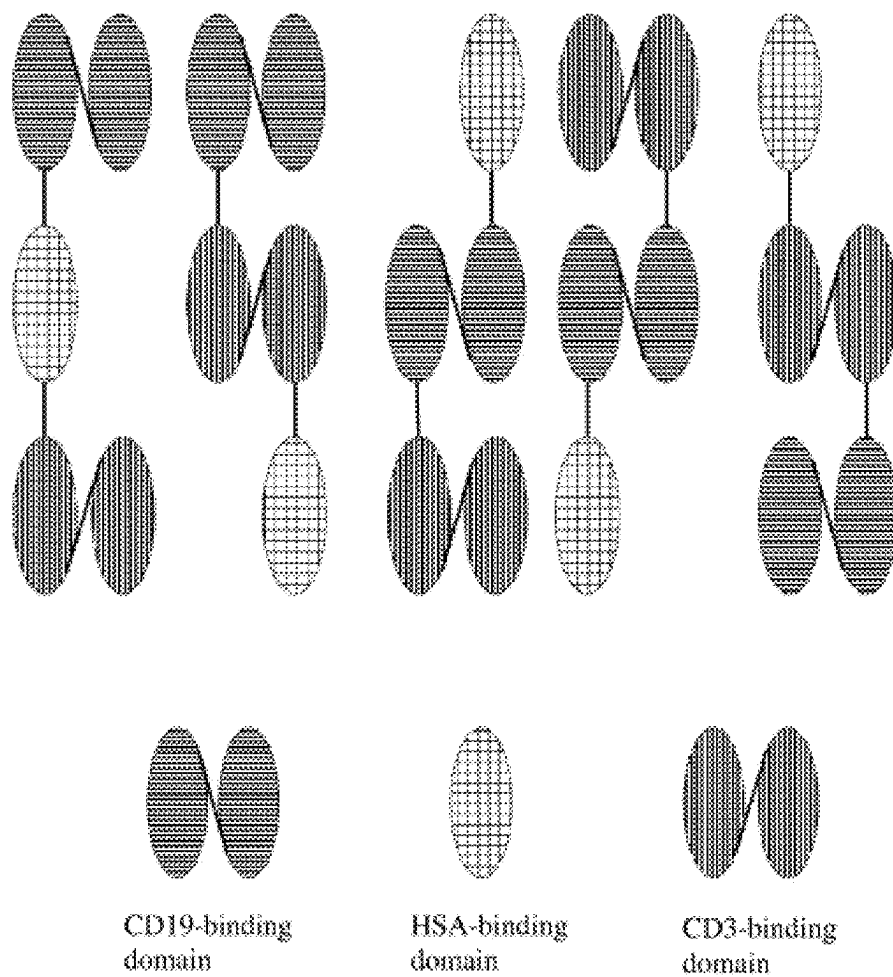


FIG. 2

1

MULTI-SPECIFIC BINDING PROTEINS AND METHODS OF USE THEREOF

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a U.S. national stage application, filed under 35 U.S.C. § 371, of International Application No. PCT/US2019/036177, filed on Jun. 7, 2019, which claims the benefit of and priority to U.S. Provisional Patent Application No. 62/681,784, filed on Jun. 7, 2018, the disclosure of each of which is incorporated by reference herein in its entirety for all purposes.

INCORPORATION BY REFERENCE OF SEQUENCE LISTING

This application contains a Sequence listing that has been submitted in a computer readable format and is hereby incorporated by reference in its entirety. The ASCII text file, created on Nov. 15, 2023, is named 67254US01_CLN-978_Sequence_Listing.txt, and is 373777 bytes in size.

FIELD OF THE INVENTION

The invention relates to multi-specific binding proteins that bind CD19, CD3, and serum albumin. The invention also relates to pharmaceutical compositions comprising these multi-specific binding proteins, expression vectors and host cells for making these multi-specific binding proteins, and methods of use of these multi-specific binding proteins in treating hematologic cancers.

BACKGROUND

Bispecific molecules available under the trade name BiTE® from Micromet AG (bispecific T-cell engager) constructs are recombinant protein constructs made from two flexibly linked antibody-derived binding domains. One binding domain of BiTE® constructs is specific for a selected tumor-associated surface antigen on target cells, and the second binding domain is specific for CD3, a subunit of the T cell receptor complex on T cells. By this design, BiTE® constructs can transiently connect T cells with target cells and, at the same time, potentially activate the inherent cytolytic potential of T cells against target cells.

The CD3 receptor complex is a protein complex composed of four polypeptide chains. In mammals, the complex contains a CD3γ (gamma) chain, a CD3δ (delta) chain, and two CD3ε (epsilon) chains. The CD3γ (gamma), CD3δ (delta), and CD3ε (epsilon) chains are highly related cell-surface proteins of the immunoglobulin superfamily containing a single extracellular immunoglobulin domain. These chains associate with the T cell receptor (TCR) to form a TCR-CD3 complex and to generate an activation signal in T lymphocytes upon antigen engagement. About 95% of T cells express αβ TCR, which contains an α (alpha) chain and a β (beta) chain. Two TCRξ (zeta) chains are also present in the TCR-CD3 complex. The αβ TCR is responsible for recognizing antigens presented by a major histocompatibility complex (MHC). When the TCR engages with antigenic peptide and MHC complex, the T lymphocyte is activated through a series of biochemical events mediated by associated enzymes, co-receptors, specialized adaptor molecules, and activated or released transcription factors.

Earlier BiTE® constructs bind conformational epitopes of CD3 and typically are species-specific (see, PCT Publication

2

No. WO2008119567A2). Improved BiTE® constructs, such as blinatumomab (also called AMG 103; see, PCT Publication No. WO1999054440A1) and solitomab (also called AMG 110; see, PCT Publication No. WO2005040220A1), bind context-independent epitopes at the N-terminus of CD3ε chain (e.g., amino acid residues 1-27 of human CD3ε extracellular domain) and show cross-species specificity for human, *Callithrix jacchus*, *Saguinus Oedipus*, and *Saimiri sciureus* CD3ε chain (see id.). These constructs do not nonspecifically activate T cells to the same degree as observed with the earlier BiTE® constructs, and are therefore believed to bear a lower risk of side effects (see, Brischwein et al. (2007) J. Immunother., 30 (8): 798-807).

BiTE® constructs are believed to suffer from rapid clearance from the body. Therefore, whilst they are able to rapidly penetrate many areas of the body, and are quick to produce and easier to handle, their in vivo applications may be limited by their brief persistence in vivo. Prolonged administration by continuous intravenous infusions may be required to achieve therapeutic effects of blinatumomab and solitomab because of their short in vivo half-life. However, such continuous intravenous infusions are inconvenient for patients and may increase the costs of treatment.

Although significant developments have been made in constructing multi-specific binding proteins, there remains a need for new and useful multi-specific binding proteins for treating cancer that have adequate therapeutic efficacy, a format straightforward to manufacture, and favorable pharmacokinetic properties such as a longer half-life.

SUMMARY OF THE INVENTION

The multi-specific binding proteins disclosed herein comprise a first domain that binds CD19 (e.g., human CD19), a second domain that binds CD3 (e.g., human and/or Macaca CD3), and a third domain that binds serum albumin (e.g., human serum albumin (HSA)). These domains are linked in certain manners for favorable therapeutic efficacy and in vivo half-life. The multi-specific binding proteins can be used to stimulate an immune response against a cell expressing CD19. As a result, the multi-specific binding proteins can be used to treat a disease or disorder associated with aberrant cells expressing CD19, such as certain hematologic B-cell cancers.

In one aspect, the present disclosure provides a multi-specific binding protein comprising: (a) a first antigen-binding site that binds human CD19; (b) a second antigen-binding site that binds human CD3; and (c) a third antigen-binding site that binds human serum albumin (HSA).

In certain embodiments, the multi-specific binding protein comprises a single polypeptide chain. In certain embodiments, the third antigen-binding site is not positioned between the first antigen-binding site and the second antigen-binding site in the polypeptide chain.

In certain embodiments, the third antigen-binding site is positioned N-terminal to both the first antigen-binding site and the second antigen-binding site in the polypeptide chain. In certain embodiments, the third antigen-binding site is positioned N-terminal to the first antigen-binding site, and the first antigen-binding site is positioned N-terminal to the second antigen-binding site in the polypeptide chain. In certain embodiments, the third antigen-binding site is positioned N-terminal to the second antigen-binding site, and the second antigen-binding site is positioned N-terminal to the first antigen-binding site in the polypeptide chain.

In certain embodiments, the third antigen-binding site is positioned C-terminal to both the first antigen-binding site

3

and the second antigen-binding site in the polypeptide chain. In certain embodiments, the first antigen-binding site is positioned N-terminal to the second antigen-binding site, and the second antigen-binding site is positioned N-terminal to the third antigen-binding site in the polypeptide chain. In certain embodiments, the second antigen-binding site is positioned N-terminal to the first antigen-binding site, and the first antigen-binding site is positioned N-terminal of the third antigen-binding site in the polypeptide chain.

In certain embodiments, the first antigen-binding site is positioned N-terminal to the third antigen-binding site, and the third antigen-binding site is positioned N-terminal to the second antigen-binding site in the polypeptide chain. In certain embodiments, the second antigen-binding site is positioned N-terminal to the third antigen-binding site, and the third antigen-binding site is positioned N-terminal binding protein the first antigen-binding site in the polypeptide chain.

In certain embodiments, the first antigen-binding site comprises a single-chain variable fragment (scFv) or a single-domain antibody (sdAb). In certain embodiments, the first antigen-binding site comprises a single-chain variable fragment (scFv). In certain embodiments, the first antigen-binding site binds human CD19 with a dissociation constant (K_D) equal to or lower than 20 nM (namely, binding equal to or stronger than 20 nM). In certain embodiments, the first antigen-binding site has a melting temperature of at least 60° C.

In certain embodiments, the first antigen-binding site comprises a heavy chain variable domain (VH) comprising complementarity determining regions HCDR1, HCDR2, and HCDR3, and a light chain variable domain (VL) comprising complementarity determining regions LCDR1, LCDR2, and LCDR3, wherein (i) the HCDR1, HCDR2, HCDR3, LCDR1, LCDR2, and LCDR3 comprise the amino acid sequences set forth in SEQ ID NOs: 3, 4, 5, 6, 7, and 8, respectively; and/or (ii) the VH comprises an amino acid sequence at least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%) identical to SEQ ID NO: 1, and the VL comprises an amino acid sequence at least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%) identical to SEQ ID NO: 2.

In certain embodiments, the VH comprises the amino acid sequence set forth in SEQ ID NO: 1, and the VL comprises the amino acid sequence set forth in SEQ ID NO: 2.

In certain embodiments, the first antigen-binding site comprises a VH comprising complementarity determining regions HCDR1, HCDR2, and HCDR3, and a VL comprising complementarity determining regions LCDR1, LCDR2, and LCDR3, wherein (i) the HCDR1, HCDR2, HCDR3, LCDR1, LCDR2, and LCDR3 comprise the amino acid sequences set forth in SEQ ID NOs: 12, 13, 14, 15, 16, and 17, respectively; and/or (ii) the VH comprises an amino acid sequence at least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%) identical to SEQ ID NO: 10, and the VL comprises an amino acid sequence at least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%) identical to SEQ ID NO: 11. In certain embodiments,

4

the VH comprises the amino acid sequence set forth in SEQ ID NO: 10, and the VL comprises the amino acid sequence set forth in SEQ ID NO: 11.

In certain embodiments, the second antigen-binding site comprises an scFv or an sdAb. In certain embodiments, the second antigen-binding site comprises an scFv. In certain embodiments, the second antigen-binding site binds human CD3ε. In certain embodiments, the second antigen-binding site binds human CD3 with a K_D equal to or lower than 10 nM (namely, binding equal to or stronger than 10 nM). In certain embodiments, the second antigen-binding site binds human CD3ε with a K_D equal to or lower than 10 nM. In certain embodiments, the second antigen-binding site has a melting temperature of at least 60° C.

In certain embodiments, the second antigen-binding site comprises a VH comprising complementarity determining regions HCDR1, HCDR2, and HCDR3, and a VL comprising complementarity determining regions LCDR1, LCDR2, and LCDR3, wherein (i) the HCDR1, HCDR2, HCDR3, LCDR1, LCDR2, and LCDR3 comprise the amino acid sequences set forth in SEQ ID NOs: 99, 100, 101, 102, 103, and 104, respectively; and/or (ii) the VH comprises an amino acid sequence at least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%) identical to SEQ ID NO: 97, and the VL comprises an amino acid sequence at least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%) identical to SEQ ID NO: 98. In certain embodiments, the VH comprises the amino acid sequence set forth in SEQ ID NO: 97, and the VL comprises the amino acid sequence set forth in SEQ ID NO: 98.

In certain embodiments, the third antigen-binding site comprises an scFv or an sdAb. In certain embodiments, the third antigen-binding site comprises an sdAb. In certain embodiments, the third antigen-binding site binds HSA with a K_D equal to or lower than 20 nM (namely, binding equal to or stronger than 20 nM). In certain embodiments, the third antigen-binding site has a melting temperature of at least 60° C.

In certain embodiments, the third antigen-binding site comprises a VH comprising complementarity determining regions HCDR1, HCDR2, and HCDR3, wherein (i) the HCDR1, HCDR2, and HCDR3 comprise the amino acid sequences set forth in SEQ ID NOs: 122 or 123, 124 or 125, and 126, respectively; and/or (ii) the VH comprises an amino acid sequence at least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%) identical to SEQ ID NO: 121. In certain embodiments, the VH comprises the amino acid sequence set forth in SEQ ID NO: 121.

In certain embodiments, at least two adjacent antigen-binding sites are connected by a peptide linker. In certain embodiments, each of the adjacent antigen-binding sites in the multi-specific binding protein are connected by a peptide linker. In certain embodiments, the peptide linker comprises the amino acid sequence of SEQ ID NO: 298, 299, or 302. In certain embodiments, the peptide linker consists of the amino acid sequence of SEQ ID NO: 298, 299, or 302.

In certain embodiments, the multi-specific binding protein does not comprise an antibody Fc region.

In certain embodiments, the molecular weight of the multi-specific binding protein is at least 65 kD. In certain embodiments, the molecular weight of the multi-specific

5

binding protein is in the range of 50-90 kD, 50-80 kD, 50-70 kD, 50-60 kD, 60-90 kD, 60-80 kD, 60-70 kD, 65-90 kD, 65-80 kD, 65-70 kD, 70-90 kD, or 70-80 kD.

In certain embodiments, the serum half-life of the multi-specific binding protein is at least 24, 36, 48, or 60 hours.

In another aspect, the present disclosure provides a pharmaceutical composition comprising a multi-specific binding protein disclosed herein and a pharmaceutically acceptable carrier.

In another aspect, the present disclosure provides an isolated polynucleotide encoding a multi-specific binding protein disclosed herein. In another aspect, the present disclosure provides a vector comprising the polynucleotide. In another aspect, the present disclosure provides a recombinant host cell comprising the polynucleotide or the vector.

In another aspect, the present disclosure provides a method of producing a multi-specific binding protein, the method comprising culturing a host cell disclosed herein under suitable conditions that allow expression of the multi-specific binding protein. In certain embodiments, the method further comprises isolating the multi-specific binding protein. In certain embodiments, the method further comprises formulating the isolated multi-specific binding protein with a pharmaceutically acceptable carrier.

In another aspect, the present disclosure provides a method of stimulating an immune response against a cell expressing CD19, the method comprising exposing the cell and a T lymphocyte to a multi-specific binding protein or pharmaceutical composition disclosed herein.

In another aspect, the present disclosure provides a method of treating a hematologic cancer in a subject in need thereof, the method comprising administering to the subject an effective amount of a multi-specific binding protein or pharmaceutical composition disclosed herein. In certain embodiments, the hematologic cancer is a B-cell hematologic malignancy.

In another aspect, the present disclosure provides a complex comprising a T cell expressing CD3, a B cell expressing CD19, and a multi-specific binding protein disclosed herein, wherein the T cell and the B cell are simultaneously bound by the multi-specific binding protein. In certain embodiments, the complex further comprises HSA.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a schematic representation of six domain arrangements of single-chain multi-specific binding proteins. The CD19 binding domain in the form of a single-domain antibody (sdAb), the CD3 binding domain in the form of a single-chain variable fragment (scFv), and the HSA binding domain in the form of an sdAb are linked in different orientations. The top of each construct represents the N-terminus and the bottom of each construct represents the C-terminus of a given polypeptide chain.

FIG. 2 is a schematic representation of six domain arrangements of single-chain multi-specific binding proteins. The CD19 binding domain in the form of an scFv, the CD3 binding domain in the form of an scFv, and the HSA binding domain in the form of an sdAb are linked in different orientations. The top of each construct represents the N-terminus and the bottom of each construct represents the C-terminus of a given polypeptide chain.

DETAILED DESCRIPTION

The multi-specific binding proteins disclosed herein comprise a first domain that binds CD19 (e.g., human CD19), a

6

second domain that binds CD3 (e.g., human CD3), and a third domain that binds serum albumin (e.g., HSA). These domains are linked in certain manners for favorable therapeutic efficacy and in vivo half-life. Also provided are pharmaceutical compositions comprising the multi-specific binding proteins, methods of treating a disease or disorder using the multi-specific binding proteins or pharmaceutical compositions, and methods of producing the multi-specific binding proteins. Various aspects of the invention are set forth below in sections; however, aspects of the invention described in one particular section are not to be limited to any particular section.

To facilitate an understanding of the present invention, a number of terms and phrases are defined below.

The term "multi-specific binding protein" refers to a multi-specific molecule in which the structure and/or function is/are based on the structure and/or function of an antibody, e.g., a full-length or whole immunoglobulin molecule, or based on the heavy chain variable domain (VH) and/or light chain variable domain (VL) of an antibody, and/or single chain variants thereof. A multi-specific binding protein is hence capable of binding to its specific target or antigen. Furthermore, any one of the binding domains of a multi-specific binding protein according to the invention comprises the minimum structural requirements of an antibody which allow for the target binding. This minimum requirement may be, e.g., defined by the presence of at least the three heavy chain CDRs (i.e. CDR1, CDR2 and CDR3 of the VH domain) and/or the three light chain CDRs (i.e. CDR1, CDR2 and CDR3 of the VL domain). An alternative approach to defining the minimal structural requirements of an antibody is defining the epitope of a specific target to which the antibody binds, or by referring to a known antibody with which the antibody competes to bind to the same epitope that the known antibody binds. The antibodies on which the constructs according to the invention are based include for example monoclonal, recombinant, chimeric, deimmunized, humanized and human antibodies.

Any one of the binding domains of a multi-specific binding protein according to the invention may comprise the above referred groups of CDRs. Those CDRs may be comprised in the framework of a VH and/or VL. Fd fragments, for example, have two VH domains and often retain some antigen-binding function of the intact antigen-binding domain. Additional examples for formats of antibody fragments, antibody variants or binding domains include: (1) a Fab fragment, a monovalent fragment having the VL, VH, CL and CH1 domains; (2) a F(ab')₂ fragment, a bivalent fragment having two Fab fragments linked by a disulfide bridge at the hinge region; (3) an Fd fragment having the two VH and CH1 domains; (4) an Fv fragment having the VL and VH domains of a single arm of an antibody; (5) a dAb fragment (Ward et al., (1989) Nature 341:544-546), which has a VH domain; (6) an isolated complementarity determining region (CDR); and (7) a single chain Fv (scFv), which may be derived, for example, from an scFv-library. Exemplary formats of multi-specific binding proteins according to the invention are described in, e.g., WO2000006605A2, WO2005040220A1, WO2008119567A2, WO2010037838A2, WO2013026837A1, WO 2013026833A1, US20140308285A1, US20140302037A1, WO2014144722A2, WO2014151910A1, and WO2015048272A1.

Multi-specific binding proteins according to the invention may also comprise modified fragments of antibodies, also called antibody variants, such as di-scFv or bi(s)-scFv,

scFv-Fc, scFv-zipper, scFab, Fab₂, Fab₃, diabodies, single chain diabodies, tandem diabodies (Tandab's), tandem di-scFv, tandem tri-scFv, "multibodies" such as triabodies or tetrabodies, or single-domain antibodies such as nanobodies or single variable domain antibodies comprising a single variable domain, which might be VH (also called VHH in the context of an sdAb) or VL, that specifically bind an antigen or epitope independently of other V regions or domains.

As used herein, the terms "single-chain Fv," "single-chain antibody," and "scFv" refer to a single-polypeptide-chain antibody fragment that comprise the variable regions from both the heavy and light chains, but lack the constant regions. Generally, a single-chain antibody further comprises a peptide linker connecting the VH and VL domains which enables it to form the desired structure to bind to antigen. Single chain antibodies are discussed in detail by Pluckthun in *The Pharmacology of Monoclonal Antibodies*, vol. 113, Rosenberg and Moore eds. Springer-Verlag, New York, pp. 269-315 (1994). Various methods of generating single chain antibodies are known, including those described in U.S. Pat. Nos. 4,694,778 and 5,260,203; International Patent Application Publication No. WO 88/01649; Bird (1988) *Science* 242:423-442; Huston et al. (1988) *Proc. Natl. Acad. Sci. USA* 85:5879-5883; Ward et al. (1989) *Nature* 334:54454; Skerra et al. (1988) *Science* 242:1038-1041. In specific embodiments, single-chain antibodies can also be bispecific, multispecific, human, humanized and/or synthetic.

Furthermore, the "multi-specific binding protein" described herein can be a monovalent, bivalent or polyvalent/multivalent construct. Moreover, the "multi-specific binding protein" described herein can include a molecule consisting of only one polypeptide chain, or a molecule consisting of more than one polypeptide chain, wherein the chains can be either identical (homodimers, homotrimers or homo oligomers) or different (heterodimer, heterotrimer or heterooligomer). Examples for the above identified antibodies and the variants or derivatives thereof are described, for example, in Harlow and Lane, *Antibodies a laboratory manual*, CSHL Press (1988); Using Antibodies: a laboratory manual, CSHL Press (1999); Kontermann and Dibel, *Antibody Engineering*, Springer, 2nd ed. 2010; and Little, *Recombinant Antibodies for Immunotherapy*, Cambridge University Press 2009.

The domains of the multi-specific binding protein of the present invention may be connected through one or more peptide bonds and/or peptide linkers. The term "peptide linker" comprises in accordance with the present invention an amino acid sequence linking two domains. The peptide linkers can also be used to fuse the third domain to the other domains of the multi-specific binding protein of the invention. An essential technical feature of such peptide linker is that it does not comprise any polymerization activity. Among the suitable peptide linkers are those described in U.S. Pat. Nos. 4,751,180 and 4,935,233 or WO198809344A1.

The multi-specific binding proteins of the present invention may be in vitro generated multi-specific binding proteins. The term "in vitro generated multi-specific binding protein" refers to a multi-specific binding protein according to the above definition where all or part of the variable region (e.g., at least one CDR) is generated by non-immune cell selection, e.g., an in vitro phage display, protein chip or any other method in which candidate sequences can be tested for their ability to bind to an antigen. The multi-specific binding proteins of the present invention may also

be generated by genomic rearrangement in an immune cell in an animal. A "recombinant antibody" is an antibody made through the use of recombinant DNA technology or genetic engineering.

The multi-specific binding protein of the invention may be monoclonal. The term "monoclonal," as used herein, means that the proteins obtained from a population are substantially homogeneous, i.e., the individual proteins in the population are identical except for naturally occurring mutations and/or post-translation modifications (e.g., isomerizations, amidations) that may be present. In the context of antibodies, monoclonal antibodies are highly specific, being directed against a single antigenic side or determinant on the antigen, in contrast to conventional (polyclonal) antibody preparations which typically include different antibodies directed against different determinants (or epitopes). The modifier "monoclonal" indicates the character of the antibody as being obtained from a substantially homogeneous population of antibodies, and is not to be construed as requiring production of the antibody by any particular method.

The multi-specific binding protein of the invention or one or more antigen-binding site thereof may be affinity matured. In immunology, affinity maturation is the process by which B cells produce antibodies with increased affinity for antigen during the course of an immune response. With repeated exposures to the same antigen, a host will produce antibodies of successively greater affinities. Like the natural prototype, the in vitro affinity maturation is based on the principles of mutation and selection. Two or three rounds of mutation and selection using display methods such as phage display can result in antibody fragments with affinities in the low nanomolar range.

An amino acid substitution variation can be introduced into the multi-specific binding proteins by substituting one or more hypervariable region residues of a parent antibody (e.g., a humanized or human antibody). Generally, the resulting variant(s) selected for further development will have improved biological properties relative to the parent antibody from which they are generated. A convenient way for generating such substitutional variants involves affinity maturation using phage display. Briefly, several hypervariable region sides (e.g., 6-7 sides) are mutated to generate all possible amino acid substitutions at each side. The antibody variants thus generated are displayed in a monovalent fashion from filamentous phage particles as fusions to the gene III product of M13 packaged within each particle. The phage-displayed variants are then screened for their biological activity (e.g., binding affinity) as herein disclosed. In order to identify candidate hypervariable region sides for modification, alanine scanning mutagenesis can be performed to identify hypervariable region residues contributing significantly to antigen binding. Alternatively, or additionally, it may be beneficial to analyze a crystal structure of the antigen-antibody complex to identify contact points between the binding domains. Such contact residues and neighboring residues are candidates for substitution according to the techniques elaborated herein. Once such variants are generated, the panel of variants is subjected to screening as described herein and antibodies with superior properties in one or more relevant assays may be selected for further development.

The multi-specific binding proteins of the present invention specifically can comprise "chimeric" antibodies (immunoglobulins) or fragments thereof in which a portion of the heavy and/or light chain is identical with or homologous to corresponding sequences in antibodies derived from a par-

ticular species or belonging to a particular antibody class or subclass, while the remainder of the chain(s) is/are identical with or homologous to corresponding sequences in antibodies derived from another species or belonging to another antibody class or subclass, so long as they exhibit the desired biological activity (U.S. Pat. No. 4,816,567; Morrison et al. (1984) Proc. Natl. Acad. Sci. U.S.A., 81:6851-55). Chimeric antibodies of interest herein include "primatized" antibodies comprising variable domain antigen-binding sequences derived from a non-human primate (e.g., Old World Monkey, Ape etc.) or human constant region sequences. A variety of approaches for making chimeric antibodies have been described. See e.g., Morrison et al. (1985) Proc. Natl. Acad. Sci. U.S.A., 81:6851; Takeda et al. (1985) Nature, 314:452; U.S. Pat. Nos. 4,816,567; 4,816,397; European Patent No. EP0171496; European Patent Application Publication No. EP0173494; and U.K. Patent No. GB2177096.

The term "binding domain" or "domain that binds (an antigen)" characterizes in connection with the present invention a domain which (specifically) binds to or interacts with a given target epitope or a given target side on the target molecules (antigens), e.g. CD19, serum albumin, and CD3, respectively. The structure and function of the first binding domain, the second binding domain, and/or the third binding domain can be based on the structure and/or function of an antibody, e.g. of a full-length or whole immunoglobulin molecule. A binding domain can be drawn from the VH and/or VL or VHH domain of an antibody or fragment thereof. For example, a binding domain can include three light chain CDRs (i.e., CDR1, CDR2 and CDR3 of the VL domain) and/or three heavy chain CDRs (i.e., CDR1, CDR2 and CDR3 of the VH domain). A binding domain can also include VHH CDRs (i.e., CDR1, CDR2 and CDR3 of the VHH region).

The terms "variable domain" and "variable region" are used interchangeably and refer to the portions of the antibody or immunoglobulin domains that exhibit variability in their sequence and that are involved in determining the specificity and binding affinity of a particular antibody. Variability is not evenly distributed throughout the variable domains of antibodies; it is concentrated in sub-domains of each of the heavy and light chain variable regions. These sub-domains are called "hypervariable regions" or "complementarity determining regions" (CDRs). The more conserved (i.e., non-hypervariable) portions of the variable domains are called the "framework" regions (FRM or FR) and provide a scaffold for the six CDRs in three dimensional space to form an antigen-binding surface.

In the present invention, any one of the binding domains of the multi-specific binding protein may comprise a single domain antibody (sdAb). A single domain antibody comprises a single, monomeric antibody variable domain which is able to bind selectively to a specific antigen, independently of other variable regions or domains. The first single domain antibodies were engineered from heavy chain antibodies found in camelids, and these are called VHH fragments. Cartilaginous fishes also have heavy chain antibodies (IgNAR) from which single domain antibodies called V_{NAR} fragments can be obtained. An alternative approach is to split the dimeric variable domains from common immunoglobulins e.g., from humans or rodents into monomers, hence obtaining VH or VL as a single domain antibody. Although most research into single domain antibodies is currently based on heavy chain variable domains, nanobodies derived from light chains have also been shown to bind

specifically to target epitopes. Examples of single domain antibodies include nanobodies and single variable domain antibodies.

As used herein, the term "antigen-binding site" refers to the part of an immunoglobulin molecule or a derivative or variant thereof that participates in antigen binding. In human antibodies, the antigen binding site is formed by amino acid residues of the N-terminal variable ("V") regions of the heavy ("H") and light ("L") chains. Three highly divergent stretches within the V regions of the heavy and light chains are referred to as "hypervariable regions" which are interposed between more conserved flanking stretches known as "framework regions," or "FR." Thus the term "FR" refers to amino acid sequences which are naturally found between and adjacent to hypervariable regions in immunoglobulins. In a human antibody molecule, the three hypervariable regions of a light chain and the three hypervariable regions of a heavy chain are disposed relative to each other in three dimensional space to form an antigen-binding surface. The antigen-binding surface is complementary to the three-dimensional surface of a bound antigen, and the three hypervariable regions of each of the heavy and light chains are referred to as "complementarity-determining regions," or "CDRs." In certain animals, such as camels and cartilaginous fish, the antigen-binding site is formed by a single antibody chain providing a "single domain antibody." Antigen-binding sites can exist in an intact antibody, in an antigen-binding fragment of an antibody that retains the antigen-binding surface, or in a recombinant polypeptide such as an scFv, using a peptide linker to connect the heavy chain variable domain to the light chain variable domain in a single polypeptide.

The terms "a" and "an" as used herein mean "one or more" and include the plural unless the context is inappropriate.

As used herein, the terms "subject" and "patient" refer to an organism to be treated by the methods and compositions described herein. Such organisms preferably include, but are not limited to, mammals (e.g., murines, simians, equines, bovines, porcines, canines, felines, and the like), and more preferably include humans.

As used herein, the term "effective amount" refers to the amount of a compound (e.g., a compound of the present invention) sufficient to effect beneficial or desired results. An effective amount can be administered in one or more administrations, applications or dosages and is not intended to be limited to a particular formulation or administration route. As used herein, the term "treating" includes any effect, e.g., lessening, reducing, modulating, ameliorating or eliminating, that results in the improvement of the condition, disease, disorder, and the like, or ameliorating a symptom thereof.

As used herein, the term "pharmaceutical composition" refers to the combination of an active agent with a carrier, inert or active, making the composition especially suitable for diagnostic or therapeutic use in vivo or ex vivo.

As used herein, the term "pharmaceutically acceptable carrier" refers to any of the standard pharmaceutical carriers, such as a phosphate buffered saline solution, water, emulsions (e.g., such as an oil/water or water/oil emulsions), and various types of wetting agents. The compositions also can include stabilizers and preservatives. For examples of carriers, stabilizers and adjuvants, see e.g., Martin, Remington's Pharmaceutical Sciences, 15th Ed., Mack Publ. Co., Easton, PA (1975).

Throughout the description, where compositions are described as having, including, or comprising specific components, or where processes and methods are described as

having, including, or comprising specific steps, it is contemplated that, additionally, there are compositions of the present invention that consist essentially of, or consist of, the recited components, and that there are processes and methods according to the present invention that consist essentially of, or consist of, the recited processing steps.

As a general matter, compositions specifying a percentage are by weight unless otherwise specified. Further, if a variable is not accompanied by a definition, then the previous definition of the variable controls.

I. Multi-Specific Binding Proteins

In one aspect, the present disclosure provides a multi-specific binding protein that comprises a first domain that binds CD19 (e.g., human CD19); a second domain that binds CD3 (e.g., human and/or Macaca CD3), such as CD3 ϵ (epsilon), CD3 δ (delta), and/or CD3 γ (gamma); and a third domain that binds serum albumin (e.g., HSA).

In certain embodiments, the first domain is a first antigen-binding site that binds CD19. In certain embodiments, the first antigen-binding site comprises an antibody heavy chain variable domain (VH). In certain embodiments, the first antigen-binding site comprises an antibody light chain variable domain (VL). In certain embodiments, the first antigen-binding site comprises a VH and a VL. In certain embodiments, the first antigen-binding site comprises an sdAb. In certain embodiments, the first antigen-binding site comprises an scFv.

In certain embodiments, the second domain is a second antigen-binding site that binds CD3. In certain embodiments, the second antigen-binding site comprises a VH. In certain embodiments, the second antigen-binding site comprises a VL. In certain embodiments, the second antigen-binding site comprises a VH and a VL. In certain embodiments, the second antigen-binding site comprises an sdAb. In certain embodiments, the second antigen-binding site comprises an scFv.

In certain embodiments, the third domain is a third antigen-binding site that binds serum albumin. In certain embodiments, the third antigen-binding site comprises a VH. In certain embodiments, the third antigen-binding site comprises a VL region. In certain embodiments, the third antigen-binding site comprises a VH region and a VL region. In certain embodiments, the third antigen-binding site comprises an sdAb. In certain embodiments, the third antigen-binding site comprises an scFv.

Alternatively, it is also contemplated that one or more of the binding domains may not comprise an antigen-binding site. For example, U.S. Patent Application Publication No. US20130316952A1 discloses a polypeptide that binds serum albumin having the amino acid sequence of LKEAKEKAIEELKKAGITSDYYFDLINKAKTVEGVNALKDEILKA (SEQ ID NO: 282). Additional exemplary polypeptides that bind HSA are described in Dennis et al. (2002) *J. Biol. Chem.*, 277:35035-43; Jacobs et al. (2015) *Protein Eng. Des. Sel.*, 28:385-93; and Zorzi et al. (2017) *Nat. Commun.*, 8: 16092.

In certain embodiments, the multi-specific binding protein further comprises an antibody Fc region. The presence of an Fc region may increase the serum half-life of the multi-specific binding protein. Depending on the specific Fc subtype and variant used, the Fc region may also alter the activity (e.g., cytotoxic activity) of the multi-specific binding protein.

In other embodiments, the multi-specific binding protein does not comprise an antibody Fc region. The absence of Fc contributes to a smaller size of the multi-specific binding protein, which can exhibit improved tissue penetration and

pharmacokinetic properties. In certain embodiments, the multi-specific binding proteins consists of or consists essentially of the first, second, and third antigen-binding sites and the linkers between them. In certain embodiments, the multi-specific binding proteins consists essentially of the first, second, and third antigen-binding sites.

In certain embodiments, the multi-specific binding protein binds CD19, CD3, and/or serum albumin monovalently. The exclusion of additional binding domains reduces the risk of non-specific immune cell activation and decreases the size of the multi-specific binding protein.

A. First Antigen-Binding Site

The first antigen-binding site of the multi-specific binding protein binds CD19 (e.g., human CD19).

CD19, also known as B-cell surface antigen B4 or Leu-12, is a transmembrane protein expressed on B lymphocytes and follicular dendritic cells. CD19 is a co-receptor for the B-cell antigen receptor complex on B lymphocytes (see, Carter et al. (2002) *Science*, 256:105-07; van Zelm et al. (2006) *N. Eng. J. Med.*, 354:1901-12). It associates with CD21, CD81, and Leu-13 and potentiates B cell receptor (BCR) signaling. Together with the BCR, CD19 modulates intrinsic and antigen receptor-induced signaling thresholds critical for clonal expansion of B cells and humoral immunity. Upon activation, the cytoplasmic tail of CD19 becomes phosphorylated, which leads to binding by Src-family kinases and recruitment of PI-3 kinase.

CD19 is a human B-cell surface marker that is expressed from early stages of pre-B cell development through terminal differentiation into plasma cells. It is also expressed on many non-Hodgkin lymphoma (NHL) cells and certain leukemias. Antibodies that bind CD19 have been developed and tested in clinical studies against cancers of lymphoid origin such as B-cell malignancies (see, e.g., Hekman et al. (1991) *Cancer Immunol. Immunother.*, 32:364-72; Vlasfeld et al. (1995) *Cancer Immunol. Immunother.*, 40:37-47; Corny et al. (1995) *J. Immunother. Emphasis Tumor Immunol.*, 18:231-41; and Manzke et al. (2001) *Int. J. Cancer*, 91:516-22). Furthermore, a BiTE® construct called blinatumomab has been developed for clinical use.

The first antigen-binding site that binds CD19 can be derived from, for example, MT-103 (a single-chain bispecific CD19/CD3 antibody; see, Hoffman et al. (2005) *Int. J. Cancer*, 115:98-104; Schlereth et al. (2006) *Cancer Immunol. Immunother.* 55:503-14), a CD19/CD16 diabody (see, Schlenzka et al. (2004) *Anti-cancer Drugs* 15:915-19; Kipriyanov et al. (2002) *J. Immunol.* 169:137-44), BU12-saporin (see, Flavell et al. (1995) *Br. J. Cancer* 72:1373-79), and anti-CD19-idarubicin (see, Rowland et al. (1993) *Cancer Immunol. Immunother.* 55:503-14). Additional exemplary antigen-binding sites that bind CD19, from which the instant first antigen-binding site may be derived, are disclosed in U.S. Patent Application Publication Nos. US20170174786A1, US20090042291A1, US20160046730A1, US20070154473A1, US20090142349A1, US20180142018A1, US20090136526A1, US20060257398A1, and US20180230225A1, and PCT Publication No. WO2019057100A1.

A first antigen-binding site that binds CD19 can include a VH comprising three complementarity regions (HCDR1, HCDR2, and HCDR3) and/or a VL comprising three complementarity regions (LCDR1, LCDR2, and LCDR3). Table 1 summarizes, for each variable region, the CDRs of the variable region and scFv constructs based on the given heavy and light chain variable regions. The first antigen-binding site can be derived from the exemplary variable domain and CDR sequences as listed in Table 1.

TABLE 1

Sequences of Exemplary First Antigen-Binding Sites		
Antigen-Binding Site	VH and HCDRs	VL and LCDRs
Roche mAb	QVQLVQSGAEVKKPGASVKVSCKASGY	DIVMTQTPLSLSVTPGQPASISCKSS
	TFTDYIMHWVRQAPGQGLEWMGYINPY	QSLETSTGTTYLNWYLQKPGQSPQ
	NDGSKYTEKFQGRVTMTSDTSISTAYME	LLIYRVSKRFSVPDRFSGSGSGTDF
	LSRLRSDDTAVYYCARGTYYYGPQLFD	TLKISRVEAEDVGYYVCLQLLEDPY
	YWGGTTVTVSS (SEQ ID NO: 1)	TFGGGTKLEIK (SEQ ID NO: 2)
	HCDR1: DYIMH (SEQ ID NO: 3)	LCDR1: KSSQSLETSTGTTYLN (SEQ ID NO: 6)
	HCDR2: YINPYNDGSKYTEKFOG (SEQ ID NO: 4)	LCDR2: RVSKRFS (SEQ ID NO: 7)
	HCDR3: GTYYYGPQLEDY (SEQ ID NO: 5)	LCDR3: LQLLEDPTYT (SEQ ID NO: 8)
	scFv:	
	QVQLVQSGAEVKKPGASVKVSCKASGYTFTDYIMHWVRQAPGQGLEWMGYIN PYNDGSKYTEKFQGRVTMTSDTSISTAYMELSRRLRSDDTAVYYCARGTYYYGPQ LPDYWGQGTTVTVSSGGGGSGGGSGGGSDIVMTQTPLSLSVTPGQPASISCKS QSLETSTGTTYLNWYLQKPGQSPQLLIYRVSKRFSVPDRFSGSGSGTDFTLKIS RVEAEDVGYYVCLQLLEDPTYTFGGGTKLEIK (SEQ ID NO: 9)	
SG mAb	QVQLQESGPGLVKPSQTLSTCTVSGGSI	EIVLTQSPATLSLSPGERATLSCSAS
	STSGMGVGVIRQHPGKLEWIGHIWD	SSVSYMHWYQQKPGQAPRLLIYDT
	DDKRYNPALKSRVTISVDTSKNQFSLKL	SKLASGIPARFSGSGSGTDFTLTISSL
	SSVTAADTAVYYCARMELWSYFYDYW	EPEDFAVYYCFQGSVYPFTFGQGTK
	GQGTLLTVSS (SEQ ID NO: 10)	LEIKR (SEQ ID NO: 11)
	HCDR1: TSGMGVG (SEQ ID NO: 12)	LCDR1: SASSSVSYMH (SEQ ID NO: 15)
	HCDR2: HIWDDDDKRYNPALKS (SEQ ID NO: 13)	LCDR2: DTSKLAS (SEQ ID NO: 16)
	HCDR3: MELWSYFYDY (SEQ ID NO: 14)	LCDR3: FQGSVYPFT (SEQ ID NO: 17)
	scFv:	
	QVQLQESGPGLVKPSQTLSTCTVSGGSI STSGMGVGVIRQHPGKLEWIGHIWD DDDKRYNPALKSRVTISVDTSKNQFSLKLSSVTAADTAVYYCARMELWSYFY DYWGQGTLLTVSSGGGGSGGGSGGGSEIVLTQSPATLSLSPGERATLSCSASS SVSYMHWYQQKPGQAPRLLIYDTSKLASGIPARFSGSGSGTDFTLTISSLEPEDFA VYYCFQGSVYPFTFGQGTKLEIKR (SEQ ID NO: 18)	
Xencor mAb	EVQLVESGGGLVKGPGSLKLSCAASGYT	DIVMTQSPATLSLSPGERATLSCRSS
	FTSYVMHWVRQAPGKLEWIGYINPYN	KSLQNVNGNTYLYWFQKPGQSPQ
	DGTYNEKQGRVTISDDKSISTAYMELS	LLIYRMSNLNSGVPDRFSGSGSGTE
	SLRSEDYAMYYCARGTYYYGTRVFDYW	FTLTISSELEPEDFAVYYCMQHLEYPI
	GQGTLLTVSS (SEQ ID NO: 19)	TFGAGTKLEIK (SEQ ID NO: 20)
	HCDR1: SYVMH (SEQ ID NO: 21)	LCDR1: RSSKSLQNVNGNTYLY (SEQ ID NO: 24)
	HCDR2: WIGYINPYNDGTKY (SEQ ID NO: 22)	LCDR2: RMSNLNS (SEQ ID NO: 25)
	HCDR3: GTYYYGTRVEDY (SEQ ID NO: 23)	LCDR3: MQHLEYPI (SEQ ID NO: 26)
	scFv:	
	EVQLVESGGGLVKGPGSLKLSCAASGYT FTSYVMHWVRQAPGKLEWIGYINPYN DGTYNEKQGRVTISDDKSISTAYMELS SLRSEDYAMYYCARGTYYYGTRVFDYW GQGTLLTVSS (SEQ ID NO: 19) HCDR1: SYVMH (SEQ ID NO: 21) HCDR2: WIGYINPYNDGTKY (SEQ ID NO: 22) HCDR3: GTYYYGTRVEDY (SEQ ID NO: 23)	DIVMTQSPATLSLSPGERATLSCRSS KSLQNVNGNTYLYWFQKPGQSPQ LLIYRMSNLNSGVPDRFSGSGSGTE FTLTISSELEPEDFAVYYCMQHLEYPI TFGAGTKLEIK (SEQ ID NO: 20) LCDR1: RSSKSLQNVNGNTYLY (SEQ ID NO: 24) LCDR2: RMSNLNS (SEQ ID NO: 25) LCDR3: MQHLEYPI (SEQ ID NO: 26)
Abbvie mAb	QVQLQQSGAELVRPGSSVKISCKASGYA	DILTQTSPASLAVSLGQRATISCKAS
	FSSYWMNWVKQRPGQGLEWIGQIWP	QSVVDYDGD SYLNWYQQIPGQPKL
	DGDTNYNGKFKGKATLTADESSSTAYM	LIYDASNLVSGIPPRFSGSGSGTDF
	QLSSLASSEDYAVYFCARRETTTVGRYYY	LNHHPVEKVDAAATYHCQSTEDPW
	AMDYWGQGTSTVTVSS (SEQ ID NO: 27)	TFGGGTKLEIK (SEQ ID NO: 28)
	HCDR1: SYWMN (SEQ ID NO: 29)	LCDR1: KASQSVVDYDGD SYLN (SEQ ID NO: 32)
	HCDR2: QIWPGDGDTNYNGKFKG (SEQ ID NO: 30)	LCDR2: DASNLVS (SEQ ID NO: 33)
	HCDR3: RETTTVGRYYYAMDY (SEQ ID NO: 31)	LCDR3: QQSTEDPWT (SEQ ID NO: 34)
	scFv:	
	QVQLQQSGAELVRPGSSVKISCKASGYA FSSYWMNWVKQRPGQGLEWIGQIWP DGDTNYNGKFKGKATLTADESSSTAYM QLSSLASSEDYAVYFCARRETTTVGRYYY AMDYWGQGTSTVTVSS (SEQ ID NO: 27) HCDR1: SYWMN (SEQ ID NO: 29) HCDR2: QIWPGDGDTNYNGKFKG (SEQ ID NO: 30) HCDR3: RETTTVGRYYYAMDY (SEQ ID NO: 31)	DILTQTSPASLAVSLGQRATISCKAS QSVVDYDGD SYLNWYQQIPGQPKL LIYDASNLVSGIPPRFSGSGSGTDF LNHHPVEKVDAAATYHCQSTEDPW TFGGGTKLEIK (SEQ ID NO: 28) LCDR1: KASQSVVDYDGD SYLN (SEQ ID NO: 32) LCDR2: DASNLVS (SEQ ID NO: 33) LCDR3: QQSTEDPWT (SEQ ID NO: 34)
Immuno medics mAb	QVQLQQSGAEVKKPGSSVKVSCKASGY	DIQLTQSPSSLSASVGDRTITCKAS
	AFSSYWMNWVRQRPQGLEWIGQIWP	QSVVDYDGD SYLNWYQQIPGKAPKL
	DGDTNYNGKFKGRATITADESTNTAYM	LIYDASNLVSGIPPRFSGSGSGTDT
	ELSSLRSEDYAFYSCARRETTTVGRYYY	FTISSLPEDIATYHCQSTEDPWT
	AMDYWGQGTSTVTVSS (SEQ ID NO: 35)	GGGKTLQIKR (SEQ ID NO: 36)
	HCDR1: SYWMN (SEQ ID NO: 29)	LCDR1: KASQSVVDYDGD SYLN (SEQ ID NO: 32)
	HCDR2: QIWPGDGDTNYNGKFKG (SEQ ID NO: 30)	LCDR2: DASNLVS (SEQ ID NO: 33)
	HCDR3: RETTTVGRYYYAMDY (SEQ ID NO: 31)	LCDR3: QQSTEDPWT (SEQ ID NO: 34)
	scFv:	
	QVQLQQSGAEVKKPGSSVKVSCKASGY AFSSYWMNWVRQRPQGLEWIGQIWP DGDTNYNGKFKGRATITADESTNTAYM ELSSLRSEDYAFYSCARRETTTVGRYYY AMDYWGQGTSTVTVSS (SEQ ID NO: 35) HCDR1: SYWMN (SEQ ID NO: 29) HCDR2: QIWPGDGDTNYNGKFKG (SEQ ID NO: 30) HCDR3: RETTTVGRYYYAMDY (SEQ ID NO: 31)	DIQLTQSPSSLSASVGDRTITCKAS QSVVDYDGD SYLNWYQQIPGKAPKL LIYDASNLVSGIPPRFSGSGSGTDT FTISSLPEDIATYHCQSTEDPWT GGGKTLQIKR (SEQ ID NO: 36) LCDR1: KASQSVVDYDGD SYLN (SEQ ID NO: 32) LCDR2: DASNLVS (SEQ ID NO: 33) LCDR3: QQSTEDPWT (SEQ ID NO: 34)
Merck mAb	QVQLQPGAEVKKPGASVKVSCKTSGY	QIVLTQSPATLSASPGEKATMTCSA
	TFTSNWMHWVKTPGKLEWIGEIDPS	SSGVNMYHWYQQKPGTSPKRWIY
	DSYTNYNQKFDGKAKLTVDKSSSTAYM	DDTKTASGVPARFSGSGSGTSYSLT
	EVSDLTAEDSATYYCARGSNPYYYAMD	ISSMEAEADAATYCHQRGSYTFGG
	YWGGTSTVTVSS (SEQ ID NO: 37)	GTKLEIK (SEQ ID NO: 38)
	HCDR1: SNWMH (SEQ ID NO: 39)	LCDR1: SASSGVNMYH (SEQ ID NO: 42)
	HCDR2: EIDPSDSYTN (SEQ ID NO: 40)	
	scFv:	
	QVQLQPGAEVKKPGASVKVSCKTSGY TFTSNWMHWVKTPGKLEWIGEIDPS DSYTNYNQKFDGKAKLTVDKSSSTAYM EVSDLTAEDSATYYCARGSNPYYYAMD YWGGTSTVTVSS (SEQ ID NO: 37) HCDR1: SNWMH (SEQ ID NO: 39) HCDR2: EIDPSDSYTN (SEQ ID NO: 40)	QIVLTQSPATLSASPGEKATMTCSA SSGVNMYHWYQQKPGTSPKRWIY DDTKTASGVPARFSGSGSGTSYSLT ISSMEAEADAATYCHQRGSYTFGG GTKLEIK (SEQ ID NO: 38) LCDR1: SASSGVNMYH (SEQ ID NO: 42)

TABLE 1-continued

Sequences of Exemplary First Antigen-Binding Sites		
Antigen-Binding Site	VH and HCDRs	VL and LCDRs
	HCDR3: GSNPYYYAMDY (SEQ ID NO: 41)	LCDR2: DTDKTAS (SEQ ID NO: 43) LCDR3: HQRGSYT (SEQ ID NO: 44)
Medarex mAb 21D4a	EVQLVQSGAEVKKPGESLKISCKGSGYS FSSSWIGWVRQMPGKGLEWMGIIPDDSD DTRYSPSPFQGGVITISADKSIRTAYLQWSS LKASDTAMYICARHVTMIWGVIIIDFWG QGTLLTVSS (SEQ ID NO: 45) HCDR1: SSWIG (SEQ ID NO: 47) HCDR2: IIYPDDSDTRYSPSPFQG (SEQ ID NO: 48) HCDR3: HVTMIWGVIIIDF (SEQ ID NO: 49)	AIQLTQSPSSLSASVGDRTITCRAS QGISSALAWYQQKPGKAPKLLIYD ASSLESQVPSRFRSGSGSGTDFTLTIS LQPEDFATYYCQQFNSTPFTFGPGT KVDIK (SEQ ID NO: 46) LCDR1: RASQGISSALA (SEQ ID NO: 50) LCDR2: DASSLES (SEQ ID NO: 51) LCDR3: QQFNSTPFT (SEQ ID NO: 52)
Medarex mAb 21D4	EVQLVQSGAEVKKPGESLKISCKGSGYS FSSSWIGWVRQMPGKGLEWMGIIPDDSD DTRYSPSPFQGGVITISADKSIRTAYLQWSS LKASDTAMYICARHVTMIWGVIIIDFWG QGTLLTVSS (SEQ ID NO: 45) HCDR1: SSWIG (SEQ ID NO: 47) HCDR2: IIYPDDSDTRYSPSPFQG (SEQ ID NO: 48) HCDR3: HVTMIWGVIIIDF (SEQ ID NO: 49)	AIQLTQSPSSLSASVGDRTITCRAS QGISSALAWYQQKPGKAPKLLIYD ASSLESQVPSRFRSGSGSGTDFTLTIS LQPEDFATYYCQQFNSTPFTFGQG TKLEIK (SEQ ID NO: 53) LCDR1: RASQGISSALA (SEQ ID NO: 50) LCDR2: DASSLES (SEQ ID NO: 51) LCDR3: QQFNSTPYT (SEQ ID NO: 54)
Medarex mAb 47G4	QVQLVQSGAEVKKPGSSVKVCKDSGG TFSSY AISWVRQAPGGGLEWMGGIIPFG TTNYAQPFQGRVTITADESTSTAYMELS SLRSEDATVYYCAREAAADWLDPWG QGTLLTVSS (SEQ ID NO: 55) HCDR1: SYAIS (SEQ ID NO: 57) HCDR2: GIIPFGTTNYAQPFQG (SEQ ID NO: 58) HCDR3: EAAADWLDP (SEQ ID NO: 59)	EIVLTQSPGTLSPGERATLSCRAS QSVSSSYLAWYQQKPGQAPRLIY GASSRATGIPDRFSGSGSGTDFTLTIT SRLEPEDFAVYYCQQYGSSRFTFGP GTKVDIK (SEQ ID NO: 56) LCDR1: RASQVSSSYLA (SEQ ID NO: 60) LCDR2: GASSRAT (SEQ ID NO: 61) LCDR3: QQYGSSRFT (SEQ ID NO: 62)
Medarex mAb 27F3	EVQLVQSGAEVKKPGESLKISCKGSGYS FTSYWIAWVRQMPGKGLEWMGIIPGD SDTRYSPSPFQGGVITISADKSISTAYLQWS SLKASDTAMYICARQYSSGWDSSYG MGVWGGGTTVTSS (SEQ ID NO: 63) HCDR1: SYWIA (SEQ ID NO: 65) HCDR2: IIYPGDSDTRYSPSPFQG (SEQ ID NO: 66) HCDR3: QGYSSGWDSSYGMGV (SEQ ID NO: 67)	AIQLTQSPSSLSASVGDRTITCRAS QGISSALAWYQQKPGKAPKLLIYD ASSLESQVPSRFRSGSGSGTDFTLTIS LQPEDFATYYCQQFNSTPFTFGQG TKLEIK (SEQ ID NO: 64) LCDR1: RASQGISSALA (SEQ ID NO: 50) LCDR2: DASSLES (SEQ ID NO: 51) LCDR3: QQFNSTPYT (SEQ ID NO: 54)
Medarex mAb 3C10	QVQLVQSGAEVKKPGSSVKVCKASGG TFSSYTINWVRQAPGGGLEWMGGIIPFG IPNYAQKPFQGRVTITADESTNTAYMELS SLRAEDTAVYYCARAGSGGADYSYGM VWGGTAVTVSS (SEQ ID NO: 68) HCDR1: SYTIN (SEQ ID NO: 70) HCDR2: GIIPFGIPNYAQKPFQG (SEQ ID NO: 71) HCDR3: ASGGGADYSYGMV (SEQ ID NO: 72)	DIQMTQSPSSLSASVGDRTITCRAS QGISSWLAHYQQKPEKAPKSLIYA ASSLQSGVPSRFRSGSGSGTDFTLTIS SLQPEDFATYYCQQYKRYPTFTFGQ GTKLEIK (SEQ ID NO: 69) LCDR1: RASQGISSWLA (SEQ ID NO: 73) LCDR2: AASSLQS (SEQ ID NO: 74) LCDR3: QQYKRYPT (SEQ ID NO: 75)
Medarex mAb 5G7	EVQLVQSGAEVKKPGESLNISCKGSGYS FTSYWIGWVRQMPGKGLEWMGIIPGD SDTRYSPSPFQGGVITISADKSINTAYLQWS SLKASDTAMYICARGVSMIWGVIMDV WGQGTITVTSS (SEQ ID NO: 76) HCDR1: SYWIG (SEQ ID NO: 78) HCDR2: IIYPGDSDTRYSPSPFQG (SEQ ID NO: 66) HCDR3: GVSMIWGVIMDV (SEQ ID NO: 79)	AIQLTQSPSSLSASVGDRTITCRAS QGISSALAWYQQKPGKAPKLLIYD ASSLESQVPSRFRSGSGSGTDFTLTIS LQPEDFATYYCQQFNSTPFTFGQG TKVEIK (SEQ ID NO: 77) LCDR1: RASQGISSALA (SEQ ID NO: 50) LCDR2: DASSLES (SEQ ID NO: 51) LCDR3: QQFNSTPWT (SEQ ID NO: 80)
Medarex mAb 13F1	EVQLVQSGAEVKKPGESLKISCKGSGYT FTNYWIAWVRQMPGKGLEWMGIIPGD SDTRYSPSPFQGGVITISADKSISTAYLQWS GLKASDTAMYICARQYSSGWRSSYG MGVWGGGTTVTSS (SEQ ID NO: 81)	AIQLTQSPSSLSASVGDRTITCRAS QGISSALAWYQQKPGKAPKLLIYD ASSLESQVPSRFRSGSGSGTDFTLTIS LQPEDFATYYCQQFNSTPFTFGQG TKLEIK (SEQ ID NO: 82)

TABLE 1-continued

Sequences of Exemplary First Antigen-Binding Sites		
Antigen-Binding Site	VH and HCDRs	VL and LCDRs
	HCDR1: NYWIA (SEQ ID NO: 83) HCDR2: IIYPGDS DTRYSPSPFQG (SEQ ID NO: 66) HCDR3: QGYSSGWRSYYG (SEQ ID NO: 84)	LCDR1: RASQGISSALA (SEQ ID NO: 50) LCDR2: DASSLES (SEQ ID NO: 51) LCDR3: QQFNYPHT (SEQ ID NO: 85)
Medarex mAb 46E8	EVQLVQSGAEVKKPGESLKISCKGSGYT FTNYWIAWVRQMPGKGLEWMGIIYPGD SDTRYSPSPFQGVTTISADKSISTAYLQWS GLKASDTAMYYCARQGYSSGWRSYYG MGVWGQGTIVTVSS (SEQ ID NO: 314) HCDR1: NYWIA (SEQ ID NO: 83) HCDR2: IIYPGDS DTRYSPSPFQG (SEQ ID NO: 66) HCDR3: QGYSSGWRSYYG (SEQ ID NO: 84)	AIQLTQSPSSLSASVGDRTITCRAS QGISSALAWYQQKPGKAPKLLIYD ASSLESQVPSRFSGSGSGTDFLTITSS LQPEDFATYYCQQFNYPHTFGQG TKLEIK (SEQ ID NO: 315) LCDR1: RASQGISSALA (SEQ ID NO: 50) LCDR2: DASSLES (SEQ ID NO: 51) LCDR3: QQFNYPHT (SEQ ID NO: 85)
Novimmune mAb	EVQLVQSGAEVKKPGESLKISCKGSGYS FTSYWIGWVRQMPGKGLEWMGIIYPGD SDTRYSPSPFQGVTTISADKSISTAYLQWS SLKASDTAMYYCARGVSGIYNLHGFDI WGQGTIVTVSS (SEQ ID NO: 86) HCDR1: GYSFTSYW (SEQ ID NO: 88) HCDR2: IYPGDS D (SEQ ID NO: 89) HCDR3: ARGVSGIYNLHGFDI (SEQ ID NO: 90)	DIQMTQSPSSLSASVGDRTITCRAS QSISSYLNWYQQKPGKAPKLLIYAA SSLQSGVPSRFSGSGSGTDFLTITSS QPEDFATYYCQGRFGSPFTFGQGT KVEIK (SEQ ID NO: 87) LCDR1: QSISSY (SEQ ID NO: 91) LCDR2: AAS (SEQ ID NO: 92) LCDR3: QQGRFGSPFT (SEQ ID NO: 93)
Eureka mAb-1	QVQLVETGGGLVQPGGSLRLSCAASGFT FSSYAMSWVRQAPGKGLEWVSAISGSG GSTYYADSVKGRFTISRDNKNTLYLQM NSLRAEDTAVYYCARYYSLDYWGQ GTLVTVSS (SEQ ID NO: 94) scFv: QTVVVTQEPSVSAAPGQKVTISCSGSSSNIGNNYVSWYQQLPGTAPKLLIYDNNKR PSGIPDREGSGSGTSATLGITGLQTGDEADYYCGTWDSLSAGVFGTGKTLTVL GSRGGGGGGGGGGGGGSLMAQVQLVETGGGLVQPGGSLRLSCAASGFTFSSY AMSWVRQAPGKGLEWVSAISGSGGSTYYADSVKGRFTISRDNKNTLYLQMNS LRAEDTAVYYCARYYSLDYWGQGTIVTVSS (SEQ ID NO: 96)	QTVVVTQEPSVSAAPGQKVTISCSGS SSNIGNNYVSWYQQLPGTAPKLLIY DNNKRPSGIPDRFSGSGSGTSATLGI TGLQTGDEADYYCGTWDSLSAGV FGTGKTLTVLGS (SEQ ID NO: 95)
Eureka mAb-2	QVQLVESGGGLVQPGGSLRLSCAASGFT FSSYAMSWVRQAPGKGLEWVSGISASG GSTYYADSVKGRFTISRDNKNTLYLQM NSLRAEDTAVYYCARYYSLQIDSWGQG TLVTVSS (SEQ ID NO: 210) scFv: QSVLTQPPSVSAAPGQKVTISCSGSSSNIGNNYVSWYRQLPGTAPKLLIYENNNKR SGIPDRFSGSGSGTSATLGITGLQTGDEADYYCGTWDSLSRAGVFGTGKTVLVG SRGGGGGGGGGGGGGSLMAQVQLVESGGGLVQPGGSLRLSCAASGFTFSSY MSWVRQAPGKGLEWVSGISASGGSTYYADSVKGRFTISRDNKNTLYLQMNSL RAEDTAVYYCARYYSLQIDSWGQGTIVTVSS (SEQ ID NO: 212)	QSVLTQPPSVSAAPGQKVTISCSGS SNIGNNYVSWYRQLPGTAPKLLIY NNKRPSGIPDRFSGSGSGTSATLGI TGLQTGDEADYYCGTWDSLSRAGV GTGKTVLV (SEQ ID NO: 211)
Eureka mAb-3	EVQLVQSGAEVKKPGATVKISCKVSGYT FTDYYMHVWVQAPGKGLEWMGLVDPE DGETIYAEKFGQGRVTITADTSTDATYME LSSLRSED TAVYYCATGIYSRPLGYWGQ GTLVTVSS (SEQ ID NO: 213) scFv: QSVLTQPPSVSAAPGQKVTISCSGSSSNIGNNTVNWYQQLPGTAPKLLIYNNQRP SGVPDRFSGSGSGTSASLAISGLQSEDEADYYCAWDDSLNGHVFGGTGKTVLV LGSRRGGGGGGGGGGGSLMAEVQLVQSGAEVKKPGATVKISCKVSGYTFT DYYMHVWVQAPGKGLEWMGLVDPEGETIYAEKFGQGRVTITADTSTDATYME LSSLRSED TAVYYCATGIYSRPLGYWGQGTIVTVSS (SEQ ID NO: 215)	QSVLTQPPSVSAAPGQKVTISCSGS SNIGNNTVNWYQQLPGTAPKLLIY NNKRPSGVPDRFSGSGSGTSASLAIS GLQSEDEADYYCAWDDSLNGHV VFGGTGKTVLV (SEQ ID NO: 214)
Eureka mAb-4	EVQLVETGGGLVQPGGSLRLSCAASGFT FSSYAMSWVRQAPGKGLEWVSAISGSG GSTYYADSVKGRFTISRDNKNTLYLQM NSLRAEDTAVYYCARSDGKHFWQQYD AWGQGTIVTVSS (SEQ ID NO: 216) scFv: SYVLTQPPSVSAAPGQKVTISCSGSSSNIGNHTVNWYQQLPGTAPKLLIYNNQRP SGVPDRFSGSGSGTSASLAISGLQSEDEADYYCAWDDSLNGYVFGGTGKTVLV LGSRRGGGGGGGGGGGSLMAEVQLVETGGGLVQPGGSLRLSCAASGFTFSSY AMSWVRQAPGKGLEWVSAISGSGGSTYYADSVKGRFTISRDNKNTLYLQMNS LRAEDTAVYYCARSDGKHFWQQYDAGWQGTIVTVSS (SEQ ID NO: 218)	SYVLTQPPSVSAAPGQKVTISCSGS SNIGNHTVNWYQQLPGTAPKLLIY NNKRPSGVPDRFSGSGSGTSASLAIS GLQSEDEADYYCAWDDSLNGYV FGTGKTVLV (SEQ ID NO: 217)

TABLE 1-continued

Sequences of Exemplary First Antigen-Binding Sites		
Antigen-Binding Site	VH and HCDRs	VL and LCDRs
Eureka mAb-5	EVQLVESGGGLVQPGGSLRLSCAASGFT VSSNYMSWVRQAPGKGLEWVSAISGSG GSTYYADSVKGRFTISRDN SKNTLYLQM NSLRAEDTAVYYCARMNIDYWGGQTLV TVSS (SEQ ID NO: 219) scFv: DIQLTQSPSSL SAYVGDRVITITCRASQGITNSLAWYQQKPGKAPKLLHAAASRL SGVPSRFGSGSGGTDFLTITISSLPEDFAVYYCQHYLGTPYSFGQGTKEIKRSRG GGGSGGGSGGGGSLMAEVLVESGGGLVQPGGSLRLSCAASGFTVSSNYMS WVRQAPGKGLEWVSAISGSGGSTYYADSVKGRFTISRDN SKNTLYLQMNLSLRAE DTAVYYCARMNIDYWGGQTLVTVSS (SEQ ID NO: 221)	DIQLTQSPSSL SAYVGDRVITITCRAS QGITNSLAWYQQKPGKAPKLLHAA ASRLSGVPSRFGSGSGGTDFLTITIS SLQPEDFAVYYCQHYLGTPYSFGQ GTKVEIK (SEQ ID NO: 220)
Eureka mAb-6	EVQLVQSGAEVKRPGESLTISCKGSEYSF ASYWITWVRQMPGKGLEWMGRIDPDS YTNYSFQGHVTISADKSISTAYLQWSS LKASDTAIYYCARPFQYDYGYSDAFDI WGQGTMTVTVSS (SEQ ID NO: 222) scFv: EIVLTQSPSSL SAYVGDRVITISCRASQSVSRFLNMYQQKPGKAPKLLIYGVSTLER GVPSRFGSGSGGTDFLTITISSLPEDFAVYYCQESYIIPITFGGGTKLEIKRSRGGG GSGGGSGGGGSLMAEVLVQSGAEVKRPGESLTISCKGSEYSFASYWITWVR QMPGKGLEWMGRIDPDSYTNYSFQGHVTISADKSISTAYLQWSSSLKASDTAI YYCARPFQYDYGYSDAFDIWGQGTMTVTVSS (SEQ ID NO: 224)	EIVLTQSPSSL SAYVGDRVITISCRAS QSVSRFLNMYQQKPGKAPKLLIY VSTLERGVPSRFGSGSGGTDFLTITIS SLQPEDFAVYYCQESYIIPITFGGGT KLEIK (SEQ ID NO: 223)
Eureka mAb-7	QMQLVQSGAEVKKAGSSVKVSCETSGG TFSSSSVNWRQAPGQGLEWMGGIPIV GTPNYAQKQDRVTITTA VESTFTA YMEL SGLRSED TAVYYCARGGYRDMVWVG RGTTVTVSS (SEQ ID NO: 225) scFv: EIVMTQSPSLSLVTPGEPASISCRSSQSLDLSNGFNSLDWYLQKPGQSPQLLIHLGS DRASGVDPDRSGSGGTDFTLKISRVEAEDVGIYYCMQSLQIPTFGQGTKEIKRS RGSGGGSGGGGSLMAEQMQLVQSGAEVKKAGSSVKVSCETSGGTFSSSS VNWRQAPGQGLEWMGGIPIVGTNYAQKQDRVTITAVESTFTAYMELSGLR SEDTAVYYCARGGYRDMVWGRGTTTVTVSS (SEQ ID NO: 227)	EIVMTQSPSLSLVTPGEPASISCRSS QSLDLSNGFNSLDWYLQKPGQSPQ LLIHLGSDRASGVDPDRFGSGSGGT FTLKISRVEAEDVGIYYCMQSLQIPT FGQGTKEIK (SEQ ID NO: 226)
Eureka mAb-8	EVQLVESGGGLIQPGGSLRLSCAASGFT VSSNYMSWVRQAPGKGLEWVSVIYSGG STYYADSVKGRFTISRDN SKNTLYLQMN SLRAEDTAVYYCARGGFGAEFDYWGGQ TLVTVSS (SEQ ID NO: 228) scFv: SYELTQPPSASGTPGQRTVITSCSGSSNIGSNYVYQQLPGTAPKLLIYRNNQRP SGVPDRFSGSKSGTSASLAISGLRSEDEADYYCAWDDSLSGYVFGTGTKVTVL GSRGGGGSGGGGSLMAEVLVESGGGLIQPGGSLRLSCAASGFTVSSN YMSWVRQAPGKGLEWVSVIYSGGSTYYADSVKGRFTISRDN SKNTLYLQMNLSL RAEDTAVYYCARGGFGAEFDYWGGQTLVTVSS (SEQ ID NO: 230)	SYELTQPPSASGTPGQRTVITSCSGSS SNIGSNYVYQQLPGTAPKLLIYR NNQRP SGVPDRFSGSKSGTSASLAIS GLRSEDEADYYCAWDDSLSGYVF GTGKVTVL (SEQ ID NO: 229)
Eureka mAb-9	EVQLVESGGGLIQPGGSLRLSCAASGFT VSSNYMSWVRQAPGKGLEWVSVIYSGG STYYADSVKGRFTISRDN SKNTLYLQMN SLRAEDTAVYYCARGGISDDYGGSGSY DNWGGTGLVTVSS (SEQ ID NO: 231) scFv: SYVLTQPPSVSVSPGQTASITCSGDKLGDKYASWYQQKPGQSPVLVIYQDNKRPS GIPERFSGSGSGNTATLTISGTQAMDEADYYCQAWDDSTEDVFGPTKVTVLGSR GGGGGSGGGGSLMAEVLVESGGGLIQPGGSLRLSCAASGFTVSSNYMS WVRQAPGKGLEWVSVIYSGGSTYYADSVKGRFTISRDN SKNTLYLQMNLSLRAE DTAVYYCARGGISDDYGGSGSYDNWGGTGLVTVSS (SEQ ID NO: 233)	SYVLTQPPSVSVSPGQTASITCSGDK LGDKYASWYQQKPGQSPVLVIYQD NKRPSGIPERFSGSGSGNTATLTISG TQAMDEADYYCQAWDDSTEDVFG PGTKVTVL (SEQ ID NO: 232)
Eureka mAb-10	EVQLVESGGGLVQPGGSLRLSCAASGFT VSSNYMSWVRQAPGKGLEWVSVIYSGG STYYADSVKGRFTISRDN SKNTLYLQMN SLRAEDTAVYYCARERGMGYAFDIWQ GTMTVTVSS (SEQ ID NO: 234) scFv: DIQLTQSPSSL SAYVGDRVITITCRASQSISSYLNWYQQKPGKAPKLLIYAASSLQS GVPSRFGSGSGGTDFLTITISSLPEDFAVYYCQSYSTPFTFGGGTKVEIKRSRG GSGGGGGSGGGGSLMAEVLVESGGGLVQPGGSLRLSCAASGFTVSSNYMSW VRQAPGKGLEWVSVIYSGGSTYYADSVKGRFTISRDN SKNTLYLQMNLSLRAEDT AVYYCARERGMGYAFDIWGGTMTVTVSS (SEQ ID NO: 236)	DIQLTQSPSSL SAYVGDRVITITCRAS QSISSYLNWYQQKPGKAPKLLIYAA SSLQSGVPSRFGSGSGGTDFLTITISSL QPEDFAVYYCQSYSTPFTFGGGTK VEIK (SEQ ID NO: 235)

TABLE 1-continued

Sequences of Exemplary First Antigen-Binding Sites		
Antigen-Binding Site	VH and HCDRs	VL and LCDRs
Eureka mAb-11	QLLQLQESGPGLVKPSSETLSLTCSVSGVS MSENYWSWIRQPPGKRLEWIGCAHYTG DTHYNPSLKGRVTISLDTSMNQFSLRLN SVTAADTAVVYCASYHPFNYWGQGTIV TVSS (SEQ ID NO: 237) scFv: DIQMTQSPSSLSASVGDRTITCRASQGIGSYLA WYQQKPKGKAPKLLIYPASTLOS GVPSRFSGSGSGTEFTLTISLQPEDFATYYCQQLNSLFGQGRLEIKRSRGGGGS GGGSGGGGSLMAQLQLQESGPGLVKPSSETLSLTCSVSGVSMSENYWSWIRQP PGKRLIEWIGCAHYTG DTHYNPSLKGRVTISLDTSMNQFSLRLNSVTAADTAVVY CASYHPFNYWGQGTIVTVSS (SEQ ID NO: 239)	DIQMTQSPSSLSASVGDRTITCRAS QGIGSYLAWYQQKPKGKAPKLLIYP ASTLQSGVPSRFSGSGSGTEFTLTIS SLQPEDFATYYCQQLNSLFGQGRLE EIK (SEQ ID NO: 238)
Eureka mAb-12	EVQLVQSGAEVRRPGATVKISCKVSGYT FNDIFYLHWVRQAPGKGLEWMGRIDPED GKTRYAEKFGRLTITADTSTDITLYMQL GGLTSDDTAVVYCTTDWGYSSSLREEDI WYDCWGQGTIVTVSS (SEQ ID NO: 240) scFv: QAVLTQPPSASGTPGQRTVITCSGSSSNIGTKTVNHWYQVLPGTAPKLLIYSNYRRP SGVPDRFSGSGSGTSASLAISGLQSDDEADYYCALWDDSLDGYVFGTGKVTVL GSRGGGSGGGGSGGGGSLMAEVQLVQSGAEVRRPGATVKISCKVSGYTEND FYLHWVRQAPGKGLEWMGRIDPEDGKTRYAEKFGRLTITADTSTDITLYMQLG GLTSDDTAVVYCTTDWGYSSSLREEDIWYDCWGQGTIVTVSS (SEQ ID NO: 242)	QAVLTQPPSASGTPGQRTVITCSGSS SNIGTKTVNHWYQVLPGTAPKLLIYS NYRRPGVDRFSGSGSGTSASLAIS GLQSDDEADYYCALWDDSLDGYV FGTGKVTVL (SEQ ID NO: 241)
Eureka mAb-13	EVQLVQSGAEVKKPGSSVKVSKASGG TFSYVAISWVRQAPGQGLEWMGGIIPFG TANYAQKFGQRTITADESTSTAYMELS SLRSEDVAVVYCARDYGYGDYDAFDI WGQGTIVTVSS (SEQ ID NO: 243) scFv: SYELTQPPSVSVAPGKTARITCGNNIGSKSVHWYQQKPGQAPVLIYYSDRPS GIPERFSGSGNSGNTATLTISRVEAGDEADYYCQVWDDSSSDHYVFGTGKVTVLG RGGGGSGGGGSGGGGSLMAEVQLVQSGAEVKKPGSSVKVSKASGGTFSYVAI SWVRQAPGQGLEWMGGIIPFGTANYAQKFGQRTITADESTSTAYMELSSLRSE DTAVVYCARDYGYGDYDAFDIWGQGTIVTVSS (SEQ ID NO: 245)	SYELTQPPSVSVAPGKTARITCGGN NIGSKSVHWYQQKPGQAPVLIYY DSDRPSGIPERFSGSGNSGNTATLTIS RVEAGDEADYYCQVWDDSSSDHYV FGTGKVTVL (SEQ ID NO: 244)
Eureka mAb-14	EVQLVQSGAEVKKPGESLKISCKGSGYS FTSYWIWVRQMPGKGLEWMGGIIPGD SDTRYSPSQGQVTISADKSISTAYLQWS SLKASDTAMYICARVVGTIYSMQYDV WGQGTIVTVSS (SEQ ID NO: 246) scFv: SYVLTQPPSVSVAPGKTARITCGNNIGSKSVHWYQQRPQAPVLIYYDDSDRP SGIPERFSGSGNSGNTATLTISRVEAGDEADYSCQVWDDSSSDHYVFGTGKVTVLG SRGGGSGGGGSGGGGSLMAEVQLVQSGAEVKKPGESLKISCKGSGYSFTSYW IGWVRQMPGKGLEWMGGIIPGDSDTRYSPSQGQVTISADKSISTAYLQWSSLKA SDTAMYYICARVVGTIYSMQYDVWGQGTIVTVSS (SEQ ID NO: 248)	SYVLTQPPSVSVAPGKTARITCGGN NIGSKSVHWYQQRPQAPVLIYYD DSDRPSGIPERFSGSGNSGNTATLTIS RVEAGDEADYSCQVWDDSSSDHYV GPGTKVTVL (SEQ ID NO: 247)
Eureka mAb-15	EVQLVQSGAEVKKPGESLKISCKGSGYS FTSYWIWVRQMPGKGLEWMGGIIPGD SDTRYSPSQGQVTISADKSISTAYLQWS SLKASDTAMYICARVVGWQGMYP RSNWYNMDSWGQGTIVTVSS (SEQ ID NO: 249) scFv: LPVLTQPPSVSVAPGKTARITCGNNIGSKSVHWYQQKPGQAPVLIYYDDSDRP SGIPERFSGSGNSGNTATLTISRVEAGDEADYYCQVWDDSSDYVFGGGTKLTVLG SRGGGSGGGGSGGGGSLMAEVQLVQSGAEVKKPGESLKISCKGSGYSFTSYW IGWVRQMPGKGLEWMGGIIPGDSDTRYSPSQGQVTISADKSISTAYLQWSSLKA SDTAMYYICARVVGWQGMYP RSNWYNMDSWGQGTIVTVSS (SEQ ID NO: 251)	LPVLTQPPSVSVAPGKTARITCGGN NIGSKSVHWYQQKPGQAPVLIYYD DSDRPSGIPERFSGSGNSGNTATLTIS RVEAGDEADYYCQVWDDSSDYV FGGGTKLTVL (SEQ ID NO: 250)
Eureka mAb-16	EVQLVQSGAEVKKPGESLKISCKGSGYS FTSYWIWVRQMPGKGLEWMGGIIPGD SDTRYSPSQGQVTISADKSISTAYLQWS SLKASDTAMYICARWSSTWDSMYMDY WGQGTIVTVSS (SEQ ID NO: 252) scFv: QAVLTQPPSVSEAPRQRTVITCSGSSSNVGNNAVNWYQQVPGKAPKLLIYYDDL LSSGVSDRFSGSGSGTSASLAISGLQSEDEADYYCAAWDDSLNGPVFGGGTKLTV LGSRRGGGSGGGGSGGGGSLMAEVQLVQSGAEVKKPGESLKISCKGSGYSFTS YWIGWVRQMPGKGLEWMGGIIPGDSDTRYSPSQGQVTISADKSISTAYLQWSS LKASDTAMYYICARWSSTWDSMYMDYWGQGTIVTVSS (SEQ ID NO: 254)	QAVLTQPPSVSEAPRQRTVITCSGSS SNVGNNAVNWYQQVPGKAPKLLI YYDDLSSGVSDRFSGSGSGTSASLA AISGLQSEDEADYYCAAWDDSLNG PVFGGGTKLTVL (SEQ ID NO: 253)

TABLE 1-continued

Sequences of Exemplary First Antigen-Binding Sites		
Antigen-Binding Site	VH and HCDRs	VL and LCDRs
	GTLVTVSS (SEQ ID NO: 356)	TKLEIK (SEQ ID NO: 357)
	HCDR1: GYAFTSYNMY (SEQ ID NO: 358)	LCDR1: SASSTVNYMH (SEQ ID NO: 361)
	HCDR2: YIDPYNADTTYNQKFKG (SEQ ID NO: 359)	LCDR2: STSNLAS (SEQ ID NO: 362)
	HCDR3: TAYAMDY (SEQ ID NO: 360)	LCDR3: HQWSSYPYT (SEQ ID NO: 363)
Legend	QVKLEESGGELVQPGGFLRLSCAASGNI	N/A
mAb	FSINRMGWYRQAPGKQRAVFASITVRGI	
	TNYADSVKGRFTISVDKSKNTIYLQMNA	
	LKPEDTAVYYCNAVSSNRDPDYWGQGT	
	QVTVSS (SEQ ID NO: 364)	
	HCDR1: INRMG (SEQ ID NO: 365)	
	HCDR2: SITVRGITNYADSVKG (SEQ ID NO: 366)	
	HCDR3: VSSNRDPDY (SEQ ID NO: 367)	

Where the VL and LCDR sequences are noted as "N/A," the antigen-binding site is an sdAb having a VH (e.g., VHH) only.

In certain embodiments, the first antigen-binding site comprises a VH comprising complementarity determining regions HCDR1, HCDR2, and HCDR3, and a VL comprising complementarity determining regions LCDR1, LCDR2, and LCDR3. In certain embodiments, the VH comprises an amino acid sequence at least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%) identical to the VH of an antibody provided in Table 1, and the VL comprises an amino acid sequence at least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%) identical to the VL of the antibody provided in Table 1. In certain embodiments, the VH comprises the HCDR1, HCDR2, and HCDR3 sequences of the antibody provided in Table 1, and the VL comprises the LCDR1, LCDR2, and LCDR3 sequences of the antibody provided in Table 1. In certain embodiments, the VH comprises the amino acid sequence of the VH of an antibody provided in Table 1, and the VL comprises the amino acid sequence of the VL of the antibody provided in Table 1.

In certain embodiments, the VH of the first antigen-binding site comprises an amino acid sequence at least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%) identical to SEQ ID NO: 1, and the VL of the first antigen-binding site comprises an amino acid sequence at least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%) identical to SEQ ID NO: 2. In certain embodiments, the VH comprises HCDR1, HCDR2, and HCDR3 sequences set forth in SEQ ID NOs: 3, 4, and 5, respectively, and the VL comprises LCDR1, LCDR2, and LCDR3 sequences set forth in SEQ ID NOs: 6, 7, and 8, respectively. In certain embodiments, the VH comprises the amino acid sequence set forth in SEQ ID NO: 1, and the VL comprises the amino acid sequence set forth in SEQ ID NO: 2.

In certain embodiments, the VH of the first antigen-binding site comprises an amino acid sequence at least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%,

at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%) identical to SEQ ID NO: 10, and the VL of the first antigen-binding site comprises an amino acid sequence at least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%) identical to SEQ ID NO: 11. In certain embodiments, the VH comprises HCDR1, HCDR2, and HCDR3 sequences set forth in SEQ ID NOs: 12, 13, and 14, respectively, and the VL comprises LCDR1, LCDR2, and LCDR3 sequences set forth in SEQ ID NOs: 15, 16, and 17, respectively. In certain embodiments, the VH comprises the amino acid sequence set forth in SEQ ID NO: 10, and the VL comprises the amino acid sequence set forth in SEQ ID NO: 11.

In certain embodiments, the VH of the first antigen-binding site comprises an amino acid sequence at least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%) identical to SEQ ID NO: 19, and the VL of the first antigen-binding site comprises an amino acid sequence at least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%) identical to SEQ ID NO: 20. In certain embodiments, the VH comprises HCDR1, HCDR2, and HCDR3 sequences set forth in SEQ ID NOs: 21, 22, and 23, respectively, and the VL comprises LCDR1, LCDR2, and LCDR3 sequences set forth in SEQ ID NOs: 24, 25, and 26, respectively. In certain embodiments, the VH comprises the amino acid sequence set forth in SEQ ID NO: 19, and the VL comprises the amino acid sequence set forth in SEQ ID NO: 20.

In certain embodiments, the VH of the first antigen-binding site comprises an amino acid sequence at least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%) identical to SEQ ID NO: 27, and the VL of the first antigen-binding site comprises an amino acid sequence at least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least

94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%) identical to SEQ ID NO: 28. In certain embodiments, the VH comprises HCDR1, HCDR2, and HCDR3 sequences set forth in SEQ ID NOs: 29, 30, and 31, respectively, and the VL comprises LCDR1, LCDR2, and LCDR3 sequences set forth in SEQ ID NOs: 32, 33, and 34, respectively. In certain embodiments, the VH comprises the amino acid sequence set forth in SEQ ID NO: 27, and the VL comprises the amino acid sequence set forth in SEQ ID NO: 28.

In certain embodiments, the VH of the first antigen-binding site comprises an amino acid sequence at least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%) identical to SEQ ID NO: 35, and the VL of the first antigen-binding site comprises an amino acid sequence at least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%) identical to SEQ ID NO: 36. In certain embodiments, the VH comprises HCDR1, HCDR2, and HCDR3 sequences set forth in SEQ ID NOs: 29, 30, and 31, respectively, and the VL comprises LCDR1, LCDR2, and LCDR3 sequences set forth in SEQ ID NOs: 32, 33, and 34, respectively. In certain embodiments, the VH comprises the amino acid sequence set forth in SEQ ID NO: 35, and the VL comprises the amino acid sequence set forth in SEQ ID NO: 36.

In certain embodiments, the VH of the first antigen-binding site comprises an amino acid sequence at least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%) identical to SEQ ID NO: 37, and the VL of the first antigen-binding site comprises an amino acid sequence at least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%) identical to SEQ ID NO: 38. In certain embodiments, the VH comprises HCDR1, HCDR2, and HCDR3 sequences set forth in SEQ ID NOs: 39, 40, and 41, respectively, and the VL comprises LCDR1, LCDR2, and LCDR3 sequences set forth in SEQ ID NOs: 42, 43, and 44, respectively. In certain embodiments, the VH comprises the amino acid sequence set forth in SEQ ID NO: 37, and the VL comprises the amino acid sequence set forth in SEQ ID NO: 38.

In certain embodiments, the VH of the first antigen-binding site comprises an amino acid sequence at least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%) identical to SEQ ID NO: 45, and the VL of the first antigen-binding site comprises an amino acid sequence at least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%) identical to SEQ ID NO: 46. In certain embodiments, the VH comprises HCDR1, HCDR2, and HCDR3 sequences set forth in SEQ ID NOs: 47, 48, and 49, respectively, and the VL comprises LCDR1, LCDR2, and LCDR3 sequences set forth in SEQ ID NOs: 50, 51, and 52, respectively. In certain embodiments, the VH comprises the amino acid sequence set forth in SEQ ID NO: 45, and the VL comprises the amino acid sequence set forth in SEQ ID NO: 46.

In certain embodiments, the VH of the first antigen-binding site comprises an amino acid sequence at least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%) identical to SEQ ID NO: 86, and the VL of the first antigen-binding site comprises an amino acid sequence at least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%) identical to SEQ ID NO: 87. In certain embodiments, the VH comprises HCDR1, HCDR2, and HCDR3 sequences set forth in SEQ ID NOs: 88, 89, and 90, respectively, and the VL comprises LCDR1, LCDR2, and LCDR3 sequences set forth in SEQ ID NOs: 91, 92, and 93, respectively. In certain embodiments, the VH comprises the amino acid sequence set forth in SEQ ID NO: 86, and the VL comprises the amino acid sequence set forth in SEQ ID NO: 87.

Such antigen-binding site may take the form of scFv. In certain embodiments, the first antigen-binding site comprises an amino acid sequence at least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%) identical to an scFv sequence provided in Table 1. In certain embodiments, the first antigen-binding site comprises an amino acid sequence at least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%) identical to SEQ ID NO: 9, 18, 96, 212, 215, 218, 221, 224, 227, 230, 233, 236, 239, 242, 245, 248, 251, 254, or 257.

In other embodiments, the first antigen-binding site comprises an sdAb comprising a VH comprising complementarity determining regions HCDR1, HCDR2, and HCDR3. In certain embodiments, the VH comprises an amino acid sequence at least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%) identical to the VH of an sdAb antibody provided in Table 1. In certain embodiments, the VH comprises the HCDR1, HCDR2, and HCDR3 sequences of the antibody provided in Table 1. In certain embodiments, the VH comprises the amino acid sequence of the VH of an sdAb provided in Table 1.

In certain embodiments, the first antigen-binding site competes for binding CD19 (e.g., human CD19) with an antibody or antigen-binding fragment thereof comprising the VH, VL and/or scFv sequences provided in Table 1.

In certain embodiments, the first antigen-binding site of the multi-specific binding protein binds CD19 (e.g., human CD19) with a dissociation constant (K_D) of about 10 pM—about 1 μ M. The K_D can be measured by a method known in the art. In certain embodiments, the K_D is measured by SPR to CD19 or an extracellular fragment thereof immobilized on a chip. In certain embodiments, the K_D is measured by flow cytometry to CD19 expressed on the surface of cells, for example, following the method described in Example 5 below.

In certain embodiments, the first antigen-binding site binds CD19 with a K_D of lower than or equal to 20 nM, 15 nM, 10 nM, 9 nM, 8 nM, 7 nM, 6 nM, 5 nM, 4 nM, 3 nM, 2 nM, 1 nM, 0.9 nM, 0.8 nM, 0.7 nM, 0.6 nM, 0.5 nM, 0.4 nM, 0.3 nM, 0.2 nM, 0.1 nM, 90 pM, 80 pM, 70 pM, 60 pM, 50 pM, 40 pM, 30 pM, 20 pM, or 10 pM. For example, in certain embodiments, the first antigen-binding site binds

CD19 with a K_D of about 10 pM-about 1 nM, about 10 pM-about 0.9 nM, about 10 pM-about 0.8 nM, about 10 pM-about 0.7 nM, about 10 pM-about 0.6 nM, about 10 pM-about 0.5 nM, about 10 pM-about 0.4 nM, about 10 pM-about 0.3 nM, about 10 pM-about 0.2 nM, about 10 pM-about 0.1 nM, about 10 pM-about 50 pM, 0.1 nM-about 10 nM, about 0.1 nM-about 9 nM, about 0.1 nM-about 8 nM, about 0.1 nM-about 7 nM, about 0.1 nM-about 6 nM, about 0.1 nM-about 5 nM, about 0.1 nM-about 4 nM, about 0.1 nM-about 3 nM, about 0.1 nM-about 2 nM, about 0.1 nM-about 1 nM, about 0.1 nM-about 0.5 nM, about 0.5 nM-about 10 nM, about 1 nM-about 10 nM, about 2 nM-about 10 nM, about 3 nM-about 10 nM, about 4 nM-about 10 nM, about 5 nM-about 10 nM, about 6 nM-about 10 nM, about 7 nM-about 10 nM, about 8 nM-about 10 nM, or about 9 nM-about 10 nM.

In certain embodiments, the first antigen-binding site binds CD19 with a K_D greater than or equal to 10 nM, 20 nM, 30 nM, 40 nM, 50 nM, 60 nM, 70 nM, 80 nM, 90 nM, or 100 nM. In certain embodiments, the first antigen-binding site binds CD19 with a K_D of about 10 nM-about 1000 nM, about 10 nM-about 900 nM, about 10 nM-about 800 nM, about 10 nM-about 700 nM, about 10 nM-about 600 nM, about 10 nM-about 500 nM, about 10 nM-about 400 nM, about 10 nM-about 300 nM, about 10 nM-about 200 nM, about 10 nM-about 100 nM, about 10 nM-about 50 nM, about 50 nM-about 1000 nM, about 100 nM-about 1000 nM, about 200 nM-about 1000 nM, about 300 nM-about 1000 nM, about 400 nM-about 1000 nM, about 500 nM-about 1000 nM, about 600 nM-about 1000 nM, about 700 nM-about 1000 nM, about 800 nM-about 1000 nM, or about 900 nM-about 1000 nM.

It is understood that the binding affinity to CD19 of the first antigen-binding site alone may be different from the binding affinity of the same antigen-binding site in the context of the multi-specific binding protein disclosed herein, possibly due to the conformational restraint from the other domains. The context-dependent binding affinity is described in subsection I.G titled "Binding Affinity."

Melting temperature represents the thermostability of the antigen-binding site and can be measured by differential scanning fluorimetry, for example, as described in Durowoju et al. (2017) J. Vis. Exp. (121): 55262. The thermostability of an antibody or fragment thereof may be enhanced by grafting CDRs onto stable frameworks, introducing non-canonical disulfide bonds, and other mutagenesis, as described in McConnell et al. (2014) MABs, 6 (5): 1274-82; and Goldman et al. (2017) Front. Immunol., 8:865.

In certain embodiments, the first antigen-binding site has a melting temperature of at least 50° C., at least 55° C., at least 56° C., at least 57° C., at least 58° C., at least 59° C., at least 60° C., at least 61° C., at least 62° C., at least 63° C., at least 64° C., at least 65° C., at least 70° C., at least 75° C., or at least 80° C. In certain embodiments, the first antigen-binding site has a melting temperature in the range of 50-80° C., 50-70° C., 50-65° C., 50-60° C., 50-55° C., 55-70° C., 55-65° C., 55-60° C., 56-65° C., 56-60° C., 57-65° C., 57-60° C., 58-65° C., 58-60° C., 59-65° C., 59-60° C., 60-80° C., 60-75° C., 60-70° C., 60-65° C., 65-80° C., 65-75° C., 65-70° C., 70-80° C., or 70-75° C.

B. Second Antigen-Binding Site

The second antigen-binding site of the multi-specific binding protein binds CD3 (e.g., human CD3 and/or Macaca CD3). In certain embodiments, the second antigen-binding site binds CD3 ϵ (epsilon). In certain embodiments, the

second antigen-binding site binds CD3 δ (delta). In certain embodiments, the second antigen-binding site binds CD3 γ (gamma).

In certain embodiments, the second antigen-binding site of the multi-specific binding protein binds an epitope at the N-terminus of CD3 ϵ chain. In certain embodiments, the second antigen-binding site binds an epitope localized in amino acid residues 1-27 of human CD3 ϵ extracellular domain. This epitope or a homologous variant thereof is also present in certain non-human primates. Accordingly, in certain embodiments, the second antigen-binding site binds CD3 in different primates, for example, human, new world primates (such as *Callithrix jacchus*, *Saguinus Oedipus*, or *Saimiri sciureus*), old world primates (such as baboons and macaques), gibbons, and non-human homininae. *Callithrix jacchus* and *Saguinus oedipus* are new world primates belonging to the family of Callitrichidae, while *Saimiri sciureus* is a new world primate belonging to the family of Cebidae. In certain embodiments, the second antigen-binding site binds human CD3 ϵ and/or Macaca CD3 ϵ . In certain embodiments, the second antigen-binding site further binds *Callithrix jacchus*, *Saguinus Oedipus*, and/or *Saimiri sciureus* CD3 ϵ .

The second antigen-binding site that binds an extracellular epitope of human and/or Macaca CD3 can be derived from, for example, muromonab-CD3 (OKT3), oteizumab (TRX4), teplizumab (MGA031), visilizumab (Nuvion), SP34, X35, VIT3, BMA030 (BW264/56), CLB-T3/3, CRIS7, YTH12.5, F1 11-409, CLB-T3.4.2, TR-66, WT32, SPv-T3b, 11D8, XIII-141, XIII-46, XIII-87, 12F6, T3/RW2-8C8, T3/RW2-4B6, OKT3D, M-T301, SMC2, F101.01, UCHT-1, WT-31, and the antibodies described in WO2008119567A2. For example, the second binding domain optionally can include a VL domain comprising CDR-L1, CDR-L2 and CDR-L3 selected from: (a) CDR-L1 as depicted in SEQ ID NO: 27 of WO2008119567A2, CDR-L2 as depicted in SEQ ID NO: 28 of WO2008119567A2 and CDR-L3 as depicted in SEQ ID NO: 29 of WO2008119567A2; (b) CDR-L1 as depicted in SEQ ID NO: 117 of WO2008119567A2, CDR-L2 as depicted in SEQ ID NO: 118 of WO2008119567A2 and CDR-L3 as depicted in SEQ ID NO: 119 of WO2008119567A2; and (c) CDR-L1 as depicted in SEQ ID NO: 153 of WO2008119567A2, CDR-L2 as depicted in SEQ ID NO: 154 of WO2008119567A2 and CDR-L3 as depicted in SEQ ID NO: 155 of WO2008119567A2. Alternatively, one or more amino acid mutations can be introduced in (a), (b) or (c) group of CDR-L1, CDR-L2 and CDR-L3, and the second binding domain can include any of the mutated groups of CDR-L1, CDR-L2 and CDR-L3.

For example, the second binding domain can include a VL domain comprising CDR-L1, CDR-L2 and CDR-L3 selected from: (a) CDR-H1 as depicted in SEQ ID NO: 12 of WO2008119567A2, CDR-H2 as depicted in SEQ ID NO: 13 of WO2008119567A2 and CDR-H3 as depicted in SEQ ID NO: 14 of WO2008119567A2; (b) CDR-H1 as depicted in SEQ ID NO: 30 of WO2008119567A2, CDR-H2 as depicted in SEQ ID NO: 31 of WO2008119567A2 and CDR-H3 as depicted in SEQ ID NO: 32 of WO2008119567A2; (c) CDR-H1 as depicted in SEQ ID NO: 48 of WO2008119567A2, CDR-H2 as depicted in SEQ ID NO: 49 of WO2008119567A2 and CDR-H3 as depicted in SEQ ID NO: 50 of WO2008119567A2; (d) CDR-H1 as depicted in SEQ ID NO: 66 of WO2008119567A2, CDR-H2 as depicted in SEQ ID NO: 67 of WO2008119567A2 and CDR-H3 as depicted in SEQ ID NO: 68 of WO2008119567A2; (e) CDR-H1 as depicted in SEQ ID

31

NO: 84 of WO2008119567A2, CDR-H2 as depicted in SEQ ID NO: 85 of WO2008119567A2 and CDR-H3 as depicted in SEQ ID NO: 86 of WO2008119567A2; (f) CDR-H1 as depicted in SEQ ID NO: 102 of WO2008119567A2, CDR-H2 as depicted in SEQ ID NO: 103 of WO2008119567A2 and CDR-H3 as depicted in SEQ ID NO: 104 of WO2008119567A2; (g) CDR-H1 as depicted in SEQ ID NO: 120 of WO2008119567A2, CDR-H2 as depicted in SEQ ID NO: 121 of WO2008119567A2 and CDR-H3 as depicted in SEQ ID NO: 122 of WO2008119567A2; (h) CDR-H1 as depicted in SEQ ID NO: 138 of WO2008119567A2, CDR-H2 as depicted in SEQ ID NO: 139 of WO2008119567A2 and CDR-H3 as depicted in SEQ ID NO: 140 of WO2008119567A2; (i) CDR-H1 as depicted in SEQ ID NO: 156 of WO2008119567A2, CDR-H2 as depicted in SEQ ID NO: 157 of WO2008119567A2 and CDR-H3 as depicted in SEQ ID NO: 158 of WO2008119567A2; and (j) CDR-H1 as depicted in SEQ ID NO: 174 of WO2008119567A2, CDR-H2 as depicted in SEQ ID NO: 175 of WO2008119567A2 and CDR-H3 as depicted in SEQ ID NO: 176 of WO2008119567A2. Alternatively, one or more amino acid mutations can be introduced in (a), (b), (c), (d), (e), (f), (g), (h), (i), and (j) group of CDR-H1, CDR-H2 and CDR-H3, and the second binding domain can include any of the mutated groups of CDR-H1, CDR-H2 and CDR-H3. The referenced sequences disclosed in WO2008119567A2 are incorporated by reference herein.

Alternatively, the second domain can be derived from existing CD3 antibodies, for example, muromonab-CD3 (OKT3) as depicted in WO2008101154, otelexizumab (TRX4) as depicted in WO2007145941A2, teplizumab (MGA031) as depicted in WO2013040164A1, visilizumab (Nuvion) as depicted in WO2004052397A1, SP34 as depicted in WO2015181098A1, X35 as depicted in WO2015006749A2, VIT3 as depicted in WO2015006749A2, BMA030 (BW264/56) as depicted in

32

WO2015006749A2, CLB-T3/3 as depicted in WO2004106381A1, CRIS7 as depicted in WO2004106381A1, YTH12.5 as depicted in WO2004106383A1, Fl 11-409 as depicted in WO2012084895A2, CLB-T3.4.2 as depicted in WO2004106381A1, TR-66 as depicted in WO2013158856A2, WT32 as depicted in WO2004106381A1, SPv-T3b as depicted in WO2004106383A1, 11D8 as depicted in WO2004106381A1, XIII-141 as depicted in WO2004106381A1, XIII-46 as depicted in WO2004106381A1, XIII-87 as depicted in WO2004106381A1, 12F6 as depicted in WO2004106381A1, T3/RW2-8C8 as depicted in WO2004106381A1, T3/RW2-4B6 as depicted in WO2004106381A1, OKT3D as depicted in WO2004106381A1, M-T301 as depicted in WO2004106381A1, SMC2 as depicted in WO2004106381A1, F101.01 as depicted in WO2004106381A1, UCHT-1 as depicted in WO2000041474A2 and WT-31 as depicted in WO2016085889A1. The referenced sequences above disclosed in WO2008101154, WO2007145941A2, WO2013040164A1, WO2015181098A1, WO2004106381A1, WO2012084895A2, WO2000041474A2 and WO2016085889A1, are incorporated by reference herein.

A second antigen-binding site that binds CD3 can include a VH comprising three complementarity regions (HCDR1, HCDR2, and HCDR3) and/or a VL comprising three complementarity regions (LCDR1, LCDR2, and LCDR3). Table 2 summarizes, for each variable region, the CDRs of the variable region and scFv constructs based on the given heavy and light chain variable regions. The second antigen-binding site can be derived from the exemplary variable domain and CDR sequences as listed in Table 2.

TABLE 2

Sequences of Exemplary Second Antigen-Binding Sites		
Antigen-Binding Site	VH and HCDRs	VL and LCDRs
Adimab Ab325	QVQLVQSGAEVKKPGASVKVSKASGF NIKDYIMHWVRQAPGQRLIEWMGWIDL ENANTIYDAKFGQGRVTITRDTASTAYM ELSSLRSEDTAVYYCARDAYGRFYDV WGQGTTLVTSS (SEQ ID NO: 97) HCDR1: FNIKDYIMH (SEQ ID NO: 99) HCDR2: WIDLENANTIYDAKFGQ (SEQ ID NO: 100) HCDR3: ARDAYGRFYDV (SEQ ID NO: 101) scFv: DIVMTQSPDLSAVSLGERATINCKSSQSLNARTGKNYLAWYQQKPGQPPLLIY WASTRESGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCKQSYSRRTFGGGTKVE IKGGGGSGGGSGGGSGGGSGVQLVQSGAEVKKPGASVKVSKASGFNIKD YIMHWVRQAPGQRLIEWMGWIDLENANTIYDAKFGQGRVTITRDTASTAYMELS SLRSEDTAVYYCARDAYGRFYDVWGQGTTLVTSS (SEQ ID NO: 105)	DIVMTQSPDLSAVSLGERATINCKS SQSLNARTGKNYLAWYQQKPGQP PKLLIYWASTRESGVPDRFSGSGSG TDFTLTISSLQAEDVAVYYCKQSYS RRTFGGGTKVEIK (SEQ ID NO: 98) LCDR1: KSSQSLNARTGKNYLA (SEQ ID NO: 102) LCDR2: WASTRES (SEQ ID NO: 103) LCDR3: KQSYSRRT (SEQ ID NO: 104)
Harpoon Ab 2B2	EVQLVESGGGLVQPGGSLKLSCAASGFT FNKYAINWVRQAPGKLEWVARIRSKY NNYATYYADQVKDRFTISRDDSKNTAY LQMNNLKTEDTAVYYCVRHANFGNSYI SYWAYWGQGTTLVTSS (SEQ ID NO: 106) HCDR1: GFTFNKYAIN (SEQ ID NO: 108) HCDR2: RIRSKYNNYATYYADQVK (SEQ ID NO: 109) HCDR3: HANFGNSYISYWAY (SEQ ID	QTVVTQEPSTLVSPGGTIVTLTCASS TGAVTSGNYPNWVQQKPGQAPRG LIGGTKFLVPGTPARFSGSLGGA ALTLSGVQPEDEAEYYCTLWYSNR WVFGGGTKLTVL (SEQ ID NO: 107) LCDR1: ASSTGAVTSGNYPN (SEQ ID NO: 111) LCDR2: GTKFLVP (SEQ ID NO: 112) LCDR3: TLWYSNRWV (SEQ ID NO: 113)

TABLE 2-continued

Sequences of Exemplary Second Antigen-Binding Sites		
Antigen-Binding Site	VH and HCDRs	VL and LCDRs
	NO: 110) scFv: EVQLVESGGGLVQPGGSLKLSCAASGFTFNKYAINWVRQAPGKGLEWVARIRSK YNNYATYYADQVKDRFTISRDDSKNTAYLQMNNLKTEDTAVYYCVRHANFGN SYISYWAYWGQGLVTVSSGGGGSGGGSGGSGTQVVTQEPSTLVSPGGTVTL TCASSTGAVTSGNYPNWVQKPGQAPRGLIGGTKFLVPGTPARFSGSLLGKAA LTLSGVQPEDEAEYYCTLWYSNRWVFGGKLTVL (SEQ ID NO: 114)	
Adimab mAb393	QVQLVQSGAEVKKPGASVKVSCKASGF NIKDYIMHWVRQAPGQRLLEWMGWIDL ENANTIYDAKFQGRVTITRDTASTAYM ELSSLRSEDTAVYYCARDGYGRFYDV WGQGLVTVSS (SEQ ID NO: 115) HCDR1: FNIKDYIMH (SEQ ID NO: 99) HCDR2: WIDLENANTIYDAKFQ (SEQ ID NO: 100) HCDR3: ARDSYGRFYDV (SEQ ID NO: 368)	DIVMTQSPDSLAVSLGERATINCKS SQSLLSRTGKNYLAWYQQKPGQP PKLLIYWASTRESGVPDRFSGSGSG TDFTLTISSLQAEDVAVYYCKQSYS RRTFGGKTKVEIK (SEQ ID NO: 116) LCDR1: KSSQSLNARTGKNYLA (SEQ ID NO: 102) LCDR2: WASTRES (SEQ ID NO: 103) LCDR3: KQSYSRRT (SEQ ID NO: 104)
Adimab mAb333	QVQLVQSGAEVKKPGASVKVSCKASGF NIKDYIMHWVRQAPGQRLLEWMGWIDL ENANTIYDAKFQGRVTITRDTASTAYM ELSSLRSEDTAVYYCARDGYGRFYDL WGQGLVTVSS (SEQ ID NO: 117) HCDR1: FNIKDYIMH (SEQ ID NO: 99) HCDR2: WIDLENANTIYDAKFOG (SEQ ID NO: 100) HCDR3: ARDVYGRFYDL (SEQ ID NO: 369)	DIVMTQSPDSLAVSLGERATINCKS SQSLLESRTGKNYLAWYQQKPGQP PKLLIYWASTRESGVPDRFSGSGSG TDFTLTISSLQAEDVAVYYCKQSYS RRTFGGKTKVEIK (SEQ ID NO: 118) LCDR1: KSSQSLNARTGKNYLA (SEQ ID NO: 102) LCDR2: WASTRES (SEQ ID NO: 103) LCDR3: KQSYSRRT (SEQ ID NO: 104)
Adimab mAb334	QVQLVQSGAEVKKPGASVKVSCKASGF NIKDYIMHWVRQAPGQRLLEWMGWIDL ENANTIYDAKFQGRVTITRDTASTAYM ELSSLRSEDTAVYYCARDGYGYFYDV WGQGLVTVSS (SEQ ID NO: 186) HCDR1: FNIKDYIMH (SEQ ID NO: 99) HCDR2: WIDLENANTIYDAKFQ (SEQ ID NO: 100) HCDR3: ARDAYGGFYDV (SEQ ID NO: 370)	DIVMTQSPDSLAVSLGERATINCKS SQSLLSRTGKNYLAWYQQKPGQP PKLLIYWASTRESGVPDRFSGSGSG TDFTLTISSLQAEDVAVYYCKQSYS RRTFGGKTKVEIK (SEQ ID NO: 187) LCDR1: KSSQSLNARTGKNYLA (SEQ ID NO: 102) LCDR2: WASTRES (SEQ ID NO: 103) LCDR3: KQSYSRRT (SEQ ID NO: 104)
Adimab mAb404	EVQLLESGGGLVQPGGSLRLSCAASGFT FDTYAMNWVRQAPGKGLEWVARIRSK YNNYATYYADSVKDRFTISRDDSKSTLY LQMESLRAEDTAVYYCVRHGNFGNYAV SWFAHWGQGLVTVSS (SEQ ID NO: 371) HCDR1: FTFTDYAMN (SEQ ID NO: 373) HCDR2: RIRSKYNNYATYYADSVKD (SEQ ID NO: 374) HCDR3: VRHGNFGNYAVSWFAH (SEQ ID NO: 375)	QTVVTQEPSTLSVSPGGTVTLTCGSS TGAVTTSNYANWVQQTGQAPRG LIGGTDKRAPGVPDRFSGSLGDKA ALTITGAQAEDEADYYCALWYSNH WVFGGKTKLTVL (SEQ ID NO: 372) LCDR1: GSSTGAVTTSNYAN (SEQ ID NO: 376) LCDR2: GTDKRAP (SEQ ID NO: 377) LCDR3: ALWYSNHWV (SEQ ID NO: 378)
Adimab mAb405	EVQLLESGGGLVQPGGSLRLSCAASGFT FDTYAMNWVRQAPGKGLEWVARIRSK YNNYATYYADSVKDRFTISRDDSKSTLY LQMESLRAEDTAVYYCVRHGSFGNHIVS WFAHWGQGLVTVSS (SEQ ID NO: 379) HCDR1: FTFTDYAMN (SEQ ID NO: 373) HCDR2: RIRSKYNNYATYYADSVKD (SEQ ID NO: 374) HCDR3: VRHGSFGNHIVSWFAH (SEQ ID NO: 399)	QTVVTQEPSTLSVSPGGTVTLTCGSS TGAVTTSNYANWVQQTGQAPRG LIGGTDKRAPGVPDRFSGSLGDKA ALTITGAQAEDEADYYCALWYSNH WVFGGKTKLTVL (SEQ ID NO: 372) LCDR1: GSSTGAVTTSNYAN (SEQ ID NO: 376) LCDR2: GTDKRAP (SEQ ID NO: 377) LCDR3: ALWYSNHWV (SEQ ID NO: 378)
Adimab mAb-1	QVQLVQSGAEVKKPGASVKVSCKASGF NIKDYIMHWVRQAPGQRLLEWMGWIDL ENGNTIYDAKFQGRVTITRDTASTAYM ELSSLRSEDTAVYYCARDGYGRFYDV WGQGLVTVIS (SEQ ID NO: 188)	DIVMTQSPDSLAVSLGERATINCKS SQSLLSRTGKNYLAWYQQKPGQP PKLLIYWASTRESGVPDRFSGSGSG TDFTLTISSLQAEDVAVYYCKQSYS RRTFGGKTKVEIK (SEQ ID NO: 189)

TABLE 2-continued

Sequences of Exemplary Second Antigen-Binding Sites		
Antigen-Binding Site	VH and HCDRs	VL and LCDRs
Adimab mAb-2	QVQLVQSGAEVKKPGASVKVSCKASGF NIKDYMHVWRQAPGQRLWGMGIDL ENANTIYDAKPFQGRVTITRDTSASTAYM ELSSLRSEDTAVYYCARDVYGRYLDV WGQGTLLTVSS (SEQ ID NO: 190)	DIVMTQSPDLSAVSLGERATINCKS SQSLNLRNTRKNYLAWYQQKPGQP PKLLIYWASTRESGVPDRFSGSGSG TDFTLTISSLQAEDVAVYYCKQSYS RRTFGGGTKVEIK (SEQ ID NO: 191)
Adimab mAb-3	QVQLVQSGAEVKKPGASVKVSCKASGF NIKDYMHVWRQAPGQRLWGMGIDL ENGNTIYDAKPFQGRVTITRDTSASTAYM ELSSLRSEDTAVYYCARDAYGRYFYDV WGQGTLLTVSS (SEQ ID NO: 192)	DIVMTQSPDLSAVSLGERATINCRS SQSLNLRNTRKNYLAWYQQKPGQP PKLLIYWASTRESGVPDRFSGSGSG TDFTLTISSLQAEDVAVYYCKQSYS RRTFGGGTKVEIK (SEQ ID NO: 193)
Adimab mAb-4	QVQLVQSGAEVKKPGASVKVSCKASGF NIKDYMHVWRQAPGQRLWGMGIDL EEGNTIYDAKPFQGRVTITRDTSASTAYM ELSSLRSEDTAVYYCARDAYGRYFYDV WGQGTLLTVSS (SEQ ID NO: 194)	DIVMTQSPDLSAVSLGERATINCKS SQSLNLRNTRKNYLAWYQQKPGQP PKLLIYWASTRESGVPDRFSGSGTG TDFTLTISSLQAEDVAVYYCKQSYS RRTFGGGTKVEIK (SEQ ID NO: 195)
Adimab mAb-5	QVQLVQSGAEVKKPGASVKVSCKASGF NIKDYMHVWRQAPGQRLWGMGIDL ENANTIYDAKPFQGRVTITRDTSASTAYM ELSSLRSEDTAVYYCARDNYGGYFYDV WGQGTLLTVSS (SEQ ID NO: 196)	DIVMTQSPDLSAVPLGERATINCKS SQSLNLRNTRKNYLAWYQQKPGQP PKLLIYWASTRESGVPDRFSGSGSG TDFTLTISSLQAEDVAVYYCKQSYS RRTFGGGTKVEIK (SEQ ID NO: 197)
Adimab mAb-6	QVQLVQSGAEVKKPGASVKVSCKASGF NIKDYMHVWRQAPGQRLWGMGIDL ENGNTIYDAKPFQGRVTITRDTSASTAYM ELSSLRSEDTA VYYCARDGYGRYFYDV WGQGTLLTVSS (SEQ ID NO: 198)	DIVMTQSPDLSAVSLGERATINCKS SQSLNLRNTRKNYLAWYQQKPGQP PKLLIYWASTRESGVPDRFSGSGSG TDFTLTISSLQAEDVAVYYCKQSYS LRTFGGGTKVEIK (SEQ ID NO: 199)
Adimab mAb-7	QVQLVQSGAEVKKPGASVKVSCKASGF NIKDYMHVWRQAPGQRLWGMGIDL ENGNTIYDAKPFQGRVTITRDTSASTAYM ELSSLRSEDTAVYYCARDGYGRYFYDV WGQGTLLTVSS (SEQ ID NO: 200)	DIVMTQSPDLSAVSLGERATINCKS SQSLNLRNTRKNYLAWYQQKPGQP PKLLIYWASTRESGVPDRFSGSGSG TDFTLTISSLQAEDVAVYYCKQSYN LRTFGGGTKVEIK (SEQ ID NO: 201)
Adimab mAb-8	QVQLVQSGAEVKKPGASVKVSCKASGF NIKDYMHVWRQAPGQRLWGMGIDL ENGNTIYDAKPFQGRVTITRDTSASTAYM ELSSLRSEDTAVYYCAREGYGRYFYDV WGQGTLLTVSS (SEQ ID NO: 202)	DIVMTQSPDLSAVSLGERATINCKS SQSLNLRNTRKNYLAWYQQKPGQP PKLLIYWASTRESGVPDRFSGSGSG TDFTLTISSLQAEDVAVYYCKQSYF RRAFGGGKVEIK (SEQ ID NO: 203)
Adimab mAb-9	QVQLVQSGAEVKKPGASVKVSCKASGF NIKDYMHVWRQAPGQRLWGMGIDL ENGNTIYDAKPFQGRVTITRDTSASTAYM ELSSLRSEDTAVYYCARDGYGRYYDV WGQGTLLTVSS (SEQ ID NO: 204)	DIVMTQSPDLSAVSLGERATINCKS SQSLNLRNTRKNYLAWYQQKPGQP PKLLIYWASTRESGVPDRFSGSGSG TDFTLTISSLQAEDVAVYYCKQSYN LRTFGGGTKLEIK (SEQ ID NO: 205)
Adimab mAb-10	QVQLVQSGAEVKKPGASVKVSCKASGF NIKDYMHVWRQAPGQRLWGMGIDL ENGNTIYQKPFQGRVTITRDTSASTAYM ELSSLRSEDTAVYYCARDGYGRYFYDV WGQGTLLTVSS (SEQ ID NO: 206)	DIVMTQSPDLSAVSLGERATINCKS SQSLNLRNTRKNYLAWYQQKPGQS PKLLIYWASTRESGVPDRFTGSGSG TDFTLTISSLQAEDVAVYYCKQSYS LRTFGGGTKVEIK (SEQ ID NO: 207)
Adimab mAb-11	QVQLVQSGAEVKKPGASVKVSCKASGF NIKDYMHVWRQAPGQRLWGMGIDL ENGNTIYDAKPFQGRVTITRDTSASTAYM ELSSLRSEDTAVYYCARDGYGRYFYDY WGQGTLLTVSS (SEQ ID NO: 208)	DIVMTQSPDLSAVSLGERATINCKS SQSLLESRTGKNYLAWYQQKPGQP PKLLIYWASTRESGVPDRFSGSGSG TDFTLTISSLQAEDVAVYYCKQSYS LRTFGGGTKVEIK (SEQ ID NO: 209)
CD3 binding domain in blinatum omab	DIKLQSQGAELARPGASVKMSCKTSGYT FTRYTMHWVKQRPGGLEWIGYINPSR GYTNYNQKFKDKATLTTDKSSSTAYMQ LSSLTSEDSAVYYCARYYDDHYCLDYW GQGTLLTVSSVE (SEQ ID NO: 119) scFv: DIKLQSQGAELARPGASVKMSCKTSGYTFTRYTMHWVKQRPGGLEWIGYINPS RGYTNYNQKFKDKATLTTDKSSSTAYMQLSSLTSEDSAVYYCARYYDDHYCLD YWQGTLLTVSSVEGSGSGSGSGSGGVDIQLTQSPAIMASAPGEKVTMTCTCR ASSSVSYMNYQQKSGTSPKRWIYDTSKVASGVYRFSGSGSGTSSYSLTISSMEA EDAATYYCQQWSSNPLTFGAGTKLELK (SEQ ID NO: 258)	VDDIQLTQSPAIMASAPGEKVTMTCT RASSSVSYMNYQQKSGTSPKRWI YDTSKVASGVYRFSGSGSGTSSYSL TISSMEAEDAATYYCQQWSSNPLTF GAGTKLELK (SEQ ID NO: 120)

TABLE 2-continued

Sequences of Exemplary Second Antigen-Binding Sites		
Antigen-Binding Site	VH and HCDRs	VL and LCDRs
Novimmune	QVQLVESGGGVVQPGKSLRLSCAASGFK FSGYGMHWVRQAPGKLEWVAVIYD GSKKYYVDSVKGRFTISRDNKNTLYLQ MNSLRAEDTAVYYCARQMGYWHFDLW GRGTLVTVSS (SEQ ID NO: 259)	EIVLTQSPATLSLSPGERATLSCRAS QSVSSYLAWYQQKPGQAPRLLIYD ASNRRATGIPARFSGSGSGTDFLTIS SLEPEDFAVYYCQQRSNWPPLTFG GGTKVEIK (SEQ ID NO: 260)
28F11	HCDR1: GYGMH (SEQ ID NO: 261) HCDR2: VIWYDGSKKYYVDSVK (SEQ ID NO: 262) HCDR3: QMGYWHFDL (SEQ ID NO: 263)	LCDR1: RASQSVSSYLA (SEQ ID NO: 264) LCDR2: DASNRAT (SEQ ID NO: 265) LCDR3: QQRSNWPPLT (SEQ ID NO: 266)
Novimmune 27H5	EVQLLESGGGLVQPGGSLRLSCAASGFT FSSFPMAWVRQAPGKLEWVSTISTSGG RTYYRDSVKGRFTISRDNKNTLYLQMN SLRAEDTAVYYCAKFRQYSGGFDYWQ GTLVTVSS (SEQ ID NO: 267)	DFMLTQPHSVSESFGTKTIIISCWYQ QRPGRAPTTVIFGVDRFSGSIDRSS NSASLTISGLQTEADYYCFGGGT KLTVLGQPKAAPSVTLFPPSSEELQ (SEQ ID NO: 268)
Glaxo mAb	EVQLLESGGGLVQPGGSLRLSCAASGFT FSSFPMAWVRQAPGKLEWVSTISTSGG RTYYRDSVKGRFTISRDNKNTLYLQMN SLRAEDTAVYYCAKFRQYSGGFDYWQ GTLVTVSS (SEQ ID NO: 269)	DIQLTQPNVSTSLGSTVKLSCTLSS GNIENNVVHWYQLYEGRSPTMIY DDDKRPDGVDRFSGSIDRSSNSAF LTIHNVAIEDEAIYFCHSYVSSFNVF GGGTKLTVLR (SEQ ID NO: 270)
Eureka mAb	DVQLVQSGAEVKKPGASVKVSCKASGY TFTRYTMHWVRQAPGKLEWIGYINPS RGYTNVADSVKGRFTITTDKSTSTAYME LSSLRSEDATYYCARYYDDHYCLDYW GQGTTVTVSS (SEQ ID NO: 271) scFv: DVQLVQSGAEVKKPGASVKVSCKASGYTFTRYTMHWVRQAPGKLEWIGYINP SRGYTNVADSVKGRFTITTDKSTSTAYMELSSLRSEDATYYCARYYDDHYCLD YWQGTTVTVSSSGEGSTGSGSGSGSGGADDIVLTQSPATLSLSPGERATLSCRA SQSVSYMNWYQQKPKGAPKRWIYDTSKVASGVPARFSGSGSGTDSLINSLEA EDAATYYCQWSSNPLTFGGGKVEIK (SEQ ID NO: 273)	DIVLTQSPATLSLSPGERATLSCRAS QSVSYMNWYQQKPKGAPKRWIYD TSKVASGVPARFSGSGSGTDSLINSLEA EDAATYYCQWSSNPLTFGGGKVEIK (SEQ ID NO: 272)
Muromon ab	QVQLVQSGGGVVQPGKSLRLSCKASGY TFTRYTMHWVRQAPGKLEWIGYINPS RGYTNVQKVKDRFTISRDNKNTAFQ MDSLRPEDTGVYFCARYYDDHYCLDY WGQGTPTVTVSS (SEQ ID NO: 274)	DDIQTQSPSSLSASVGDRTITCS ASSSVSYMNWYQQTPGKAPKRWIY DTSKLASGVPSRFSGSGSGTDTFTI SSLQPEDATYYCQWSSNPLTFGGG GKTLQIT (SEQ ID NO: 275)
MacroGenics mAb humanized OKT3	QVQLVQSGGGVVQPGKSLRLSCKASGY TFTRYTMHWVRQAPGKLEWIGYINPS RGYTNVQKVKDRFTISRDNKNTAFQ MDSLRPEDTGVYFCARYYDDHYCLDY WGQGTPTVTVSS (SEQ ID NO: 276)	DIQTQSPSSLSASVGDRTITCSAS SSSVSYMNWYQQTPGKAPKRWIYD TSKLASGVPSRFSGSGSGTDTFTI SSLQPEDATYYCQWSSNPLTFGGG TKLQITR (SEQ ID NO: 277)
Roche CH2527	EVQLLESGGGLVQPGGSLRLSCAASGFT FSTYAMNWRQAPGKLEWVSRIRSKY NNYATYYADSVKGRFTISRDNKNTLYL QMNSLRAEDTAVYYCVRHGNFGNSYVS WFAYWGQGTTLVTVSS (SEQ ID NO: 278)	QAVVTQEPSTLTVSPGGTVTLTCGSS TGAVTTSNANWVQKPGQAFRG LIGGTNRKAPGTPARFSGSLGGKA ALTLSGAQPEDEAEYYCALWYSNL WVFGGGTKLTVL (SEQ ID NO: 279)
Regeneron mAb (anti-CD3/anti-CD20)	EVQLVESGGGLVQPGKSLRLSCAASGFT FDDYTMHWVRQAPGKLEWVSGISWN SGSIGYADSVKGRFTISRDNKNTLYLQ MNSLRAEDTAVYYCAIDNSGYGHYY GMDVWGQGTTLVTVAS (SEQ ID NO: 280)	AEIVMTQSPATLSVSPGERATLSCR ASQSVSSNLAWYQQKPGQAPRLLI YGASTRATGIPARFSGSGSGTEFTLT ISSLQSEDFAVYYCQHYINWPLTFG GGTKVEIK (SEQ ID NO: 281)
WuXi WBP331 1_2.166.4 8-z1	QVQLVQSGAEVKKPGSSVKVSCKASGY SFTTYIHVWRQAPGKLEWVGWIFPG NDNIKYSEKFKGRVTITADKSTSTAYME LSSLRSEDATYYCAIDNSVSIYYFDYW QGTLVTVSS (SEQ ID NO: 380) HCDR1: GYSFTTYIYH (SEQ ID NO: 382) HCDR2: WIFPGNDNIKYSEKFKG (SEQ ID NO: 383) HCDR3: DSVSIYYFDY (SEQ ID NO: 384)	DIVMTQSPDSLAVSLGERATINCKS SQSLNLSRTRKNYLAWYQQKPGQP PKLLIYWASTRSGVDRFSGSGSG TDFTLTISLQAEADVAVYYCTQSFIL RTFGGGTKVEIK (SEQ ID NO: 381) LCDR1: KSSQSLNLSRTRKNYLA (SEQ ID NO: 385) LCDR2: WASTRKS (SEQ ID NO: 386) LCDR3: TQSFILRT (SEQ ID NO: 387)
WuXi WBP331 1_2.306.4-z1	QVQLVQSGAEVKKPGSSVKVSCKASGF AFTDYIHWVRQAPGKLEWVGWISPG NVNTKYNEFKGRVTITADKSTSTAYM ELSSLRSEDATYYCARDGYSIYYFDY	DIVMTQSPDSLAVSLGERATINCKS SQSLNLSRTRKNYLAWYQQKPGQP PKLLIYWASTRSGVDRFSGSGSG TDFTLTISLQAEADVAVYYCTQSH

TABLE 2-continued

Sequences of Exemplary Second Antigen-Binding Sites		
Antigen-Binding Site	VH and HCDRs	VL and LCDRs
	WGQGTILVTVSS (SEQ ID NO: 388) HCDR1: GFAFTDYIIH (SEQ ID NO: 390) HCDR2: WISPGNVNTKYNEFKG (SEQ ID NO: 391) HCDR3: DGYSLYYFDY (SEQ ID NO: 392)	LRTFGGGTKVEIK (SEQ ID NO: 389) LCDR1: KSSQSLNSRTRKNYLA (SEQ ID NO: 385) LCDR2: WASTRQS (SEQ ID NO: 393) LCDR3: TQSHLTLT (SEQ ID NO: 394)
ADLQ mAb-1	MAESGGGSVQTGGSLRLSCAYTASSVC MAWFRQAPGKEREGVAVTREGLTQGTGY ADSVKGRFAISQDYAKKTLTYLQMSSLKP EDTARYYCAARPTSPCTVDGELLASTYN YWGQGTQVTV (SEQ ID NO: 395)	N/A
ADLQ mAb-2	MAESGGGSVQTGGSLRLSCAYTASSVC MAWFRQAPGKEREGVAVTREGLTQGTGY ADSVKGRFAISQDYAKKTLTYLQMSSLKP EDTARYYCAARPTSPCTVDGELLASTYD YWGQGTQVTV (SEQ ID NO: 396)	N/A
ADLQ mAb-3	MAESGGGSVQTGGSLRLSCAYTASSVC MAWFRQAPGKEREGVAVTREGLTQGTGY ADSVKGRFAISQDYAKKTLTYLQMSSLKP EDTARYYCAARPTSPCTVDGELLASTYN YWGQGTQVTV (SEQ ID NO: 397)	N/A
ADLQ mAb-4	MAESGGGSVQTGGSLRLSCAYTASSVC MAWFRQAPGKEREGVAVTREGLTQGTGY ADSVKGRFAISQDYAKKTLTYLQMSSLKP EDTARYYCAARPTSPCTVDGELLASTYD YWGQGTQVTV (SEQ ID NO: 398)	N/A

Where the VL and Lcdr sequences are noted as "N/A," the antigen-binding site is an sdAb having a VH (e.g., VHH) only.

In certain embodiments, the second antigen-binding site comprises a VH comprising complementarity determining regions HCDR1, HCDR2, and HCDR3, and a VL comprising complementarity determining regions LCDR1, LCDR2, and LCDR3. In certain embodiments, the VH comprises an amino acid sequence at least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%) identical to the VH of an antibody provided in Table 2, and the VL comprises an amino acid sequence at least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%) identical to the VL of the antibody provided in Table 2. In certain embodiments, the VH comprises the HCDR1, HCDR2, and HCDR3 sequences of the antibody provided in Table 2, and the VL comprises the LCDR1, LCDR2, and LCDR3 sequences of the antibody provided in Table 2. In certain embodiments, the VH comprises the amino acid sequence of the VH of an antibody provided in Table 2, and the VL comprises the amino acid sequence of the VL of the antibody provided in Table 2.

In certain embodiments, the VH of the second antigen-binding site comprises an amino acid sequence at least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%) identical to SEQ ID NO: 97, and the VL of the second antigen-binding site comprises an amino acid sequence at

least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%) identical to SEQ ID NO: 98. In certain embodiments, the VH comprises HCDR1, HCDR2, and HCDR3 sequences set forth in SEQ ID NOS: 99, 100, 101, respectively, and the VL comprises LCDR1, LCDR2, and LCDR3 sequences set forth in SEQ ID NOS: 102, 103, and 104, respectively. In certain embodiments, the VH comprises the amino acid sequence set forth in SEQ ID NO: 97, and the VL comprises the amino acid sequence set forth in SEQ ID NO: 98.

In certain embodiments, the VH of the second antigen-binding site comprises an amino acid sequence at least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%) identical to SEQ ID NO: 106, and the VL of the second antigen-binding site comprises an amino acid sequence at least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%) identical to SEQ ID NO: 107. In certain embodiments, the VH comprises HCDR1, HCDR2, and HCDR3 sequences set forth in SEQ ID NOS: 108, 109, and 110, respectively, and the VL comprises LCDR1, LCDR2, and LCDR3 sequences set forth in SEQ ID NOS: 111, 112, and 113, respectively. In certain embodiments, the VH comprises the amino acid sequence set forth in SEQ ID NO: 106, and the VL comprises the amino acid sequence set forth in SEQ ID NO: 107.

Such antigen-binding site may take the form of scFv. In certain embodiments, the second antigen-binding site comprises an amino acid sequence at least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%) identical to an scFv sequence provided in Table 2. In certain embodiments, the second antigen-binding site comprises an amino acid sequence at least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%) identical to SEQ ID NO: 105 or 114.

In other embodiments, the second antigen-binding site comprises an sdAb comprising a VH comprising complementarity determining regions HCDR1, HCDR2, and HCDR3. In certain embodiments, the VH comprises an amino acid sequence at least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%) identical to the VH of an sdAb antibody provided in Table 2. In certain embodiments, the VH comprises the HCDR1, HCDR2, and HCDR3 sequences of the antibody provided in Table 2. In certain embodiments, the VH comprises the amino acid sequence of the VH of an sdAb provided in Table 2.

In certain embodiments, the second antigen-binding site competes for binding CD3 (e.g., human CD3 and/or Macaca CD3) with an antibody or antigen-binding fragment thereof comprising the VH, VL and/or scFv sequences provided in Table 2.

In certain embodiments, the second antigen-binding site of the multi-specific binding protein binds CD3 (e.g., human CD3 and/or Macaca CD3) with a dissociation constant (K_D) of about 0.1 nM-about 1 μ M. The K_D can be measured by a method known in the art. In certain embodiments, the K_D is measured by SPR to CD3 or an extracellular fragment thereof immobilized on a chip. In certain embodiments, the K_D is measured by flow cytometry to CD3 expressed on the surface of cells, for example, following the method described in Example 5 below.

In certain embodiments, the second antigen-binding site binds CD3 with a K_D of lower than or equal to 20 nM, 15 nM, 10 nM, 9 nM, 8 nM, 7 nM, 6 nM, 5 nM, 4 nM, 3 nM, 2 nM, 1 nM, 0.9 nM, 0.8 nM, 0.7 nM, 0.6 nM, 0.5 nM, 0.4 nM, 0.3 nM, 0.2 nM, 0.1 nM, 90 pM, 80 pM, 70 pM, 60 pM, 50 pM, 40 pM, 30 pM, 20 pM, or 10 pM. For example, in certain embodiments, the first antigen-binding site binds CD3 with a K_D of about 10 pM-about 1 nM, about 10 pM-about 0.9 nM, about 10 pM-about 0.8 nM, about 10 pM-about 0.7 nM, about 10 pM-about 0.6 nM, about 10 pM-about 0.5 nM, about 10 pM-about 0.4 nM, about 10 pM-about 0.3 nM, about 10 pM-about 0.2 nM, about 10 pM-about 0.1 nM, about 10 pM-about 50 pM, 0.1 nM-about 10 nM, about 0.1 nM-about 9 nM, about 0.1 nM-about 8 nM, about 0.1 nM-about 7 nM, about 0.1 nM-about 6 nM, about 0.1 nM-about 5 nM, about 0.1 nM-about 4 nM, about 0.1 nM-about 3 nM, about 0.1 nM-about 2 nM, about 0.1 nM-about 1 nM, about 0.1 nM-about 0.5 nM, about 0.5 nM-about 10 nM, about 1 nM-about 10 nM, about 2 nM-about 10 nM, about 3 nM-about 10 nM, about 4 nM-about

10 nM, about 5 nM-about 10 nM, about 6 nM-about 10 nM, about 7 nM-about 10 nM, about 8 nM-about 10 nM, or about 9 nM-about 10 nM.

In certain embodiments, the second antigen-binding site binds CD3 with a K_D greater than or equal to 10 nM, 20 nM, 30 nM, 40 nM, 50 nM, 60 nM, 70 nM, 80 nM, 90 nM, or 100 nM. In certain embodiments, the second antigen-binding site binds CD3 with a K_D of about 10 nM-about 1000 nM, about 10 nM-about 900 nM, about 10 nM-about 800 nM, about 10 nM-about 700 nM, about 10 nM-about 600 nM, about 10 nM-about 500 nM, about 10 nM-about 400 nM, about 10 nM-about 300 nM, about 10 nM-about 200 nM, about 10 nM-about 100 nM, about 10 nM-about 50 nM, about 50 nM-about 1000 nM, about 100 nM-about 1000 nM, about 200 nM-about 1000 nM, about 300 nM-about 1000 nM, about 400 nM-about 1000 nM, about 500 nM-about 1000 nM, about 600 nM-about 1000 nM, about 700 nM-about 1000 nM, about 800 nM-about 1000 nM, or about 900 nM-about 1000 nM.

It is understood that the binding affinity to CD3 of the second antigen-binding site alone may be different from the binding affinity of the same antigen-binding site in the context of the multi-specific binding protein disclosed herein, possibly due to the conformational restraint from the other domains. The context-dependent binding affinity is described in the subsection I.G titled "Binding Affinity."

In certain embodiments, the second antigen-binding site has a melting temperature of at least 50° C., at least 55° C., at least 56° C., at least 57° C., at least 58° C., at least 59° C., at least 60° C., at least 61° C., at least 62° C., at least 63° C., at least 64° C., at least 65° C., at least 70° C., at least 75° C., or at least 80° C. In certain embodiments, the second antigen-binding site has a melting temperature in the range of 50-80° C., 50-70° C., 50-65° C., 50-60° C., 50-55° C., 55-70° C., 55-65° C., 55-60° C., 56-65° C., 56-60° C., 57-65° C., 57-60° C., 58-65° C., 58-60° C., 59-65° C., 59-60° C., 60-80° C., 60-75° C., 60-70° C., 60-65° C., 65-80° C., 65-75° C., 65-70° C., 70-80° C., or 70-75° C.

C. Third Antigen-Binding Site

The third antigen-binding site of the multi-specific binding protein binds serum albumin (e.g., HSA). In certain embodiments, the third antigen-binding site has a higher binding affinity to a human serum albumin than to a mouse serum albumin. In certain embodiments, the third antigen-binding site does not bind the D-III domain of HSA. In certain embodiments, the third antigen-binding site extends the serum half-life of the multi-specific binding protein.

The third antigen-binding site that binds serum albumin can be derived from, for example, the antigen-binding sites disclosed in U.S. Pat. No. 8,188,223, and PCT Publication Nos. WO2017085172, and WO2018050833.

A third antigen-binding site that binds HSA can include a VH comprising three complementarity regions (HCDR1, HCDR2, and HCDR3) and/or a VL comprising three complementarity regions (LCDR1, LCDR2, and LCDR3). Table 3 summarizes, for each variable region, the CDRs of the variable region and scFv constructs based on the given heavy and light chain variable regions. The third antigen-binding site can be derived from the exemplary variable domain and CDR sequences as listed in Table 3.

TABLE 3

Sequences of Exemplary Third Antigen-Binding Sites		
Antigen-Binding Site	VH and HCDRs	VL and LCDRs
Ablynx ALB8	EVQLVESGGGLVQP GNSLR LSCAASGFT FSSFGMSWVRQAPGKGLEWVSSISGSGS DTLYADSVKGRFTISRDN AKTTLYLQMN SLRPEDTAVYYCTIGGSLSRSSQGT LVT SS (SEQ ID NO: 121) HCDR1: SFGMS (SEQ ID NO: 122) or GFTFSSFGMS (SEQ ID NO: 123) HCDR2: SISGSGDTL (SEQ ID NO: 124) or SISGSGSDTLYADSVKG (SEQ ID NO: 125) HCDR3: GGSLSR (SEQ ID NO: 126)	N/A
Ablynx ALB3	EVQLVESGGGLVQP GGSRL LSCAASGFT FRSFGMSWVRQAPGKEPEWVSSISGSGS DTLYADSVKGRFTISRDN AKTTLYLQMN SLKPEDTAVYYCTIGGSLSRSSQGTQVT VSS (SEQ ID NO: 127) HCDR1: SFGMS (SEQ ID NO: 122) or GFTFRSFGMS (SEQ ID NO: 128) HCDR2: SISGSGDTL (SEQ ID NO: 124) or SISGSGSDTLYADSVKG (SEQ ID NO: 125) HCDR3: GGSLSR (SEQ ID NO: 126)	N/A
Ablynx ALB4	EVQLVESGGGLVQP GGSRL LSCAASGFT FSSFGMSWVRQAPGKEPEWVSSISGSGS DTLYADSVKGRFTISRDN AKTTLYLQMN SLKPEDTAVYYCTIGGSLSRSSQGTQVT VSS (SEQ ID NO: 129) HCDR1: SFGMS (SEQ ID NO: 122) or GFTFSSFGMS (SEQ ID NO: 123) HCDR2: SISGSGDTL (SEQ ID NO: 124) or SISGSGSDTLYADSVKG (SEQ ID NO: 125) HCDR3: GGSLSR (SEQ ID NO: 126)	N/A
Ablynx ALB5	EVQLVESGGGLVQP GGSRL LSCAASGFT FRSFGMSWVRQAPGKGLEWVSSISGSGS DTLYADSVKGRFTISRDN AKTTLYLQMN SLKPEDTAVYYCTIGGSLSRSSQGTQVT VSS (SEQ ID NO: 130) HCDR1: SFGMS (SEQ ID NO: 122) or GFTFRSFGMS (SEQ ID NO: 128) HCDR2: SISGSGDTL (SEQ ID NO: 124) or SISGSGSDTLYADSVKG (SEQ ID NO: 125) HCDR3: GGSLSR (SEQ ID NO: 126)	N/A
Ablynx ALB6	EVQLVESGGGLVQP GNSLR LSCAASGFT FRSFGMSWVRQAPGKGLEWVSSISGSGS DTLYADSVKGRFTISRDN AKTTLYLQMN SLKPEDTAVYYCTIGGSLSRSSQGT LVT SS (SEQ ID NO: 131) HCDR1: SFGMS (SEQ ID NO: 122) or GFTFRSFGMS (SEQ ID NO: 128) HCDR2: SISGSGDTL (SEQ ID NO: 124) or SISGSGSDTLYADSVKG (SEQ ID NO: 125) HCDR3: GGSLSR (SEQ ID NO: 126)	N/A
Ablynx ALB7	EVQLVESGGGLVQP GNSLR LSCAASGFT FRSFGMSWVRQAPGKGLEWVSSISGSGS DTLYADSVKGRFTISRDN AKTTLYLQMN SLRPEDTAVYYCTIGGSLSRSSQGT LVT SS (SEQ ID NO: 132) HCDR1: SFGMS (SEQ ID NO: 122) or GFTFRSFGMS (SEQ ID NO: 128) HCDR2: SISGSGDTL (SEQ ID NO: 124) or SISGSGSDTLYADSVKG (SEQ ID NO: 125) HCDR3: GGSLSR (SEQ ID NO: 126)	N/A
Ablynx ALB9	EVQLVESGGGLVQP GNSLR LSCAASGFT FSSFGMSWVRQAPGKGLEWVSSISGSGS	N/A

TABLE 3-continued

Sequences of Exemplary Third Antigen-Binding Sites		
Antigen-Binding Site	VH and HCDRs	VL and LCDRs
	DTLYADSVKGRFTISRDNKNTLYLQM NSLRPEDTAVYYCTIGGSLRSSQGTLV VSS (SEQ ID NO: 133) HCDR1: SFGMS (SEQ ID NO: 122) or GFTFSSFGMS (SEQ ID NO: 123) HCDR2: SISGSGSDTL (SEQ ID NO: 124) or SISGSGSDTLYADSVKG (SEQ ID NO: 125) HCDR3: GGSLSR (SEQ ID NO: 126)	
Ablynx ALB10	EVQLVESGGGLVQPGNSLRSLSCAASGFT FSSFGMSWVRQAPGKGLEWVSSISGSGS DTLYADSVKGRFTISRDNKNTLYLQM NSLRPEDTAVYYCTIGGSLRSSQGTLV TVSS (SEQ ID NO: 134) HCDR1: SFGMS (SEQ ID NO: 122) or GFTFSSFGMS (SEQ ID NO: 123) HCDR2: SISGSGSDTL (SEQ ID NO: 124) or SISGSGSDTLYADSVKG (SEQ ID NO: 125) HCDR3: GGSLSR (SEQ ID NO: 126)	N/A
Ablynx ALB23	EVQLLESGGGLVQPGGSLRLSCAASGFT FRSFGMSWVRQAPGKGPWVSSISGSGS DTLYADSVKGRFTISRDNKNTLYLQMN SLRPEDTAVYYCTIGGSLRSSQGTLVTV SS (SEQ ID NO: 135) HCDR1: SFGMS (SEQ ID NO: 122) or GFTFRSFGMS (SEQ ID NO: 128) HCDR2: SISGSGSDTL (SEQ ID NO: 124) or SISGSGSDTLYADSVKG (SEQ ID NO: 125) HCDR3: GGSLSR (SEQ ID NO: 126)	N/A
Ablynx mAb-1	EVQLVESGGGVVQPGNSLRSLSCAASGFT FSSFGMSWVRQAPGKGLEWVSSISGSGS DTLYADSVKGRFTISRDNKNTLYLQMN SLRPEDTAVYYCTIGGSLRSSQGTLVTV SSA (SEQ ID NO: 167) HCDR1: SFGMS (SEQ ID NO: 122) or GFTFSSFGMS (SEQ ID NO: 123) HCDR2: SISGSGSDTL (SEQ ID NO: 124) or SISGSGSDTLYADSVKG (SEQ ID NO: 125) HCDR3: GGSLSR (SEQ ID NO: 126)	N/A
Ablynx mAb-2	EVQLVESGGGVVQPGGSLRLSCAASGFT FRSFGMSWVRQAPGKGPWVSSISGSGS DTLYADSVKGRFTISRDNKNTLYLQMN SLRPEDTAVYYCTIGGSLRSSQGTLVTV SSA (SEQ ID NO: 168) HCDR1: SFGMS (SEQ ID NO: 122) or GFTFRSFGMS (SEQ ID NO: 128) HCDR2: SISGSGSDTL (SEQ ID NO: 124) or SISGSGSDTLYADSVKG (SEQ ID NO: 125) HCDR3: GGSLSR (SEQ ID NO: 126)	N/A
Ablynx mAb-3	EVQLVESGGGLVQPGNSLRSLSCAASGFT FSSFGMSWVRQAPGKGLEWVSSISGSGS DTLYADSVKGRFTISRDNKNTLYLQMN SLRPEDTAVYYCTIGGSLRSSQGTLVKV SSA (SEQ ID NO: 169) HCDR1: SFGMS (SEQ ID NO: 122) or GFTFSSFGMS (SEQ ID NO: 123) HCDR2: SISGSGSDTL (SEQ ID NO: 124) or SISGSGSDTLYADSVKG (SEQ ID NO: 125) HCDR3: GGSLSR (SEQ ID NO: 126)	N/A
Ablynx mAb-4	EVQLVESGGGVVQPGNSLRSLSCAASGFT FSSFGMSWVRQAPGKGLEWVSSISGSGS DTLYADSVKGRFTISRDNKNTLYLQMN SLRPEDTAVYYCTIGGSLRSSQGTLVTV	N/A

TABLE 3-continued

Sequences of Exemplary Third Antigen-Binding Sites		
Antigen-Binding Site	VH and HCDRs	VL and LCDRs
	SS (SEQ ID NO: 170) HCDR1: SFGMS (SEQ ID NO: 122) or GFTFSSFGMS (SEQ ID NO: 123) HCDR2: SISGSGSDL (SEQ ID NO: 124) or SISGSGSDTLYADSVKG (SEQ ID NO: 125) HCDR3: GGSLSR (SEQ ID NO: 126)	
Ablynx mAb-5	EVQLVESGGGVVQPGNSLRSLSCAASGFT FSSFGMSWVRQAPGKGLEWVSSISGSGS DTLYADSVKGRFTISRDNKTTLYLQMN SLRPEDTALYYCTIGGSLSRSSQGLTVTV SSA (SEQ ID NO: 171) HCDR1: SFGMS (SEQ ID NO: 122) or GFTFSSFGMS (SEQ ID NO: 123) HCDR2: SISGSGSDL (SEQ ID NO: 124) or SISGSGSDTLYADSVKG (SEQ ID NO: 125) HCDR3: GGSLSR (SEQ ID NO: 126)	N/A
Ablynx mAb-6	EVQLVESGGGVVQPGNSLRSLSCAASGFT FSSFGMSWVRQAPGKGLEWVSSISGSGS DTLYADSVKGRFTISRDNKTTLYLQMN SLRPEDTALYYCTIGGSLSRSSQGLTVTV SSAA (SEQ ID NO: 172) HCDR1: SFGMS (SEQ ID NO: 122) or GFTFSSFGMS (SEQ ID NO: 123) HCDR2: SISGSGSDL (SEQ ID NO: 124) or SISGSGSDTLYADSVKG (SEQ ID NO: 125) HCDR3: GGSLSR (SEQ ID NO: 126)	N/A
Ablynx mAb-7	EVQLVESGGGVVQPGNSLRSLSCAASGFT FSSFGMSWVRQAPGKGLEWVSSISGSGS DTLYADSVKGRFTISRDNKTTLYLQMN SLRPEDTALYYCTIGGSLSRSSQGLTVTV SSAAA (SEQ ID NO: 173) HCDR1: SFGMS (SEQ ID NO: 122) or GFTFSSFGMS (SEQ ID NO: 123) HCDR2: SISGSGSDL (SEQ ID NO: 124) or SISGSGSDTLYADSVKG (SEQ ID NO: 125) HCDR3: GGSLSR (SEQ ID NO: 126)	N/A
Ablynx mAb-8	EVQLVESGGGVVQPGNSLRSLSCAASGFT FSSFGMSWVRQAPGKGLEWVSSISGSGS DTLYADSVKGRFTISRDNKTTLYLQMN SLRPEDTALYYCTIGGSLSRSSQGLTVTV SSG (SEQ ID NO: 174) HCDR1: SFGMS (SEQ ID NO: 122) or GFTFSSFGMS (SEQ ID NO: 123) HCDR2: SISGSGSDL (SEQ ID NO: 124) or SISGSGSDTLYADSVKG (SEQ ID NO: 125) HCDR3: GGSLSR (SEQ ID NO: 126)	N/A
Ablynx mAb-9	EVQLVESGGGVVQPGNSLRSLSCAASGFT FSSFGMSWVRQAPGKGLEWVSSISGSGS DTLYADSVKGRFTISRDNKTTLYLQMN SLRPEDTALYYCTIGGSLSRSSQGLTVTV SSGG (SEQ ID NO: 175) HCDR1: SFGMS (SEQ ID NO: 122) or GFTFSSFGMS (SEQ ID NO: 123) HCDR2: SISGSGSDL (SEQ ID NO: 124) or SISGSGSDTLYADSVKG (SEQ ID NO: 125) HCDR3: GGSLSR (SEQ ID NO: 126)	N/A
Ablynx mAb-10	EVQLVESGGGVVQPGNSLRSLSCAASGFT FSSFGMSWVRQAPGKGLEWVSSISGSGS DTLYADSVKGRFTISRDNKTTLYLQMN SLRPEDTALYYCTIGGSLSRSSQGLTVTV SSGGG (SEQ ID NO: 176) HCDR1: SFGMS (SEQ ID NO: 122) or	N/A

TABLE 3-continued

Sequences of Exemplary Third Antigen-Binding Sites		
Antigen-Binding Site	VH and HCDRs	VL and LCDRs
	GFTFSSFGMS (SEQ ID NO: 123) HCDR2: SISGSGDTL (SEQ ID NO: 124) or SISGSGSDTLYADSVKG (SEQ ID NO: 125) HCDR3: GGSLSR (SEQ ID NO: 126)	
Ablynx PMP6A6	AVQLVESGGGLVQPQNSLRSLSCAASGFT FRSFGMSWVRQAPGKEPEWVSSISGSGS DTLYADSVKGRFTISRDNAKNTLYLQMN SLKPEDTAVYYCTIGGSLRSRSGTQVT VSS (SEQ ID NO: 136) HCDR1: SFGMS (SEQ ID NO: 122) or GFTFRSFGMS (SEQ ID NO: 128) HCDR2: SISGSGDTL (SEQ ID NO: 124) or SISGSGSDTLYADSVKG (SEQ ID NO: 125) HCDR3: GGSLSR (SEQ ID NO: 126)	N/A
Ablynx PMP6C1	AVQLVDSGGGLVQPGSLRLSCAASGFS FGSFGMSWVRQYPGKEPEWVSINGRG DDTRYADSVKGRFSISRDNAKNTLYLQ MNSLKPEDTAEYYCTIGRSVRSRSTQGT QVTVSS (SEQ ID NO: 137) HCDR1: SFGMS (SEQ ID NO: 122) or GFSFGSFGMS (SEQ ID NO: 138) HCDR2: SINGRGDDTR (SEQ ID NO: 139) or SINGRGDDTRYADSVKG (SEQ ID NO: 140) HCDR3: GRSVRS (SEQ ID NO: 141)	N/A
Ablynx PMP6G8	AVQLVESGGGLVQPGSLRLTCTASGFT FRSFGMSWVRQAPGKDQEWVSAISADS STKNYADSVKGRFTISRDNAKKMLYLE MNSLKPEDTAVYYCVIGRGPSSPGTQV TVSS (SEQ ID NO: 142) HCDR1: SFGMS (SEQ ID NO: 122) or GFTFRSFGMS (SEQ ID NO: 128) HCDR2: AISADSSTKN (SEQ ID NO: 143) or AISADSSTKNYADSVKG (SEQ ID NO: 144) HCDR3: GRGSP (SEQ ID NO: 145)	N/A
Ablynx PMP6A5	QVQLAESGGGLVQPGSLRLTCTASGFT FGSFGMSWVRQAPGEGLEWVSAISADSS DKRYADSVKGRFTISRDNAKKMLYLEM NSLKSEDTAVYYCVIGRGPASPGTQVT VSS (SEQ ID NO: 146) HCDR1: SFGMS (SEQ ID NO: 122) or GFTFGSFGMS (SEQ ID NO: 147) HCDR2: AISADSSDKR (SEQ ID NO: 148) or AISADSSDKRYADSVKG (SEQ ID NO: 149) HCDR3: GRGSP (SEQ ID NO: 145)	N/A
Ablynx PMP6G7	QVQLVESGGGLVQPGSLRLSCAASGFT FSNYWMYWRVAPGKGLERISRDISTG GGSYYADSVKGRFTISRDNAKNTLYLQ MNSLKPEDTALYYCAKDREAQVDTLDF DYRGQGTQVTVSS (SEQ ID NO: 150) HCDR1: NYWMY (SEQ ID NO: 151) or GFTFSNYWMY (SEQ ID NO: 152) HCDR2: RDISTGGGYSY (SEQ ID NO: 153) or RDISTGGGYSYADSVKG (SEQ ID NO: 154) HCDR3: DREAQVDTLDFDY (SEQ ID NO: 155)	N/A
Ablynx PMP6A8	AVQLVESGGGLVQGGSLRLACAASERI FDLNLMGWYRQPGNERELVATCITVG DSTNYADSVKGRFTISMDYTKQTVYLH MNSLRPEDTGLYYCKIRRTWHSSELWGQ GTQVTVSS (SEQ ID NO: 156) HCDR1: LNLMG (SEQ ID NO: 157) or SERIFDLNLMG (SEQ ID NO: 158)	N/A

TABLE 3-continued

Sequences of Exemplary Third Antigen-Binding Sites		
Antigen-Binding Site	VH and HCDRs	VL and LCDRs
	HCDR2: TCITVGDSTN (SEQ ID NO: 159) or TCITVGDSTNYADSVKG (SEQ ID NO: 160) HCDR3: RRTWHSEL (SEQ ID NO: 161)	
Ablynx PMP6C1	EVQLVESGGGLVQEGGSLRLCAASERI WDINLLGWYRQPGNERELVATITVGD STSYADSVKGRFTISRQYDKNTLYLQMN SLRPEDTGLYYCKIRRTWHSELWGQGT QVTVSS (SEQ ID NO: 162) HCDR1: INLLG (SEQ ID NO: 163) or SERIWDINLLG (SEQ ID NO: 164) HCDR2: TITVGDSTS (SEQ ID NO: 165) or TITVGDSTSYADSVKG (SEQ ID NO: 166) HCDR3: RRTWHSEL (SEQ ID NO: 161)	N/A
Harpoon mAb-1	EVQLVESGGGLVQPGNSLRSLCAASGFT FSKFGMSWVRQAPGKLEWVSSISGSGR DTLYAESVKGRFTISRDNKTTLYLQMN SLRPEDTAVYYCTIGGSLSVSSQGTLLTV SS (SEQ ID NO: 177) HCDR1: GFTFSKFGMS (SEQ ID NO: 178) HCDR2: SISGSGRDTLYAESVK (SEQ ID NO: 179) HCDR3: GGSLSV (SEQ ID NO: 180)	N/A
Harpoon mAb-2	EVQLVESGGGLVQPGNSLRSLCAASGFT FSRFGMSWVRQAPGKLEWVSSISGSGS DTLYAESVKGRFTISRDNKTTLYLQMN SLRPEDTAVYYCTIGGSLSRSSQGTLLTV SS (SEQ ID NO: 181) HCDR1: GFTFSRFGMS (SEQ ID NO: 182) HCDR2: SISGSGSDTLYAESVK (SEQ ID NO: 183) HCDR3: GGSLSR (SEQ ID NO: 126)	N/A
Harpoon mAb-3	EVQLVESGGGLVQPGNSLRSLCAASGFT FSKFGMSWVRQAPGKLEWVSSISGSGT DTLYAESVKGRFTISRDNKTTLYLQMN SLRPEDTAVYYCTIGGSLSRSSQGTLLTV SS (SEQ ID NO: 184) HCDR1: GFTFSKFGMS (SEQ ID NO: 178) HCDR2: SISGSGDTLYAESVK (SEQ ID NO: 185) HCDR3: GGSLSR (SEQ ID NO: 126)	N/A
Domantis DOM7h- 22	EVQLLESGGGLVQPGGSLRLCAASGFT FSKYWMSWVRQAPGKLEWVSSIDFM GPHTYYADSVKGRFTISRDNKNTLYLQ MNSLRAEDTAVYYCAKGRSMLPMKG KFDYWGQGTLLTVSS (SEQ ID NO: 283)	N/A
Domantis DOM7h- 26	EVQLLESGGGLVQPGGSLRLSCTASGFT FDEYNMSWVRQAPGKLEWVSTILPHG DRTYADSVKGRFTISRDNKNTLYLQ NSLRAEDTAVYYCAQDPLYRFDYWGQ GTLTVSS (SEQ ID NO: 284)	
Domantis DOM7h- 2	N/A	DIQMTQSPSSLSASVGDRVTITCRAS QKIATYLNWYQQKPGKAPKLLIYR SSSLQSAVPSRFSGSGSGTVFTLTIS LQPEDFATYYCQQTAYVPPFTFGQG TKVEIKR (SEQ ID NO: 285)
Domantis DOM7h- 8	N/A	DIQMTQSPSSLSASVGDRVTITCRAS QSISSYLNWYQQKPGKAPKLLIYR SPLQSGVPSRFSGSGSGTDFTLTISS LPEDFATYYCQQTIRVPPFTFGQGT KVEIKR (SEQ ID NO: 286)

TABLE 3-continued

Sequences of Exemplary Third Antigen-Binding Sites		
Antigen-Binding Site	VH and HCDRs	VL and LCDRs
MSA21	QVQLQESGGGLVQPGGSLRLSCEASGFT FSRFGMTWVRQAPGKGVWVSGISSLG DSTLYADSVKGRFTISRDNKNTLYLQM NSLKPEDTAVYYCTIGGSLNPGGQGTQV TVSS (SEQ ID NO: 287)	N/A
UCB mAb-1	EVQLLESGGGLVQPGGSLRLSCAVSGID LSNYAINWVRQAPGKCLEWIGI IWASGT TFYATWAKGRFTISRDNKNTVYLQMN SLRAEDTAVYYCARTVPGYSTAPYFDL WGQGLTLTVSS (SEQ ID NO: 288)	DIQMTQSPSSVSASVGDRTITCQSS PSVWSNFLSWYQQKPKAPKLLIY EASKLTSGVPSRFSGSGSGTDFTLTIT SSLQPEDFATYYCGGYSISDTTF GCGTKVEIKRT (SEQ ID NO: 289)
UCB mAb-2	EVQLLESGGGLVQPGGSLRLSCAVSGID LSNYAINWVRQAPGKCLEWIGI IWASGT TFYATWAKGRFTISRDNKNTVYLQMN SLRAEDTAVYYCARTVPGYSTAPYFDL WGQGLTLTVSS (SEQ ID NO: 290)	DIQMTQSPSSVSASVGDRTITCQSS PSVWSNFLSWYQQKPKAPKLLIY EASKLTSGVPSRFSGSGSGTDFTLTIT SSLQPEDFATYYCGGYSISDTTF GGGTKVEIKRT (SEQ ID NO: 291)

Where the VL and LCDR sequences are noted as "N/A," the antigen-binding site is an sdAb having a VH (e.g., VHH) only. Where the VH and HCDR sequences are noted as "N/A," the antigen-binding site is an sdAb having a VL only.

In certain embodiments, the third antigen-binding site comprises a VH comprising complementarity determining regions HCDR1, HCDR2, and HCDR3, and a VL comprising complementarity determining regions LCDR1, LCDR2, and LCDR3. In certain embodiments, the VH comprises an amino acid sequence at least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%) identical to the VH of an antibody provided in Table 3, and the VL comprises an amino acid sequence at least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%) identical to the VL of the antibody provided in Table 3. In certain embodiments, the VH comprises the HCDR1, HCDR2, and HCDR3 sequences of the antibody provided in Table 3, and the VL comprises the LCDR1, LCDR2, and LCDR3 sequences of the antibody provided in Table 3. In certain embodiments, the VH comprises the amino acid sequence of the VH of an antibody provided in Table 3, and the VL comprises the amino acid sequence of the VL of the antibody provided in Table 3.

In other embodiments, the third antigen-binding site comprises an sdAb comprising a VH comprising complementarity determining regions HCDR1, HCDR2, and HCDR3. In certain embodiments, the VH comprises an amino acid sequence at least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%) identical to the VH of an sdAb antibody provided in Table 3. In certain embodiments, the VH comprises the HCDR1, HCDR2, and HCDR3 sequences of the antibody provided in Table 3. In certain embodiments, the VH comprises the amino acid sequence of the VH of an sdAb provided in Table 3.

In certain embodiments, the VH of the third antigen-binding site comprises an amino acid sequence at least 60% (e.g., at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%) identical to SEQ ID NO: 121. In certain embodiments,

the VH comprises HCDR1, HCDR2, and HCDR3 sequences set forth in SEQ ID NOs: 122 or 123, 124 or 125, and 126, respectively. In certain embodiments, the VH comprises HCDR1, HCDR2, and HCDR3 sequences set forth in SEQ ID NOs: 122, 125, and 126, respectively. In certain embodiments, the VH comprises HCDR1, HCDR2, and HCDR3 sequences set forth in SEQ ID NOs: 123, 124, and 126, respectively. In certain embodiments, the VH comprises the amino acid sequence set forth in SEQ ID NO: 121.

In certain embodiments, the third antigen-binding site competes for binding serum albumin (e.g., HSA) with an antibody or antigen-binding fragment thereof comprising the VH, VL and/or scFv sequences provided in Table 3.

In certain embodiments, the third antigen-binding site of the multi-specific binding protein binds serum albumin (e.g., HSA) with a dissociation constant (K_D) of about 0.1 nM-about 100 μ M. The K_D can be measured by a method known in the art. In certain embodiments, the K_D is measured by SPR to serum albumin or a fragment thereof immobilized on a chip.

In certain embodiments, the third antigen-binding site binds serum albumin with a K_D of lower than or equal to 20 nM, 15 nM, 10 nM, 9 nM, 8 nM, 7 nM, 6 nM, 5 nM, 4 nM, 3 nM, 2 nM, 1 nM, 0.9 nM, 0.8 nM, 0.7 nM, 0.6 nM, 0.5 nM, 0.4 nM, 0.3 nM, 0.2 nM, 0.1 nM, 90 pM, 80 pM, 70 pM, 60 pM, 50 pM, 40 pM, 30 pM, 20 pM, or 10 pM. For example, in certain embodiments, the third antigen-binding site binds serum albumin with a K_D of about 10 pM-about 1 nM, about 10 pM-about 0.9 nM, about 10 pM-about 0.8 nM, about 10 pM-about 0.7 nM, about 10 pM-about 0.6 nM, about 10 pM-about 0.5 nM, about 10 pM-about 0.4 nM, about 10 pM-about 0.3 nM, about 10 pM-about 0.2 nM, about 10 pM-about 0.1 nM, about 10 pM-about 50 pM, 0.1 nM-about 10 nM, about 0.1 nM-about 9 nM, about 0.1 nM-about 8 nM, about 0.1 nM-about 7 nM, about 0.1 nM-about 6 nM, about 0.1 nM-about 5 nM, about 0.1 nM-about 4 nM, about 0.1 nM-about 3 nM, about 0.1 nM-about 2 nM, about 0.1 nM-about 1 nM, about 0.1 nM-about 0.5 nM, about 0.5 nM-about 10 nM, about 1 nM-about 10 nM, about 2 nM-about 10 nM, about 3 nM-about 10 nM, about 4 nM-about 10 nM, about 5 nM-about 10 nM, about 6 nM-about 10 nM,

about 7 nM-about 10 nM, about 8 nM-about 10 nM, about 9 nM-about 10 nM, about 1 nM-about 15 nM, about 2 nM-about 15 nM, about 3 nM-about 15 nM, about 4 nM-about 15 nM, about 5 nM-about 15 nM, about 6 nM-about 15 nM, about 7 nM-about 15 nM, about 8 nM-about 15 nM, about 9 nM-about 15 nM, about 10 nM-about 15 nM, about 11 nM-about 15 nM, about 12 nM-about 15 nM, about 13 nM-about 15 nM, about 14 nM-about 15 nM, about 1 nM-about 20 nM, about 2 nM-about 20 nM, about 3 nM-about 20 nM, about 4 nM-about 20 nM, about 5 nM-about 20 nM, about 6 nM-about 20 nM, about 7 nM-about 20 nM, about 8 nM-about 20 nM, about 9 nM-about 20 nM, about 10 nM-about 20 nM, about 11 nM-about 20 nM, about 12 nM-about 20 nM, about 13 nM-about 20 nM, about 14 nM-about 20 nM, about 15 nM-about 20 nM, about 16 nM-about 20 nM, about 17 nM-about 20 nM, about 18 nM-about 20 nM, or about 19 nM-about 20 nM.

In certain embodiments, the third antigen-binding site binds serum albumin with a K_D greater than or equal to 10 nM, 20 nM, 30 nM, 40 nM, 50 nM, 60 nM, 70 nM, 80 nM, 90 nM, or 100 nM. In certain embodiments, the third antigen-binding site binds serum albumin with a K_D of about 10 nM-about 1000 nM, about 10 nM-about 900 nM, about 10 nM-about 800 nM, about 10 nM-about 700 nM, about 10 nM-about 600 nM, about 10 nM-about 500 nM, about 10 nM-about 400 nM, about 10 nM-about 300 nM, about 10 nM-about 200 nM, about 10 nM-about 100 nM, about 10 nM-about 50 nM, about 50 nM-about 1000 nM, about 100 nM-about 1000 nM, about 200 nM-about 1000 nM, about 300 nM-about 1000 nM, about 400 nM-about 1000 nM, about 500 nM-about 1000 nM, about 600 nM-about 1000 nM, about 700 nM-about 1000 nM, about 800 nM-about 1000 nM, or about 900 nM-about 1000 nM.

It is understood that the binding affinity to serum albumin of the third antigen-binding site alone may be different from the binding affinity of the same antigen-binding site in the context of the multi-specific binding protein disclosed herein, possibly due to the conformational restraint from the other domains. The context-dependent binding affinity is described in subsection I.G titled "Binding Affinity."

In certain embodiments, the third antigen-binding site has a melting temperature of at least 50° C., at least 55° C., at least 56° C., at least 57° C., at least 58° C., at least 59° C., at least 60° C., at least 61° C., at least 62° C., at least 63° C., at least 64° C., at least 65° C., at least 70° C., at least 75° C., or at least 80° C. In certain embodiments, the third antigen-binding site has a melting temperature in the range of 50-80° C., 50-70° C., 50-65° C., 50-60° C., 50-55° C., 55-70° C., 55-65° C., 55-60° C., 56-65° C., 56-60° C., 57-65° C., 57-60° C., 58-65° C., 58-60° C., 59-65° C., 59-60° C., 60-80° C., 60-75° C., 60-70° C., 60-65° C., 65-80° C., 65-75° C., 65-70° C., 70-80° C., or 70-75° C.

D. Humanization and Deimmunization

The first, second, and/or third antigen-binding sites of the multi-specific binding protein disclosed herein may be humanized, for example, from one or more antigen-binding sites disclosed above, to optimize the immunogenicity and binding properties of the multi-specific binding protein, thereby enhancing the therapeutic index of the multi-specific binding protein.

"Humanized" antibodies or fragments thereof (e.g., Fv, Fab, Fab', F(ab')₂, sdAb, scFv or other antigen-binding subsequences of antibodies) contain mostly human sequences but also (a) minimal sequence(s) derived from non-human immunoglobulin(s). For the most part, humanized antibodies are human immunoglobulins (recipient antibody) in which residues from one or more hypervariable

regions (CDRs) of the recipient are replaced by residues from one or more hypervariable region of a non-human species (donor antibody) such as rodent (e.g., mouse, rat, or hamster), rabbit, or camelid (e.g., llama) having the desired specificity, affinity, and capacity. In some instances, Fv framework region (FR) residues of the human immunoglobulin are replaced by corresponding non-human residues. Furthermore, "humanized antibodies" as used herein may also comprise residues which are found neither in the recipient antibody nor the donor antibody. These modifications are made to further refine and optimize antibody performance. The humanized antibody may also comprise at least a portion of an immunoglobulin constant region (Fc), typically that of a human immunoglobulin. For further details, see Jones et al. (1986) *Nature*, 321:522-25; Reichmann et al. (1988) *Nature*, 332:323-29; and Presta (1992) *Curr. Op. Struct. Biol.*, 2:593-96.

Humanized antibodies may be produced using transgenic animals such as mice that express human heavy and light chain genes, but are incapable of expressing the endogenous mouse immunoglobulin heavy and light chain genes. Winter describes an exemplary CDR grafting method that may be used to prepare the humanized antibodies described herein (see, U.S. Pat. No. 5,225,539). All of the CDRs of a particular human antibody may be replaced with at least a portion of a non-human CDR, or only some of the CDRs may be replaced with non-human CDRs. It is only necessary to replace the number of CDRs required for binding of the humanized antibody to a predetermined antigen.

A humanized antibody can be optimized by the introduction of conservative substitutions, consensus sequence substitutions, germline substitutions and/or back mutations. Such altered immunoglobulin molecules can be made by any of several techniques known in the art, (e.g., Teng et al., *Proc. Natl. Acad. Sci. U.S.A.*, 80:7308-7312, 1983; Kozbor et al., *Immunology Today*, 4:7279, 1983; Olsson et al., *Meth. Enzymol.*, 92:3-16, 1982, and EP 239 400).

An antibody or fragment thereof may also be modified by specific deletion of human T cell epitopes in a process called "deimmunization." Methods of deimmunization have been disclosed, for example, in WO1998052976A1 or WO2000034317A2. Briefly, the heavy and light chain variable domains of an antibody can be analyzed for peptides that bind to MHC class II; these peptides represent potential T cell epitopes. For detection of potential T cell epitopes, a computer modeling approach termed "peptide threading" can be applied, and in addition a database of human MHC class II binding peptides can be searched for motifs present in the VH and VL sequences. These motifs bind to any of the 18 major MHC class II DR allotypes, and thus constitute potential T cell epitopes. Potential T cell epitopes detected can be eliminated by substituting small numbers of amino acid residues in the variable domains, or preferably, by single amino acid substitutions. Typically, conservative substitutions are made. Often, but not exclusively, an amino acid common to a position in human germline antibody sequences may be used. Human germline sequences are disclosed e.g., in Tomlinson, et al. (1992) *J. Mol. Biol.* 227:776-798; Cook, G. P. et al. (1995) *Immunol. Today* Vol. 16 (5): 237-242; and Tomlinson et al. (1995) *EMBO J.* 14:4628-38. The V BASE directory provides a comprehensive directory of human immunoglobulin variable region sequences (compiled by Tomlinson, L. A. et al. MRC Centre for Protein Engineering, Cambridge, UK). These sequences can be used as a source of human sequence, e.g., for

framework regions and CDRs. Consensus human framework regions can also be used, for example as described in U.S. Pat. No. 6,300,064.

E. Construct Formats

The first, second, and third antigen-binding sites may take various forms. In certain embodiments, the first, second, and/or third antigen-binding sites comprises two antibody variable domains (e.g., a VH and a VL). The VH and the VL can be mutated to introduce a disulfide bond (e.g., between H44 and L100) that stabilizes the antigen-binding site (see, Zhao et al. (2010) *Int. J. Mol. Sci.*, 12 (1): 1-11). In certain

embodiments, the first, second, and/or third antigen-binding sites comprises a single antibody variable domain (e.g., an sdAb).
In an antigen-binding site that contains a VH and a VL, the VH and the VL can be linked to form an scFv. The VH can be positioned N-terminal or C-terminal to the VL. The VH and the VL are typically linked through a linker, such as a peptide linker. Exemplary sequences of peptide linkers are provided in subsection I.F titled "Linkers." In certain embodiments, the VH of an antigen-binding domain is connected to the VL of the antigen-binding domain through a peptide linker having an amino acid sequence listed in Table 4. In particular embodiments, the VH of an antigen-binding domain is connected to the VL of the antigen-binding domain through a peptide linker having the amino acid sequence of SEQ ID NO: 298, 299, or 302, wherein the VH is positioned N-terminal to the VL. In other particular embodiments, the VH of an antigen-binding domain is connected to the VL of the antigen-binding domain through a peptide linker having the amino acid sequence of SEQ ID NO: 298, 299, or 302, wherein the VH is positioned C-terminal to the VL.

Alternatively, the VH and the VL may be present on separate polypeptide chains, and the formation of a VH-VL complex may be facilitated by additional domains, such as antibody constant regions CH1 and CL. Accordingly, in certain embodiments, the multi-specific binding protein comprises an Fab comprising a VH and a VL disclosed herein.

In certain embodiments, a multi-specific binding protein of the present invention comprises a first antigen-binding site comprising a single antibody variable domain, a second antigen-binding site comprising a single antibody variable domain, and a third antigen-binding site comprising a single antibody variable domain. In certain embodiments, the multi-specific binding protein comprises a first antigen-binding site in an sdAb format, a second antigen-binding site in an sdAb format, and a third antigen-binding site in an sdAb format.

In certain embodiments, a multi-specific binding protein of the present invention comprises a first antigen-binding site comprising a single antibody variable domain, a second antigen-binding site comprising a single antibody variable domain, and a third antigen-binding site comprising two antibody variable domains. In certain embodiments, the multi-specific binding protein comprises a first antigen-binding site in an sdAb format, a second antigen-binding site in an sdAb format, and a third antigen-binding site in an scFv format.

In certain embodiments, a multi-specific binding protein of the present invention comprises a first antigen-binding site comprising a single antibody variable domain, a second antigen-binding site comprising two antibody variable domains, and a third antigen-binding site comprising a single antibody variable domain. In certain embodiments, the multi-specific binding protein comprises a first antigen-

binding site in an sdAb format, a second antigen-binding site in an scFv format, and a third antigen-binding site in an sdAb format.

In certain embodiments, a multi-specific binding protein of the present invention comprises a first antigen-binding site comprising a single antibody variable domain, a second antigen-binding site comprising two antibody variable domains, and a third antigen-binding site comprising two antibody variable domains. In certain embodiments, the multi-specific binding protein comprises a first antigen-binding site in an sdAb format, a second antigen-binding site in an scFv format, and a third antigen-binding site in an scFv format.

In certain embodiments, a multi-specific binding protein of the present invention comprises a first antigen-binding site comprising two antibody variable domains, a second antigen-binding site comprising a single antibody variable domain, and a third antigen-binding site comprising a single antibody variable domain. In certain embodiments, the multi-specific binding protein comprises a first antigen-binding site in an scFv format, a second antigen-binding site in an sdAb format, and a third antigen-binding site in an sdAb format.

In certain embodiments, a multi-specific binding protein of the present invention comprises a first antigen-binding site comprising two antibody variable domains, a second antigen-binding site comprising a single antibody variable domain, and a third antigen-binding site comprising two antibody variable domains. In certain embodiments, the multi-specific binding protein comprises a first antigen-binding site in an scFv format, a second antigen-binding site in an sdAb format, and a third antigen-binding site in an scFv format.

In certain embodiments, a multi-specific binding protein of the present invention comprises a first antigen-binding site comprising two antibody variable domains, a second antigen-binding site comprising two antibody variable domains, and a third antigen-binding site comprising a single antibody variable domain. In certain embodiments, the multi-specific binding protein comprises a first antigen-binding site in an scFv format, a second antigen-binding site in an scFv format, and a third antigen-binding site in an sdAb format.

In certain embodiments, a multi-specific binding protein of the present invention comprises a first antigen-binding site comprising two antibody variable domains, a second antigen-binding site comprising two antibody variable domains, and a third antigen-binding site comprising two antibody variable domains. In certain embodiments, the multi-specific binding protein comprises a first antigen-binding site in an scFv format, a second antigen-binding site in an scFv format, and a third antigen-binding site in an scFv format.

The three antigen-binding sites of the multi-specific binding protein can be linked in any one of the following orientations in an amino-to-carboxyl direction:

- (i) the first antigen-binding site (CD19 binding domain)-the second antigen-binding site (CD3 binding domain)-the third antigen-binding site (serum albumin binding domain);
- (ii) the first antigen-binding site (CD19 binding domain)-the third antigen-binding site (serum albumin binding domain)-the second antigen-binding site (CD3 binding domain);

- (iii) the second antigen-binding site (CD3 binding domain)-the first antigen-binding site (CD19 binding domain)-the third antigen-binding site (serum albumin binding domain);
- (iv) the second antigen-binding site (CD3 binding domain)-the third antigen-binding site (serum albumin binding domain)-the first antigen-binding site (CD19 binding domain);
- (v) the third antigen-binding site (serum albumin binding domain)-the first antigen-binding site (CD19 binding domain)-the second antigen-binding site (CD3 binding domain); and
- (vi) the third antigen-binding site (serum albumin binding domain)-the second antigen-binding site (CD3 binding domain)-the first antigen-binding site (CD19 binding domain), wherein the dashes above represent a peptide bond and/or a linker (e.g., peptide linker).

In certain embodiments, the third antigen-binding site is not positioned between the first antigen-binding site and the second antigen-binding site. In certain embodiments, the third antigen-binding site is positioned N-terminal to both the first antigen-binding site and the second antigen-binding site or C-terminal to both the first antigen-binding site and the second antigen-binding site. In certain embodiments, the third antigen-binding site is positioned N-terminal to both the first antigen-binding site and the second antigen-binding site. In certain embodiments, the third antigen-binding site is positioned C-terminal to both the first antigen-binding site and the second antigen-binding site.

The position (N-terminal or C-terminal) of one antigen-binding site relative to another is determined under the definitions of "N-terminal" and "C-terminal" as known in the art if a single polypeptide chain comprises both antigen-binding sites. It is understood that if an antigen-binding site comprises two separate polypeptide chains, its position (N-terminal or C-terminal) relative to another antigen-binding site (either having a single polypeptide chain or two polypeptide chains) can be similarly determined if a single polypeptide chain comprises at least one polypeptide chain of the former and at least one polypeptide chain of the latter. It is further understood that if antigen-binding site A is N-terminal to antigen-binding site B and antigen-binding site B is N-terminal to antigen-binding site C, it is deemed that antigen-binding site A is positioned N-terminal to antigen-binding site C even if antigen-binding sites A and C are not present in any single, common polypeptide chain. More complex structures of multi-specific binding proteins are also contemplated, some of which may have orientations difficult to characterize using the terms of "N-terminal" and "C-terminal" as described above due to, for example, different relative positions of two antigen-binding sites on one polypeptide chain versus another polypeptide chain, or the presence of a loop structure.

According to the present invention, the multi-specific binding proteins and its constituent binding domains are in the form of one or more polypeptides. Such polypeptides may include proteinaceous parts and non-proteinaceous parts (e.g., chemical linkers or chemical cross-linking agents such as glutaraldehyde). In certain embodiments, a multi-specific binding protein of the present invention includes a first antigen-binding site, a second antigen-binding site, and a third antigen-binding site, all of which are linked together to form a single polypeptide chain. In certain embodiments, the first, second, and third antigen-binding sites take the forms of scFv and/or sdAb, for example, in a combination as described above, to form a single polypeptide chain.

F. Linkers

As noted above, the antigen-binding sites of the multi-specific binding proteins of the present invention can be linked through a peptide bond or a linker (e.g., peptide linker). In certain embodiments, at least two adjacent antigen-binding sites are connected by a linker (e.g., peptide linker). In certain embodiments, each two adjacent antigen-binding sites are connected by a linker (e.g., peptide linker).

In certain embodiments, the three antigen-binding sites of the multi-specific binding protein can be linked by linkers (e.g., peptide linkers) denoted as L_1 and L_2 in any one of the following orientations in an amino-to-carboxyl direction:

- (i) the first antigen-binding site (CD19 binding domain)- L_1 -the second antigen-binding site (CD3 binding domain)- L_2 -the third antigen-binding site (serum albumin binding domain);
- (ii) the first antigen-binding site (CD19 binding domain)- L_1 -the third antigen-binding site (serum albumin binding domain)- L_2 -the second antigen-binding site (CD3 binding domain);
- (iii) the second antigen-binding site (CD3 binding domain)- L_1 -the first antigen-binding site (CD19 binding domain)- L_2 -the third antigen-binding site (serum albumin binding domain);
- (iv) the second antigen-binding site (CD3 binding domain)- L_1 -the third antigen-binding site (serum albumin binding domain)- L_2 -the first antigen-binding site (CD19 binding domain);
- (v) the third antigen-binding site (serum albumin binding domain)- L_1 -the first antigen-binding site (CD19 binding domain)- L_2 -the second antigen-binding site (CD3 binding domain); and
- (vi) the third antigen-binding site (serum albumin binding domain)- L_1 -the second antigen-binding site (CD3 binding domain)- L_2 -the first antigen-binding site (CD19 binding domain). It is appreciated that in a given construct, either L_1 or L_2 may be replaced with a peptide bond.

It is understood that if a single polypeptide chain comprises two adjacent antigen-binding sites, the peptide linker connecting the two antigen-binding sites represents the amino acid sequence between them. If an antigen-binding site comprises two separate polypeptide chains, one of which is present in a single, common polypeptide as an adjacent antigen-binding site or a polypeptide chain thereof, the peptide linker connecting the two antigen-binding sites represents the amino acid sequence between them in the common, single polypeptide.

In certain embodiments, the linkers L_1 and L_2 are peptide linkers. Suitable lengths of L_1 and L_2 can be independently selected. For example, in certain embodiments, L_1 and/or L_2 are about 50 or less amino acid residues in length. In certain embodiments, L_1 consists of about 50 or less amino acid residues. In certain embodiments, L_1 consists of about 20 or less amino acid residues. In certain embodiments, L_2 consists of about 50 or less amino acid residues. In certain embodiments, L_2 consists of about 20 or less amino acid residues. In certain embodiments, L_1 and L_2 independently consist of about 50 or less amino acid residues. In certain embodiments, L_1 and L_2 independently consist of about 20 or less amino acid residues.

In some embodiments, peptide linkers L_1 and L_2 have an optimized length and/or amino acid composition. In some embodiments, L_1 and L_2 are of the same length and have the same amino acid composition. In other embodiments, L_1 and L_2 are different. In certain embodiments, L_1 and/or L_2 are "short," i.e., consist of 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11 or 12 amino acid residues. Thus, in certain instances, the linkers

61

consist of about 12 or less amino acid residues. In certain embodiments, L_1 and/or L_2 are “long,” e.g., consist of 15, 20 or 25 amino acid residues. In some embodiments, L_1 and/or L_2 consist of about 3 to about 15, for example 8, 9 or 10 contiguous amino acid residues.

Regarding the amino acid composition of L_1 and L_2 , peptides are selected with properties that confer flexibility to multi-specific binding protein of the present invention, do not interfere with the binding domains as well as resist cleavage from proteases. For example, glycine and serine residues generally provide protease resistance. Examples of the linkers suitable for linking the domains in the multi-specific binding protein include but are not limited to $(GS)_n$, $(GGS)_n$, $(GGGS)_n$, $(GGSG)_n$, $(GGSGG)_n$, and $(GGGGS)_n$, wherein n is 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15,

62

16, 17, 18, 19, or 20. In some embodiments, L_1 and/or L_2 are independently selected from the peptide sequences listed in table 4. In some embodiments, L_1 and/or L_2 are independently selected from SEQ ID NOs: 292, 293, 294, 295, 296, 297, 298, 299, 300, 301, or 302. In some embodiments, L_1 and/or L_2 are independently selected from SEQ ID NOs: 298, 299, and 302. In some embodiments, L_1 and/or L_2 comprise the amino acid sequence of SEQ ID NO: 298, 299, or 302. In some embodiments, L_1 and/or L_2 consist of the amino acid sequence of 298, 299, or 302. In some embodiments, L_1 and L_2 each comprise the amino acid sequence of SEQ ID NO: 298, 299, or 302. In some embodiments, L_1 and L_2 each consist of the amino acid sequence of SEQ ID NO: 298, 299, or 302.

TABLE 4

Sequences of Exemplary Peptide Linkers			
Linker	SEQ ID NO	Length	Amino Acid Sequence
$(GS)_{10}$	292	20	GS GSGSGSGSGSGSGSGSGSGS
$(GGS)_{10}$	293	30	GGS GGS GGS GGS GGS GGS GGS GGS GGS GGS
$(GGGS)_{10}$	294	40	GGGS GGS GGS GGS GGS GGS GGS GGS GGS GGS GGGS
$(GGSG)_{10}$	295	40	GGS GGS GGS GGS GGS GGS GGS GGS GGS GGS GGSG
$(GGSGG)_{10}$	296	50	GGS GGS GGS GGS GGS GGS GGS GGS GGS GGS GGSGGGSGGGSGG
$(GGGGS)_{10}$	297	50	GGGS GGS GGS GGS GGS GGS GGS GGS GGS GGS GGGS GGS GGS GGS
$(GGGGS)_4$	298	20	GGGS GGS GGS GGS GGS GGS
$(GGGGS)_3$	299	15	GGGS GGS GGS GGS GGS
$(GGGGS)_{20}$	300	100	GGGS GGS GGS GGS GGS GGS GGS GGS GGS GGS GGGS GGS GGS GGS GGS GGS GGS GGS GGS GGS GGGS GGS GGS GGS GGS GGS GGS GGS GGS GGS
$(GGSGG)_{20}$	301	100	GGS GGS
Assymetrical linker	302	9	GGGS GGS

A linker, such as a peptide linker disclosed herein, can also be used to connect the VH and VL of an scFv, as mentioned in subsection I.E titled "Construct Formats."

G. Binding Affinity

In certain embodiments, the multi-specific binding protein binds CD19 (e.g., human CD19), CD3 (e.g., human CD3 and/or Macaca CD3), and/or serum albumin (e.g., HSA) with a K_D of about 0.1 nM-about 100 μ M. The K_D can be measured by a method known in the art, such as by SPR or by flow cytometry as described in Example 5 below.

In certain embodiments, the multi-specific binding protein binds CD19, CD3, and/or serum albumin with a K_D of lower than or equal to 20 nM, 15 nM, 10 nM, 9 nM, 8 nM, 7 nM, 6 nM, 5 nM, 4 nM, 3 nM, 2 nM, 1 nM, 0.9 nM, 0.8 nM, 0.7 nM, 0.6 nM, 0.5 nM, 0.4 nM, 0.3 nM, 0.2 nM, 0.1 nM, 90 pM, 80 pM, 70 pM, 60 pM, 50 pM, 40 pM, 30 pM, 20 pM, or 10 pM. For example, in certain embodiments, the multi-specific binding protein binds CD19, CD3, and/or serum albumin with a K_D of about 10 pM-about 1 nM, about 10 pM-about 0.9 nM, about 10 pM-about 0.8 nM, about 10 pM-about 0.7 nM, about 10 pM-about 0.6 nM, about 10 pM-about 0.5 nM, about 10 pM-about 0.4 nM, about 10 pM-about 0.3 nM, about 10 pM-about 0.2 nM, about 10 pM-about 0.1 nM, about 10 pM-about 50 pM, 0.1 nM-about 10 nM, about 0.1 nM-about 9 nM, about 0.1 nM-about 8 nM, about 0.1 nM-about 7 nM, about 0.1 nM-about 6 nM, about 0.1 nM-about 5 nM, about 0.1 nM-about 4 nM, about 0.1 nM-about 3 nM, about 0.1 nM-about 2 nM, about 0.1 nM-about 1 nM, about 0.1 nM-about 0.5 nM, about 0.5 nM-about 10 nM, about 1 nM-about 10 nM, about 2 nM-about 10 nM, about 3 nM-about 10 nM, about 4 nM-about 10 nM, about 5 nM-about 10 nM, about 6 nM-about 10 nM, about 7 nM-about 10 nM, about 8 nM-about 10 nM, or about 9 nM-about 10 nM.

In certain embodiments, the multi-specific binding protein binds CD19, CD3, and/or serum albumin with a K_D greater than or equal to 10 nM, 20 nM, 30 nM, 40 nM, 50 nM, 60 nM, 70 nM, 80 nM, 90 nM, or 100 nM. In certain embodiments, the multi-specific binding protein binds CD19, CD3, and/or serum albumin with a K_D of about 10 nM-about 1000 nM, about 10 nM-about 900 nM, about 10 nM-about 800 nM, about 10 nM-about 700 nM, about 10 nM-about 600 nM, about 10 nM-about 500 nM, about 10 nM-about 400 nM, about 10 nM-about 300 nM, about 10 nM-about 200 nM, about 10 nM-about 100 nM, about 10 nM-about 50 nM, about 50 nM-about 1000 nM, about 100 nM-about 1000 nM, about 200 nM-about 1000 nM, about 300 nM-about 1000 nM, about 400 nM-about 1000 nM, about 500 nM-about 1000 nM, about 600 nM-about 1000 nM, about 700 nM-about 1000 nM, about 800 nM-about 1000 nM, or about 900 nM-about 1000 nM.

In certain embodiments, the K_D of binding to CD19 or CD3 is measured in the absence of serum albumin (e.g., HSA). In certain embodiments, the K_D of binding to CD19 or CD3 is measured in substantial absence of serum albumin (e.g., HSA). In certain embodiments, the K_D of binding to CD19 or CD3 is measured in the presence of serum albumin (e.g., HSA), for example, in the presence of about 10 mg/mL, 15 mg/mL, 20 mg/mL, 25 mg/mL, 30 mg/mL, 35 mg/mL, 40 mg/mL, 45 mg/mL, or 50 mg/mL serum albumin (e.g., HSA).

In certain embodiments, the multi-specific binding protein of the present disclosure binds CD19, CD3, and/or serum albumin with a similar K_D value to that of the respective antigen-binding site alone or a monoclonal antibody having the same antigen-binding site. In certain embodiments, the K_D value of the multi-specific binding protein to CD19,

CD3, and/or serum albumin is increased by no more than 1.5 fold, 2 fold, 3 fold, 4 fold, 5 fold, 6 fold, 7 fold, 8 fold, 9 fold, 10 fold, 15 fold, 20 fold, 25 fold, 30 fold, 35 fold, 40 fold, 45 fold, or 50 fold compared to that of the respective antigen-binding site alone or a monoclonal antibody having the same antigen-binding site.

In certain embodiments, the multi-specific binding protein of the present disclosure binds CD19 and/or CD3 with a similar K_D value in the presence of serum albumin to that in the absence or substantial absence of serum albumin. In certain embodiments, the K_D value of the multi-specific binding protein for binding CD19 and/or CD3 in the presence of serum albumin is increased by no more than 1.5 fold, 2 fold, 3 fold, 4 fold, 5 fold, 6 fold, 7 fold, 8 fold, 9 fold, 10 fold, 15 fold, 20 fold, 25 fold, 30 fold, 35 fold, 40 fold, 45 fold, or 50 fold compared to that in the absence or substantial absence of serum albumin.

H. Therapeutic Activities of the Constructs

The multi-specific binding protein disclosed herein is designed to simultaneously bind B cells and T cells. Recruitment of T cells facilitates lysis of the B cells involving cytolytic synapse formation and delivery of perforin and granzymes. The engaged T cells are capable of serial target cell lysis, and are not affected by immune escape mechanisms interfering with peptide antigen processing and presentation, or clonal T cell differentiation; see, for example, WO2007042261A2. Accordingly, binding of the multi-specific binding proteins to the target B cells destroys the target cells and/or impairs the progression of B cell related diseases.

Cytotoxicity mediated by multi-specific binding proteins of the invention can be measured in various ways in vitro. Effector cells can be e.g., stimulated enriched (human) CD8 positive T cells or unstimulated (human) peripheral blood mononuclear cells (PBMC). If the target cells are of macaque origin or express or are transfected with macaque target cell surface antigen which is bound by the first domain, the effector cells should also be of macaque origin such as a macaque T cell line, e.g., 4119LnPx. The target cells should express CD19, e.g., human or macaque CD19. The target cells can be a cell line (such as CHO) which is stably or transiently transfected with CD19. Alternatively, the target cells can be a cell line naturally expressing CD19, such as B lymphocytes. The effector to target cell (E:T) ratio is usually about 10:1, but can also vary. Killing of the target cells can be measured in a ^{51}Cr -release assay (incubation time of about 18 hours) or in a FACS-based cytotoxicity assay (incubation time of about 48 hours). Other methods of measuring cell death are well-known to the skilled person, such as MTT or MTS assays, ATP-based assays including bioluminescent assays, the sulforhodamine B (SRB) assay, WST assay, clonogenic assay and the ECIS technology.

In certain embodiments, the cytotoxic activity mediated by the multi-specific binding protein disclosed herein is measured in a cell-based cytotoxicity assay described above. It is represented by the EC_{50} value, which corresponds to the half maximal effective concentration (concentration of the multi-specific binding protein which induces a cytotoxic response halfway between the baseline and maximum). In certain embodiments, the EC_{50} value of the multi-specific binding proteins is ≤ 5000 pM, for example, ≤ 4000 pM, ≤ 3000 pM, ≤ 2000 pM, ≤ 1000 pM, ≤ 500 pM, ≤ 400 pM, ≤ 300 pM, ≤ 200 pM, ≤ 100 pM, ≤ 50 pM, ≤ 20 pM, ≤ 10 pM, ≤ 5 pM, ≤ 4 pM, ≤ 3 pM, ≤ 2 pM, or ≤ 1 pM.

It is understood that an EC_{50} value is generally lower when stimulated/enriched CD8⁺ T cells are used as effector cells, compared with unstimulated PBMC. It is further

understood that the EC_{50} value is generally lower when the target cells express a high level of the target cell surface antigen compared with a low level of the target antigen. For example, when stimulated/enriched human $CD8^+$ T cells are used as effector cells (and either target cell surface antigen transfected cells such as CHO cells or target cell surface antigen positive human cell lines are used as target cells), the EC_{50} value of multi-specific binding protein is ≤ 1000 pM, for example, ≤ 500 pM, ≤ 250 pM, ≤ 100 pM, ≤ 50 pM, ≤ 10 pM, or ≤ 5 pM. When human PBMCs are used as effector cells, the EC_{50} value of the multi-specific binding protein is ≤ 5000 pM, for example, ≤ 4000 pM, ≤ 2000 pM, ≤ 1000 pM, ≤ 500 pM, ≤ 200 pM, ≤ 150 pM, ≤ 100 pM, ≤ 50 pM, ≤ 10 pM, or ≤ 5 pM. When a macaque T cell line such as LnPx4119 is used as effector cells, and a macaque target cell surface antigen transfected cell line such as CHO cells is used as target cell line, the EC_{50} value of the multi-specific binding protein is ≤ 2000 pM, for example, ≤ 1500 pM, ≤ 1000 pM, ≤ 500 pM, ≤ 300 pM, ≤ 250 pM, ≤ 100 pM, ≤ 50 pM, ≤ 10 pM, or ≤ 5 pM.

Accordingly, in certain embodiments, the EC_{50} value is measured using stimulated/enriched human $CD8^+$ T cells as effector cells. In certain embodiments, the EC_{50} value is measured using human PBMCs as effector cells. In certain embodiments, the EC_{50} value is measured using a macaque T cell line such as LnPx4119 as effector cells and cells (e.g., CHO cells) engineered to express macaque CD19 as target cells.

In certain embodiments, the multi-specific binding protein of the present invention does not induce or mediate lysis of cells that do not express CD19. The term “does not induce lysis” or “does not mediate lysis,” or grammatical equivalents thereof, means that the multi-specific binding protein, at a concentration of up to 500 nM, does not induce or mediate lysis of more than 30%, for example, no more than 30%, 25%, 20%, 15%, 10%, 9%, 8%, 7%, 6% or 5% of cells that do not express CD19, whereby lysis of a cell line that expresses CD19 is set to be 100%.

In certain embodiments, a multi-specific binding protein disclosed herein is more effective in killing CD19-expressing cells (e.g., cancer cells) than the corresponding respective anti-CD19 or anti-CD3 monoclonal antibody at the same molar concentration. In certain embodiments, the multi-specific binding protein is more effective in killing CD19-expressing cells (e.g., cancer cells) than a combination of the corresponding respective anti-CD19 and anti-CD3 monoclonal antibodies each at the same molar concentration.

The cytotoxic activity of the multi-specific binding protein can be measured in the presence or absence of serum albumin (e.g., HSA). In certain embodiments, the cytotoxic activity disclosed above is measured in the absence of serum albumin (e.g., HSA). In certain embodiments, the cytotoxic activity disclosed above is measured in substantial absence of serum albumin (e.g., HSA). In certain embodiments, the cytotoxic activity disclosed above is measured in the presence of serum albumin (e.g., HSA), for example, in the presence of about 10 mg/mL, 15 mg/mL, 20 mg/mL, 25 mg/mL, 30 mg/mL, 35 mg/mL, 40 mg/mL, 45 mg/mL, or 50 mg/mL serum albumin (e.g., HSA).

In certain embodiments, the multi-specific binding protein of the present disclosure kills CD19-expressing cells with a similar EC_{50} value in the presence of serum albumin to that in the absence or substantial absence of serum albumin. In certain embodiments, the EC_{50} value of the multi-specific binding protein for killing CD19-expressing cells in the presence of serum albumin is increased by no more than 1.5 fold, 2 fold, 3 fold, 4 fold, 5 fold, 6 fold, 7 fold, 8 fold, 9 fold,

10 fold, 15 fold, 20 fold, 25 fold, 30 fold, 35 fold, 40 fold, 45 fold, or 50 fold compared to that in the absence or substantial absence of serum albumin. It is understood that the presence of serum albumin (e.g., about 10 mg/mL, 15 mg/mL, 20 mg/mL, 25 mg/mL, 30 mg/mL, 35 mg/mL, 40 mg/mL, 45 mg/mL, or 50 mg/mL serum albumin) may also alter the EC_{50} value of a multi-specific binding protein nonspecifically. The nonspecific effect can be assessed by comparing the EC_{50} values of a control protein, which does not contain a serum albumin binding domain, in the presence and absence of serum albumin. In certain embodiments, the fold change is offset by the nonspecific effect of serum albumin on a control protein, such as a bispecific protein that binds CD19 and CD3.

I. Construct Size

In certain embodiments, the molecular weight of the multi-specific binding protein is from about 40 kD to about 100 kD. In certain embodiments, the molecular weight of the multi-specific binding protein is at least 60 kD, at least 65 kD, at least 70 kD, at least 75 kD, at least 80 kD, at least 85 kD, at least 90 kD, or at least 95 kD. It is understood that smaller size generally contributes to faster diffusion and tissue penetration, but size reduction may not be as critical for the purpose of treating the indications with substantial presence of target cells (e.g., cancer cells) in the blood.

In certain embodiments, the molecular weight of the multi-specific binding protein is from about 40 kD to about 90 kD, from about 40 kD to about 80 kD, from about 40 kD to about 70 kD, from about 40 kD to about 60 kD, from about 40 kD to about 50 kD, from about 50 kD to about 100 kD, from about 50 kD to about 90 kD, from about 50 kD to about 80 kD, from about 50 kD to about 70 kD, from about 50 kD to about 60 kD, from about 60 kD to about 100 kD, from about 60 kD to about 90 kD, from about 60 kD to about 80 kD, from about 60 kD to about 70 kD, from about 60 kD to about 60 kD, from about 60 kD to about 50 kD, from about 65 kD to about 100 kD, from about 65 kD to about 90 kD, from about 65 kD to about 80 kD, from about 65 kD to about 70 kD, from about 65 kD to about 60 kD, from about 65 kD to about 50 kD, from about 70 kD to about 100 kD, from about 70 kD to about 90 kD, from about 70 kD to about 80 kD, from about 70 kD to about 70 kD, from about 70 kD to about 60 kD, from about 70 kD to about 50 kD, from about 75 kD to about 100 kD, from about 75 kD to about 90 kD, from about 75 kD to about 80 kD, from about 75 kD to about 70 kD, from about 75 kD to about 60 kD, from about 75 kD to about 50 kD, from about 80 kD to about 100 kD, from about 80 kD to about 90 kD, from about 80 kD to about 80 kD, from about 80 kD to about 70 kD, from about 80 kD to about 60 kD, from about 80 kD to about 50 kD, from about 85 kD to about 100 kD, from about 85 kD to about 90 kD, from about 85 kD to about 80 kD, from about 85 kD to about 70 kD, from about 85 kD to about 60 kD, from about 85 kD to about 50 kD, from about 90 kD to about 100 kD, from about 90 kD to about 90 kD, from about 90 kD to about 80 kD, from about 90 kD to about 70 kD, from about 90 kD to about 60 kD, from about 90 kD to about 50 kD, from about 95 kD to about 100 kD, from about 95 kD to about 90 kD, from about 95 kD to about 80 kD, from about 95 kD to about 70 kD, from about 95 kD to about 60 kD, from about 95 kD to about 50 kD, or about 100 kD.

J. Serum Half-Life

Fusion proteins have been developed to increase the in vivo half-life of a small protein, particularly an antibody fragment. For example, fusion with a heterodimeric antibody Fc region, such as an Fc with one or more mutations that extend the in vivo half-life, is described in U.S. Patent Application Publication Nos. US20140302037A1, US20140308285A1, and PCT Publication Nos. WO2014144722A2, WO2014151910A1 and WO2015048272A1. An alternative strategy is fusion with human serum albumin (HSA) or an HSA-binding peptide (see, e.g., PCT Publication Nos. WO2013128027A1 and WO2014140358A1). The neonatal Fc receptor (FcRn) appears to be involved in prolonging the life-span of albumin in circulation (see, Chaudhury et al. (2003) J. Exp. Med., 3:315-22). Albumin and IgG bind noncooperatively to distinct sites of FcRn and form a tri-molecular (see id.).

Binding of human FcRn to HSA and to human IgG is pH dependent, stronger at acidic pH and weaker at neutral or physiological pH (see id.). This observation suggests that proteins and protein complexes containing albumin, similar to those containing IgG (particularly Fc), are protected from degradation through pH-sensitive interaction with FcRn (see id.). Using surface plasmon resonance (SPR) to measure the capacity of individual HSA domains to bind immobilized soluble human FcRn, it has been shown that FcRn and albumin interact via the D-III domain of albumin in a pH-dependent manner, on a site distinct from the IgG binding site (see, Chaudhury et al. (2006) *Biochemistry* 45:4983-90 and PCT Publication No. WO2008068280A1).

The present disclosure provides multi-specific binding proteins with extended half-life. In certain embodiments, the multi-specific binding protein has a serum half-life of at least 24, 36, 48, 60, 72, 84, or 96 hours. In certain embodiments, the multi-specific binding protein has a serum half-life of at least about 50 hours. In certain embodiments, the multi-specific binding protein has a serum half-life of at least about 100 hours. Methods of measuring serum half-life are known in the art, and exemplary methods are described in Example 4. In certain embodiments, the serum half-life is measured in a non-human primate. In certain embodiments, the serum half-life is measured in human.

In certain embodiments, 50 hours after intravenous administration to a subject, the serum concentration of the multi-specific binding protein is at least 10%, at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, or at least 90% of the serum concentration of the multi-specific binding protein 1 hour after the administration in said subject.

In certain embodiments, the multi-specific binding protein has a serum half-life that is at least 20% longer than a control multi-specific binding protein, wherein the control multi-specific binding protein includes a first domain identical to the first antigen-binding site of the multi-specific binding protein, a second domain identical to the second antigen-binding site of the multi-specific binding protein, but not a third domain identical or substantially identical to the third antigen-binding site of the multi-specific binding protein. In certain embodiments, the control multi-specific binding protein is identical to the multi-specific binding protein but for the absence of the third antigen-binding site. In certain embodiments, the serum half-life of the multi-specific binding protein is at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, or at least 90% longer than the serum half-life of the control multi-specific binding protein. In certain embodiments, the serum half-life of the multi-specific binding protein is longer than the serum half-life of the control multi-specific binding protein by at least 2 fold, at least 3 fold, at least 4 fold, at least 5 fold, at least 6 fold, at least 7 fold, at least 8 fold, at least 9 fold, or at least 10 fold.

II. Methods of Preparation

The multi-specific binding proteins described above can be made using recombinant DNA technology well known to a skilled person in the art. For example, one or more isolated polynucleotides encoding the multi-specific binding protein can be ligated to other appropriate nucleotide sequences, including, for example, constant region coding sequences, and expression control sequences, to produce conventional gene expression constructs (i.e., expression vectors) encoding the desired multi-specific binding proteins. Production of defined gene constructs is within routine skill in the art.

Nucleic acids encoding desired multi-specific binding proteins can be incorporated (ligated) into expression vec-

tors, which can be introduced into host cells through conventional transfection or transformation techniques. Exemplary host cells are *E. coli* cells, Chinese hamster ovary (CHO) cells, human embryonic kidney 293 (HEK 293) cells, HeLa cells, baby hamster kidney (BHK) cells, monkey kidney cells (COS), human hepatocellular carcinoma cells (e.g., Hep G2), and myeloma cells that do not otherwise produce IgG protein. Transformed host cells can be grown under conditions that permit the host cells to express the genes that encode the multi-specific binding proteins.

Specific expression and purification conditions will vary depending upon the expression system employed. For example, if a gene is to be expressed in *E. coli*, it is first cloned into an expression vector by positioning the engineered gene downstream from a suitable bacterial promoter, e.g., Trp or Tac, and a prokaryotic signal sequence. The expressed protein may be secreted. The expressed protein may accumulate in refractile or inclusion bodies, which can be harvested after disruption of the cells by French press or sonication. The refractile bodies then are solubilized, and the protein may be refolded and/or cleaved by methods known in the art.

If the engineered gene is to be expressed in eukaryotic host cells, e.g., CHO cells, it is first inserted into an expression vector containing a suitable eukaryotic promoter, a secretion signal, a poly A sequence, and a stop codon. Optionally, the vector or gene construct may contain enhancers and introns. In embodiments involving fusion proteins comprising an antibody or portion thereof, the expression vector optionally contains sequences encoding all or part of a constant region, enabling an entire, or a part of, a heavy or light chain to be expressed. The gene construct can be introduced into eukaryotic host cells using conventional techniques.

The multi-specific binding protein disclosed herein may comprise a single polypeptide chain. In this instance, a host cell can be transfected with a single vector expressing the polypeptide (e.g., containing an expression control sequence operably linked to a nucleotide sequence encoding the polypeptide). Alternatively, multi-specific binding protein disclosed herein may comprise two or more polypeptides. In this instance, a host cell can be co-transfected with more than one expression vector, for example, one expression vector expressing each polypeptide. A host cell can also be transfected with a single expression vector that expresses the two or more polypeptides. For example, the coding sequences of the two or more polypeptides can be operably linked to different expression control sequences (e.g., promoter, enhancer, and/or internal ribosome entry site (IRES)). The coding sequences of the two or more polypeptides can also be separated by a ribosomal skipping sequence or self-cleaving sequence, such as a 2A peptide.

In certain embodiments, in order to express a multi-specific binding protein, an N-terminal signal sequence is included in the protein construct. Exemplary N-terminal signal sequences include signal sequences from interleukin-2, CD-5, IgG kappa light chain, trypsinogen, serum albumin, and prolactin.

After transfection, single clones can be isolated for cell bank generation using methods known in the art, such as limited dilution, ELISA, FACS, microscopy, or Clonepix. Clones can be cultured under conditions suitable for bioreactor scale-up and maintained expression of the multi-specific binding proteins.

The multi-specific binding proteins can be isolated and purified using methods known in the art including centrifugation, depth filtration, cell lysis, homogenization, freeze-

thawing, affinity purification, gel filtration, ion exchange chromatography, hydrophobic interaction exchange chromatography, and mixed-mode chromatography.

III. Pharmaceutical Compositions

The present disclosure also features pharmaceutical compositions that contain a therapeutically effective amount of the multi-specific binding proteins described herein. The composition can be formulated for use in a variety of drug delivery systems. One or more physiologically acceptable excipients or carriers can also be included in the composition for proper formulation. Suitable formulations for use in the present disclosure are found in Remington's Pharmaceutical Sciences, Mack Publishing Company, Philadelphia, Pa., 17th ed., 1985. For a brief review of methods for drug delivery, see, e.g., Langer (Science 249:1527-1533, 1990).

In certain embodiments, a pharmaceutical composition may contain formulation materials for modifying, maintaining or preserving, for example, the pH, osmolality, viscosity, clarity, color, isotonicity, odor, sterility, stability, rate of dissolution or release, adsorption or penetration of the composition. In such embodiments, suitable formulation materials include, but are not limited to, amino acids (such as glycine, glutamine, asparagine, arginine or lysine); antimicrobials; antioxidants (such as ascorbic acid, sodium sulfite or sodium hydrogen-sulfite); buffers (such as borate, bicarbonate, Tris-HCl, citrates, phosphates or other organic acids); bulking agents (such as mannitol or glycine); chelating agents (such as ethylenediamine tetraacetic acid (EDTA)); complexing agents (such as caffeine, polyvinylpyrrolidone, beta-cyclodextrin or hydroxypropyl-beta-cyclodextrin); fillers; monosaccharides; disaccharides; and other carbohydrates (such as glucose, mannose or dextrans); proteins (such as serum albumin, gelatin or immunoglobulins); coloring, flavoring and diluting agents; emulsifying agents; hydrophilic polymers (such as polyvinylpyrrolidone); low molecular weight polypeptides; salt-forming counterions (such as sodium); preservatives (such as benzalkonium chloride, benzoic acid, salicylic acid, thimerosal, phenethyl alcohol, methylparaben, propylparaben, chlorhexidine, sorbic acid or hydrogen peroxide); solvents (such as glycerin, propylene glycol or polyethylene glycol); sugar alcohols (such as mannitol or sorbitol); suspending agents; surfactants or wetting agents (such as pluronics, PEG, sorbitan esters, polysorbates such as polysorbate 20, polysorbate, triton, tromethamine, lecithin, cholesterol, tyloxapal); stability enhancing agents (such as sucrose or sorbitol); tonicity enhancing agents (such as alkali metal halides, preferably sodium or potassium chloride, mannitol sorbitol); delivery vehicles; diluents; excipients and/or pharmaceutical adjuvants (see, Remington's *Pharmaceutical Sciences*, 18th ed. (Mack Publishing Company, 1990)).

In certain embodiments, a pharmaceutical composition may contain nanoparticles, e.g., polymeric nanoparticles, liposomes, or micelles (See Anselmo et al. (2016) BIOENG. TRANSL. MED. 1:10-29).

In certain embodiments, a pharmaceutical composition may contain a sustained- or controlled-delivery formulation. Techniques for formulating sustained- or controlled-delivery means, such as liposome carriers, bio-erodible microparticles or porous beads and depot injections, are also known to those skilled in the art. Sustained-release preparations may include, e.g., porous polymeric microparticles or semi-permeable polymer matrices in the form of shaped articles, e.g., films, or microcapsules. Sustained release matrices may include polyesters, hydrogels, polylactides, copolymers of L-glutamic acid and gamma ethyl-L-glutamate, poly (2-hydroxyethyl-inethacrylate), ethylene vinyl acetate, or poly-D

(-)-3-hydroxybutyric acid. Sustained release compositions may also include liposomes that can be prepared by any of several methods known in the art.

Pharmaceutical compositions containing a multi-specific binding protein disclosed herein can be presented in a dosage unit form and can be prepared by any suitable method. A pharmaceutical composition should be formulated to be compatible with its intended route of administration. Examples of routes of administration are intravenous (IV), intradermal, inhalation, transdermal, topical, transmuscular, intrathecal and rectal administration. In certain embodiments, a recombinant human sialidase, a recombinant human sialidase fusion protein, or an antibody conjugate disclosed herein is administered by IV infusion. In certain embodiments, a recombinant human sialidase, a recombinant human sialidase fusion protein, or an antibody conjugate disclosed herein is administered by intratumoral injection. Useful formulations can be prepared by methods known in the pharmaceutical art. For example, see *Remington's Pharmaceutical Sciences*, 18th ed. (Mack Publishing Company, 1990). Formulation components suitable for parenteral administration include a sterile diluent such as water for injection, saline solution, fixed oils, polyethylene glycols, glycerin, propylene glycol or other synthetic solvents; antibacterial agents such as benzyl alcohol or methyl parabens; antioxidants such as ascorbic acid or sodium bisulfite; chelating agents such as EDTA; buffers such as acetates, citrates or phosphates; and agents for the adjustment of tonicity such as sodium chloride or dextrose.

For intravenous administration, suitable carriers include physiological saline, bacteriostatic water, Cremophor® EL™ (BASF®, Parsippany, NJ) or phosphate buffered saline (PBS). The carrier should be stable under the conditions of manufacture and storage, and should be preserved against microorganisms. The carrier can be a solvent or dispersion medium containing, for example, water, ethanol, polyol (for example, glycerol, propylene glycol, and liquid polyethylene glycol), and suitable mixtures thereof.

An intravenous drug delivery formulation may be contained in a syringe, pen, or bag. In certain embodiments, the bag may be connected to a channel comprising a tube and/or a needle. In certain embodiments, the formulation may be a lyophilized formulation or a liquid formulation. In certain embodiments, the formulation may freeze-dried (lyophilized) and contained in about 12-60 vials. In certain embodiments, the formulation may be freeze-dried and 45 mg of the freeze-dried formulation may be contained in one vial. In certain embodiments, the about 40 mg-about 100 mg of freeze-dried formulation may be contained in one vial. In certain embodiments, freeze dried formulation from 12, 27, or 45 vials are combined to obtained a therapeutic dose of the protein in the intravenous drug formulation. In certain embodiments, the formulation may be a liquid formulation and stored as about 250 mg/vial to about 1,000 mg/vial. In certain embodiments, the formulation may be a liquid formulation and stored as about 600 mg/vial. In certain embodiments, the formulation may be a liquid formulation and stored as about 250 mg/vial.

These compositions may be sterilized by conventional sterilization techniques, or may be sterile filtered. The resulting aqueous solutions may be packaged for use as-is, or lyophilized, the lyophilized preparation being combined with a sterile aqueous carrier prior to administration. The pH of the preparations typically will be between 3 and 11, more preferably between 5 and 9 or between 6 and 8, and most preferably between 7 and 8, such as 7 to 7.5. The resulting compositions in solid form may be packaged in multiple

single dose units, each containing a fixed amount of the above-mentioned agent or agents. The composition in solid form can also be packaged in a container for a flexible quantity.

In certain embodiments, the present disclosure provides a formulation with an extended shelf life including the protein of the present disclosure, in combination with mannitol, citric acid monohydrate, sodium citrate, disodium phosphate dihydrate, sodium dihydrogen phosphate dihydrate, sodium chloride, polysorbate 80, water, and sodium hydroxide.

In certain embodiments, an aqueous formulation is prepared including the protein of the present disclosure in a pH-buffered solution. The buffer of this invention may have a pH ranging from about 4 to about 8, e.g., from about 4.5 to about 6.0, or from about 4.8 to about 5.5, or may have a pH of about 5.0 to about 5.2. Ranges intermediate to the above recited pH's are also intended to be part of this disclosure. For example, ranges of values using a combination of any of the above recited values as upper and/or lower limits are intended to be included. Examples of buffers that will control the pH within this range include acetate (e.g. sodium acetate), succinate (such as sodium succinate), gluconate, histidine, citrate and other organic acid buffers.

In certain embodiments, the formulation includes a buffer system which contains citrate and phosphate to maintain the pH in a range of about 4 to about 8. In certain embodiments the pH range may be from about 4.5 to about 6.0, or from about pH 4.8 to about 5.5, or in a pH range of about 5.0 to about 5.2. In certain embodiments, the buffer system includes citric acid monohydrate, sodium citrate, disodium phosphate dihydrate, and/or sodium dihydrogen phosphate dihydrate. In certain embodiments, the buffer system includes about 1.3 mg/ml of citric acid (e.g., 1.305 mg/ml), about 0.3 mg/ml of sodium citrate (e.g., 0.305 mg/ml), about 1.5 mg/ml of disodium phosphate dihydrate (e.g., 1.53 mg/ml), about 0.9 mg/ml of sodium dihydrogen phosphate dihydrate (e.g., 0.86), and about 6.2 mg/ml of sodium chloride (e.g., 6.165 mg/ml). In certain embodiments, the buffer system includes 1-1.5 mg/ml of citric acid, 0.25 to 0.5 mg/ml of sodium citrate, 1.25 to 1.75 mg/ml of disodium phosphate dihydrate, 0.7 to 1.1 mg/ml of sodium dihydrogen phosphate dihydrate, and 6.0 to 6.4 mg/ml of sodium chloride. In certain embodiments, the pH of the formulation is adjusted with sodium hydroxide.

A polyol, which acts as a tonicifier and may stabilize the multi-specific binding protein, may also be included in the formulation. The polyol is added to the formulation in an amount which may vary with respect to the desired isotonicity of the formulation. In certain embodiments, the aqueous formulation may be isotonic. The amount of polyol added may also be altered with respect to the molecular weight of the polyol. For example, a lower amount of a monosaccharide (e.g., mannitol) may be added, compared to a disaccharide (such as trehalose). In certain embodiments, the polyol which may be used in the formulation as a tonicity agent is mannitol. In certain embodiments, the mannitol concentration may be about 5 to about 20 mg/ml. In certain embodiments, the concentration of mannitol may be about 7.5 to 15 mg/ml. In certain embodiments, the concentration of mannitol may be about 10-14 mg/ml. In certain embodiments, the concentration of mannitol may be about 12 mg/ml. In certain embodiments, the polyol sorbitol may be included in the formulation.

A detergent or surfactant may also be added to the formulation. Exemplary detergents include nonionic detergents such as polysorbates (e.g., polysorbates 20, 80 etc.) or poloxamers (e.g., poloxamer 188). The amount of detergent

added is such that it reduces aggregation of the formulated antibody and/or minimizes the formation of particulates in the formulation and/or reduces adsorption. In certain embodiments, the formulation may include a surfactant which is a polysorbate. In certain embodiments, the formulation may contain the detergent polysorbate 80 or Tween 80. Tween 80 is a term used to describe polyoxyethylene (20) sorbitanmonooleate (see Fiedler, *Lexikon der Hilfsstoffe*, Editio Cantor Verlag Aulendorf, 4th ed., 1996). In certain embodiments, the formulation may contain between about 0.1 mg/mL and about 10 mg/mL of polysorbate 80, or between about 0.5 mg/mL and about 5 mg/mL. In certain embodiments, about 0.1% polysorbate 80 may be added in the formulation.

In embodiments, the protein product of the present disclosure is formulated as a liquid formulation. The liquid formulation may be presented at a 10 mg/mL concentration in either a USP/Ph Eur type I 50R vial closed with a rubber stopper and sealed with an aluminum crimp seal closure. The stopper may be made of elastomer complying with USP and Ph Eur. In certain embodiments, the liquid formulation may be diluted with 0.9% saline solution.

In certain embodiments, the liquid formulation of the disclosure may be prepared as a 10 mg/mL concentration solution in combination with a sugar at stabilizing levels. In certain embodiments the liquid formulation may be prepared in an aqueous carrier. In certain embodiments, a stabilizer may be added in an amount no greater than that which may result in a viscosity undesirable or unsuitable for intravenous administration. In certain embodiments, the sugar may be disaccharides, e.g., sucrose. In certain embodiments, the liquid formulation may also include one or more of a buffering agent, a surfactant, and a preservative.

In certain embodiments, the pH of the liquid formulation may be set by addition of a pharmaceutically acceptable acid and/or base. In certain embodiments, the pharmaceutically acceptable acid may be hydrochloric acid. In certain embodiments, the base may be sodium hydroxide.

The aqueous carrier of interest herein is one which is pharmaceutically acceptable (safe and non-toxic for administration to a human) and is useful for the preparation of a liquid formulation. Illustrative carriers include sterile water for injection (SWFI), bacteriostatic water for injection (BWFI), a pH buffered solution (e.g., phosphate-buffered saline), sterile saline solution, Ringer's solution or dextrose solution.

A preservative may be optionally added to the formulations herein to reduce bacterial action. The addition of a preservative may, for example, facilitate the production of a multi-use (multiple-dose) formulation.

The multi-specific binding protein may be lyophilized to produce a lyophilized formulation including the proteins and a lyoprotectant. The lyoprotectant may be sugar, e.g., disaccharides. In certain embodiments, the lyoprotectant may be sucrose or maltose. The lyophilized formulation may also include one or more of a buffering agent, a surfactant, a bulking agent, and/or a preservative.

The amount of sucrose or maltose useful for stabilization of the lyophilized drug product may be in a weight ratio of at least 1:2 protein to sucrose or maltose. In certain embodiments, the protein to sucrose or maltose weight ratio may be of from 1:2 to 1:5. In certain embodiments, the pH of the formulation, prior to lyophilization, may be set by addition of a pharmaceutically acceptable acid and/or base. In certain embodiments the pharmaceutically acceptable acid may be hydrochloric acid. In certain embodiments, the pharmaceutically acceptable base may be sodium hydroxide. Before

lyophilization, the pH of the solution containing the protein of the present disclosure may be adjusted between 6 to 8. In certain embodiments, the pH range for the lyophilized drug product may be from 7 to 8.

Actual dosage levels of the active ingredients in the pharmaceutical compositions of this invention may be varied so as to obtain an amount of the active ingredient which is effective to achieve the desired therapeutic response for a particular patient, composition, and mode of administration, without being toxic to the patient.

The specific dose can be a uniform dose for each patient, for example, 50-5,000 mg of protein. Alternatively, a patient's dose can be tailored to the approximate body weight or surface area of the patient. Other factors in determining the appropriate dosage can include the disease or condition to be treated or prevented, the severity of the disease, the route of administration, and the age, sex and medical condition of the patient. Further refinement of the calculations necessary to determine the appropriate dosage for treatment is routinely made by those skilled in the art, especially in light of the dosage information and assays disclosed herein. The dosage can also be determined through the use of known assays for determining dosages used in conjunction with appropriate dose-response data. An individual patient's dosage can be adjusted as the progress of the disease is monitored. Blood levels of the targetable construct or complex in a patient can be measured to see if the dosage needs to be adjusted to reach or maintain an effective concentration. Pharmacogenomics may be used to determine which targetable constructs and/or complexes, and dosages thereof, are most likely to be effective for a given individual (Schmitz et al., *Clinica Chimica Acta* 308:43-53, 2001; Steimer et al., *Clinica Chimica Acta* 308:33-41, 2001).

In general, dosages based on body weight are from about 0.01 µg to about 100 mg per kg of body weight, such as about 0.01 µg to about 100 mg/kg of body weight, about 0.01 µg to about 50 mg/kg of body weight, about 0.01 µg to about 10 mg/kg of body weight, about 0.01 µg to about 1 mg/kg of body weight, about 0.01 µg to about 100 µg/kg of body weight, about 0.01 µg to about 50 µg/kg of body weight, about 0.01 µg to about 10 µg/kg of body weight, about 0.01 µg to about 1 µg/kg of body weight, about 0.01 µg to about 0.1 µg/kg of body weight, about 0.1 µg to about 100 mg/kg of body weight, about 0.1 µg to about 50 mg/kg of body weight, about 0.1 µg to about 10 mg/kg of body weight, about 0.1 µg to about 1 mg/kg of body weight, about 0.1 µg to about 100 µg/kg of body weight, about 0.1 µg to about 10 µg/kg of body weight, about 0.1 µg to about 1 µg/kg of body weight, about 1 µg to about 100 mg/kg of body weight, about 1 µg to about 50 mg/kg of body weight, about 1 µg to about 10 mg/kg of body weight, about 1 µg to about 1 mg/kg of body weight, about 1 µg to about 100 µg/kg of body weight, about 1 µg to about 50 µg/kg of body weight, about 1 µg to about 10 µg/kg of body weight, about 10 µg to about 100 mg/kg of body weight, about 10 µg to about 50 mg/kg of body weight, about 10 µg to about 10 mg/kg of body weight, about 10 µg to about 1 mg/kg of body weight, about 10 µg to about 100 µg/kg of body weight, about 10 µg to about 50 µg/kg of body weight, about 50 µg to about 100 mg/kg of body weight, about 50 µg to about 50 mg/kg of body weight, about 50 µg to about 10 mg/kg of body weight, about 50 µg to about 1 mg/kg of body weight, about 50 µg to about 100 µg/kg of body weight, about 100 µg to about 50 mg/kg of body weight, about 100 µg to about 10 mg/kg of body weight, about 100 µg to about 1 mg/kg of body weight, about 1 mg to about 100 mg/kg of body weight, about 1 mg to about 50

mg/kg of body weight, about 1 mg to about 10 mg/kg of body weight, about 10 mg to about 100 mg/kg of body weight, about 10 mg to about 50 mg/kg of body weight, about 50 mg to about 100 mg/kg of body weight.

Doses may be given once or more times daily, weekly, monthly or yearly, or even once every 2 to 20 years. Persons of ordinary skill in the art can easily estimate repetition rates for dosing based on measured residence times and concentrations of the targetable construct or complex in bodily fluids or tissues. Administration of the present invention could be intravenous, intraarterial, intraperitoneal, intramuscular, subcutaneous, intrapleural, intrathecal, intracavitary, by perfusion through a catheter or by direct intralesional injection. This may be administered once or more times daily, once or more times weekly, once or more times monthly, and once or more times annually.

IV. Therapeutic Applications

It is contemplated that the multi-specific binding proteins can be used either alone or in combination with other therapeutic agents.

A. Indications

The present disclosure provides methods for the treatment or amelioration of a proliferative disease, a tumorous disease, an inflammatory disease, an immunological disorder, an autoimmune disease, an infectious disease, a viral disease, an allergic reaction, a parasitic reaction, a graft-versus-host disease or a host-versus-graft disease in a subject in need thereof, the method comprising administration of the multi-specific binding proteins described herein.

In certain embodiments, the cancer to be treated is non-Hodgkin's lymphoma, such as a B-cell lymphoma. In certain embodiments, the non-Hodgkin's lymphoma is a B-cell lymphoma, such as a diffuse large B-cell lymphoma, primary mediastinal B-cell lymphoma, follicular lymphoma, small lymphocytic lymphoma, mantle cell lymphoma, marginal zone B-cell lymphoma, extranodal marginal zone B-cell lymphoma, nodal marginal zone B-cell lymphoma, splenic marginal zone B-cell lymphoma, Burkitt lymphoma, lymphoplasmacytic lymphoma, hairy cell leukemia, chronic lymphocytic leukemia, or primary central nervous system lymphoma. In certain other embodiments, the cancer to be treated is multiple myeloma. In certain other embodiments, the cancer to be treated is acute lymphoblastic leukemia (ALL). In certain embodiments, the ALL is relapsed/refractory adult and pediatric ALL.

B. Combination Therapies

The methods and compositions described herein can be used alone or in combination with other therapeutic agents and/or modalities. The term administered "in combination," as used herein, is understood to mean that two (or more) different treatments are delivered to the subject during the course of the subject's affliction with the disorder, such that the effects of the treatments on the patient overlap at a point in time. In certain embodiments, the delivery of one treatment is still occurring when the delivery of the second begins, so that there is overlap in terms of administration. This is sometimes referred to herein as "simultaneous" or "concurrent delivery." In other embodiments, the delivery of one treatment ends before the delivery of the other treatment begins. In certain embodiments of either case, the treatment is more effective because of combined administration. For example, the second treatment is more effective, e.g., an equivalent effect is seen with less of the second treatment, or the second treatment reduces symptoms to a greater extent, than would be seen if the second treatment were administered in the absence of the first treatment, or the analogous situation is seen with the first treatment. In certain embodi-

ments, delivery is such that the reduction in a symptom, or other parameter related to the disorder is greater than what would be observed with one treatment delivered in the absence of the other. The effect of the two treatments can be partially additive, wholly additive, or greater than additive. The delivery can be such that an effect of the first treatment delivered is still detectable when the second is delivered.

In one aspect, the present disclosure provides a method of treating a subject by the administration of a second therapeutic agent in combination with one or more of the multi-specific binding proteins described herein.

Exemplary therapeutic agents that may be used as part of a combination therapy in treating cancer, include, for example, radiation, mitomycin, tretinoin, ribomustin, gemcitabine, vincristine, etoposide, cladribine, mitobronitol, methotrexate, doxorubicin, carboquone, pentostatin, nitracrine, zinostatin, cetorelix, letrozole, raltitrexed, daunorubicin, fadrozole, fotemustine, thymalfasin, sobuzoxane, nedaplatin, cytarabine, bicalutamide, vinorelbine, vesnarinone, aminoglutethimide, amsacrine, proglumide, ellipritinium acetate, ketanserin, doxifluridine, etretinate, isotretinoin, streptozocin, nimustine, vindesine, flutamide, drogenil, butocin, carmofur, razoxane, sizofilan, carboplatin, mitolactol, tegafur, ifosfamide, prednimustine, picibanil, levamisole, teniposide, improsulfan, enocitabine, lisuride, oxymetholone, tamoxifen, progesterone, mepitiostane, epitiostanol, formestane, interferon-alpha, interferon-2 alpha, interferon-beta, interferon-gamma, colony stimulating factor-1, colony stimulating factor-2, denileukin diftitox, interleukin-2, luteinizing hormone releasing factor and variations of the aforementioned agents that may exhibit differential binding to its cognate receptor, and increased or decreased serum half-life.

An additional class of agents that may be used as part of a combination therapy in treating cancer is immune checkpoint inhibitors. The checkpoint inhibitor may, for example, be selected from a PD-1 antagonist, PD-L1 antagonist, CTLA-4 antagonist, adenosine A2A receptor antagonist, B7-H3 antagonist, B7-H4 antagonist, BTLA antagonist, KIR antagonist, LAG3 antagonist, TIM-3 antagonist, VISTA antagonist or TIGIT antagonist.

In certain embodiments, the checkpoint inhibitor is a PD-1 or PD-L1 inhibitor. PD-1 is a receptor present on the surface of T-cells that serves as an immune system checkpoint that inhibits or otherwise modulates T-cell activity at the appropriate time to prevent an overactive immune response. Cancer cells, however, can take advantage of this checkpoint by expressing ligands, for example, PD-L1, that interact with PD-1 on the surface of T-cells to shut down or modulate T-cell activity. Exemplary PD-1/PD-L1 based immune checkpoint inhibitors include antibody based therapeutics. Exemplary treatment methods that employ PD-1/PD-L1 based immune checkpoint inhibition are described in U.S. Pat. Nos. 8,728,474 and 9,073,994, and EP U.S. Pat. No. 1,537,878B1, and, for example, include the use of anti-PD-1 antibodies. Exemplary anti-PD-1 antibodies are described, for example, in U.S. Pat. Nos. 8,952,136, 8,779, 105, 8,008,449, 8,741,295, 9,205,148, 9,181,342, 9,102,728, 9,102,727, 8,952,136, 8,927,697, 8,900,587, 8,735,553, and 7,488,802. Exemplary anti-PD-1 antibodies include, for example, nivolumab (available under the tradename Opdivo® from Bristol-Myers Squibb Co.), pembrolizumab (available under the tradename Keytruda® from Merck Sharp & Dohme Corp.), PDR001 (Novartis Pharmaceuticals), and pidilizumab (CT-011, Cure Tech). Exemplary anti-PD-L1 antibodies are described, for example, in U.S. Pat. Nos. 9,273,135, 7,943,743, 9,175,082, 8,741,295,

8,552,154, and 8,217,149. Exemplary anti-PD-L1 antibodies include, for example, atezolizumab (available under the tradename Tecentriq® from Genentech®), duvalumab (AstraZeneca®), MEDI4736, avelumab, and BMS 936559 (Bristol Myers Squibb Co.).

In certain embodiments, a method or composition described herein is administered in combination with a CTLA-4 inhibitor. In the CTLA-4 pathway, the interaction of CTLA-4 on a T-cell with its ligands (e.g., CD80, also known as B7-1, and CD86) on the surface of an antigen presenting cells (rather than cancer cells) leads to T-cell inhibition. Exemplary CTLA-4 based immune checkpoint inhibition methods are described in U.S. Pat. Nos. 5,811, 097, 5,855,887, 6,051,227. Exemplary anti-CTLA-4 antibodies are described in U.S. Pat. Nos. 6,984,720, 6,682,736, 7,311,910; 7,307,064, 7,109,003, 7,132,281, 6,207,156, 7,807,797, 7,824,679, 8,143,379, 8,263,073, 8,318,916, 8,017,114, 8,784,815, and 8,883,984, International (PCT) Publication Nos. WO98/42752, WO00/37504, and WO01/14424, and European Patent No. EP 1212422 B1. Exemplary CTLA-4 antibodies include ipilimumab or tremelimumab.

In certain embodiments, a method or composition described herein is administered in combination with (i) a PD-1 or PD-L1 inhibitor, e.g., a PD-1 or PD-L1 inhibitor disclosed herein, and (ii) CTLA-4 inhibitor, e.g., a CTLA-4 inhibitor disclosed herein.

In certain embodiments, a method or composition described herein is administered in combination with an IDO inhibitor. Exemplary IDO inhibitors include 1-methyl-D-tryptophan (known as indoximod), epacadostat (INCB24360), navoximod (GDC-0919), and BMS-986205.

Yet other agents that may be used as part of a combination therapy in treating cancer are monoclonal antibody agents that target non-checkpoint targets (e.g., herceptin) and non-cytotoxic agents (e.g., tyrosine-kinase inhibitors).

Yet other categories of anti-cancer agents include, for example: (i) an inhibitor selected from an ALK Inhibitor, an ATR Inhibitor, an A2A Antagonist, a Base Excision Repair Inhibitor, a Bcr-Abl Tyrosine Kinase Inhibitor, a Bruton's Tyrosine Kinase Inhibitor, a CDC7 Inhibitor, a CHK1 Inhibitor, a Cyclin-Dependent Kinase Inhibitor, a DNA-PK Inhibitor, an Inhibitor of both DNA-PK and mTOR, a DNMT1 Inhibitor, a DNMT1 Inhibitor plus 2-chloro-deoxyadenosine, an HDAC Inhibitor, a Hedgehog Signaling Pathway Inhibitor, an IDO Inhibitor, a JAK Inhibitor, a mTOR Inhibitor, a MEK Inhibitor, a MELK Inhibitor, a MTH1 Inhibitor, a PARP Inhibitor, a Phosphoinositide 3-Kinase Inhibitor, an Inhibitor of both PARP1 and DHODH, a Proteasome Inhibitor, a Topoisomerase-II Inhibitor, a Tyrosine Kinase Inhibitor, a VEGFR Inhibitor, and a WEE1 Inhibitor; (ii) an agonist of OX40, CD137, CD40, GITR, CD27, HVEM, TNFRSF25, or ICOS; and (iii) a cytokine selected from IL-12, IL-15, GM-CSF, and G-CSF.

It is understood that the multi-specific antibody disclosed herein, which is designed to activate T lymphocytes, may cause side effects such as neurotoxicity. Accordingly, in certain embodiments, the second therapeutic agent that can be used in combination with the multi-specific binding protein comprises an agent that mitigates a side effect of the multi-specific binding protein, e.g., reduces neurotoxicity. In certain embodiments, the second therapeutic agent inhibits T cell trafficking, for example, reduces or inhibits immune cells from crossing the blood-brain barrier. Non-limiting examples of such therapeutic agents include antagonists (e.g., antagonistic antibodies) of adhesion molecules on immune cells (e.g., $\alpha 4$ integrin), such as natalizumab. In

certain embodiments, the second therapeutic agent increases the internalization of a sphingosine-1-phosphate (SIP) receptor (e.g., S1PR1 or S1PR5), such as fingolimod or ozanimod. In certain embodiments, the second therapeutic agent is a nitric oxide synthase (NOS) inhibitor, such as 5 rionopterin, cindunistat, A-84643, ONO-1714, L-NOARG, NCX-456, VAS-2381, GW-273629, NXN-462, CKD-712, KD-7040, or guanidinoethyldisulfide. In certain embodiments, the second therapeutic agent is an antagonist of CSF1 or CSF1R, such as pexidartinib, emactuzumab, cabiralizumab, LY-3022855, JNJ-40346527, or MCS110. Additional non-limiting examples of the second therapeutic agents include pentosan polysulfate, minocycline, anti-ICAM-1 antibodies, anti-P-selectin antibodies, anti-CD11a antibodies, anti-CD162 antibodies, and anti-IL-6R antibodies (e.g., tocilizumab).

The amount of the multi-specific binding protein and additional therapeutic agent and the relative timing of administration may be selected in order to achieve a desired combined therapeutic effect. For example, when administering a combination therapy to a patient in need of such administration, the therapeutic agents in the combination, or a pharmaceutical composition or compositions comprising the therapeutic agents, may be administered in any order such as, for example, sequentially, concurrently, together, simultaneously and the like. Further, for example, a multi-specific binding protein may be administered during a time when the additional therapeutic agent(s) exerts its prophylactic or therapeutic effect, or vice versa.

Throughout the description, where compositions are described as having, including, or comprising specific components, or where processes and methods are described as having, including, or comprising specific steps, it is contemplated that, additionally, there are compositions of the present invention that consist essentially of, or consist of, the recited components, and that there are processes and methods according to the present invention that consist essentially of, or consist of, the recited processing steps.

In the application, where an element or component is said to be included in and/or selected from a list of recited elements or components, it should be understood that the element or component can be any one of the recited elements or components, or the element or component can be selected from a group consisting of two or more of the recited elements or components.

Further, it should be understood that elements and/or features of a composition or a method described herein can be combined in a variety of ways without departing from the spirit and scope of the present invention, whether explicit or implicit herein. For example, where reference is made to a particular compound, that compound can be used in various embodiments of compositions of the present invention and/or in methods of the present invention, unless otherwise understood from the context. In other words, within this application, embodiments have been described and depicted in a way that enables a clear and concise application to be written and drawn, but it is intended and will be appreciated that embodiments may be variously combined or separated without parting from the present teachings and invention(s). For example, it will be appreciated that all features described and depicted herein can be applicable to all aspects of the invention(s) described and depicted herein.

It should be understood that the expression "at least one of" includes individually each of the recited objects after the expression and the various combinations of two or more of the recited objects unless otherwise understood from the context and use. The expression "and/or" in connection with

three or more recited objects should be understood to have the same meaning unless otherwise understood from the context.

The use of the term "include," "includes," "including," "have," "has," "having," "contain," "contains," or "containing," including grammatical equivalents thereof, should be understood generally as open-ended and non-limiting, for example, not excluding additional unrecited elements or steps, unless otherwise specifically stated or understood from the context.

Where the use of the term "about" is before a quantitative value, the present invention also includes the specific quantitative value itself, unless specifically stated otherwise. As used herein, the term "about" refers to a $\pm 10\%$ variation from the nominal value unless otherwise indicated or inferred.

It should be understood that the order of steps or order for performing certain actions is immaterial so long as the present invention remain operable. Moreover, two or more steps or actions may be conducted simultaneously.

The use of any and all examples, or exemplary language herein, for example, "such as" or "including," is intended merely to illustrate better the present invention and does not pose a limitation on the scope of the invention unless claimed. No language in the specification should be construed as indicating any non-claimed element as essential to the practice of the present invention.

The description above describes multiple aspects and embodiments of the invention. The patent application specifically contemplates all combinations and permutations of the aspects and embodiments.

EXAMPLES

The invention now being generally described, will be more readily understood by reference to the following examples, which are included merely for purposes of illustration of certain aspects and embodiments of the present invention, and is not intended to limit the invention.

Example 1. Production of Multi-Specific Binding Proteins

This example describes the production and purification of multi-specific binding proteins.

Nucleic acids encoding single-chain multi-specific binding proteins (see Table 5) were constructed and codon optimized for expression in human cells and cloned into a mammalian expression vector following standard procedures. Following sequence verification, the expression vectors, in the form of plasmids, were prepared in sufficient quantity for transfection using Plasmid Plus purification kits (Qiagen®). Human embryonic kidney 293 (HEK 293) cells were passaged to appropriate density for transient transfection. Cells were transiently transfected with the expression vectors and cultured for six days.

The amino acid sequences of the various multi-specific binding proteins are summarized in Table 5. Constructs tAb0027 to tAb0032 each contained an anti-CD19 scFv having the amino acid sequence set forth in SEQ ID NO: 9, an anti-CD3 scFv having the amino acid sequence set forth in SEQ ID NO: 105, and an anti-HSA sdAb having the amino acid sequence set forth in SEQ ID NO: 121. Constructs tAb0033 to tAb0038 each contained an anti-CD19 scFv having the amino acid sequence set forth in SEQ ID NO: 18, an anti-CD3 scFv having the amino acid sequence set forth in SEQ ID NO: 105, and an anti-HSA sdAb having the amino acid sequence set forth in SEQ ID NO: 121.

TABLE 5

Exemplary Multi-specific Binding Proteins		
Construct	Format	Amino Acid Sequence
tAb0027	CD19:CD3: HSA	QVQLVQSGAEVKKPGASVKVSKASGYTFTDYIMHWVRQAPGQ GLEWMGYINPYNDGSKYTEKFGQGRVTMTSDTSISTAYMELSRRLRS DDTAVYYCARGTYYYGPQLFDYWGGQTTVTVSSGGGSGGGGS GGGGSDIVMTQTPLSLSVTPGQPASISCKSSQSLETSTGTTYLNWY LQKPGQSPQLLIYRVSKRFSGVPDRFSGSGSGTDFTLKISRVEAEDV GVYYCLQLEDPTFGQGTKLEIKGGGSGGGSDIVMTQSPDSLAV VSLGERATINCKSSQSLNARTGKNYLAWYQQKPGQPPKLLIYWA STRESGVPDRFSGSGSGTDFTLTISLQAEADVAVYYCKQSYSRRTF GGGKVEIKGGGSGGGSGGGSGGGSGVQLVQSGAEVKKPG GASVKVSKASGFNIKDYIMHWVRQAPGQRLIEWMGWIDLENAN TIYDAKFQGRVTITRDTASTAYMELSSLRSEDTAVYYCARDAYG RYFYDVGQGTTLVTVSSGGGSGGGSEVQLVESGGGLVQPGNSL RLSCAASGFTFSFGMSWVRQAPGKGLEWVSSIISGSGSDTLYADS VKGRFTISRDNAKTTLTLQMNLSLRPEDTAVYYCTIGGSLSRSSQGT LVTVSSHSHHHHHHHH (SEQ ID NO: 303)
tAb0029	CD3:CD19: HSA	DIVMTQSPDSLAVSLGERATINCKSSQSLNARTGKNYLAWYQQK PGQPPKLLIYWASTRESGVPDRFSGSGSGTDFTLTISLQAEADVAV YYCKQSYSRRTFSGGKVEIKGGGSGGGSGGGSGGGSGVQV LVQSGAEVKKPGASVKVSKASGFNIKDYIMHWVRQAPGQRLIE WMGWIDLENANTIIYDAKFQGRVTITRDTASTAYMELSSLRSEDT AVYYCARDAYGRIFYDVGQGTTLVTVSSGGGSGGGSEVQLVQ SGAEVKKPGASVKVSKASGYTFTDYIMHWVRQAPGQGLEWMG YINPYNDGSKYTEKFGQGRVTMTSDTSISTAYMELSRRLSDDTAVY YCARGTYYYGPQLFDYWGGQTTVTVSSGGGSGGGSGGGSGDI VMTQTPLSLSVTPGQPASISCKSSQSLETSTGTTYLNWYLQKPGQS PQLLIYRVSKRFSGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCL QLEDPTFGQGTKLEIKGGGSGGGSEVQLVESGGGLVQPGNSL RLSCAASGFTFSFGMSWVRQAPGKGLEWVSSIISGSGSDTLYADS VKGRFTISRDNAKTTLTLQMNLSLRPEDTAVYYCTIGGSLSRSSQGT LVTVSSHSHHHHHHHH (SEQ ID NO: 304)
tAb0030	CD3:HSA: CD19	DIVMTQSPDSLAVSLGERATINCKSSQSLNARTGKNYLAWYQQK PGQPPKLLIYWASTRESGVPDRFSGSGSGTDFTLTISLQAEADVAV YYCKQSYSRRTFSGGKVEIKGGGSGGGSGGGSGGGSGVQV LVQSGAEVKKPGASVKVSKASGFNIKDYIMHWVRQAPGQRLIE WMGWIDLENANTIIYDAKFQGRVTITRDTASTAYMELSSLRSEDT AVYYCARDAYGRIFYDVGQGTTLVTVSSGGGSGGGSEVQLVE SGGGLVQPGNSLRLSCAASGFTFSFGMSWVRQAPGKGLEWVSSI SGSGSDTLYADSVKGRFTISRDNAKTTLTLQMNLSLRPEDTAVYYC TIGGSLSRSSQGTTLVTVSSGGGSGGGSGVQLVQSGAEVKKPGAS VKVSKASGYTFTDYIMHWVRQAPGQGLEWMGYINPYNDGSKY TEKFGQGRVTMTSDTSISTAYMELSRRLSDDTAVYYCARGTYYYGP QLFDYWGGQTTVTVSSGGGSGGGSGGGSDIVMTQTPLSLSV TPGQPASISCKSSQSLETSTGTTYLNWYLQKPGQSPQLLIYRVSKRF SGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCLQLEDPTFGQG TKLEIKHHHHHHHHH (SEQ ID NO: 305)
tAb0031	HSA:CD3: CD19	EVQLVESGGGLVQPGNSLRLSCAASGFTFSFGMSWVRQAPGKGL EWVSSIISGSGSDTLYADSVKGRFTISRDNAKTTLTLQMNLSLRPEDT AVYYCTIGGSLSRSSQGTTLVTVSSGGGSGGGSDIVMTQSPDSLAV SLGERATINCKSSQSLNARTGKNYLAWYQQKPGQPPKLLIYWAS TRESGVPDRFSGSGSGTDFTLTISLQAEADVAVYYCKQSYSRRTFG GGKVEIKGGGSGGGSGGGSGGGSGVQLVQSGAEVKKPG ASVKVSKASGFNIKDYIMHWVRQAPGQRLIEWMGWIDLENANT IYDAKFQGRVTITRDTASTAYMELSSLRSEDTAVYYCARDAYGR YFYDVGQGTTLVTVSSGGGSGGGSGVQLVQSGAEVKKPGASV KVSCKASGYTFTDYIMHWVRQAPGQGLEWMGYINPYNDGSKYT EKFQGRVTMTSDTSISTAYMELSRRLSDDTAVYYCARGTYYYGPQ LFDYWGGQTTVTVSSGGGSGGGSGGGSDIVMTQTPLSLSVTP GQPASISCKSSQSLETSTGTTYLNWYLQKPGQSPQLLIYRVSKRFS GVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCLQLEDPTFGQG TKLEIKHHHHHHHHH (SEQ ID NO: 306)
tAb0032	HSA:CD19: CD3	EVQLVESGGGLVQPGNSLRLSCAASGFTFSFGMSWVRQAPGKGL EWVSSIISGSGSDTLYADSVKGRFTISRDNAKTTLTLQMNLSLRPEDT AVYYCTIGGSLSRSSQGTTLVTVSSGGGSGGGSGVQLVQSGAEVK KPGASVKVSKASGYTFTDYIMHWVRQAPGQGLEWMGYINPYN DGSKYTEKFGQGRVTMTSDTSISTAYMELSRRLSDDTAVYYCARGT YYYGPQLFDYWGGQTTVTVSSGGGSGGGSGGGSDIVMTQTPL LSLSVTPGQPASISCKSSQSLETSTGTTYLNWYLQKPGQSPQLLIYR VSKRFSGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCLQLEDPT FGQGTKLEIKGGGSGGGSDIVMTQSPDSLAVSLGERATINCKSS

TABLE 5-continued

Exemplary Multi-specific Binding Proteins		
Construct	Format	Amino Acid Sequence
		QSLNARTGKNYLAWYQQKPGQPPKLLIYWASTRESGVPDRESGS GSGTDFTLTISLQAEADVAVYYCKQSYSRRTFGGGTKVEIKGGGG SGGGGSGGGSGGGGSQVQLVQSGAEVKKPGASVKVSKASGEN IKDYIMHWVRQAPGQRLEWMGWIDLENANTIYDAKFQGRVTITR DTSASTAYMELSSLRSEDVAVYYCARDAYGRIFYDVWGQGTTLVT VSSHHHHHHHHH (SEQ ID NO: 307)
tAb0033	CD19:CD3: HSA	QVQLQESGPGLVKPSQTLSTCTVSGGSISTSGMGVGVIRQHPGK GLEWIGHIWDDDKRYNPALKSRVTISVDTSKNQFSLKLSVTAA DTAVYYCARMELWSYFDYWGGTTLVTVSSGGGSGGGSGGG GSEIVLTQSPATLSLSPGERATLSCASSSVSYMHYQQKPGQAPR LLIYDTSKLASGIPARFSGSGSGTDFTLTISLLEPEDFAVYYCFQGSV YPFTFGQGTKLEIKRGGGSGGGSDIVMTQSPDSLAVSLGERATIN CKSSQSLNARTGKNYLAWYQQKPGQPPKLLIYWASTRESGVPD RFSGSGSGTDFTLTISLQAEADVAVYYCKQSYSRRTFGGGTKVEIK GGGSGGGSGGGSGGGGSQVQLVQSGAEVKKPGASVKVSK ASGFNIKDYIMHWVRQAPGQRLEWMGWIDLENANTIYDAKFQ RVTITRDTASASTAYMELSSLRSEDVAVYYCARDAYGRIFYDVWG QGTTLVTVSSGGGSGGGSEVQLVESGGGLVQPGNSLRSLCAASGF TFSFPGMSWVRQAPGKGLEWVSSISGSGSDTLYADSVKGRFTISR D NAKTTLYLQMNSLRPEDTAVYYCTIGGSLSRSSQGTTLVTVSSHHH HHHHHH (SEQ ID NO: 308)
tAb0034	CD19:HSA: CD3	QVQLQESGPGLVKPSQTLSTCTVSGGSISTSGMGVGVIRQHPGK GLEWIGHIWDDDKRYNPALKSRVTISVDTSKNQFSLKLSVTAA DTAVYYCARMELWSYFDYWGGTTLVTVSSGGGSGGGSGGG GSEIVLTQSPATLSLSPGERATLSCASSSVSYMHYQQKPGQAPR LLIYDTSKLASGIPARFSGSGSGTDFTLTISLLEPEDFAVYYCFQGSV YPFTFGQGTKLEIKRGGGSGGGSEVQLVESGGGLVQPGNSLRSL CAASGFTFSFPGMSWVRQAPGKGLEWVSSISGSGSDTLYADSVK RFTISRDNAKTTLYLQMNSLRPEDTAVYYCTIGGSLSRSSQGTTLV TVSSGGGSGGGSDIVMTQSPDSLAVSLGERATINCKSSQSLNART GKNYLAWYQQKPGQPPKLLIYWASTRESGVPDRFSGSGSGTDFTL TISLQAEADVAVYYCKQSYSRRTFGGGTKVEIKGGGSGGGSGG GSGGGGSQVQLVQSGAEVKKPGASVKVSKASGFNIKDYIMH WVRQAPGQRLEWMGWIDLENANTIYDAKFQGRVTITRDTASASTA YMELSSLRSEDVAVYYCARDAYGRIFYDVWGQGTTLVTVSSHHH HHHHHH (SEQ ID NO: 309)
tAb0035	CD3:CD19: HSA	DIVMTQSPDSLAVSLGERATINCKSSQSLNARTGKNYLAWYQQK PGQPPKLLIYWASTRESGVPDRESGSGSGTDFTLTISLQAEADVAV YYCKQSYSRRTFGGGTKVEIKGGGSGGGSGGGSGGGGSQVQ LVQSGAEVKKPGASVKVSKASGFNIKDYIMHWVRQAPGQRLE WMGWIDLENANTIYDAKFQGRVTITRDTASASTAYMELSSLRSED AVYYCARDAYGRIFYDVWGQGTTLVTVSSGGGSGGGGSQVQLQE SGPGLVKPSQTLSTCTVSGGSISTSGMGVGVIRQHPGKGLEWIGH IWDDDKRYNPALKSRVTISVDTSKNQFSLKLSVTAADTAVYYC ARMELWSYFDYWGGTTLVTVSSGGGSGGGSGGGSGGGSEIVLTQ SPATLSLSPGERATLSCASSSVSYMHYQQKPGQAPRLLIYDTSK LASGIPARFSGSGSGTDFTLTISLLEPEDFAVYYCFQGSVYPFTFGQ GTKLEIKRGGGSGGGSEVQLVESGGGLVQPGNSLRSLCAASGFT FSSFGMSWVRQAPGKGLEWVSSISGSGSDTLYADSVKGRFTISR D NAKTTLYLQMNSLRPEDTAVYYCTIGGSLSRSSQGTTLVTVSSHHH HHHHHH (SEQ ID NO: 310)
tAb0036	CD3:HSA: CD19	DIVMTQSPDSLAVSLGERATINCKSSQSLNARTGKNYLAWYQQK PGQPPKLLIYWASTRESGVPDRFSGSGSGTDFTLTISLQAEADVAV YYCKQSYSRRTFGGGTKVEIKGGGSGGGSGGGSGGGGSQVQ LVQSGAEVKKPGASVKVSKASGFNIKDYIMHWVRQAPGQRLE WMGWIDLENANTIYDAKFQGRVTITRDTASASTAYMELSSLRSED AVYYCARDAYGRIFYDVWGQGTTLVTVSSGGGSGGGSEVQLVE SGGGLVQPGNSLRSLCAASGFTFSFPGMSWVRQAPGKGLEWVSSI SSGSGSDTLYADSVKGRFTISRDNAKTTLYLQMNSLRPEDTAVYYC TIGGSLSRSSQGTTLVTVSSGGGSGGGGSQVQLQESGPGLVKPSQTL SLTCTVSGGSISTSGMGVGVIRQHPGKGLEWIGHIWDDDKRYN PALKSRVTISVDTSKNQFSLKLSVTAADTAVYYCARMELWSYFD YWGQGTTLVTVSSGGGSGGGSGGGSGGGSEIVLTQSPATLSLSPGE RATLSCASSSVSYMHYQQKPGQAPRLLIYDTSKLASGIPARFSG SSGSGTDFTLTISLLEPEDFAVYYCFQGSVYPFTFGQGTKLEIKRHHH HHHHHH (SEQ ID NO: 311)
tAb0037	HSA:CD3: CD19	EVQLVESGGGLVQPGNSLRSLCAASGFTFSFPGMSWVRQAPGKGL EWVSSISGSGSDTLYADSVKGRFTISRDNAKTTLYLQMNSLRPEDT AVYYCTIGGSLSRSSQGTTLVTVSSGGGSGGGSDIVMTQSPDSLAV SLGERATINCKSSQSLNARTGKNYLAWYQQKPGQPPKLLIYWAS TRESGVPDRESGSGSGTDFTLTISLQAEADVAVYYCKQSYSRRTFG

TABLE 5-continued

Exemplary Multi-specific Binding Proteins		
Construct	Format	Amino Acid Sequence
		GGTKVEIKGGGSGGGGSGGGGSGGGGSGVQLVQSGAEVKKPG ASVKVSCKASGFNIKDYMHVVRQAPGQRLWGMWIDLENANT IYDAKFQGRVTITRDTASTAYMELSSLRSEDVAVYCARDAYGR YFYDVWGQGLTVTVSSGGGSGGGGSGVQLQESGPGLVKPSQTLS LTCTVSGGSISTSGMGVWIRQHPGKLEWIGHIWWDDDKRYNP ALKSRVTISVDTSKNQFSLKLSVTAADTAVYVCARMELWSYFDF YWGQGLTVTVSSGGGSGGGGSGGGGSEIVLTQSPATLSLSPGER ATLSCSASSSVSYMHVYQKPGQAPRLLIYDTSKLASGIPARFSGS GSGTDFTLTISLEPEDFAVYVCFQGSVYPFTFGQGTKEIKRHHH HHHHHHH (SEQ ID NO: 312)
tAb0038	HSA: CD19v CD3	EVQLVESGGGLVQPGNSLRSLSCAASGFTFSSFGMSWVRQAPGKGL EHWSSISGSGDLYADSVKGRFTISRDAKNTTLVYLQMNSLRPEDT AVYYCTIGGSLRSRSGQGLTVTVSSGGGSGGGGSGVQLQESGPGLV KPSQTLSTCTVSGGSISTSGMGVWIRQHPGKLEWIGHIWWDD DKRYNPALKSRVTISVDTSKNQFSLKLSVTAADTAVYVCARMEL WSYFDFYWGQGLTVTVSSGGGSGGGGSGGGGSEIVLTQSPATL SLSPGERATLSCSASSSVSYMHVYQKPGQAPRLLIYDTSKLASGI PARFSGSGTDFTLTISLEPEDFAVYVCFQGSVYPFTFGQGTKE IKRGGGSGGGSDIVMTQSPDSLAVSLGERATINCKSSQSLNART GKNYLAWYQKPGQPPKLLIYWASTRESGVDRFSGSGSGTDFTL TISLQAEDVAVYVCKQSYRRTFGGQTKVEIKGGGSGGGGSGG GGSGGGGSGVQLVQSGAEVKKPGASVKVSCKASGFNIKDYMH VVRQAPGQRLWGMWIDLENANT IYDAKFQGRVTITRDTASTA YMELSSLRSEDVAVYCARDAYGRYFYDVWGQGLTVTVSSHHH HHHHHHH (SEQ ID NO: 313)
tAb0042	CD3: HSA: CD19	EVQLVESGGGLVQPGGSLKLSAASGFTFNKYAINWVRQAPGKG LEWVARIRSKYNNYATYYADQVKDRFTISRDDSKNTAYLQMNNL KTEDTAVYYCVRHANFGNSYISYWAYWGQGLTVTVSSGGGSGG GGGSGGGGSGTQVVTQEPSTLVSPGGTVTLTCASTGAVTSGNYPN WVQKPGQAPRGLIGGTFKFLVPGTPARFSGSLGGAALTLGSGVQ PEDEAEYYCTLWYSNRWVFGGQTKLTVLGGGSGGGSEVQLVES GGGLVQPGNSLRSLSCAASGFTFSSFGMSWVRQAPGKLEWVSSI SGSGDLYADSVKGRFTISRDAKNTTLVYLQMNSLRPEDTAVYYCT IGGSLRSRSGQGLTVTVSSGGGSGGGGSGVQLQESGPGLVKPSQTLS LTCTVSGGSISTSGMGVWIRQHPGKLEWIGHIWWDDDKRYNP ALKSRVTISVDTSKNQFSLKLSVTAADTAVYVCARMELWSYFDF YWGQGLTVTVSSGGGSGGGGSGGGGSEIVLTQSPATLSLSPGER ATLSCSASSSVSYMHVYQKPGQAPRLLIYDTSKLASGIPARFSGS GSGTDFTLTISLEPEDFAVYVCFQGSVYPFTFGQGTKEIKRHHH HHHHHHH (SEQ ID NO: 406)

The cultures were harvested by centrifugation at 4000 rpm, and the supernatant filtered through a 0.22 mm filter. The multi-specific binding proteins, which carried a 10×His tag at the C-terminus, were purified in two steps. The first step was Nickel affinity chromatography with elution using PBS containing 400 mM imidazole. The second step was size exclusion chromatography with elution in PBS (phosphate buffered saline) pH7.2. Multi-specific binding protein concentrations were determined by UV spectroscopy, and the protein samples were concentrated when necessary. The purity of the proteins was determined by sodium dodecyl sulphate polyacrylamide gel electrophoresis (SDS-PAGE) and high performance liquid chromatography (HPLC). Specifically, HPLC was performed on an Agilent® 1100 series instrument using MabPac size exclusion column run in PBS at 0.2 mL/min. The fractions with an elution time of about 225-240 minutes were collected for further characterization.

As noted above, the constructs produced contained an anti-CD19 scFv having the amino acid sequence set forth in SEQ ID NO: 9 or 18. The binding affinity of the two CD19 binding domains to CD19 were measured by SPR using a monomeric CD19 extracellular domain and a dimeric CD19 extracellular domain fused with human IgG1 Fc. Binding kinetic parameters were measured using an instrument (available under the tradename ForteBio™ from Pall Corporation®) generally as previously described (see, Estep et

al. (2013) MABs, 5 (2): 270-78). When measured with the monomeric CD19 protein, the K_D value of the CD19 binding domain having the sequence of SEQ ID NO: 9 was 7 nM, and the K_D value of the CD19 binding domain having the sequence of SEQ ID NO: 18 was 11 nM. When measured with the dimeric CD19 protein, the K_D value of the CD19 binding domain having the sequence of SEQ ID NO: 9 was 5 nM, and the K_D value of the CD19 binding domain having the sequence of SEQ ID NO: 18 was 15 nM.

Example 2. Multi-Specific Binding Proteins Induce T Cell Cytotoxicity Against CD19⁺ Target Cells

This example describes the cytotoxic activity of multi-specific binding proteins.

The T cell redirection activity of multi-specific binding proteins and BiTE® proteins were evaluated using a cell model available under the trade name KILR® Raji from Eurofins DiscoverX®. Briefly, pan T cells were isolated from primary human PBMCs from a single healthy donor by negative selection using a commercial kit (e.g. available under the name Easy Sep Human T Cell Enrichment Kit, StemCell Technologies®). T cells were maintained in RPMI 1640 medium supplemented with 10% serum and 300 IU/mL IL-2 to expand T cells. The harvested T cells were washed twice to remove any serum.

KILR® Raji cells, which expressed CD19 on the surface, were used as target cells. To opsonize the target cells, each multi-specific binding protein or BiTE® (see Table 5) was incubated with the target cells for 30 minutes at 37° C. in RPMI 1640 medium supplemented with 5% heat inactivated low IgG fetal bovine serum and penicillin-streptomycin-glutamine. The proteins were added in serial dilution at 10 different doses, with each dose run in duplicate. Human serum albumin was added to the medium of certain samples at a final concentration of 15 mg/mL. Selective proteins were also evaluated with KILR® SKOV3 cells, which were CD19-negative, as negative controls.

After opsonization, the target cells were incubated with the pan T cells at an effector-to-target (E:T) ratio of 10:1 for 6 hours at 37° C. Killing of the KILR® Raji cells resulted in release of a labeled housekeeping protein from these cells into the medium, which was quantified by addition of a KILR® detection reagent (Eurofins DiscoverX®). The luminescence signals from all wells were read on an Envision plate reader. Spontaneous release and total lysis controls were included on each plate to allow calculation of percent killing.

Percent killing was calculated from the luminescence signal values using the following formula: % killing=(value from test protein sample-mean value from spontaneous release control)/(mean value from total lysis control-mean value from spontaneous release control)×100. The EC50 values were calculated from the percent killing by fitting with a dose-response curve using software available under the trade name GraphPad® from GraphPad Software, LLC.

Table 6 lists the EC50 values of T cell-redirected killing in the absence and presence of human serum albumin for exemplary multi-specific binding proteins and comparator anti-CD19 BiTE® protein. No substantial killing was observed with the CD19-negative KILR® SKOV3 cells.

TABLE 6

Cytotoxic Activity of Multi-specific Binding Proteins				
Construct	Format	EC50 (pg/mL)		Fold change
		-HSA	+HSA	
tAb0027	CD19:CD3:HSA	2.33	67.9	29.1
tAb0029	CD3:CD19:HSA	5.48	188.1	34.3
tAb0030	CD3:HSA:CD19	3.17	141.7	44.7
tAb0031	HSA:CD3:CD19	6.55	189.3	28.9
tAb0032	HSA:CD19:CD3	5.71	88.4	15.5
tAb0033	CD19:CD3:HSA	20.4	1294	63.4
tAb0034	CD19:HSA:CD3	59.3	2376	40.1
tAb0035	CD3:CD19:HSA	51.4	2184	42.5
tAb0036	CD3:HSA:CD19	103.8	4195	40.4
tAb0037	HSA:CD3:CD19	175.1	2151	12.3
tAb0038	HSA:CD19:CD3	52.6	288	5.5
tAb0042	CD3:HSA:CD19	69.9	13290	190.1
blinatumomab	CD19:CD3	2854	10750	3.8

As shown in Table 6, the multi-specific binding proteins containing the anti-CD19 scFv having the amino acid sequence of SEQ ID NO: 9 showed stronger cytotoxic activity than those containing the anti-CD19 scFv having the amino acid sequence of SEQ ID NO: 18, regardless of the construct format, CD3 binding domain, HSA binding domain, and the presence or absence of HSA in the assay medium. From this data, it is contemplated that constructs containing this anti-CD19 scFv with the higher binding affinity to CD19 will demonstrate stronger therapeutic activity than constructs containing the other anti-CD19 scFv with the lower binding affinity.

Furthermore, all the multi-specific binding proteins tested showed lower EC₅₀ value (namely, stronger ability to induce cytotoxicity) in the absence of HSA than in the presence of HSA. Without wishing to be bound by theory, it appears that the presence of HSA causes a change in the protein complex, which was specific to the multi-specific binding proteins containing an HSA binding domain, rather than a nonspecific effect as observed with blinatumomab. The ratio of the EC₅₀ value in the presence of HSA to the EC₅₀ value in the absence of HSA, also called “fold change” herein, was used to assess the effect of HSA on the potential therapeutic activity of the multi-specific binding protein. As shown in Table 6, the construct formats with the HSA binding domain N-terminal to both the CD19 binding domain and the CD3 binding domain (namely, tAb0031, tAb0032, tAb0037, and tAb0038) showed lower fold changes than the other construct formats, regardless of which CD19 binding domain was used in the construct.

Furthermore, among the constructs having the same CD19 binding domain, CD3 binding domain, and HSA binding domain, the constructs in the CD19:CD3:HSA format (i.e., the CD19 binding domain positioned N-terminal to the CD3 binding domain, and the CD3 binding domain positioned N-terminal to the HSA binding domain), namely, tAb0027 and tAb0033, showed the lowest or second lowest EC₅₀ values both in the absence and in the presence of HSA.

Example 3. Cytotoxicity of Multi-Specific Binding Proteins Against CD19⁺ Target Cells

This example provides alternative methods for determining the cytotoxic activity of a multi-specific binding protein.

The multi-specific binding proteins disclosed herein can be evaluated in in vitro assays on their mediation of T cell dependent cytotoxicity to B cell antigen positive target cells. For example, the CD19-binding multi-specific binding protein disclosed herein is evaluated in in vitro assays on its mediation of T cell dependent cytotoxicity to CD19⁺ target cells.

Fluorescence labeled CD19⁺ MEC-1 cells (a CD19⁺ human chronic B cell leukemia cell line) are incubated with isolated PBMC of random donors or CB15 T-cells (standardized T-cell line) as effector cells in the presence of the CD19-binding multi-specific binding protein. After incubation for 4 hours at 37° C. in a humidified incubator, the release of the fluorescent dye from the target cells into the supernatant is determined in a spectrofluorimeter. Target cells incubated without the CD19-binding multi-specific binding protein and target cells totally lysed by the addition of saponin at the end of the incubation serve as negative and positive controls, respectively. Based on the measured remaining living target cells, the percentage of specific cell lysis can be calculated according to the following formula: $[1 - (\text{number of living targets}_{\text{sample}} / \text{number of living targets}_{\text{spontaneous}})] \times 100\%$. Sigmoidal dose response curves and EC₅₀ values are calculated by non-linear regression/4-parameter logistic fit using software available under the trade name GraphPad® from GraphPad Software, LLC. The lysis values obtained for a given multi-specific binding protein concentration are used to calculate sigmoidal dose-response curves by 4 parameter logistic fit analysis using the Prism software. It is expected that the target cell lysis rate induced by CD19-binding multi-specific binding protein is higher than the target cell lysis rate induced by similar constructs lacking either a CD19-binding domain or a CD3-binding domain.

Alternatively, a human T-cell dependent cellular cytotoxicity (TDCC) assay is used to measure the ability of the multi-specific binding protein to direct T cells to kill tumor cells (Nazarian et al. 2015, *J. Biomol. Screen*, 20:519-27). In this assay, T cells and target cancer cell line cells are mixed together at a 10:1 ratio in a 384 wells plate, and varying amounts of the multi-specific binding proteins are added. After 48 hours, the T cells are washed away leaving attached to the plate target cells that were not killed by the T cells. To quantitate the remaining viable cells, a luminescent cell viability assay (available under the tradename CellTiter-GLO® from Promega®) is used. It is contemplated that the killing rate of B-cell antigen expressing cancer cell induced by CD19-binding multi-specific binding protein will be higher than that induced by similar constructs lacking either a CD19-binding domain or a CD3-binding domain and/or other negative control molecules.

Example 4. Pharmacokinetics of Multi-Specific Binding Proteins with HSA Binding Domain

This example is designed to determine the pharmacokinetics of multi-specific binding proteins.

Multi-specific binding proteins containing a domain that binds CD19, a domain that binds CD3, and a domain that binds serum albumin are tested in the cynomolgus monkey in the context of pharmacokinetic (PK) studies to evaluate the serum elimination time of the multi-specific binding protein.

The multi-specific binding proteins are administered as intravenous bolus or intravenous infusion. The multi-specific binding proteins are administered in a dose-linear, pharmacokinetic relevant range of 0.5 µg/kg to 3 µg/kg, 6 µg/kg, 12 µg/kg, and 15 µg/kg, respectively. For purposes of comparability, the serum concentrations of the multi-specific binding proteins are does-normalized and molecular weight-normalized (described in nmol).

For each multi-specific binding protein, a group of at least two to three animals are used. Blood samples are collected and serum is prepared for determination of serum concentrations of the multi-specific binding proteins. Serum multi-specific binding protein levels are measured using an immunoassay. The assay is performed by capturing the multi-specific binding protein via its CD19-binding domain, while an antibody directed against the CD3-binding domain of the multi-specific binding protein is used for detection. The serum concentration-time profiles are used to determine PK parameters using known analytical methods such as those described in Ritschel W A and Kearns G L, 1999, IN: *Handbook Of Basic Pharmacokinetics Including Clinical Applications*, 5th edition, American Pharmaceutical Assoc., Washington, D.C. and software available under the trade name WinNonlin® professional V. 3.1 from Certara, L.P.).

Alternatively, the serum half-life of the various multi-specific binding proteins containing the serum albumin binding domain is compared to that of control constructs capable of binding CD19 and CD3 but lacking a serum albumin binding domain by including in the experiment another cynomolgus monkey group that receives the control constructs. Additional domains can be included such that the control constructs are similar in size to the multi-specific binding proteins.

It is expected that CD19-binding multi-specific binding protein will have significantly longer serum half-life compared to similar constructs capable of binding CD19 and

CD3 but lacking a serum albumin binding domain and/or other negative control molecules.

Example 5. Determination of Antigen Affinity by Flow Cytometry

This example is designed to determine the affinity of a multi-specific binding protein to an antigen.

Various multi-specific binding proteins disclosed herein are tested for their binding affinities to human CD3⁺ cells and the corresponding B cell surface antigen positive cells, such as human CD19⁺ cells. The multi-specific binding proteins are also tested for their binding affinities to cynomolgus CD3⁺ cells and the corresponding B cell surface antigen positive cells, such as cynomolgus CD19⁺ cells.

CD3⁺ and CD19⁺ cells are incubated with 100 µL of serial dilutions of the multi-specific binding protein. After washing three times with FACS buffer the cells are incubated with 0.1 mL of 10 µg/mL mouse monoclonal anti-idiotypic antibody in the same buffer for 45 mins on ice. After a second washing cycle, the cells are incubated with 0.1 mL of 15 µg/mL FITC-conjugated goat anti-mouse IgG antibodies under the same conditions as before. As a control, cells are incubated with the anti-His IgG followed by the FITC-conjugated goat anti-mouse IgG antibodies without the multi-specific binding protein. The cells are then washed again and resuspended in 0.2 mL of FACS buffer containing 2 µg/mL propidium iodide (PI) in order to exclude dead cells. The fluorescence of 1×10⁴ living cells is measured using a commercially available flow cytometer and software. Mean fluorescence intensities of the cell samples are calculated using software available under the trade names CXP software (Beckman-Coulter®, Krefeld, Germany) or Incyte™ software (Merck Millipore, Schwalbach, Germany). K_D values for one-site binding can be calculated using normalized fluorescence intensity values with known computational equations such as those supplied under the trade name GraphPad Prism® software from GraphPad® Software, La Jolla Calif. USA. CD3 binding affinity and cross-reactivity are evaluated in titration and flow cytometric experiments on CD3⁺ Jurkat cells and the cynomolgus CD3⁺ HSC-F cell line. CD19 binding and cross-reactivity are assessed on the human CD19⁺ tumor cell lines. The K_D ratio of cross-reactivity can be calculated using the K_D values determined on the CHO cell lines expressing either recombinant human or recombinant cynomolgus antigens.

Example 6. Cytokine Production Induced by Multi-Specific Binding Proteins

This example is designed to determine the ability of a multi-specific binding protein to induce cytokine production from immune cells.

Assays available under the trade name AlphaLISA® (Perkin Elmer®) for TNFα and Interferon γ are used to obtain evidence that T cells are activated by the multi-specific binding proteins of current invention, such as CD19-binding multi-specific binding protein, in the presence of target cells, such as CD19⁺ B cells. For this assay, primary human T cells and human tumor cells expressing B cell surface antigen are incubated in the presence of the CD19-binding multi-specific binding protein as described under cytotoxicity assays. After 48 hours of incubation, 2 microliter aliquots of the assay supernatants are analyzed according to the manufacturer's instructions. It is contemplated that the TNFα or Interferon γ level induced by CD19-binding multi-specific binding protein is higher than

that induced by similar constructs lacking either a CD19-binding domain or a CD3-binding domain and/or other negative control molecules.

Example 7. Identification of scFv Variants that Bind Human CD3ε

This example is designed to identify variants of the antigen-binding sites disclosed herein that bind human CD3ε.

The binding properties of the parental CD3ε binding construct to biotin-CD3ε and to biotin-HSA are characterized. To construct the anti-CD3ε scFv phage libraries, a single substitution library is provided for the heavy chain CDR1, heavy chain CDR2, heavy chain CDR3, light chain CDR1, light chain CDR2, and light chain CDR3 domains. Residues are varied one at a time via mutagenesis. For selection of clones and determination of binding affinity, single substitution libraries are bound to biotinylated human CD3ε, washed, eluted, and counted. Biotinylated cynomolgus CD3ε is used as the round 1 selection target, and washed for 4 hours after combinatorial phage binding from the two independent libraries (~2× selection). Biotinylated human CD3ε is used as the round 2 selection target, and washed for 3 hours after binding of both libraries (<2× selection). PCRred inserts from the second round of selection are sub-cloned into the pcDNA3.4 His6 expression vector. 180 clones are picked and DNA is purified, sequenced, and

transfected into an expression system available under the trade name Expi293® from Thermo Fisher®. A panel of sixteen clones with a range of affinities for human CD3ε are selected for more precise determination of the parameters such as the dissociation constant (K_D), the dissociation rate (k_d or k_{off}), and the association rate (k_a or k_{on}).

INCORPORATION BY REFERENCE

All publications and patents cited throughout the text of this specification (including all patents, patent applications, scientific publications, manufacturer's specifications, instructions, etc.), whether supra or infra, are hereby incorporated by reference in their entirety for all purposes. To the extent the material incorporated by reference contradicts or is inconsistent with this specification, the specification will supersede any such material.

EQUIVALENTS

The invention may be embodied in other specific forms without departing from the spirit or essential characteristics thereof. The foregoing embodiments are therefore to be considered in all respects illustrative rather than limiting the invention described herein. Scope of the invention is thus indicated by the appended claims rather than by the foregoing description, and all changes that come within the meaning and range of equivalency of the claims are intended to be embraced therein.

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Met Glu Leu Ser Arg Leu Arg Ser Asp Asp Thr Ala Val Tyr Tyr Cys
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35 40 45

Pro Gln Leu Leu Ile Tyr Arg Val Ser Lys Arg Phe Ser Gly Val Pro
50 55 60

Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Lys Ile
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Arg Val Ser Lys Arg Phe Ser
 1 5

<210> SEQ ID NO 8

<211> LENGTH: 9

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 8

Leu Gln Leu Leu Glu Asp Pro Tyr Thr
 1 5

<210> SEQ ID NO 9

<211> LENGTH: 248

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 9

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala
 1 5 10 15

Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Asp Tyr
 20 25 30

Ile Met His Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Met
 35 40 45

Gly Tyr Ile Asn Pro Tyr Asn Asp Gly Ser Lys Tyr Thr Glu Lys Phe
 50 55 60

Gln Gly Arg Val Thr Met Thr Ser Asp Thr Ser Ile Ser Thr Ala Tyr
 65 70 75 80

Met Glu Leu Ser Arg Leu Arg Ser Asp Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Ala Arg Gly Thr Tyr Tyr Tyr Gly Pro Gln Leu Phe Asp Tyr Trp Gly
 100 105 110

Gln Gly Thr Thr Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly
 115 120 125

Gly Gly Ser Gly Gly Gly Gly Ser Asp Ile Val Met Thr Gln Thr Pro
 130 135 140

Leu Ser Leu Ser Val Thr Pro Gly Gln Pro Ala Ser Ile Ser Cys Lys
 145 150 155 160

Ser Ser Gln Ser Leu Glu Thr Ser Thr Gly Thr Thr Tyr Leu Asn Trp
 165 170 175

Tyr Leu Gln Lys Pro Gly Gln Ser Pro Gln Leu Leu Ile Tyr Arg Val

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180	185	190
Ser Lys Arg Phe Ser Gly Val	Pro Asp Arg Phe Ser Gly Ser Gly Ser	
195	200	205
Gly Thr Asp Phe Thr Leu Lys Ile Ser Arg Val	Glu Ala Glu Asp Val	
210	215	220
Gly Val Tyr Tyr Cys Leu Gln Leu Leu Glu Asp	Pro Tyr Thr Phe Gly	
225	230	235
Gln Gly Thr Lys Leu Glu Ile Lys		
245		

<210> SEQ ID NO 10
 <211> LENGTH: 120
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 10

Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Gln
1 5 10 15
Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Gly Ser Ile Ser Thr Ser
20 25 30
Gly Met Gly Val Gly Trp Ile Arg Gln His Pro Gly Lys Gly Leu Glu
35 40 45
Trp Ile Gly His Ile Trp Trp Asp Asp Asp Lys Arg Tyr Asn Pro Ala
50 55 60
Leu Lys Ser Arg Val Thr Ile Ser Val Asp Thr Ser Lys Asn Gln Phe
65 70 75 80
Ser Leu Lys Leu Ser Ser Val Thr Ala Ala Asp Thr Ala Val Tyr Tyr
85 90 95
Cys Ala Arg Met Glu Leu Trp Ser Tyr Tyr Phe Asp Tyr Trp Gly Gln
100 105 110
Gly Thr Leu Val Thr Val Ser Ser
115 120

<210> SEQ ID NO 11
 <211> LENGTH: 107
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 11

Glu Ile Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Leu Ser Pro Gly
1 5 10 15
Glu Arg Ala Thr Leu Ser Cys Ser Ala Ser Ser Ser Val Ser Tyr Met
20 25 30
His Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile Tyr
35 40 45
Asp Thr Ser Lys Leu Ala Ser Gly Ile Pro Ala Arg Phe Ser Gly Ser
50 55 60
Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Glu Pro Glu
65 70 75 80
Asp Phe Ala Val Tyr Tyr Cys Phe Gln Gly Ser Val Tyr Pro Phe Thr
85 90 95
Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys Arg

-continued

100

105

<210> SEQ ID NO 12
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 12

Thr Ser Gly Met Gly Val Gly
1 5

<210> SEQ ID NO 13
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 13

His Ile Trp Trp Asp Asp Asp Lys Arg Tyr Asn Pro Ala Leu Lys Ser
1 5 10 15

<210> SEQ ID NO 14
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 14

Met Glu Leu Trp Ser Tyr Tyr Phe Asp Tyr
1 5 10

<210> SEQ ID NO 15
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 15

Ser Ala Ser Ser Ser Val Ser Tyr Met His
1 5 10

<210> SEQ ID NO 16
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 16

Asp Thr Ser Lys Leu Ala Ser
1 5

<210> SEQ ID NO 17
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic

-continued

peptide

<400> SEQUENCE: 17

Phe Gln Gly Ser Val Tyr Pro Phe Thr
 1 5

<210> SEQ ID NO 18

<211> LENGTH: 242

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 18

Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Gln
 1 5 10 15

Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Gly Ser Ile Ser Thr Ser
 20 25 30

Gly Met Gly Val Gly Trp Ile Arg Gln His Pro Gly Lys Gly Leu Glu
 35 40 45

Trp Ile Gly His Ile Trp Trp Asp Asp Asp Lys Arg Tyr Asn Pro Ala
 50 55 60

Leu Lys Ser Arg Val Thr Ile Ser Val Asp Thr Ser Lys Asn Gln Phe
 65 70 75 80

Ser Leu Lys Leu Ser Ser Val Thr Ala Ala Asp Thr Ala Val Tyr Tyr
 85 90 95

Cys Ala Arg Met Glu Leu Trp Ser Tyr Tyr Phe Asp Tyr Trp Gly Gln
 100 105 110

Gly Thr Leu Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly
 115 120 125

Gly Ser Gly Gly Gly Gly Ser Glu Ile Val Leu Thr Gln Ser Pro Ala
 130 135 140

Thr Leu Ser Leu Ser Pro Gly Glu Arg Ala Thr Leu Ser Cys Ser Ala
 145 150 155 160

Ser Ser Ser Val Ser Tyr Met His Trp Tyr Gln Gln Lys Pro Gly Gln
 165 170 175

Ala Pro Arg Leu Leu Ile Tyr Asp Thr Ser Lys Leu Ala Ser Gly Ile
 180 185 190

Pro Ala Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr
 195 200 205

Ile Ser Ser Leu Glu Pro Glu Asp Phe Ala Val Tyr Tyr Cys Phe Gln
 210 215 220

Gly Ser Val Tyr Pro Phe Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile
 225 230 235 240

Lys Arg

<210> SEQ ID NO 19

<211> LENGTH: 121

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 19

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Lys Pro Gly Gly
 1 5 10 15

-continued

Ser Leu Lys Leu Ser Cys Ala Ala Ser Gly Tyr Thr Phe Thr Ser Tyr
 20 25 30

Val Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile
 35 40 45

Gly Tyr Ile Asn Pro Tyr Asn Asp Gly Thr Lys Tyr Asn Glu Lys Phe
 50 55 60

Gln Gly Arg Val Thr Ile Ser Ser Asp Lys Ser Ile Ser Thr Ala Tyr
 65 70 75 80

Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Met Tyr Tyr Cys
 85 90 95

Ala Arg Gly Thr Tyr Tyr Tyr Gly Thr Arg Val Phe Asp Tyr Trp Gly
 100 105 110

Gln Gly Thr Leu Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 20
 <211> LENGTH: 112
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 20

Asp Ile Val Met Thr Gln Ser Pro Ala Thr Leu Ser Leu Ser Pro Gly
 1 5 10 15

Glu Arg Ala Thr Leu Ser Cys Arg Ser Ser Lys Ser Leu Gln Asn Val
 20 25 30

Asn Gly Asn Thr Tyr Leu Tyr Trp Phe Gln Gln Lys Pro Gly Gln Ser
 35 40 45

Pro Gln Leu Leu Ile Tyr Arg Met Ser Asn Leu Asn Ser Gly Val Pro
 50 55 60

Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile
 65 70 75 80

Ser Ser Leu Glu Pro Glu Asp Phe Ala Val Tyr Tyr Cys Met Gln His
 85 90 95

Leu Glu Tyr Pro Ile Thr Phe Gly Ala Gly Thr Lys Leu Glu Ile Lys
 100 105 110

<210> SEQ ID NO 21
 <211> LENGTH: 5
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 21

Ser Tyr Val Met His
 1 5

<210> SEQ ID NO 22
 <211> LENGTH: 14
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 22

Trp Ile Gly Tyr Ile Asn Pro Tyr Asn Asp Gly Thr Lys Tyr

-continued

1 5 10

<210> SEQ ID NO 23
 <211> LENGTH: 12
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 23

Gly Thr Tyr Tyr Tyr Gly Thr Arg Val Phe Asp Tyr
 1 5 10

<210> SEQ ID NO 24
 <211> LENGTH: 16
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 24

Arg Ser Ser Lys Ser Leu Gln Asn Val Asn Gly Asn Thr Tyr Leu Tyr
 1 5 10 15

<210> SEQ ID NO 25
 <211> LENGTH: 7
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 25

Arg Met Ser Asn Leu Asn Ser
 1 5

<210> SEQ ID NO 26
 <211> LENGTH: 9
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 26

Met Gln His Leu Glu Tyr Pro Ile Thr
 1 5

<210> SEQ ID NO 27
 <211> LENGTH: 124
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 27

Gln Val Gln Leu Gln Gln Ser Gly Ala Glu Leu Val Arg Pro Gly Ser
 1 5 10 15

Ser Val Lys Ile Ser Cys Lys Ala Ser Gly Tyr Ala Phe Ser Ser Tyr
 20 25 30

Trp Met Asn Trp Val Lys Gln Arg Pro Gly Gln Gly Leu Glu Trp Ile
 35 40 45

Gly Gln Ile Trp Pro Gly Asp Gly Asp Thr Asn Tyr Asn Gly Lys Phe

-continued

50	55	60
Lys Gly Lys Ala Thr	Leu Thr Ala Asp Glu Ser Ser Ser Thr Ala Tyr	
65	70	75 80
Met Gln Leu Ser Ser	Leu Ala Ser Glu Asp Ser Ala Val Tyr Phe Cys	
	85 90 95	
Ala Arg Arg Glu Thr Thr Thr Val Gly Arg Tyr Tyr Tyr Ala Met Asp		
100	105	110
Tyr Trp Gly Gln Gly Thr Ser Val Thr Val Ser Ser		
115	120	

<210> SEQ ID NO 28
 <211> LENGTH: 111
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 28

Asp Ile Leu Leu Thr Gln Thr Pro Ala Ser Leu Ala Val Ser Leu Gly
1 5 10 15
Gln Arg Ala Thr Ile Ser Cys Lys Ala Ser Gln Ser Val Asp Tyr Asp
20 25 30
Gly Asp Ser Tyr Leu Asn Trp Tyr Gln Gln Ile Pro Gly Gln Pro Pro
35 40 45
Lys Leu Leu Ile Tyr Asp Ala Ser Asn Leu Val Ser Gly Ile Pro Pro
50 55 60
Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Asn Ile His
65 70 75 80
Pro Val Glu Lys Val Asp Ala Ala Thr Tyr His Cys Gln Gln Ser Thr
85 90 95
Glu Asp Pro Trp Thr Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys
100 105 110

<210> SEQ ID NO 29
 <211> LENGTH: 5
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 29

Ser Tyr Trp Met Asn
1 5

<210> SEQ ID NO 30
 <211> LENGTH: 17
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 30

Gln Ile Trp Pro Gly Asp Gly Asp Thr Asn Tyr Asn Gly Lys Phe Lys
1 5 10 15

Gly

<210> SEQ ID NO 31
 <211> LENGTH: 15

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<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 31

Arg Glu Thr Thr Thr Val Gly Arg Tyr Tyr Tyr Ala Met Asp Tyr
1 5 10 15

<210> SEQ ID NO 32
<211> LENGTH: 15
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 32

Lys Ala Ser Gln Ser Val Asp Tyr Asp Gly Asp Ser Tyr Leu Asn
1 5 10 15

<210> SEQ ID NO 33
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 33

Asp Ala Ser Asn Leu Val Ser
1 5

<210> SEQ ID NO 34
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 34

Gln Gln Ser Thr Glu Asp Pro Trp Thr
1 5

<210> SEQ ID NO 35
<211> LENGTH: 124
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 35

Gln Val Gln Leu Gln Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ser
1 5 10 15

Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Ala Phe Ser Ser Tyr
20 25 30

Trp Met Asn Trp Val Arg Gln Arg Pro Gly Gln Gly Leu Glu Trp Ile
35 40 45

Gly Gln Ile Trp Pro Gly Asp Gly Asp Thr Asn Tyr Asn Gly Lys Phe
50 55 60

Lys Gly Arg Ala Thr Ile Thr Ala Asp Glu Ser Thr Asn Thr Ala Tyr
65 70 75 80

-continued

Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Phe Tyr Ser Cys
85 90 95

Ala Arg Arg Glu Thr Thr Thr Val Gly Arg Tyr Tyr Tyr Ala Met Asp
100 105 110

Tyr Trp Gly Gln Gly Thr Thr Val Thr Val Ser Ser
115 120

<210> SEQ ID NO 36
<211> LENGTH: 112
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 36

Asp Ile Gln Leu Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1 5 10 15

Asp Arg Val Thr Ile Thr Cys Lys Ala Ser Gln Ser Val Asp Tyr Asp
20 25 30

Gly Asp Ser Tyr Leu Asn Trp Tyr Gln Gln Ile Pro Gly Lys Ala Pro
35 40 45

Lys Leu Leu Ile Tyr Asp Ala Ser Asn Leu Val Ser Gly Ile Pro Pro
50 55 60

Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Tyr Thr Phe Thr Ile Ser
65 70 75 80

Ser Leu Gln Pro Glu Asp Ile Ala Thr Tyr His Cys Gln Gln Ser Thr
85 90 95

Glu Asp Pro Trp Thr Phe Gly Gly Gly Thr Lys Leu Gln Ile Lys Arg
100 105 110

<210> SEQ ID NO 37
<211> LENGTH: 120
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 37

Gln Val Gln Leu Glu Gln Pro Gly Ala Glu Val Val Lys Pro Gly Ala
1 5 10 15

Ser Val Lys Val Ser Cys Lys Thr Ser Gly Tyr Thr Phe Thr Ser Asn
20 25 30

Trp Met His Trp Val Lys Gln Thr Pro Gly Lys Gly Leu Glu Trp Ile
35 40 45

Gly Glu Ile Asp Pro Ser Asp Ser Tyr Thr Asn Tyr Asn Gln Lys Phe
50 55 60

Asp Gly Lys Ala Lys Leu Thr Val Asp Lys Ser Ser Ser Thr Ala Tyr
65 70 75 80

Met Glu Val Ser Asp Leu Thr Ala Glu Asp Ser Ala Thr Tyr Tyr Cys
85 90 95

Ala Arg Gly Ser Asn Pro Tyr Tyr Tyr Ala Met Asp Tyr Trp Gly Gln
100 105 110

Gly Thr Ser Val Thr Val Ser Ser
115 120

<210> SEQ ID NO 38
<211> LENGTH: 104

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<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 38

Gln Ile Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Ala Ser Pro Gly
1 5 10 15
Glu Lys Ala Thr Met Thr Cys Ser Ala Ser Ser Gly Val Asn Tyr Met
 20 25 30
His Trp Tyr Gln Gln Lys Pro Gly Thr Ser Pro Lys Arg Trp Ile Tyr
 35 40 45
Asp Thr Asp Lys Thr Ala Ser Gly Val Pro Ala Arg Phe Ser Gly Ser
 50 55 60
Gly Ser Gly Thr Ser Tyr Ser Leu Thr Ile Ser Ser Met Glu Ala Glu
65 70 75 80
Asp Ala Ala Thr Tyr Tyr Cys His Gln Arg Gly Ser Tyr Thr Phe Gly
 85 90 95
Gly Gly Thr Lys Leu Glu Ile Lys
 100

<210> SEQ ID NO 39
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 39

Ser Asn Trp Met His
1 5

<210> SEQ ID NO 40
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 40

Glu Ile Asp Pro Ser Asp Ser Tyr Thr Asn
1 5 10

<210> SEQ ID NO 41
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 41

Gly Ser Asn Pro Tyr Tyr Tyr Ala Met Asp Tyr
1 5 10

<210> SEQ ID NO 42
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

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<400> SEQUENCE: 42

Ser Ala Ser Ser Gly Val Asn Tyr Met His
1 5 10

<210> SEQ ID NO 43

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 43

Asp Thr Asp Lys Thr Ala Ser
1 5

<210> SEQ ID NO 44

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 44

His Gln Arg Gly Ser Tyr Thr
1 5

<210> SEQ ID NO 45

<211> LENGTH: 121

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 45

Glu Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Glu
1 5 10 15

Ser Leu Lys Ile Ser Cys Lys Gly Ser Gly Tyr Ser Phe Ser Ser Ser
20 25 30

Trp Ile Gly Trp Val Arg Gln Met Pro Gly Lys Gly Leu Glu Trp Met
35 40 45

Gly Ile Ile Tyr Pro Asp Asp Ser Asp Thr Arg Tyr Ser Pro Ser Phe
50 55 60

Gln Gly Gln Val Thr Ile Ser Ala Asp Lys Ser Ile Arg Thr Ala Tyr
65 70 75 80

Leu Gln Trp Ser Ser Leu Lys Ala Ser Asp Thr Ala Met Tyr Tyr Cys
85 90 95

Ala Arg His Val Thr Met Ile Trp Gly Val Ile Ile Asp Phe Trp Gly
100 105 110

Gln Gly Thr Leu Val Thr Val Ser Ser
115 120

<210> SEQ ID NO 46

<211> LENGTH: 107

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 46

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Ala Ile Gln Leu Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
 1 5 10 15
 Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Gly Ile Ser Ser Ala
 20 25 30
 Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
 35 40 45
 Tyr Asp Ala Ser Ser Leu Glu Ser Gly Val Pro Ser Arg Phe Ser Gly
 50 55 60
 Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
 65 70 75 80
 Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Phe Asn Ser Tyr Pro Phe
 85 90 95
 Thr Phe Gly Pro Gly Thr Lys Val Asp Ile Lys
 100 105

<210> SEQ ID NO 47
 <211> LENGTH: 5
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 47

Ser Ser Trp Ile Gly
 1 5

<210> SEQ ID NO 48
 <211> LENGTH: 17
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 48

Ile Ile Tyr Pro Asp Asp Ser Asp Thr Arg Tyr Ser Pro Ser Phe Gln
 1 5 10 15

Gly

<210> SEQ ID NO 49
 <211> LENGTH: 12
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 49

His Val Thr Met Ile Trp Gly Val Ile Ile Asp Phe
 1 5 10

<210> SEQ ID NO 50
 <211> LENGTH: 11
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 50

Arg Ala Ser Gln Gly Ile Ser Ser Ala Leu Ala
 1 5 10

-continued

<210> SEQ ID NO 51
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 51

Asp Ala Ser Ser Leu Glu Ser
1 5

<210> SEQ ID NO 52
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 52

Gln Gln Phe Asn Ser Tyr Pro Phe Thr
1 5

<210> SEQ ID NO 53
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 53

Ala Ile Gln Leu Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1 5 10 15
Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Gly Ile Ser Ser Ala
20 25 30
Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
35 40 45
Tyr Asp Ala Ser Ser Leu Glu Ser Gly Val Pro Ser Arg Phe Ser Gly
50 55 60
Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65 70 75 80
Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Phe Asn Ser Tyr Pro Tyr
85 90 95
Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys
100 105

<210> SEQ ID NO 54
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 54

Gln Gln Phe Asn Ser Tyr Pro Tyr Thr
1 5

<210> SEQ ID NO 55
<211> LENGTH: 119
<212> TYPE: PRT

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<213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 55

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ser
 1 5 10 15
 Ser Val Lys Val Ser Cys Lys Asp Ser Gly Gly Thr Phe Ser Ser Tyr
 20 25 30
 Ala Ile Ser Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Met
 35 40 45
 Gly Gly Ile Ile Pro Ile Phe Gly Thr Thr Asn Tyr Ala Gln Gln Phe
 50 55 60
 Gln Gly Arg Val Thr Ile Thr Ala Asp Glu Ser Thr Ser Thr Ala Tyr
 65 70 75 80
 Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95
 Ala Arg Glu Ala Val Ala Ala Asp Trp Leu Asp Pro Trp Gly Gln Gly
 100 105 110
 Thr Leu Val Thr Val Ser Ser
 115

<210> SEQ ID NO 56
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 56

Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly
 1 5 10 15
 Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Val Ser Ser Ser
 20 25 30
 Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu
 35 40 45
 Ile Tyr Gly Ala Ser Ser Arg Ala Thr Gly Ile Pro Asp Arg Phe Ser
 50 55 60
 Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Arg Leu Glu
 65 70 75 80
 Pro Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Gly Ser Ser Arg
 85 90 95
 Phe Thr Phe Gly Pro Gly Thr Lys Val Asp Ile Lys
 100 105

<210> SEQ ID NO 57
 <211> LENGTH: 5
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 57

Ser Tyr Ala Ile Ser
 1 5

<210> SEQ ID NO 58

-continued

<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 58

Gly Ile Ile Pro Ile Phe Gly Thr Thr Asn Tyr Ala Gln Gln Phe Gln
1 5 10 15

Gly

<210> SEQ ID NO 59
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 59

Glu Ala Val Ala Ala Asp Trp Leu Asp Pro
1 5 10

<210> SEQ ID NO 60
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 60

Arg Ala Ser Gln Ser Val Ser Ser Ser Tyr Leu Ala
1 5 10

<210> SEQ ID NO 61
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 61

Gly Ala Ser Ser Arg Ala Thr
1 5

<210> SEQ ID NO 62
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 62

Gln Gln Tyr Gly Ser Ser Arg Phe Thr
1 5

<210> SEQ ID NO 63
<211> LENGTH: 124
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

-continued

<400> SEQUENCE: 63

Glu Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Glu
 1 5 10 15
 Ser Leu Lys Ile Ser Cys Lys Gly Ser Gly Tyr Ser Phe Thr Ser Tyr
 20 25 30
 Trp Ile Ala Trp Val Arg Gln Met Pro Gly Lys Gly Leu Glu Trp Met
 35 40 45
 Gly Ile Ile Tyr Pro Gly Asp Ser Asp Thr Arg Tyr Ser Pro Ser Phe
 50 55 60
 Gln Gly Gln Val Thr Ile Ser Ala Asp Lys Ser Ile Ser Thr Ala Tyr
 65 70 75 80
 Leu Gln Trp Ser Ser Leu Lys Ala Ser Asp Thr Ala Met Tyr Tyr Cys
 85 90 95
 Ala Arg Gln Gly Tyr Ser Ser Gly Trp Asp Ser Tyr Tyr Gly Met Gly
 100 105 110
 Val Trp Gly Gln Gly Thr Thr Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 64

<211> LENGTH: 107

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 64

Ala Ile Gln Leu Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
 1 5 10 15
 Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Gly Ile Ser Ser Ala
 20 25 30
 Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
 35 40 45
 Tyr Asp Ala Ser Ser Leu Glu Ser Gly Val Pro Ser Arg Phe Ser Gly
 50 55 60
 Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
 65 70 75 80
 Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Phe Asn Ser Tyr Pro Tyr
 85 90 95
 Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys
 100 105

<210> SEQ ID NO 65

<211> LENGTH: 5

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 65

Ser Tyr Trp Ile Ala
 1 5

<210> SEQ ID NO 66

<211> LENGTH: 17

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic

-continued

peptide

<400> SEQUENCE: 66

Ile Ile Tyr Pro Gly Asp Ser Asp Thr Arg Tyr Ser Pro Ser Phe Gln
 1 5 10 15

Gly

<210> SEQ ID NO 67

<211> LENGTH: 15

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 67

Gln Gly Tyr Ser Ser Gly Trp Asp Ser Tyr Tyr Gly Met Gly Val
 1 5 10 15

<210> SEQ ID NO 68

<211> LENGTH: 123

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 68

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ser
 1 5 10 15

Ser Val Lys Val Ser Cys Lys Ala Ser Gly Gly Thr Phe Ser Ser Tyr
 20 25 30

Thr Ile Asn Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Met
 35 40 45

Gly Gly Ile Ile Pro Ile Phe Gly Ile Pro Asn Tyr Ala Gln Lys Phe
 50 55 60

Gln Gly Arg Val Thr Ile Thr Ala Asp Glu Ser Thr Asn Thr Ala Tyr
 65 70 75 80

Met Glu Leu Ser Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Ala Arg Ala Ser Gly Gly Ser Ala Asp Tyr Ser Tyr Gly Met Asp Val
 100 105 110

Trp Gly Gln Gly Thr Ala Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 69

<211> LENGTH: 107

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 69

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
 1 5 10 15

Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Gly Ile Ser Ser Trp
 20 25 30

Leu Ala Trp Tyr Gln Gln Lys Pro Glu Lys Ala Pro Lys Ser Leu Ile
 35 40 45

Tyr Ala Ala Ser Ser Leu Gln Ser Gly Val Pro Ser Arg Phe Ser Gly

-continued

50	55	60
Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro		
65	70	75 80
Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Tyr Lys Arg Tyr Pro Tyr		
	85	90 95
Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys		
	100	105

<210> SEQ ID NO 70
 <211> LENGTH: 5
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide
 <400> SEQUENCE: 70

Ser Tyr Thr Ile Asn
1 5

<210> SEQ ID NO 71
 <211> LENGTH: 17
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide
 <400> SEQUENCE: 71

Gly Ile Ile Pro Ile Phe Gly Ile Pro Asn Tyr Ala Gln Lys Phe Gln
1 5 10 15

Gly

<210> SEQ ID NO 72
 <211> LENGTH: 14
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide
 <400> SEQUENCE: 72

Ala Ser Gly Gly Ser Ala Asp Tyr Ser Tyr Gly Met Asp Val
1 5 10

<210> SEQ ID NO 73
 <211> LENGTH: 11
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide
 <400> SEQUENCE: 73

Arg Ala Ser Gln Gly Ile Ser Ser Trp Leu Ala
1 5 10

<210> SEQ ID NO 74
 <211> LENGTH: 7
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide
 <400> SEQUENCE: 74

-continued

Ala Ala Ser Ser Leu Gln Ser
1 5

<210> SEQ ID NO 75
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 75

Gln Gln Tyr Lys Arg Tyr Pro Tyr Thr
1 5

<210> SEQ ID NO 76
<211> LENGTH: 121
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 76

Glu Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Glu
1 5 10 15
Ser Leu Asn Ile Ser Cys Lys Gly Ser Gly Tyr Ser Phe Thr Ser Tyr
20 25 30
Trp Ile Gly Trp Val Arg Gln Met Pro Gly Lys Gly Leu Glu Trp Met
35 40 45
Gly Ile Ile Tyr Pro Gly Asp Ser Asp Thr Arg Tyr Ser Pro Ser Phe
50 55 60
Gln Gly Gln Val Thr Ile Ser Ala Asp Lys Ser Ile Asn Thr Ala Tyr
65 70 75 80
Leu Gln Trp Ser Ser Leu Lys Ala Ser Asp Thr Ala Met Tyr Tyr Cys
85 90 95
Ala Arg Gly Val Ser Met Ile Trp Gly Val Ile Met Asp Val Trp Gly
100 105 110
Gln Gly Thr Thr Val Thr Val Ser Ser
115 120

<210> SEQ ID NO 77
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 77

Ala Ile Gln Leu Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1 5 10 15
Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Gly Ile Ser Ser Ala
20 25 30
Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
35 40 45
Tyr Asp Ala Ser Ser Leu Glu Ser Gly Val Pro Ser Arg Phe Ser Gly
50 55 60
Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65 70 75 80

-continued

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Phe Asn Ser Tyr Pro Trp
 85 90 95

Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys
 100 105

<210> SEQ ID NO 78
 <211> LENGTH: 5
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 78

Ser Tyr Trp Ile Gly
 1 5

<210> SEQ ID NO 79
 <211> LENGTH: 12
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 79

Gly Val Ser Met Ile Trp Gly Val Ile Met Asp Val
 1 5 10

<210> SEQ ID NO 80
 <211> LENGTH: 9
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 80

Gln Gln Phe Asn Ser Tyr Pro Trp Thr
 1 5

<210> SEQ ID NO 81
 <211> LENGTH: 124
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 81

Glu Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Glu
 1 5 10 15

Ser Leu Gln Ile Ser Cys Lys Gly Ser Gly Tyr Thr Phe Thr Asn Tyr
 20 25 30

Trp Ile Ala Trp Val Arg Gln Met Pro Gly Lys Gly Leu Glu Trp Met
 35 40 45

Gly Ile Ile Tyr Pro Gly Asp Ser Asp Thr Arg Tyr Ser Pro Ser Phe
 50 55 60

Gln Gly Gln Val Thr Ile Ser Ala Asp Lys Ser Ile Ser Thr Ala Tyr
 65 70 75 80

Leu Gln Trp Ser Gly Leu Lys Ala Ser Asp Thr Ala Met Tyr Tyr Cys
 85 90 95

Ala Arg Gln Gly Tyr Ser Ser Gly Trp Arg Ser Tyr Tyr Gly Met Gly
 100 105 110

-continued

Val Trp Gly Gln Gly Thr Thr Val Thr Val Ser Ser
115 120

<210> SEQ ID NO 82
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 82

Ala Ile Gln Leu Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1 5 10 15

Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Gly Ile Ser Ser Ala
20 25 30

Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
35 40 45

Tyr Asp Ala Ser Ser Leu Glu Ser Gly Val Pro Ser Arg Phe Ser Gly
50 55 60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65 70 75 80

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Phe Asn Ser Tyr Pro His
85 90 95

Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys
100 105

<210> SEQ ID NO 83
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 83

Asn Tyr Trp Ile Ala
1 5

<210> SEQ ID NO 84
<211> LENGTH: 15
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 84

Gln Gly Tyr Ser Ser Gly Trp Arg Ser Tyr Tyr Gly Met Gly Val
1 5 10 15

<210> SEQ ID NO 85
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 85

Gln Gln Phe Asn Ser Tyr Pro His Thr
1 5

-continued

<210> SEQ ID NO 86
 <211> LENGTH: 122
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 86

Glu Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Glu
 1 5 10 15
 Ser Leu Lys Ile Ser Cys Lys Gly Ser Gly Tyr Ser Phe Thr Ser Tyr
 20 25 30
 Trp Ile Gly Trp Val Arg Gln Met Pro Gly Lys Gly Leu Glu Trp Met
 35 40 45
 Gly Ile Ile Tyr Pro Gly Asp Ser Asp Thr Arg Tyr Ser Pro Ser Phe
 50 55 60
 Gln Gly Gln Val Thr Ile Ser Ala Asp Lys Ser Ile Ser Thr Ala Tyr
 65 70 75 80
 Leu Gln Trp Ser Ser Leu Lys Ala Ser Asp Thr Ala Met Tyr Tyr Cys
 85 90 95
 Ala Arg Gly Val Ser Gly Ile Tyr Asn Leu His Gly Phe Asp Ile Trp
 100 105 110
 Gly Gln Gly Thr Leu Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 87
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 87

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
 1 5 10 15
 Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Ser Ile Ser Ser Tyr
 20 25 30
 Leu Asn Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
 35 40 45
 Tyr Ala Ala Ser Ser Leu Gln Ser Gly Val Pro Ser Arg Phe Ser Gly
 50 55 60
 Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
 65 70 75 80
 Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Gly Arg Phe Gly Ser Pro
 85 90 95
 Phe Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys
 100 105

<210> SEQ ID NO 88
 <211> LENGTH: 8
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 88

Gly Tyr Ser Phe Thr Ser Tyr Trp
 1 5

-continued

<210> SEQ ID NO 89
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 89

Ile Tyr Pro Gly Asp Ser Asp Thr
1 5

<210> SEQ ID NO 90
<211> LENGTH: 15
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 90

Ala Arg Gly Val Ser Gly Ile Tyr Asn Leu His Gly Phe Asp Ile
1 5 10 15

<210> SEQ ID NO 91
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 91

Gln Ser Ile Ser Ser Tyr
1 5

<210> SEQ ID NO 92
<211> LENGTH: 3
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 92

Ala Ala Ser
1

<210> SEQ ID NO 93
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 93

Gln Gln Gly Arg Phe Gly Ser Pro Phe Thr
1 5 10

<210> SEQ ID NO 94
<211> LENGTH: 117
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

-continued

<400> SEQUENCE: 94

Gln Val Gln Leu Val Glu Thr Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
 20 25 30
 Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45
 Ser Ala Ile Ser Gly Ser Gly Gly Ser Thr Tyr Tyr Ala Asp Ser Val
 50 55 60
 Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
 65 70 75 80
 Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95
 Ala Arg Tyr Tyr Tyr Ser Arg Leu Asp Tyr Trp Gly Gln Gly Thr Leu
 100 105 110
 Val Thr Val Ser Ser
 115

<210> SEQ ID NO 95

<211> LENGTH: 113

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 95

Gln Thr Val Val Thr Gln Glu Pro Ser Val Ser Ala Ala Pro Gly Gln
 1 5 10 15
 Lys Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Ile Gly Asn Asn
 20 25 30
 Tyr Val Ser Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu Leu
 35 40 45
 Ile Tyr Asp Asn Asn Lys Arg Pro Ser Gly Ile Pro Asp Arg Phe Ser
 50 55 60
 Gly Ser Lys Ser Gly Thr Ser Ala Thr Leu Gly Ile Thr Gly Leu Gln
 65 70 75 80
 Thr Gly Asp Glu Ala Asp Tyr Tyr Cys Gly Thr Trp Asp Ser Ser Leu
 85 90 95
 Ser Ala Gly Val Phe Gly Thr Gly Thr Lys Leu Thr Val Leu Gly Ser
 100 105 110
 Arg

<210> SEQ ID NO 96

<211> LENGTH: 249

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 96

Gln Thr Val Val Thr Gln Glu Pro Ser Val Ser Ala Ala Pro Gly Gln
 1 5 10 15
 Lys Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Ile Gly Asn Asn
 20 25 30
 Tyr Val Ser Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu Leu

-continued

35	40	45
Ile Tyr Asp Asn Asn Lys Arg	Pro Ser Gly Ile	Pro Asp Arg Phe Ser
50	55	60
Gly Ser Lys Ser Gly Thr Ser	Ala Thr Leu Gly	Ile Thr Gly Leu Gln
65	70	75
Thr Gly Asp Glu Ala Asp Tyr	Tyr Cys Gly Thr	Trp Asp Ser Ser Leu
85	90	95
Ser Ala Gly Val Phe Gly Thr	Gly Thr Lys Leu Thr	Val Leu Gly Ser
100	105	110
Arg Gly Gly Gly Gly Ser Gly	Gly Gly Gly Ser Gly	Gly Gly Gly Ser
115	120	125
Leu Glu Met Ala Gln Val Gln	Leu Val Glu Thr	Gly Gly Gly Leu Val
130	135	140
Gln Pro Gly Gly Ser Leu Arg	Leu Ser Cys Ala	Ala Ser Gly Phe Thr
145	150	155
Phe Ser Ser Tyr Ala Met Ser	Trp Val Arg Gln	Ala Pro Gly Lys Gly
165	170	175
Leu Glu Trp Val Ser Ala Ile	Ser Gly Ser Gly	Gly Ser Thr Tyr Tyr
180	185	190
Ala Asp Ser Val Lys Gly Arg	Phe Thr Ile Ser	Arg Asp Asn Ser Lys
195	200	205
Asn Thr Leu Tyr Leu Gln Met	Asn Ser Leu Arg	Ala Glu Asp Thr Ala
210	215	220
Val Tyr Tyr Cys Ala Arg Tyr	Tyr Tyr Tyr Ser	Arg Leu Asp Tyr Trp Gly
225	230	235
Gln Gly Thr Leu Val Thr Val	Ser Ser	
245		

<210> SEQ ID NO 97

<211> LENGTH: 119

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 97

Gln Val Gln Leu Val Gln Ser	Gly Ala Glu Val Lys Lys	Pro Gly Ala
1	5	10
Ser Val Lys Val Ser Cys Lys	Ala Ser Gly Phe Asn Ile	Lys Asp Tyr
20	25	30
Tyr Met His Trp Val Arg Gln	Ala Pro Gly Gln Arg	Leu Glu Trp Met
35	40	45
Gly Trp Ile Asp Leu Glu Asn	Ala Asn Thr Ile Tyr	Asp Ala Lys Phe
50	55	60
Gln Gly Arg Val Thr Ile Thr	Arg Asp Thr Ser	Ala Ser Thr Ala Tyr
65	70	75
Met Glu Leu Ser Ser Leu Arg	Ser Glu Asp Thr	Ala Val Tyr Tyr Cys
85	90	95
Ala Arg Asp Ala Tyr Gly Arg	Tyr Phe Tyr Asp	Val Trp Gly Gln Gly
100	105	110
Thr Leu Val Thr Val Ser Ser		
115		

<210> SEQ ID NO 98

<211> LENGTH: 112

-continued

<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 98

Asp Ile Val Met Thr Gln Ser Pro Asp Ser Leu Ala Val Ser Leu Gly
1 5 10 15
Glu Arg Ala Thr Ile Asn Cys Lys Ser Ser Gln Ser Leu Leu Asn Ala
20 25 30
Arg Thr Gly Lys Asn Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln
35 40 45
Pro Pro Lys Leu Leu Ile Tyr Trp Ala Ser Thr Arg Glu Ser Gly Val
50 55 60
Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr
65 70 75 80
Ile Ser Ser Leu Gln Ala Glu Asp Val Ala Val Tyr Tyr Cys Lys Gln
85 90 95
Ser Tyr Ser Arg Arg Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
100 105 110

<210> SEQ ID NO 99
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 99

Phe Asn Ile Lys Asp Tyr Tyr Met His
1 5

<210> SEQ ID NO 100
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 100

Trp Ile Asp Leu Glu Asn Ala Asn Thr Ile Tyr Asp Ala Lys Phe Gln
1 5 10 15
Gly

<210> SEQ ID NO 101
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 101

Ala Arg Asp Ala Tyr Gly Arg Tyr Phe Tyr Asp Val
1 5 10

<210> SEQ ID NO 102
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:

-continued

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 102

Lys Ser Ser Gln Ser Leu Leu Asn Ala Arg Thr Gly Lys Asn Tyr Leu
 1 5 10 15

Ala

<210> SEQ ID NO 103

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 103

Trp Ala Ser Thr Arg Glu Ser
 1 5

<210> SEQ ID NO 104

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 104

Lys Gln Ser Tyr Ser Arg Arg Thr
 1 5

<210> SEQ ID NO 105

<211> LENGTH: 251

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 105

Asp Ile Val Met Thr Gln Ser Pro Asp Ser Leu Ala Val Ser Leu Gly
 1 5 10 15

Glu Arg Ala Thr Ile Asn Cys Lys Ser Ser Gln Ser Leu Leu Asn Ala
 20 25 30

Arg Thr Gly Lys Asn Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln
 35 40 45

Pro Pro Lys Leu Leu Ile Tyr Trp Ala Ser Thr Arg Glu Ser Gly Val
 50 55 60

Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr
 65 70 75 80

Ile Ser Ser Leu Gln Ala Glu Asp Val Ala Val Tyr Tyr Cys Lys Gln
 85 90 95

Ser Tyr Ser Arg Arg Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
 100 105 110

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser Gly
 115 120 125

Gly Gly Gly Ser Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys
 130 135 140

Lys Pro Gly Ala Ser Val Lys Val Ser Cys Lys Ala Ser Gly Phe Asn
 145 150 155 160

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Ile Lys Asp Tyr Tyr Met His Trp Val Arg Gln Ala Pro Gly Gln Arg
      165                      170                      175

Leu Glu Trp Met Gly Trp Ile Asp Leu Glu Asn Ala Asn Thr Ile Tyr
      180                      185                      190

Asp Ala Lys Phe Gln Gly Arg Val Thr Ile Thr Arg Asp Thr Ser Ala
      195                      200                      205

Ser Thr Ala Tyr Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala
      210                      215                      220

Val Tyr Tyr Cys Ala Arg Asp Ala Tyr Gly Arg Tyr Phe Tyr Asp Val
      225                      230                      235                      240

Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
      245                      250

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<210> SEQ ID NO 106
<211> LENGTH: 125
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

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<400> SEQUENCE: 106

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Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1      5      10      15

Ser Leu Lys Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Asn Lys Tyr
      20      25      30

Ala Ile Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
      35      40      45

Ala Arg Ile Arg Ser Lys Tyr Asn Asn Tyr Ala Thr Tyr Tyr Ala Asp
      50      55      60

Gln Val Lys Asp Arg Phe Thr Ile Ser Arg Asp Asp Ser Lys Asn Thr
      65      70      75      80

Ala Tyr Leu Gln Met Asn Asn Leu Lys Thr Glu Asp Thr Ala Val Tyr
      85      90      95

Tyr Cys Val Arg His Ala Asn Phe Gly Asn Ser Tyr Ile Ser Tyr Trp
      100     105     110

Ala Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
      115     120     125

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<210> SEQ ID NO 107
<211> LENGTH: 109
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

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<400> SEQUENCE: 107

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Gln Thr Val Val Thr Gln Glu Pro Ser Leu Thr Val Ser Pro Gly Gly
1      5      10      15

Thr Val Thr Leu Thr Cys Ala Ser Ser Thr Gly Ala Val Thr Ser Gly
      20      25      30

Asn Tyr Pro Asn Trp Val Gln Gln Lys Pro Gly Gln Ala Pro Arg Gly
      35      40      45

Leu Ile Gly Gly Thr Lys Phe Leu Val Pro Gly Thr Pro Ala Arg Phe
      50      55      60

Ser Gly Ser Leu Leu Gly Gly Lys Ala Ala Leu Thr Leu Ser Gly Val
      65      70      75      80

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-continued

Gln Pro Glu Asp Glu Ala Glu Tyr Tyr Cys Thr Leu Trp Tyr Ser Asn
 85 90 95

Arg Trp Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
 100 105

<210> SEQ ID NO 108
 <211> LENGTH: 10
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 108

Gly Phe Thr Phe Asn Lys Tyr Ala Ile Asn
 1 5 10

<210> SEQ ID NO 109
 <211> LENGTH: 18
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 109

Arg Ile Arg Ser Lys Tyr Asn Asn Tyr Ala Thr Tyr Tyr Ala Asp Gln
 1 5 10 15

Val Lys

<210> SEQ ID NO 110
 <211> LENGTH: 14
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 110

His Ala Asn Phe Gly Asn Ser Tyr Ile Ser Tyr Trp Ala Tyr
 1 5 10

<210> SEQ ID NO 111
 <211> LENGTH: 14
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 111

Ala Ser Ser Thr Gly Ala Val Thr Ser Gly Asn Tyr Pro Asn
 1 5 10

<210> SEQ ID NO 112
 <211> LENGTH: 7
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 112

Gly Thr Lys Phe Leu Val Pro
 1 5

-continued

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<210> SEQ ID NO 113
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        peptide

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<400> SEQUENCE: 113

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Thr Leu Trp Tyr Ser Asn Arg Trp Val
1             5

```

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<210> SEQ ID NO 114
<211> LENGTH: 249
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        polypeptide

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<400> SEQUENCE: 114

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Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1             5             10             15

Ser Leu Lys Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Asn Lys Tyr
                20             25             30

Ala Ile Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
            35             40             45

Ala Arg Ile Arg Ser Lys Tyr Asn Asn Tyr Ala Thr Tyr Tyr Ala Asp
            50             55             60

Gln Val Lys Asp Arg Phe Thr Ile Ser Arg Asp Asp Ser Lys Asn Thr
65             70             75             80

Ala Tyr Leu Gln Met Asn Asn Leu Lys Thr Glu Asp Thr Ala Val Tyr
            85             90             95

Tyr Cys Val Arg His Ala Asn Phe Gly Asn Ser Tyr Ile Ser Tyr Trp
            100            105            110

Ala Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser Gly Gly Gly
            115            120            125

Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gln Thr Val Val
            130            135            140

Thr Gln Glu Pro Ser Leu Thr Val Ser Pro Gly Gly Thr Val Thr Leu
145            150            155            160

Thr Cys Ala Ser Ser Thr Gly Ala Val Thr Ser Gly Asn Tyr Pro Asn
            165            170            175

Trp Val Gln Gln Lys Pro Gly Gln Ala Pro Arg Gly Leu Ile Gly Gly
            180            185            190

Thr Lys Phe Leu Val Pro Gly Thr Pro Ala Arg Phe Ser Gly Ser Leu
            195            200            205

Leu Gly Gly Lys Ala Ala Leu Thr Leu Ser Gly Val Gln Pro Glu Asp
            210            215            220

Glu Ala Glu Tyr Tyr Cys Thr Leu Trp Tyr Ser Asn Arg Trp Val Phe
225            230            235            240

Gly Gly Gly Thr Lys Leu Thr Val Leu
            245

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<210> SEQ ID NO 115
<211> LENGTH: 119
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic

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-continued

polypeptide

<400> SEQUENCE: 115

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala
1 5 10 15

Ser Val Lys Val Ser Cys Lys Ala Ser Gly Phe Asn Ile Lys Asp Tyr
20 25 30

Tyr Met His Trp Val Arg Gln Ala Pro Gly Gln Arg Leu Glu Trp Met
35 40 45

Gly Trp Ile Asp Leu Glu Asn Ala Asn Thr Ile Tyr Asp Ala Lys Phe
50 55 60

Gln Gly Arg Val Thr Ile Thr Arg Asp Thr Ser Ala Ser Thr Ala Tyr
65 70 75 80

Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95

Ala Arg Asp Ser Tyr Gly Arg Tyr Phe Tyr Asp Val Trp Gly Gln Gly
100 105 110

Thr Leu Val Thr Val Ser Ser
115

<210> SEQ ID NO 116

<211> LENGTH: 112

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 116

Asp Ile Val Met Thr Gln Ser Pro Asp Ser Leu Ala Val Ser Leu Gly
1 5 10 15

Glu Arg Ala Thr Ile Asn Cys Lys Ser Ser Gln Ser Leu Leu Asn Ser
20 25 30

Arg Thr Gly Lys Asn Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln
35 40 45

Pro Pro Lys Leu Leu Ile Tyr Trp Ala Ser Thr Arg Glu Ser Gly Val
50 55 60

Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr
65 70 75 80

Ile Ser Ser Leu Gln Ala Glu Asp Val Ala Val Tyr Tyr Cys Lys Gln
85 90 95

Ser Tyr Ser Arg Arg Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
100 105 110

<210> SEQ ID NO 117

<211> LENGTH: 119

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 117

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala
1 5 10 15

Ser Val Lys Val Ser Cys Lys Ala Ser Gly Phe Asn Ile Lys Asp Tyr
20 25 30

Tyr Met His Trp Val Arg Gln Ala Pro Gly Gln Arg Leu Glu Trp Met
35 40 45

-continued

Gly Trp Ile Asp Leu Glu Asn Ala Asn Thr Ile Tyr Asp Ala Lys Phe
50 55 60

Gln Gly Arg Val Thr Ile Thr Arg Asp Thr Ser Ala Ser Thr Ala Tyr
65 70 75 80

Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95

Ala Arg Asp Val Tyr Gly Arg Tyr Phe Tyr Asp Leu Trp Gly Gln Gly
100 105 110

Thr Leu Val Thr Val Ser Ser
115

<210> SEQ ID NO 118
<211> LENGTH: 112
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 118

Asp Ile Val Met Thr Gln Ser Pro Asp Ser Leu Ala Val Ser Leu Gly
1 5 10 15

Glu Arg Ala Thr Ile Asn Cys Lys Ser Ser Gln Ser Leu Leu Glu Ser
20 25 30

Arg Thr Gly Lys Asn Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln
35 40 45

Pro Pro Lys Leu Leu Ile Tyr Trp Ala Ser Thr Arg Glu Ser Gly Val
50 55 60

Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr
65 70 75 80

Ile Ser Ser Leu Gln Ala Glu Asp Val Ala Val Tyr Tyr Cys Lys Gln
85 90 95

Ser Tyr Ser Arg Arg Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
100 105 110

<210> SEQ ID NO 119
<211> LENGTH: 121
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 119

Asp Ile Lys Leu Gln Ser Gly Ala Glu Leu Ala Arg Pro Gly Ala
1 5 10 15

Ser Val Lys Met Ser Cys Lys Thr Ser Gly Tyr Thr Phe Thr Arg Tyr
20 25 30

Thr Met His Trp Val Lys Gln Arg Pro Gly Gln Gly Leu Glu Trp Ile
35 40 45

Gly Tyr Ile Asn Pro Ser Arg Gly Tyr Thr Asn Tyr Asn Gln Lys Phe
50 55 60

Lys Asp Lys Ala Thr Leu Thr Thr Asp Lys Ser Ser Ser Thr Ala Tyr
65 70 75 80

Met Gln Leu Ser Ser Leu Thr Ser Glu Asp Ser Ala Val Tyr Tyr Cys
85 90 95

Ala Arg Tyr Tyr Asp Asp His Tyr Cys Leu Asp Tyr Trp Gly Gln Gly
100 105 110

-continued

Thr Thr Leu Thr Val Ser Ser Val Glu
 115 120

<210> SEQ ID NO 120
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 120

Val Asp Asp Ile Gln Leu Thr Gln Ser Pro Ala Ile Met Ser Ala Ser
 1 5 10 15

Pro Gly Glu Lys Val Thr Met Thr Cys Arg Ala Ser Ser Ser Val Ser
 20 25 30

Tyr Met Asn Trp Tyr Gln Gln Lys Ser Gly Thr Ser Pro Lys Arg Trp
 35 40 45

Ile Tyr Asp Thr Ser Lys Val Ala Ser Gly Val Pro Tyr Arg Phe Ser
 50 55 60

Gly Ser Gly Ser Gly Thr Ser Tyr Ser Leu Thr Ile Ser Ser Met Glu
 65 70 75 80

Ala Glu Asp Ala Ala Thr Tyr Tyr Cys Gln Gln Trp Ser Ser Asn Pro
 85 90 95

Leu Thr Phe Gly Ala Gly Thr Lys Leu Glu Leu Lys
 100 105

<210> SEQ ID NO 121
 <211> LENGTH: 115
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 121

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Asn
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Phe
 20 25 30

Gly Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ser Ser Ile Ser Gly Ser Gly Ser Asp Thr Leu Tyr Ala Asp Ser Val
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Thr Thr Leu Tyr
 65 70 75 80

Leu Gln Met Asn Ser Leu Arg Pro Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Thr Ile Gly Gly Ser Leu Ser Arg Ser Ser Gln Gly Thr Leu Val Thr
 100 105 110

Val Ser Ser
 115

<210> SEQ ID NO 122
 <211> LENGTH: 5
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

-continued

<400> SEQUENCE: 122

Ser Phe Gly Met Ser
1 5

<210> SEQ ID NO 123

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 123

Gly Phe Thr Phe Ser Ser Phe Gly Met Ser
1 5 10

<210> SEQ ID NO 124

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 124

Ser Ile Ser Gly Ser Gly Ser Asp Thr Leu
1 5 10

<210> SEQ ID NO 125

<211> LENGTH: 17

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 125

Ser Ile Ser Gly Ser Gly Ser Asp Thr Leu Tyr Ala Asp Ser Val Lys
1 5 10 15

Gly

<210> SEQ ID NO 126

<211> LENGTH: 6

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 126

Gly Gly Ser Leu Ser Arg
1 5

<210> SEQ ID NO 127

<211> LENGTH: 115

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 127

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Arg Ser Phe

-continued

20	25	30
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Gly Met Ser Trp Val Arg Gln Ala Pro Gly Lys Glu Pro Glu Trp Val
 35 40 45

Ser Ser Ile Ser Gly Ser Gly Ser Asp Thr Leu Tyr Ala Asp Ser Val
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Thr Thr Leu Tyr
 65 70 75 80

Leu Gln Met Asn Ser Leu Lys Pro Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Thr Ile Gly Gly Ser Leu Ser Arg Ser Ser Gln Gly Thr Gln Val Thr
 100 105 110

Val Ser Ser
 115

<210> SEQ ID NO 128
 <211> LENGTH: 10
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 128

Gly Phe Thr Phe Arg Ser Phe Gly Met Ser
 1 5 10

<210> SEQ ID NO 129
 <211> LENGTH: 115
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 129

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Phe
 20 25 30

Gly Met Ser Trp Val Arg Gln Ala Pro Gly Lys Glu Pro Glu Trp Val
 35 40 45

Ser Ser Ile Ser Gly Ser Gly Ser Asp Thr Leu Tyr Ala Asp Ser Val
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Thr Thr Leu Tyr
 65 70 75 80

Leu Gln Met Asn Ser Leu Lys Pro Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Thr Ile Gly Gly Ser Leu Ser Arg Ser Ser Gln Gly Thr Gln Val Thr
 100 105 110

Val Ser Ser
 115

<210> SEQ ID NO 130
 <211> LENGTH: 115
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 130

-continued

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Arg Ser Phe
 20 25 30
 Gly Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45
 Ser Ser Ile Ser Gly Ser Gly Ser Asp Thr Leu Tyr Ala Asp Ser Val
 50 55 60
 Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Thr Thr Leu Tyr
 65 70 75 80
 Leu Gln Met Asn Ser Leu Lys Pro Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95
 Thr Ile Gly Gly Ser Leu Ser Arg Ser Ser Gln Gly Thr Gln Val Thr
 100 105 110
 Val Ser Ser
 115

<210> SEQ ID NO 131
 <211> LENGTH: 115
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 131

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Asn
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Arg Ser Phe
 20 25 30
 Gly Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45
 Ser Ser Ile Ser Gly Ser Gly Ser Asp Thr Leu Tyr Ala Asp Ser Val
 50 55 60
 Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Thr Thr Leu Tyr
 65 70 75 80
 Leu Gln Met Asn Ser Leu Lys Pro Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95
 Thr Ile Gly Gly Ser Leu Ser Arg Ser Ser Gln Gly Thr Leu Val Thr
 100 105 110
 Val Ser Ser
 115

<210> SEQ ID NO 132
 <211> LENGTH: 115
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 132

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Asn
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Arg Ser Phe
 20 25 30
 Gly Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

-continued

Ser Ser Ile Ser Gly Ser Gly Ser Asp Thr Leu Tyr Ala Asp Ser Val
50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Thr Thr Leu Tyr
65 70 75 80

Leu Gln Met Asn Ser Leu Arg Pro Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95

Thr Ile Gly Gly Ser Leu Ser Arg Ser Ser Gln Gly Thr Leu Val Thr
100 105 110

Val Ser Ser
115

<210> SEQ ID NO 133
<211> LENGTH: 115
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 133

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Asn
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Phe
20 25 30

Gly Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45

Ser Ser Ile Ser Gly Ser Gly Ser Asp Thr Leu Tyr Ala Asp Ser Val
50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn Thr Leu Tyr
65 70 75 80

Leu Gln Met Asn Ser Leu Arg Pro Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95

Thr Ile Gly Gly Ser Leu Ser Arg Ser Ser Gln Gly Thr Leu Val Thr
100 105 110

Val Ser Ser
115

<210> SEQ ID NO 134
<211> LENGTH: 115
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 134

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Asn
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Phe
20 25 30

Gly Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45

Ser Ser Ile Ser Gly Ser Gly Ser Asp Thr Leu Tyr Ala Asp Ser Val
50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn Thr Leu Tyr
65 70 75 80

Leu Gln Met Asn Ser Leu Arg Pro Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95

-continued

Thr Ile Gly Gly Ser Leu Ser Arg Ser Gly Gln Gly Thr Leu Val Thr
 100 105 110

Val Ser Ser
 115

<210> SEQ ID NO 135
 <211> LENGTH: 115
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 135

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Arg Ser Phe
 20 25 30

Gly Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Pro Glu Trp Val
 35 40 45

Ser Ser Ile Ser Gly Ser Gly Ser Asp Thr Leu Tyr Ala Asp Ser Val
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
 65 70 75 80

Leu Gln Met Asn Ser Leu Arg Pro Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Thr Ile Gly Gly Ser Leu Ser Arg Ser Ser Gln Gly Thr Leu Val Thr
 100 105 110

Val Ser Ser
 115

<210> SEQ ID NO 136
 <211> LENGTH: 115
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 136

Ala Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Asn
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Arg Ser Phe
 20 25 30

Gly Met Ser Trp Val Arg Gln Ala Pro Gly Lys Glu Pro Glu Trp Val
 35 40 45

Ser Ser Ile Ser Gly Ser Gly Ser Asp Thr Leu Tyr Ala Asp Ser Val
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Thr Thr Leu Tyr
 65 70 75 80

Leu Gln Met Asn Ser Leu Lys Pro Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Thr Ile Gly Gly Ser Leu Ser Arg Ser Ser Gln Gly Thr Gln Val Thr
 100 105 110

Val Ser Ser
 115

<210> SEQ ID NO 137

-continued

<211> LENGTH: 116
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 137

Ala Val Gln Leu Val Asp Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1 5 10 15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Ser Phe Gly Ser Phe
20 25 30
Gly Met Ser Trp Val Arg Gln Tyr Pro Gly Lys Glu Pro Glu Trp Val
35 40 45
Ser Ser Ile Asn Gly Arg Gly Asp Asp Thr Arg Tyr Ala Asp Ser Val
50 55 60
Lys Gly Arg Phe Ser Ile Ser Arg Asp Asn Ala Lys Asn Thr Leu Tyr
65 70 75 80
Leu Gln Met Asn Ser Leu Lys Pro Glu Asp Thr Ala Glu Tyr Tyr Cys
85 90 95
Thr Ile Gly Arg Ser Val Ser Arg Ser Arg Thr Gln Gly Thr Gln Val
100 105 110
Thr Val Ser Ser
115

<210> SEQ ID NO 138
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 138

Gly Phe Ser Phe Gly Ser Phe Gly Met Ser
1 5 10

<210> SEQ ID NO 139
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 139

Ser Ile Asn Gly Arg Gly Asp Asp Thr Arg
1 5 10

<210> SEQ ID NO 140
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 140

Ser Ile Asn Gly Arg Gly Asp Asp Thr Arg Tyr Ala Asp Ser Val Lys
1 5 10 15
Gly

<210> SEQ ID NO 141

-continued

<211> LENGTH: 7
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 141

Gly Arg Ser Val Ser Arg Ser
 1 5

<210> SEQ ID NO 142
 <211> LENGTH: 114
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 142

Ala Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15

Ser Leu Arg Leu Thr Cys Thr Ala Ser Gly Phe Thr Phe Arg Ser Phe
 20 25 30

Gly Met Ser Trp Val Arg Gln Ala Pro Gly Lys Asp Gln Glu Trp Val
 35 40 45

Ser Ala Ile Ser Ala Asp Ser Ser Thr Lys Asn Tyr Ala Asp Ser Val
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Lys Met Leu Tyr
 65 70 75 80

Leu Glu Met Asn Ser Leu Lys Pro Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Val Ile Gly Arg Gly Ser Pro Ser Ser Pro Gly Thr Gln Val Thr Val
 100 105 110

Ser Ser

<210> SEQ ID NO 143
 <211> LENGTH: 10
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 143

Ala Ile Ser Ala Asp Ser Ser Thr Lys Asn
 1 5 10

<210> SEQ ID NO 144
 <211> LENGTH: 17
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 144

Ala Ile Ser Ala Asp Ser Ser Thr Lys Asn Tyr Ala Asp Ser Val Lys
 1 5 10 15

Gly

<210> SEQ ID NO 145
 <211> LENGTH: 5

-continued

<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 145

Gly Arg Gly Ser Pro
1 5

<210> SEQ ID NO 146
<211> LENGTH: 114
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 146

Gln Val Gln Leu Ala Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1 5 10 15
Ser Leu Arg Leu Thr Cys Thr Ala Ser Gly Phe Thr Phe Gly Ser Phe
20 25 30
Gly Met Ser Trp Val Arg Gln Ala Pro Gly Glu Gly Leu Glu Trp Val
35 40 45
Ser Ala Ile Ser Ala Asp Ser Ser Asp Lys Arg Tyr Ala Asp Ser Val
50 55 60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Lys Met Leu Tyr
65 70 75 80
Leu Glu Met Asn Ser Leu Lys Ser Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95
Val Ile Gly Arg Gly Ser Pro Ala Ser Gln Gly Thr Gln Val Thr Val
100 105 110
Ser Ser

<210> SEQ ID NO 147
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 147

Gly Phe Thr Phe Gly Ser Phe Gly Met Ser
1 5 10

<210> SEQ ID NO 148
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 148

Ala Ile Ser Ala Asp Ser Ser Asp Lys Arg
1 5 10

<210> SEQ ID NO 149
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:

-continued

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 149

Ala Ile Ser Ala Asp Ser Ser Asp Lys Arg Tyr Ala Asp Ser Val Lys
1 5 10 15

Gly

<210> SEQ ID NO 150

<211> LENGTH: 123

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 150

Gln Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Asn Tyr
20 25 30

Trp Met Tyr Trp Val Arg Val Ala Pro Gly Lys Gly Leu Glu Arg Ile
35 40 45

Ser Arg Asp Ile Ser Thr Gly Gly Gly Tyr Ser Tyr Tyr Ala Asp Ser
50 55 60

Val Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn Thr Leu
65 70 75 80

Tyr Leu Gln Met Asn Ser Leu Lys Pro Glu Asp Thr Ala Leu Tyr Tyr
85 90 95

Cys Ala Lys Asp Arg Glu Ala Gln Val Asp Thr Leu Asp Phe Asp Tyr
100 105 110

Arg Gly Gln Gly Thr Gln Val Thr Val Ser Ser
115 120

<210> SEQ ID NO 151

<211> LENGTH: 5

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 151

Asn Tyr Trp Met Tyr
1 5

<210> SEQ ID NO 152

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 152

Gly Phe Thr Phe Ser Asn Tyr Trp Met Tyr
1 5 10

<210> SEQ ID NO 153

<211> LENGTH: 11

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

-continued

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 153

Arg Asp Ile Ser Thr Gly Gly Gly Tyr Ser Tyr
1 5 10

<210> SEQ ID NO 154

<211> LENGTH: 18

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 154

Arg Asp Ile Ser Thr Gly Gly Gly Tyr Ser Tyr Tyr Ala Asp Ser Val
1 5 10 15

Lys Gly

<210> SEQ ID NO 155

<211> LENGTH: 13

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 155

Asp Arg Glu Ala Gln Val Asp Thr Leu Asp Phe Asp Tyr
1 5 10

<210> SEQ ID NO 156

<211> LENGTH: 117

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 156

Ala Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Gly Gly Gly
1 5 10 15

Ser Leu Arg Leu Ala Cys Ala Ala Ser Glu Arg Ile Phe Asp Leu Asn
20 25 30

Leu Met Gly Trp Tyr Arg Gln Gly Pro Gly Asn Glu Arg Glu Leu Val
35 40 45

Ala Thr Cys Ile Thr Val Gly Asp Ser Thr Asn Tyr Ala Asp Ser Val
50 55 60

Lys Gly Arg Phe Thr Ile Ser Met Asp Tyr Thr Lys Gln Thr Val Tyr
65 70 75 80

Leu His Met Asn Ser Leu Arg Pro Glu Asp Thr Gly Leu Tyr Tyr Cys
85 90 95

Lys Ile Arg Arg Thr Trp His Ser Glu Leu Trp Gly Gln Gly Thr Gln
100 105 110

Val Thr Val Ser Ser
115

<210> SEQ ID NO 157

<211> LENGTH: 5

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

-continued

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 157

Leu Asn Leu Met Gly
1 5

<210> SEQ ID NO 158

<211> LENGTH: 11

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 158

Ser Glu Arg Ile Phe Asp Leu Asn Leu Met Gly
1 5 10

<210> SEQ ID NO 159

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 159

Thr Cys Ile Thr Val Gly Asp Ser Thr Asn
1 5 10

<210> SEQ ID NO 160

<211> LENGTH: 17

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 160

Thr Cys Ile Thr Val Gly Asp Ser Thr Asn Tyr Ala Asp Ser Val Lys
1 5 10 15

Gly

<210> SEQ ID NO 161

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 161

Arg Arg Thr Trp His Ser Glu Leu
1 5

<210> SEQ ID NO 162

<211> LENGTH: 116

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 162

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Glu Gly Gly
1 5 10 15

-continued

Ser Leu Arg Leu Ala Cys Ala Ala Ser Glu Arg Ile Trp Asp Ile Asn
 20 25 30

Leu Leu Gly Trp Tyr Arg Gln Gly Pro Gly Asn Glu Arg Glu Leu Val
 35 40 45

Ala Thr Ile Thr Val Gly Asp Ser Thr Ser Tyr Ala Asp Ser Val Lys
 50 55 60

Gly Arg Phe Thr Ile Ser Arg Asp Tyr Asp Lys Asn Thr Leu Tyr Leu
 65 70 75 80

Gln Met Asn Ser Leu Arg Pro Glu Asp Thr Gly Leu Tyr Tyr Cys Lys
 85 90 95

Ile Arg Arg Thr Trp His Ser Glu Leu Trp Gly Gln Gly Thr Gln Val
 100 105 110

Thr Val Ser Ser
 115

<210> SEQ ID NO 163
 <211> LENGTH: 5
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 163

Ile Asn Leu Leu Gly
 1 5

<210> SEQ ID NO 164
 <211> LENGTH: 11
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 164

Ser Glu Arg Ile Trp Asp Ile Asn Leu Leu Gly
 1 5 10

<210> SEQ ID NO 165
 <211> LENGTH: 9
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 165

Thr Ile Thr Val Gly Asp Ser Thr Ser
 1 5

<210> SEQ ID NO 166
 <211> LENGTH: 16
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 166

Thr Ile Thr Val Gly Asp Ser Thr Ser Tyr Ala Asp Ser Val Lys Gly
 1 5 10 15

-continued

<210> SEQ ID NO 167
<211> LENGTH: 116
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 167

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Asn
1 5 10 15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Phe
20 25 30
Gly Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45
Ser Ser Ile Ser Gly Ser Gly Ser Asp Thr Leu Tyr Ala Asp Ser Val
50 55 60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Thr Thr Leu Tyr
65 70 75 80
Leu Gln Met Asn Ser Leu Arg Pro Glu Asp Thr Ala Thr Tyr Tyr Cys
85 90 95
Thr Ile Gly Gly Ser Leu Ser Arg Ser Ser Gln Gly Thr Leu Val Thr
100 105 110
Val Ser Ser Ala
115

<210> SEQ ID NO 168
<211> LENGTH: 116
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 168

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Gly
1 5 10 15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Arg Ser Phe
20 25 30
Gly Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Pro Glu Trp Val
35 40 45
Ser Ser Ile Ser Gly Ser Gly Ser Asp Thr Leu Tyr Ala Asp Ser Val
50 55 60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65 70 75 80
Leu Gln Met Asn Ser Leu Arg Pro Glu Asp Thr Ala Thr Tyr Tyr Cys
85 90 95
Thr Ile Gly Gly Ser Leu Ser Arg Ser Ser Gln Gly Thr Leu Val Thr
100 105 110
Val Ser Ser Ala
115

<210> SEQ ID NO 169
<211> LENGTH: 116
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 169

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Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Asn
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Phe
 20 25 30
 Gly Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45
 Ser Ser Ile Ser Gly Ser Gly Ser Asp Thr Leu Tyr Ala Asp Ser Val
 50 55 60
 Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Thr Thr Leu Tyr
 65 70 75 80
 Leu Gln Met Asn Ser Leu Arg Pro Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95
 Thr Ile Gly Gly Ser Leu Ser Arg Ser Ser Gln Gly Thr Leu Val Lys
 100 105 110
 Val Ser Ser Ala
 115

<210> SEQ ID NO 170
 <211> LENGTH: 115
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 170

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Asn
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Phe
 20 25 30
 Gly Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45
 Ser Ser Ile Ser Gly Ser Gly Ser Asp Thr Leu Tyr Ala Asp Ser Val
 50 55 60
 Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Thr Thr Leu Tyr
 65 70 75 80
 Leu Gln Met Asn Ser Leu Arg Pro Glu Asp Thr Ala Leu Tyr Tyr Cys
 85 90 95
 Thr Ile Gly Gly Ser Leu Ser Arg Ser Ser Gln Gly Thr Leu Val Thr
 100 105 110
 Val Ser Ser
 115

<210> SEQ ID NO 171
 <211> LENGTH: 116
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 171

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Asn
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Phe
 20 25 30
 Gly Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

-continued

Ser Ser Ile Ser Gly Ser Gly Ser Asp Thr Leu Tyr Ala Asp Ser Val
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Thr Thr Leu Tyr
 65 70 75 80

Leu Gln Met Asn Ser Leu Arg Pro Glu Asp Thr Ala Leu Tyr Tyr Cys
 85 90 95

Thr Ile Gly Gly Ser Leu Ser Arg Ser Ser Gln Gly Thr Leu Val Thr
 100 105 110

Val Ser Ser Ala
 115

<210> SEQ ID NO 172
 <211> LENGTH: 117
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 172

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Asn
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Phe
 20 25 30

Gly Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ser Ser Ile Ser Gly Ser Gly Ser Asp Thr Leu Tyr Ala Asp Ser Val
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Thr Thr Leu Tyr
 65 70 75 80

Leu Gln Met Asn Ser Leu Arg Pro Glu Asp Thr Ala Leu Tyr Tyr Cys
 85 90 95

Thr Ile Gly Gly Ser Leu Ser Arg Ser Ser Gln Gly Thr Leu Val Thr
 100 105 110

Val Ser Ser Ala Ala
 115

<210> SEQ ID NO 173
 <211> LENGTH: 118
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 173

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Asn
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Phe
 20 25 30

Gly Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ser Ser Ile Ser Gly Ser Gly Ser Asp Thr Leu Tyr Ala Asp Ser Val
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Thr Thr Leu Tyr
 65 70 75 80

Leu Gln Met Asn Ser Leu Arg Pro Glu Asp Thr Ala Leu Tyr Tyr Cys
 85 90 95

-continued

Thr Ile Gly Gly Ser Leu Ser Arg Ser Ser Gln Gly Thr Leu Val Thr
 100 105 110

Val Ser Ser Ala Ala Ala
 115

<210> SEQ ID NO 174

<211> LENGTH: 116

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 174

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Asn
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Phe
 20 25 30

Gly Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ser Ser Ile Ser Gly Ser Gly Ser Asp Thr Leu Tyr Ala Asp Ser Val
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Thr Thr Leu Tyr
 65 70 75 80

Leu Gln Met Asn Ser Leu Arg Pro Glu Asp Thr Ala Leu Tyr Tyr Cys
 85 90 95

Thr Ile Gly Gly Ser Leu Ser Arg Ser Ser Gln Gly Thr Leu Val Thr
 100 105 110

Val Ser Ser Gly
 115

<210> SEQ ID NO 175

<211> LENGTH: 117

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 175

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Asn
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Phe
 20 25 30

Gly Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ser Ser Ile Ser Gly Ser Gly Ser Asp Thr Leu Tyr Ala Asp Ser Val
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Thr Thr Leu Tyr
 65 70 75 80

Leu Gln Met Asn Ser Leu Arg Pro Glu Asp Thr Ala Leu Tyr Tyr Cys
 85 90 95

Thr Ile Gly Gly Ser Leu Ser Arg Ser Ser Gln Gly Thr Leu Val Thr
 100 105 110

Val Ser Ser Gly Gly
 115

<210> SEQ ID NO 176

-continued

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<211> LENGTH: 118
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

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<400> SEQUENCE: 176

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Glu Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Asn
1           5           10           15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Phe
           20           25           30
Gly Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
           35           40           45
Ser Ser Ile Ser Gly Ser Gly Ser Asp Thr Leu Tyr Ala Asp Ser Val
           50           55           60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Thr Thr Leu Tyr
65           70           75           80
Leu Gln Met Asn Ser Leu Arg Pro Glu Asp Thr Ala Leu Tyr Tyr Cys
           85           90           95
Thr Ile Gly Gly Ser Leu Ser Arg Ser Ser Gln Gly Thr Leu Val Thr
           100          105          110
Val Ser Ser Gly Gly Gly
           115

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<210> SEQ ID NO 177
<211> LENGTH: 115
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

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<400> SEQUENCE: 177

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Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Asn
1           5           10           15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Lys Phe
           20           25           30
Gly Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
           35           40           45
Ser Ser Ile Ser Gly Ser Gly Arg Asp Thr Leu Tyr Ala Glu Ser Val
           50           55           60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Thr Thr Leu Tyr
65           70           75           80
Leu Gln Met Asn Ser Leu Arg Pro Glu Asp Thr Ala Val Tyr Tyr Cys
           85           90           95
Thr Ile Gly Gly Ser Leu Ser Val Ser Ser Gln Gly Thr Leu Val Thr
           100          105          110
Val Ser Ser
           115

```

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<210> SEQ ID NO 178
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      peptide

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<400> SEQUENCE: 178

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-continued

Gly Phe Thr Phe Ser Lys Phe Gly Met Ser
1 5 10

<210> SEQ ID NO 179
 <211> LENGTH: 16
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 179

Ser Ile Ser Gly Ser Gly Arg Asp Thr Leu Tyr Ala Glu Ser Val Lys
1 5 10 15

<210> SEQ ID NO 180
 <211> LENGTH: 6
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 180

Gly Gly Ser Leu Ser Val
1 5

<210> SEQ ID NO 181
 <211> LENGTH: 115
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 181

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Asn
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Arg Phe
20 25 30

Gly Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45

Ser Ser Ile Ser Gly Ser Gly Ser Asp Thr Leu Tyr Ala Glu Ser Val
50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Thr Thr Leu Tyr
65 70 75 80

Leu Gln Met Asn Ser Leu Arg Pro Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95

Thr Ile Gly Gly Ser Leu Ser Arg Ser Ser Gln Gly Thr Leu Val Thr
100 105 110

Val Ser Ser
115

<210> SEQ ID NO 182
 <211> LENGTH: 10
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 182

Gly Phe Thr Phe Ser Arg Phe Gly Met Ser
1 5 10

-continued

<210> SEQ ID NO 183
 <211> LENGTH: 16
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 183

Ser Ile Ser Gly Ser Gly Ser Asp Thr Leu Tyr Ala Glu Ser Val Lys
 1 5 10 15

<210> SEQ ID NO 184
 <211> LENGTH: 115
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 184

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Asn
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Lys Phe
 20 25 30

Gly Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ser Ser Ile Ser Gly Ser Gly Thr Asp Thr Leu Tyr Ala Glu Ser Val
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Thr Thr Leu Tyr
 65 70 75 80

Leu Gln Met Asn Ser Leu Arg Pro Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Thr Ile Gly Gly Ser Leu Ser Arg Ser Ser Gln Gly Thr Leu Val Thr
 100 105 110

Val Ser Ser
 115

<210> SEQ ID NO 185
 <211> LENGTH: 16
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 185

Ser Ile Ser Gly Ser Gly Thr Asp Thr Leu Tyr Ala Glu Ser Val Lys
 1 5 10 15

<210> SEQ ID NO 186
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 186

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala
 1 5 10 15

Ser Val Lys Val Ser Cys Lys Ala Ser Gly Phe Asn Ile Lys Asp Tyr

-continued

85	90	95
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Ala Arg Asp Gly Tyr Gly Arg Tyr Phe Tyr Asp Val Trp Gly Gln Gly
 100 105 110

Thr Leu Val Thr Val Ser Ser
 115

<210> SEQ ID NO 189
 <211> LENGTH: 112
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 189

Asp	Ile	Val	Met	Thr	Gln	Ser	Pro	Asp	Ser	Leu	Ala	Val	Ser	Leu	Gly
1				5					10					15	

Glu	Arg	Ala	Thr	Ile	Asn	Cys	Lys	Ser	Ser	Gln	Ser	Leu	Leu	Asn	Ser
			20					25					30		

Arg	Thr	Arg	Lys	Asn	Tyr	Leu	Ala	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Gln
			35				40					45			

Pro	Pro	Lys	Leu	Leu	Ile	Tyr	Trp	Ala	Ser	Thr	Arg	Glu	Ser	Gly	Val
		50				55					60				

Pro	Asp	Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	Thr
65					70					75				80	

Ile	Ser	Ser	Leu	Gln	Ala	Glu	Asp	Val	Ala	Val	Tyr	Tyr	Cys	Lys	Gln
			85						90					95	

Ser	Tyr	Ser	Arg	Arg	Thr	Phe	Gly	Gly	Gly	Thr	Lys	Val	Glu	Ile	Lys
			100					105						110	

<210> SEQ ID NO 190
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 190

Gln	Val	Gln	Leu	Val	Gln	Ser	Gly	Ala	Glu	Val	Lys	Lys	Pro	Gly	Ala
1			5						10					15	

Ser	Val	Lys	Val	Ser	Cys	Lys	Ala	Ser	Gly	Phe	Asn	Ile	Lys	Asp	Tyr
			20					25					30		

Tyr	Met	His	Trp	Val	Arg	Gln	Ala	Pro	Gly	Gln	Arg	Leu	Glu	Trp	Met
		35					40					45			

Gly	Trp	Ile	Asp	Leu	Glu	Asn	Ala	Asn	Thr	Ile	Tyr	Asp	Ala	Lys	Phe
	50					55					60				

Gln	Gly	Arg	Val	Thr	Ile	Thr	Arg	Asp	Thr	Ser	Ala	Ser	Thr	Ala	Tyr
65				70					75					80	

Met	Glu	Leu	Ser	Ser	Leu	Arg	Ser	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	Cys
			85					90						95	

Ala	Arg	Asp	Val	Tyr	Gly	Arg	Tyr	Leu	Tyr	Asp	Val	Trp	Gly	Gln	Gly
			100					105						110	

Thr Leu Val Thr Val Ser Ser
 115

<210> SEQ ID NO 191
 <211> LENGTH: 112
 <212> TYPE: PRT

-continued

<213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 191

```

Asp Ile Val Met Thr Gln Ser Pro Asp Ser Leu Ala Val Ser Leu Gly
1           5           10          15
Glu Arg Ala Thr Ile Asn Cys Lys Ser Ser Gln Ser Leu Leu Asn Asn
                20           25          30
Arg Thr Arg Lys Asn Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln
                35           40          45
Pro Pro Lys Leu Leu Ile Tyr Trp Ala Ser Thr Arg Glu Ser Gly Val
                50           55          60
Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr
65           70           75          80
Ile Ser Ser Leu Gln Ala Glu Asp Val Ala Val Tyr Tyr Cys Lys Gln
                85           90          95
Ser Tyr Ser Arg Arg Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
                100          105          110

```

<210> SEQ ID NO 192

<211> LENGTH: 119

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 192

```

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala
1           5           10          15
Ser Val Lys Val Ser Cys Lys Ala Ser Gly Phe Asn Ile Lys Asp Tyr
                20           25          30
Tyr Met His Trp Val Arg Gln Ala Pro Gly Gln Arg Leu Glu Trp Met
                35           40          45
Gly Trp Ile Asp Leu Glu Asn Gly Asn Thr Ile Tyr Asp Pro Lys Phe
50           55           60
Gln Gly Arg Val Thr Ile Thr Arg Asp Thr Ser Ala Ser Thr Ala Tyr
65           70           75          80
Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
                85           90          95
Ala Arg Asp Ala Tyr Gly Arg Tyr Phe Tyr Asp Val Trp Gly Gln Gly
                100          105          110
Thr Leu Val Thr Val Ser Ser
                115

```

<210> SEQ ID NO 193

<211> LENGTH: 112

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 193

```

Asp Ile Val Met Thr Gln Ser Pro Asp Ser Leu Ala Val Ser Leu Gly
1           5           10          15
Glu Arg Ala Thr Ile Asn Cys Arg Ser Ser Gln Ser Leu Leu Asn Ser
                20           25          30

```


-continued

```

Arg Thr Arg Lys Asn Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln
   35                               40               45

Pro Pro Lys Leu Leu Ile Tyr Trp Ala Ser Thr Arg Glu Ser Gly Val
   50                               55               60

Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr
   65                               70               75               80

Ile Ser Ser Leu Gln Ala Glu Asp Val Ala Val Tyr Tyr Cys Lys Gln
           85                               90               95

Ser Tyr Ser Arg Arg Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
   100                               105               110

```

```

<210> SEQ ID NO 194
<211> LENGTH: 119
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

```

```

<400> SEQUENCE: 194

```

```

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala
 1                               5               10               15

Ser Val Lys Val Ser Cys Lys Ala Ser Gly Phe Asn Ile Lys Asp Tyr
   20                               25               30

Tyr Met His Trp Val Arg Gln Ala Pro Gly Gln Arg Leu Glu Trp Met
   35                               40               45

Gly Trp Ile Asp Leu Glu Glu Gly Asn Thr Ile Tyr Asp Ala Lys Phe
   50                               55               60

Gln Gly Arg Val Thr Ile Thr Arg Asp Thr Ser Ala Ser Thr Ala Tyr
   65                               70               75               80

Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
           85                               90               95

Ala Arg Asp Ala Tyr Gly Arg Tyr Phe Tyr Asp Val Trp Gly Gln Gly
   100                               105               110

Thr Leu Val Thr Val Ser Ser
   115

```

```

<210> SEQ ID NO 195
<211> LENGTH: 112
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

```

```

<400> SEQUENCE: 195

```

```

Asp Ile Val Met Thr Gln Ser Pro Asp Ser Leu Ala Val Ser Leu Gly
 1                               5               10               15

Glu Arg Ala Thr Ile Asn Cys Lys Ser Ser Gln Ser Leu Leu Asn Gly
   20                               25               30

Arg Thr Arg Lys Asn Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln
   35                               40               45

Pro Pro Lys Leu Leu Ile Tyr Trp Ala Ser Thr Arg Glu Ser Gly Val
   50                               55               60

Pro Asp Arg Phe Ser Gly Ser Gly Thr Gly Thr Asp Phe Thr Leu Thr
   65                               70               75               80

Ile Ser Ser Leu Gln Ala Glu Asp Val Ala Val Tyr Tyr Cys Lys Gln
           85                               90               95

```

-continued

Ser Tyr Ser Arg Arg Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
 100 105 110

<210> SEQ ID NO 196
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 196

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala
 1 5 10 15

Ser Val Lys Val Ser Cys Lys Ala Ser Gly Phe Asn Ile Lys Asp Tyr
 20 25 30

Tyr Met His Trp Val Arg Gln Ala Pro Gly Gln Arg Leu Glu Trp Met
 35 40 45

Gly Trp Ile Asp Leu Glu Asn Ala Asn Thr Ile Tyr Asp Ala Lys Phe
 50 55 60

Gln Gly Arg Val Thr Ile Thr Arg Asp Thr Ser Ala Ser Thr Ala Tyr
 65 70 75 80

Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Ala Arg Asp Asn Tyr Gly Gly Tyr Phe Tyr Asp Val Trp Gly Gln Gly
 100 105 110

Thr Leu Val Thr Val Ser Ser
 115

<210> SEQ ID NO 197
 <211> LENGTH: 112
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 197

Asp Ile Val Met Thr Gln Ser Pro Asp Ser Leu Ala Val Pro Leu Gly
 1 5 10 15

Glu Arg Ala Thr Ile Asn Cys Lys Ser Ser Gln Ser Leu Leu Asn Ser
 20 25 30

Arg Thr Arg Lys Asn Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln
 35 40 45

Pro Pro Lys Leu Leu Ile Tyr Trp Ala Ser Thr Arg Glu Ser Gly Val
 50 55 60

Pro Asp Arg Phe Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr
 65 70 75 80

Ile Ser Ser Leu Gln Ala Glu Asp Val Ala Val Tyr Tyr Cys Lys Gln
 85 90 95

Ser Tyr Ser Arg Arg Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
 100 105 110

<210> SEQ ID NO 198
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

-continued

<400> SEQUENCE: 198

```

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala
1           5           10          15
Ser Val Lys Val Ser Cys Lys Ala Ser Gly Phe Asn Ile Lys Asp Tyr
          20          25          30
Tyr Met His Trp Val Arg Gln Ala Pro Gly Gln Arg Leu Glu Trp Met
          35          40          45
Gly Trp Ile Asp Leu Glu Asn Gly Asn Thr Ile Tyr Asp Pro Lys Phe
          50          55          60
Gln Gly Arg Val Thr Ile Thr Arg Asp Thr Ser Ala Ser Thr Ala Tyr
65          70          75          80
Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
          85          90          95
Ala Arg Asp Gly Tyr Gly Arg Tyr Phe Tyr Asp Val Trp Gly Gln Gly
          100         105         110
Thr Leu Val Thr Val Ser Ser
          115

```

<210> SEQ ID NO 199

<211> LENGTH: 112

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 199

```

Asp Ile Val Met Thr Gln Ser Pro Asp Ser Leu Ala Val Ser Leu Gly
1           5           10          15
Glu Arg Ala Thr Ile Asn Cys Lys Ser Ser Gln Ser Leu Leu Asn Ser
          20          25          30
Arg Thr Arg Lys Asn Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln
          35          40          45
Pro Pro Lys Leu Leu Ile Tyr Trp Ala Ser Thr Arg Glu Ser Gly Val
          50          55          60
Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr
65          70          75          80
Ile Ser Ser Leu Gln Ala Glu Asp Val Ala Val Tyr Tyr Cys Lys Gln
          85          90          95
Ser Tyr Ser Leu Arg Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
          100         105         110

```

<210> SEQ ID NO 200

<211> LENGTH: 119

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 200

```

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala
1           5           10          15
Ser Val Lys Val Ser Cys Lys Ala Ser Gly Phe Asn Ile Lys Asp Tyr
          20          25          30
Tyr Met His Trp Val Arg Gln Ala Pro Gly Gln Arg Leu Glu Trp Met
          35          40          45

```

-continued

Gly Trp Ile Asp Leu Glu Asn Gly Asn Thr Ile Tyr Asp Pro Lys Phe
 50 55 60

Gln Gly Arg Val Thr Ile Thr Arg Asp Thr Ser Ala Ser Thr Ala Tyr
 65 70 75 80

Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Ala Arg Asp Gly Tyr Gly Arg Tyr Phe Phe Asp Val Trp Gly Gln Gly
 100 105 110

Thr Leu Val Thr Val Ser Ser
 115

<210> SEQ ID NO 201
 <211> LENGTH: 112
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 201

Asp Ile Val Met Thr Gln Ser Pro Asp Ser Leu Ala Val Ser Leu Gly
 1 5 10 15

Glu Arg Ala Thr Ile Asn Cys Lys Ser Ser Gln Ser Leu Leu Asn Ser
 20 25 30

Arg Thr Arg Lys Asn Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln
 35 40 45

Pro Pro Lys Leu Leu Ile Tyr Trp Ala Ser Thr Arg Glu Ser Gly Val
 50 55 60

Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr
 65 70 75 80

Ile Ser Ser Leu Gln Ala Glu Asp Val Ala Val Tyr Tyr Cys Lys Gln
 85 90 95

Ser Tyr Asn Leu Arg Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
 100 105 110

<210> SEQ ID NO 202
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 202

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala
 1 5 10 15

Ser Val Lys Val Ser Cys Lys Ala Ser Gly Phe Asn Ile Lys Asp Tyr
 20 25 30

Tyr Met His Trp Val Arg Gln Ala Pro Gly Gln Arg Leu Glu Trp Met
 35 40 45

Gly Trp Ile Asp Leu Glu Asn Gly Asn Thr Ile Tyr Asp Pro Lys Phe
 50 55 60

Gln Gly Arg Val Thr Ile Thr Arg Asp Thr Ser Ala Ser Thr Ala Tyr
 65 70 75 80

Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Ala Arg Glu Gly Tyr Gly Arg Tyr Phe Tyr Asp Val Trp Gly Gln Gly
 100 105 110

-continued

Thr Leu Val Thr Val Ser Ser
115

<210> SEQ ID NO 203
 <211> LENGTH: 112
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 203

Asp Ile Val Met Thr Gln Ser Pro Asp Ser Leu Ala Val Ser Leu Gly
 1 5 10 15
 Glu Arg Ala Thr Ile Asn Cys Lys Ser Ser Gln Ser Leu Leu Asn Ser
 20 25 30
 Arg Thr Arg Lys Asn Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln
 35 40 45
 Pro Pro Lys Leu Leu Ile Tyr Trp Ala Ser Thr Arg Glu Ser Gly Val
 50 55 60
 Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr
 65 70 75 80
 Ile Ser Ser Leu Gln Ala Glu Asp Val Ala Val Tyr Tyr Cys Lys Gln
 85 90 95
 Ser Tyr Phe Arg Arg Ala Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
 100 105 110

<210> SEQ ID NO 204
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 204

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala
 1 5 10 15
 Ser Val Lys Val Ser Cys Lys Ala Ser Gly Phe Asn Ile Lys Asp Tyr
 20 25 30
 Tyr Met His Trp Val Arg Gln Ala Pro Gly Gln Arg Leu Glu Trp Met
 35 40 45
 Gly Trp Ile Asp Leu Glu Asn Gly Asn Thr Ile Tyr Asp Pro Lys Phe
 50 55 60
 Gln Gly Arg Val Thr Ile Thr Arg Asp Thr Ser Ala Ser Thr Ala Tyr
 65 70 75 80
 Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95
 Ala Arg Asp Gly Tyr Gly Arg Tyr Tyr Tyr Asp Val Trp Gly Gln Gly
 100 105 110
 Thr Leu Val Thr Val Ser Ser
 115

<210> SEQ ID NO 205
 <211> LENGTH: 112
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

-continued

<400> SEQUENCE: 205

```

Asp Ile Val Met Thr Gln Ser Pro Asp Ser Leu Ala Val Ser Leu Gly
1           5           10           15
Glu Arg Ala Thr Ile Asn Cys Lys Ser Ser Gln Ser Leu Leu Asn Ser
20           25           30
Arg Thr Arg Lys Asn Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln
35           40           45
Pro Pro Lys Leu Leu Ile Tyr Trp Ala Ser Thr Arg Glu Ser Gly Val
50           55           60
Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr
65           70           75           80
Ile Ser Ser Leu Gln Ala Glu Asp Val Ala Val Tyr Tyr Cys Lys Gln
85           90           95
Ser Tyr Asn Leu Arg Thr Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys
100          105          110

```

<210> SEQ ID NO 206

<211> LENGTH: 119

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 206

```

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala
1           5           10           15
Ser Val Lys Val Ser Cys Lys Ala Ser Gly Phe Asn Ile Lys Asp Tyr
20           25           30
Tyr Met His Trp Val Arg Gln Ala Pro Gly Gln Arg Leu Glu Trp Met
35           40           45
Gly Trp Ile Asp Leu Glu Asn Gly Asn Thr Ile Tyr Gln Pro Lys Phe
50           55           60
Gln Gly Arg Val Thr Ile Thr Arg Asp Thr Ser Ala Ser Thr Ala Tyr
65           70           75           80
Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
85           90           95
Ala Arg Asp Gly Tyr Gly Arg Tyr Phe Tyr Asp Val Trp Gly Gln Gly
100          105          110
Thr Leu Val Thr Val Ser Ser
115

```

<210> SEQ ID NO 207

<211> LENGTH: 112

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 207

```

Asp Ile Val Met Thr Gln Ser Pro Asp Ser Leu Ala Val Ser Leu Gly
1           5           10           15
Glu Arg Ala Thr Ile Asn Cys Lys Ser Ser Gln Ser Leu Leu Asn Ser
20           25           30
Arg Thr Arg Lys Asn Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln
35           40           45
Ser Pro Lys Leu Leu Ile Tyr Trp Ala Ser Thr Arg Glu Ser Gly Val

```

-continued

50	55	60
Pro Asp Arg Phe Thr Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr		
65	70	75 80
Ile Ser Ser Leu Gln Ala Glu Asp Val Ala Val Tyr Tyr Cys Lys Gln		
	85	90 95
Ser Tyr Ser Leu Arg Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys		
	100	105 110

<210> SEQ ID NO 208
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 208

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala		
1	5	10 15
Ser Val Lys Val Ser Cys Lys Ala Ser Gly Phe Asn Ile Lys Asp Tyr		
	20	25 30
Tyr Met His Trp Val Arg Gln Ala Pro Gly Gln Arg Leu Glu Trp Met		
	35	40 45
Gly Trp Ile Asp Leu Glu Asn Gly Asn Thr Ile Tyr Asp Pro Lys Phe		
	50	55 60
Gln Gly Arg Val Thr Ile Thr Arg Asp Thr Ser Ala Ser Thr Ala Tyr		
65	70	75 80
Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys		
	85	90 95
Ala Arg Asp Gly Tyr Gly Arg Tyr Phe Tyr Asp Tyr Trp Gly Gln Gly		
	100	105 110
Thr Leu Val Thr Val Ser Ser		
	115	

<210> SEQ ID NO 209
 <211> LENGTH: 112
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 209

Asp Ile Val Met Thr Gln Ser Pro Asp Ser Leu Ala Val Ser Leu Gly		
1	5	10 15
Glu Arg Ala Thr Ile Asn Cys Lys Ser Ser Gln Ser Leu Leu Glu Ser		
	20	25 30
Arg Thr Gly Lys Asn Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln		
	35	40 45
Pro Pro Lys Leu Leu Ile Tyr Trp Ala Ser Thr Arg Glu Ser Gly Val		
	50	55 60
Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr		
65	70	75 80
Ile Ser Ser Leu Gln Ala Glu Asp Val Ala Val Tyr Tyr Cys Lys Gln		
	85	90 95
Ser Tyr Ser Leu Arg Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys		
	100	105 110

-continued

```

<210> SEQ ID NO 210
<211> LENGTH: 117
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

```

```

<400> SEQUENCE: 210

```

```

Gln Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1             5             10             15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
      20             25             30
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
      35             40             45
Ser Gly Ile Ser Ala Ser Gly Gly Ser Thr Tyr Tyr Ala Asp Ser Val
      50             55             60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
      65             70             75             80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
      85             90             95
Ala Arg Tyr Tyr Leu Ser Gln Ile Asp Ser Trp Gly Gln Gly Thr Leu
      100            105            110
Val Thr Val Ser Ser
      115

```

```

<210> SEQ ID NO 211
<211> LENGTH: 110
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

```

```

<400> SEQUENCE: 211

```

```

Gln Ser Val Leu Thr Gln Pro Pro Ser Val Ser Ala Ala Pro Gly Gln
 1             5             10             15
Lys Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Ile Gly Asn Asn
      20             25             30
Tyr Val Ser Trp Tyr Arg Gln Leu Pro Gly Thr Ala Pro Lys Leu Leu
      35             40             45
Ile Tyr Glu Asn Asn Lys Arg Pro Ser Gly Ile Pro Asp Arg Phe Ser
      50             55             60
Gly Ser Lys Ser Gly Thr Ser Ala Thr Leu Gly Ile Thr Gly Leu Gln
      65             70             75             80
Thr Gly Asp Glu Ala Asp Tyr Tyr Cys Gly Thr Trp Asp Ser Ser Leu
      85             90             95
Arg Ala Gly Val Phe Gly Thr Gly Thr Lys Val Thr Val Leu
      100            105            110

```

```

<210> SEQ ID NO 212
<211> LENGTH: 249
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

```

```

<400> SEQUENCE: 212

```

```

Gln Ser Val Leu Thr Gln Pro Pro Ser Val Ser Ala Ala Pro Gly Gln
 1             5             10             15

```


Lys	Val	Thr	Ile	Ser	Cys	Ser	Gly	Ser	Ser	Ser	Asn	Ile	Gly	Asn	Asn
			20				25				30				
Tyr	Val	Ser	Trp	Tyr	Arg	Gln	Leu	Pro	Gly	Thr	Ala	Pro	Lys	Leu	Leu
			35				40				45				
Ile	Tyr	Glu	Asn	Asn	Lys	Arg	Pro	Ser	Gly	Ile	Pro	Asp	Arg	Phe	Ser
			50				55				60				
Gly	Ser	Lys	Ser	Gly	Thr	Ser	Ala	Thr	Leu	Gly	Ile	Thr	Gly	Leu	Gln
65				70						75			80		
Thr	Gly	Asp	Glu	Ala	Asp	Tyr	Tyr	Cys	Gly	Thr	Trp	Asp	Ser	Ser	Leu
			85						90			95			
Arg	Ala	Gly	Val	Phe	Gly	Thr	Gly	Thr	Lys	Val	Thr	Val	Leu	Gly	Ser
			100						105			110			
Arg	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser
			115			120						125			
Leu	Glu	Met	Ala	Gln	Val	Gln	Leu	Val	Glu	Ser	Gly	Gly	Gly	Leu	Val
			130			135						140			
Gln	Pro	Gly	Gly	Ser	Leu	Arg	Leu	Ser	Cys	Ala	Ala	Ser	Gly	Phe	Thr
145				150						155			160		
Phe	Ser	Ser	Tyr	Ala	Met	Ser	Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys	Gly
			165						170			175			
Leu	Glu	Trp	Val	Ser	Gly	Ile	Ser	Ala	Ser	Gly	Gly	Ser	Thr	Tyr	Tyr
			180						185			190			
Ala	Asp	Ser	Val	Lys	Gly	Arg	Phe	Thr	Ile	Ser	Arg	Asp	Asn	Ser	Lys
			195			200						205			
Asn	Thr	Leu	Tyr	Leu	Gln	Met	Asn	Ser	Leu	Arg	Ala	Glu	Asp	Thr	Ala
			210			215						220			
Val	Tyr	Tyr	Cys	Ala	Arg	Tyr	Tyr	Leu	Ser	Gln	Ile	Asp	Ser	Trp	Gly
225				230						235			240		
Gln	Gly	Thr	Leu	Val	Thr	Val	Ser	Ser							
			245												

```
<210> SEQ ID NO 213
<211> LENGTH: 118
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide
```

<400> SEQUENCE: 213

[illegible]

-continued

115

<210> SEQ ID NO 214
 <211> LENGTH: 111
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 214

```
Gln Ser Val Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln
 1             5             10             15
Arg Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Ile Gly Ser Asn
          20             25             30
Thr Val Asn Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu Leu
          35             40             45
Ile Tyr Ser Asn Asn Gln Arg Pro Ser Gly Val Pro Asp Arg Phe Ser
          50             55             60
Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu Gln
65             70             75             80
Ser Glu Asp Glu Ala Asp Tyr Tyr Cys Ala Ala Trp Asp Asp Ser Leu
          85             90             95
Asn Gly His Val Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
          100            105            110
```

<210> SEQ ID NO 215
 <211> LENGTH: 251
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 215

```
Gln Ser Val Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln
 1             5             10             15
Arg Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Ile Gly Ser Asn
          20             25             30
Thr Val Asn Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu Leu
          35             40             45
Ile Tyr Ser Asn Asn Gln Arg Pro Ser Gly Val Pro Asp Arg Phe Ser
          50             55             60
Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu Gln
65             70             75             80
Ser Glu Asp Glu Ala Asp Tyr Tyr Cys Ala Ala Trp Asp Asp Ser Leu
          85             90             95
Asn Gly His Val Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu Gly
          100            105            110
Ser Arg Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly
          115            120            125
Ser Leu Glu Met Ala Glu Val Gln Leu Val Gln Ser Gly Ala Glu Val
          130            135            140
Lys Lys Pro Gly Ala Thr Val Lys Ile Ser Cys Lys Val Ser Gly Tyr
          145            150            155            160
Thr Phe Thr Asp Tyr Tyr Met His Trp Val Gln Gln Ala Pro Gly Lys
          165            170            175
Gly Leu Glu Trp Met Gly Leu Val Asp Pro Glu Asp Gly Glu Thr Ile
```

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180	185	190
Tyr Ala Glu Lys Phe Gln Gly Arg Val Thr Ile Thr Ala Asp Thr Ser		
195	200	205
Thr Asp Thr Ala Tyr Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr		
210	215	220
Ala Val Tyr Tyr Cys Ala Thr Gly Ile Tyr Ser Arg Pro Leu Gly Tyr		
225	230	235
Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser		
245	250	

<210> SEQ ID NO 216
 <211> LENGTH: 121
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 216

Glu Val Gln Leu Val Glu Thr Gly Gly Gly Leu Val Gln Pro Gly Gly
1 5 10 15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
20 25 30
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45
Ser Ala Ile Ser Gly Ser Gly Gly Ser Thr Tyr Tyr Ala Asp Ser Val
50 55 60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65 70 75 80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95
Ala Arg Ser Asp Gly Lys His Phe Trp Gln Gln Tyr Asp Ala Trp Gly
100 105 110
Gln Gly Thr Leu Val Thr Val Ser Ser
115 120

<210> SEQ ID NO 217
 <211> LENGTH: 110
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 217

Ser Tyr Val Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln
1 5 10 15
Arg Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Ile Gly Ser His
20 25 30
Thr Val Asn Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu Leu
35 40 45
Ile Tyr Ser Asn Asn Gln Arg Pro Ser Gly Val Pro Asp Arg Phe Ser
50 55 60
Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu Gln
65 70 75 80
Ser Glu Asp Glu Ala Asp Tyr Tyr Cys Ala Ala Trp Asp Asp Ser Leu
85 90 95
Asn Gly Tyr Val Phe Gly Thr Gly Thr Lys Val Thr Val Leu

-continued

100	105	110
<210> SEQ ID NO 218		
<211> LENGTH: 253		
<212> TYPE: PRT		
<213> ORGANISM: Artificial Sequence		
<220> FEATURE:		
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide		
<400> SEQUENCE: 218		
Ser Tyr Val Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln		
1 5 10 15		
Arg Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Ile Gly Ser His		
20 25 30		
Thr Val Asn Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu Leu		
35 40 45		
Ile Tyr Ser Asn Asn Gln Arg Pro Ser Gly Val Pro Asp Arg Phe Ser		
50 55 60		
Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu Gln		
65 70 75 80		
Ser Glu Asp Glu Ala Asp Tyr Tyr Cys Ala Ala Trp Asp Asp Ser Leu		
85 90 95		
Asn Gly Tyr Val Phe Gly Thr Gly Thr Lys Val Thr Val Leu Gly Ser		
100 105 110		
Arg Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser		
115 120 125		
Leu Glu Met Ala Glu Val Gln Leu Val Glu Thr Gly Gly Gly Leu Val		
130 135 140		
Gln Pro Gly Gly Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr		
145 150 155 160		
Phe Ser Ser Tyr Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly		
165 170 175		
Leu Glu Trp Val Ser Ala Ile Ser Gly Ser Gly Gly Ser Thr Tyr Tyr		
180 185 190		
Ala Asp Ser Val Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys		
195 200 205		
Asn Thr Leu Tyr Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala		
210 215 220		
Val Tyr Tyr Cys Ala Arg Ser Asp Gly Lys His Phe Trp Gln Gln Tyr		
225 230 235 240		
Asp Ala Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser		
245 250		
<210> SEQ ID NO 219		
<211> LENGTH: 114		
<212> TYPE: PRT		
<213> ORGANISM: Artificial Sequence		
<220> FEATURE:		
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide		
<400> SEQUENCE: 219		
Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly		
1 5 10 15		
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Val Ser Ser Asn		
20 25 30		
Tyr Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val		

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35	40	45
Ser Ala Ile Ser Gly Ser Gly Gly Ser Thr Tyr Tyr Ala Asp Ser Val		
50	55	60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr		
65	70	75
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys		
	85	90
Ala Arg Met Asn Ile Asp Tyr Trp Gly Gln Gly Thr Leu Val Thr Val		
	100	105
		110
Ser Ser		

<210> SEQ ID NO 220
 <211> LENGTH: 107
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 220

Asp Ile Gln Leu Thr Gln Ser Pro Ser Ser Leu Ser Ala Tyr Val Gly
1 5 10 15
Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Gly Ile Thr Asn Ser
20 25 30
Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Leu
35 40 45
His Ala Ala Ser Arg Leu Glu Ser Gly Val Pro Ser Arg Phe Ser Gly
50 55 60
Ser Gly Phe Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65 70 75 80
Glu Asp Phe Ala Val Tyr Tyr Cys Gln His Tyr Leu Gly Thr Pro Tyr
85 90 95
Ser Phe Gly Gln Gly Thr Lys Val Glu Ile Lys
100 105

<210> SEQ ID NO 221
 <211> LENGTH: 243
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 221

Asp Ile Gln Leu Thr Gln Ser Pro Ser Ser Leu Ser Ala Tyr Val Gly
1 5 10 15
Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Gly Ile Thr Asn Ser
20 25 30
Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Leu
35 40 45
His Ala Ala Ser Arg Leu Glu Ser Gly Val Pro Ser Arg Phe Ser Gly
50 55 60
Ser Gly Phe Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65 70 75 80
Glu Asp Phe Ala Val Tyr Tyr Cys Gln His Tyr Leu Gly Thr Pro Tyr
85 90 95
Ser Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg Ser Arg Gly Gly
100 105 110

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Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Leu Glu Met
 115 120 125

Ala Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly
 130 135 140

Gly Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Val Ser Ser
 145 150 155 160

Asn Tyr Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp
 165 170 175

Val Ser Ala Ile Ser Gly Ser Gly Gly Ser Thr Tyr Tyr Ala Asp Ser
 180 185 190

Val Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu
 195 200 205

Tyr Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr
 210 215 220

Cys Ala Arg Met Asn Ile Asp Tyr Trp Gly Gln Gly Thr Leu Val Thr
 225 230 235 240

Val Ser Ser

<210> SEQ ID NO 222

<211> LENGTH: 124

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 222

Glu Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Arg Pro Gly Glu
 1 5 10 15

Ser Leu Thr Ile Ser Cys Lys Gly Ser Glu Tyr Ser Phe Ala Ser Tyr
 20 25 30

Trp Ile Thr Trp Val Arg Gln Met Pro Gly Lys Gly Leu Glu Trp Met
 35 40 45

Gly Arg Ile Asp Pro Ser Asp Ser Tyr Thr Asn Tyr Ser Pro Ser Phe
 50 55 60

Gln Gly His Val Thr Ile Ser Ala Asp Lys Ser Ile Ser Thr Ala Tyr
 65 70 75 80

Leu Gln Trp Ser Ser Leu Lys Ala Ser Asp Thr Ala Ile Tyr Tyr Cys
 85 90 95

Ala Arg Pro Phe Gln Tyr Asp Tyr Gly Gly Tyr Ser Asp Ala Phe Asp
 100 105 110

Ile Trp Gly Gln Gly Thr Met Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 223

<211> LENGTH: 107

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 223

Glu Ile Val Leu Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
 1 5 10 15

Asp Arg Val Thr Ile Ser Cys Arg Ala Ser Gln Ser Val Ser Arg Phe
 20 25 30

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Leu Asn Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
   35                               40           45

Tyr Gly Val Ser Thr Leu Glu Arg Gly Val Pro Ser Arg Phe Ser Gly
   50                               55           60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
   65                               70           75           80

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Glu Ser Tyr Ile Ile Pro Leu
                        85                               90           95

Thr Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys
      100                               105

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<210> SEQ ID NO 224
<211> LENGTH: 253
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
                        polypeptide

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<400> SEQUENCE: 224

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Glu Ile Val Leu Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
 1           5           10           15

Asp Arg Val Thr Ile Ser Cys Arg Ala Ser Gln Ser Val Ser Arg Phe
 20           25           30

Leu Asn Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
   35                               40           45

Tyr Gly Val Ser Thr Leu Glu Arg Gly Val Pro Ser Arg Phe Ser Gly
   50                               55           60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
   65                               70           75           80

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Glu Ser Tyr Ile Ile Pro Leu
                        85                               90           95

Thr Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys Arg Ser Arg Gly Gly
      100                               105           110

Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Leu Glu Met
      115                               120           125

Ala Glu Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Arg Pro Gly
      130                               135           140

Glu Ser Leu Thr Ile Ser Cys Lys Gly Ser Glu Tyr Ser Phe Ala Ser
      145                               150           155           160

Tyr Trp Ile Thr Trp Val Arg Gln Met Pro Gly Lys Gly Leu Glu Trp
      165                               170           175

Met Gly Arg Ile Asp Pro Ser Asp Ser Tyr Thr Asn Tyr Ser Pro Ser
      180                               185           190

Phe Gln Gly His Val Thr Ile Ser Ala Asp Lys Ser Ile Ser Thr Ala
      195                               200           205

Tyr Leu Gln Trp Ser Ser Leu Lys Ala Ser Asp Thr Ala Ile Tyr Tyr
      210                               215           220

Cys Ala Arg Pro Phe Gln Tyr Asp Tyr Gly Gly Tyr Ser Asp Ala Phe
      225                               230           235           240

Asp Ile Trp Gly Gln Gly Thr Met Val Thr Val Ser Ser
      245                               250

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<210> SEQ ID NO 225
<211> LENGTH: 118
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence

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<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

<400> SEQUENCE: 225

Gln Met Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Ala Gly Ser
1          5          10          15

Ser Val Lys Val Ser Cys Glu Thr Ser Gly Gly Thr Phe Ser Ser Ser
20          25          30

Ser Val Asn Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Met
35          40          45

Gly Gly Ile Ile Pro Ile Val Gly Thr Pro Asn Tyr Ala Gln Lys Phe
50          55          60

Gln Asp Arg Val Thr Ile Thr Ala Val Glu Ser Thr Phe Thr Ala Tyr
65          70          75          80

Met Glu Leu Ser Gly Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
85          90          95

Ala Arg Gly Gly Tyr Arg Asp Tyr Met Asp Val Trp Gly Arg Gly Thr
100         105         110

Thr Val Thr Val Ser Ser
115

```

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<210> SEQ ID NO 226
<211> LENGTH: 111
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

<400> SEQUENCE: 226

Glu Ile Val Met Thr Gln Ser Pro Leu Ser Leu Ser Val Thr Pro Gly
1          5          10          15

Glu Pro Ala Ser Ile Ser Cys Arg Ser Ser Gln Ser Leu Leu Asp Ser
20          25          30

Asn Gly Phe Asn Ser Leu Asp Trp Tyr Leu Gln Lys Pro Gly Gln Ser
35          40          45

Pro Gln Leu Leu Ile His Leu Gly Ser Asp Arg Ala Ser Gly Val Pro
50          55          60

Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Lys Ile
65          70          75          80

Ser Arg Val Glu Ala Glu Asp Val Gly Ile Tyr Tyr Cys Met Gln Ser
85          90          95

Leu Gln Ile Pro Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys
100         105         110

```

```

<210> SEQ ID NO 227
<211> LENGTH: 251
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

<400> SEQUENCE: 227

Glu Ile Val Met Thr Gln Ser Pro Leu Ser Leu Ser Val Thr Pro Gly
1          5          10          15

Glu Pro Ala Ser Ile Ser Cys Arg Ser Ser Gln Ser Leu Leu Asp Ser
20          25          30

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Asn	Gly	Phe	Asn	Ser	Leu	Asp	Trp	Tyr	Leu	Gln	Lys	Pro	Gly	Gln	Ser
	35						40					45			
Pro	Gln	Leu	Leu	Ile	His	Leu	Gly	Ser	Asp	Arg	Ala	Ser	Gly	Val	Pro
	50					55					60				
Asp	Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	Lys	Ile
65					70					75					80
Ser	Arg	Val	Glu	Ala	Glu	Asp	Val	Gly	Ile	Tyr	Tyr	Cys	Met	Gln	Ser
			85						90					95	
Leu	Gln	Ile	Pro	Thr	Phe	Gly	Gln	Gly	Thr	Lys	Val	Glu	Ile	Lys	Arg
			100					105					110		
Ser	Arg	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly
		115					120					125			
Ser	Leu	Glu	Met	Ala	Gln	Met	Gln	Leu	Val	Gln	Ser	Gly	Ala	Glu	Val
	130					135					140				
Lys	Lys	Ala	Gly	Ser	Ser	Val	Lys	Val	Ser	Cys	Glu	Thr	Ser	Gly	Gly
145					150					155					160
Thr	Phe	Ser	Ser	Ser	Ser	Val	Asn	Trp	Val	Arg	Gln	Ala	Pro	Gly	Gln
			165						170						175
Gly	Leu	Glu	Trp	Met	Gly	Gly	Ile	Ile	Pro	Ile	Val	Gly	Thr	Pro	Asn
			180					185					190		
Tyr	Ala	Gln	Lys	Phe	Gln	Asp	Arg	Val	Thr	Ile	Thr	Ala	Val	Glu	Ser
		195					200					205			
Thr	Phe	Thr	Ala	Tyr	Met	Glu	Leu	Ser	Gly	Leu	Arg	Ser	Glu	Asp	Thr
	210					215					220				
Ala	Val	Tyr	Tyr	Cys	Ala	Arg	Gly	Gly	Tyr	Arg	Asp	Tyr	Met	Asp	Val
225					230					235					240
Trp	Gly	Arg	Gly	Thr	Thr	Val	Thr	Val	Ser	Ser					
			245						250						

<210> SEQ ID NO 228

<211> LENGTH: 117

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 228

Glu	Val	Gln	Leu	Val	Glu	Ser	Gly	Gly	Gly	Leu	Ile	Gln	Pro	Gly	Gly
1			5					10					15		
Ser	Leu	Arg	Leu	Ser	Cys	Ala	Ala	Ser	Gly	Phe	Thr	Val	Ser	Ser	Asn
	20					25						30			
Tyr	Met	Ser	Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Val
	35					40					45				
Ser	Val	Ile	Tyr	Ser	Gly	Gly	Ser	Thr	Tyr	Tyr	Ala	Asp	Ser	Val	Lys
	50				55					60					
Gly	Arg	Phe	Thr	Ile	Ser	Arg	Asp	Asn	Ser	Lys	Asn	Thr	Leu	Tyr	Leu
65				70					75						80
Gln	Met	Asn	Ser	Leu	Arg	Ala	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	Cys	Ala
			85						90					95	
Arg	Gly	Gly	Phe	Gly	Ala	Glu	Phe	Asp	Tyr	Trp	Gly	Gln	Gly	Thr	Leu
			100					105					110		
Val	Thr	Val	Ser	Ser											
			115												

<210> SEQ ID NO 229

-continued

<211> LENGTH: 110
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 229

```

Ser Tyr Glu Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln
1           5           10           15

Arg Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Ile Gly Ser Asn
          20           25           30

Tyr Val Tyr Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu Leu
          35           40           45

Ile Tyr Arg Asn Asn Gln Arg Pro Ser Gly Val Pro Asp Arg Phe Ser
          50           55           60

Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu Arg
65           70           75           80

Ser Glu Asp Glu Ala Asp Tyr Tyr Cys Ala Ala Trp Asp Asp Ser Leu
          85           90           95

Ser Gly Tyr Val Phe Gly Thr Gly Thr Lys Val Thr Val Leu
          100          105          110

```

<210> SEQ ID NO 230
 <211> LENGTH: 249
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 230

```

Ser Tyr Glu Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln
1           5           10           15

Arg Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Ile Gly Ser Asn
          20           25           30

Tyr Val Tyr Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu Leu
          35           40           45

Ile Tyr Arg Asn Asn Gln Arg Pro Ser Gly Val Pro Asp Arg Phe Ser
          50           55           60

Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu Arg
65           70           75           80

Ser Glu Asp Glu Ala Asp Tyr Tyr Cys Ala Ala Trp Asp Asp Ser Leu
          85           90           95

Ser Gly Tyr Val Phe Gly Thr Gly Thr Lys Val Thr Val Leu Gly Ser
          100          105          110

Arg Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
          115          120          125

Leu Glu Met Ala Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Ile
          130          135          140

Gln Pro Gly Gly Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr
          145          150          155          160

Val Ser Ser Asn Tyr Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly
          165          170          175

Leu Glu Trp Val Ser Val Ile Tyr Ser Gly Gly Ser Thr Tyr Tyr Ala
          180          185          190

Asp Ser Val Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn
          195          200          205

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Thr Leu Tyr Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val
 210 215 220

Tyr Tyr Cys Ala Arg Gly Gly Phe Gly Ala Glu Phe Asp Tyr Trp Gly
 225 230 235 240

Gln Gly Thr Leu Val Thr Val Ser Ser
 245

<210> SEQ ID NO 231

<211> LENGTH: 123

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 231

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Ile Gln Pro Gly Gly
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Val Ser Ser Asn
 20 25 30

Tyr Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ser Val Ile Tyr Ser Gly Gly Ser Thr Tyr Tyr Ala Asp Ser Val Lys
 50 55 60

Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
 65 70 75 80

Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
 85 90 95

Arg Gly Gly Ile Ser Asp Asp Tyr Tyr Gly Ser Gly Ser Tyr Asp Asn
 100 105 110

Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 232

<211> LENGTH: 107

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 232

Ser Tyr Val Leu Thr Gln Pro Pro Ser Val Ser Val Ser Pro Gly Gln
 1 5 10 15

Thr Ala Ser Ile Thr Cys Ser Gly Asp Lys Leu Gly Asp Lys Tyr Ala
 20 25 30

Ser Trp Tyr Gln Gln Lys Pro Gly Gln Ser Pro Val Leu Val Ile Tyr
 35 40 45

Gln Asp Asn Lys Arg Pro Ser Gly Ile Pro Glu Arg Phe Ser Gly Ser
 50 55 60

Asn Ser Gly Asn Thr Ala Thr Leu Thr Ile Ser Gly Thr Gln Ala Met
 65 70 75 80

Asp Glu Ala Asp Tyr Tyr Cys Gln Ala Trp Asp Ser Ser Thr Glu Asp
 85 90 95

Val Phe Gly Pro Gly Thr Lys Val Thr Val Leu
 100 105

<210> SEQ ID NO 233

-continued

<211> LENGTH: 252
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 233

```

Ser Tyr Val Leu Thr Gln Pro Pro Ser Val Ser Val Ser Pro Gly Gln
1           5           10          15

Thr Ala Ser Ile Thr Cys Ser Gly Asp Lys Leu Gly Asp Lys Tyr Ala
          20          25          30

Ser Trp Tyr Gln Gln Lys Pro Gly Gln Ser Pro Val Leu Val Ile Tyr
          35          40          45

Gln Asp Asn Lys Arg Pro Ser Gly Ile Pro Glu Arg Phe Ser Gly Ser
          50          55          60

Asn Ser Gly Asn Thr Ala Thr Leu Thr Ile Ser Gly Thr Gln Ala Met
          65          70          75          80

Asp Glu Ala Asp Tyr Tyr Cys Gln Ala Trp Asp Ser Ser Thr Glu Asp
          85          90          95

Val Phe Gly Pro Gly Thr Lys Val Thr Val Leu Gly Ser Arg Gly Gly
          100         105         110

Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Leu Glu Met
          115         120         125

Ala Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Ile Gln Pro Gly
          130         135         140

Gly Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Val Ser Ser
          145         150         155         160

Asn Tyr Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp
          165         170         175

Val Ser Val Ile Tyr Ser Gly Gly Ser Thr Tyr Tyr Ala Asp Ser Val
          180         185         190

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
          195         200         205

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
          210         215         220

Ala Arg Gly Gly Ile Ser Asp Asp Tyr Tyr Gly Ser Gly Ser Tyr Asp
          225         230         235         240

Asn Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
          245         250

```

<210> SEQ ID NO 234
 <211> LENGTH: 118
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 234

```

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1           5           10          15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Val Ser Ser Asn
          20          25          30

Tyr Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
          35          40          45

Ser Val Ile Tyr Ser Gly Gly Ser Thr Tyr Tyr Ala Asp Ser Val Lys
          50          55          60

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-continued

Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Leu
65 70 75 80

Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
85 90 95

Arg Glu Arg Gly Met Gly Tyr Ala Phe Asp Ile Trp Gly Gln Gly Thr
100 105 110

Met Val Thr Val Ser Ser
115

<210> SEQ ID NO 235

<211> LENGTH: 107

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 235

Asp Ile Gln Leu Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1 5 10 15

Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Ser Ile Ser Ser Tyr
20 25 30

Leu Asn Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
35 40 45

Tyr Ala Ala Ser Ser Leu Gln Ser Gly Val Pro Ser Arg Phe Ser Gly
50 55 60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65 70 75 80

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Ser Tyr Ser Thr Pro Phe
85 90 95

Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
100 105

<210> SEQ ID NO 236

<211> LENGTH: 247

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 236

Asp Ile Gln Leu Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1 5 10 15

Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Ser Ile Ser Ser Tyr
20 25 30

Leu Asn Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
35 40 45

Tyr Ala Ala Ser Ser Leu Gln Ser Gly Val Pro Ser Arg Phe Ser Gly
50 55 60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65 70 75 80

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Ser Tyr Ser Thr Pro Phe
85 90 95

Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys Arg Ser Arg Gly Gly
100 105 110

Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Leu Glu Met
115 120 125

-continued

Ala Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly
 130 135 140

Gly Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Val Ser Ser
 145 150 155 160

Asn Tyr Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp
 165 170 175

Val Ser Val Ile Tyr Ser Gly Gly Ser Thr Tyr Tyr Ala Asp Ser Val
 180 185 190

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
 195 200 205

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
 210 215 220

Ala Arg Glu Arg Gly Met Gly Tyr Ala Phe Asp Ile Trp Gly Gln Gly
 225 230 235 240

Thr Met Val Thr Val Ser Ser
 245

<210> SEQ ID NO 237
 <211> LENGTH: 114
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 237

Gln Leu Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Glu
 1 5 10 15

Thr Leu Ser Leu Thr Cys Ser Val Ser Gly Val Ser Met Ser Glu Asn
 20 25 30

Tyr Trp Ser Trp Ile Arg Gln Pro Pro Gly Lys Arg Leu Glu Trp Ile
 35 40 45

Gly Cys Ala His Tyr Thr Gly Asp Thr His Tyr Asn Pro Ser Leu Lys
 50 55 60

Gly Arg Val Thr Ile Ser Leu Asp Thr Ser Met Asn Gln Phe Ser Leu
 65 70 75 80

Arg Leu Asn Ser Val Thr Ala Ala Asp Thr Ala Val Tyr Tyr Cys Ala
 85 90 95

Ser Tyr His Pro Phe Asn Tyr Trp Gly Gln Gly Thr Leu Val Thr Val
 100 105 110

Ser Ser

<210> SEQ ID NO 238
 <211> LENGTH: 104
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 238

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
 1 5 10 15

Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Gly Ile Gly Ser Tyr
 20 25 30

Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
 35 40 45

-continued

Tyr Pro Ala Ser Thr Leu Gln Ser Gly Val Pro Ser Arg Phe Ser Gly
50 55 60

Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65 70 75 80

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Leu Asn Ser Leu Phe Gly
85 90 95

Gln Gly Thr Arg Leu Glu Ile Lys
100

<210> SEQ ID NO 239
<211> LENGTH: 240
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 239

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1 5 10 15

Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Gly Ile Gly Ser Tyr
20 25 30

Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
35 40 45

Tyr Pro Ala Ser Thr Leu Gln Ser Gly Val Pro Ser Arg Phe Ser Gly
50 55 60

Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65 70 75 80

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Leu Asn Ser Leu Phe Gly
85 90 95

Gln Gly Thr Arg Leu Glu Ile Lys Arg Ser Arg Gly Gly Gly Ser
100 105 110

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Leu Glu Met Ala Gln Leu
115 120 125

Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Glu Thr Leu
130 135 140

Ser Leu Thr Cys Ser Val Ser Gly Val Ser Met Ser Glu Asn Tyr Trp
145 150 155 160

Ser Trp Ile Arg Gln Pro Pro Gly Lys Arg Leu Glu Trp Ile Gly Cys
165 170 175

Ala His Tyr Thr Gly Asp Thr His Tyr Asn Pro Ser Leu Lys Gly Arg
180 185 190

Val Thr Ile Ser Leu Asp Thr Ser Met Asn Gln Phe Ser Leu Arg Leu
195 200 205

Asn Ser Val Thr Ala Ala Asp Thr Ala Val Tyr Tyr Cys Ala Ser Tyr
210 215 220

His Pro Phe Asn Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
225 230 235 240

<210> SEQ ID NO 240
<211> LENGTH: 126
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 240

-continued

Glu Val Gln Leu Val Gln Ser Gly Ala Glu Val Arg Arg Pro Gly Ala
 1 5 10 15
 Thr Val Lys Ile Ser Cys Lys Val Ser Gly Tyr Thr Phe Asn Asp Phe
 20 25 30
 Tyr Leu His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Met
 35 40 45
 Gly Arg Ile Asp Pro Glu Asp Gly Lys Thr Arg Tyr Ala Glu Lys Phe
 50 55 60
 Gln Gly Arg Leu Thr Ile Thr Ala Asp Thr Ser Thr Asp Thr Leu Tyr
 65 70 75 80
 Met Gln Leu Gly Gly Leu Thr Ser Asp Asp Thr Ala Val Tyr Tyr Cys
 85 90 95
 Thr Thr Asp Trp Gly Tyr Ser Ser Ser Leu Arg Glu Glu Asp Ile Trp
 100 105 110
 Tyr Asp Cys Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
 115 120 125

<210> SEQ ID NO 241
 <211> LENGTH: 110
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 241

Gln Ala Val Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln
 1 5 10 15
 Arg Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Ile Gly Thr Lys
 20 25 30
 Thr Val Asn Trp Tyr Gln Val Leu Pro Gly Thr Ala Pro Lys Leu Leu
 35 40 45
 Ile Tyr Ser Asn Tyr Arg Arg Pro Ser Gly Val Pro Asp Arg Phe Ser
 50 55 60
 Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu Gln
 65 70 75 80
 Ser Asp Asp Glu Ala Asp Tyr Tyr Cys Ala Leu Trp Asp Asp Ser Leu
 85 90 95
 Asp Gly Tyr Val Phe Gly Thr Gly Thr Lys Val Thr Val Leu
 100 105 110

<210> SEQ ID NO 242
 <211> LENGTH: 258
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 242

Gln Ala Val Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln
 1 5 10 15
 Arg Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Ile Gly Thr Lys
 20 25 30
 Thr Val Asn Trp Tyr Gln Val Leu Pro Gly Thr Ala Pro Lys Leu Leu
 35 40 45
 Ile Tyr Ser Asn Tyr Arg Arg Pro Ser Gly Val Pro Asp Arg Phe Ser
 50 55 60

-continued

Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu Gln
 65 70 75 80
 Ser Asp Asp Glu Ala Asp Tyr Tyr Cys Ala Leu Trp Asp Asp Ser Leu
 85 90 95
 Asp Gly Tyr Val Phe Gly Thr Gly Thr Lys Val Thr Val Leu Gly Ser
 100 105 110
 Arg Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
 115 120 125
 Leu Glu Met Ala Glu Val Gln Leu Val Gln Ser Gly Ala Glu Val Arg
 130 135 140
 Arg Pro Gly Ala Thr Val Lys Ile Ser Cys Lys Val Ser Gly Tyr Thr
 145 150 155 160
 Phe Asn Asp Phe Tyr Leu His Trp Val Arg Gln Ala Pro Gly Lys Gly
 165 170 175
 Leu Glu Trp Met Gly Arg Ile Asp Pro Glu Asp Gly Lys Thr Arg Tyr
 180 185 190
 Ala Glu Lys Phe Gln Gly Arg Leu Thr Ile Thr Ala Asp Thr Ser Thr
 195 200 205
 Asp Thr Leu Tyr Met Gln Leu Gly Gly Leu Thr Ser Asp Asp Thr Ala
 210 215 220
 Val Tyr Tyr Cys Thr Thr Asp Trp Gly Tyr Ser Ser Ser Leu Arg Glu
 225 230 235 240
 Glu Asp Ile Trp Tyr Asp Cys Trp Gly Gln Gly Thr Leu Val Thr Val
 245 250 255
 Ser Ser

<210> SEQ ID NO 243
 <211> LENGTH: 122
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 243

Glu Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ser
 1 5 10 15
 Ser Val Lys Val Ser Cys Lys Ala Ser Gly Gly Thr Phe Ser Ser Tyr
 20 25 30
 Ala Ile Ser Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Met
 35 40 45
 Gly Gly Ile Ile Pro Ile Phe Gly Thr Ala Asn Tyr Ala Gln Lys Phe
 50 55 60
 Gln Gly Arg Val Thr Ile Thr Ala Asp Glu Ser Thr Ser Thr Ala Tyr
 65 70 75 80
 Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95
 Ala Arg Asp Tyr Gly Tyr Gly Asp Tyr Gly Asp Ala Phe Asp Ile Trp
 100 105 110
 Gly Gln Gly Thr Met Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 244
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:

-continued

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 244

```

Ser Tyr Glu Leu Thr Gln Pro Pro Ser Val Ser Val Ala Pro Gly Lys
1           5           10           15

Thr Ala Arg Ile Thr Cys Gly Gly Asn Asn Ile Gly Ser Lys Ser Val
                20           25           30

His Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Val Leu Val Ile Tyr
                35           40           45

Tyr Asp Ser Asp Arg Pro Ser Gly Ile Pro Glu Arg Phe Ser Gly Ser
50           55           60

Asn Ser Gly Asn Thr Ala Thr Leu Thr Ile Ser Arg Val Glu Ala Gly
65           70           75           80

Asp Glu Ala Asp Tyr Tyr Cys Gln Val Trp Asp Ser Ser Ser Asp His
                85           90           95

Tyr Val Phe Gly Thr Gly Thr Lys Val Thr Val Leu
100           105

```

<210> SEQ ID NO 245

<211> LENGTH: 252

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 245

```

Ser Tyr Glu Leu Thr Gln Pro Pro Ser Val Ser Val Ala Pro Gly Lys
1           5           10           15

Thr Ala Arg Ile Thr Cys Gly Gly Asn Asn Ile Gly Ser Lys Ser Val
20           25           30

His Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Val Leu Val Ile Tyr
35           40           45

Tyr Asp Ser Asp Arg Pro Ser Gly Ile Pro Glu Arg Phe Ser Gly Ser
50           55           60

Asn Ser Gly Asn Thr Ala Thr Leu Thr Ile Ser Arg Val Glu Ala Gly
65           70           75           80

Asp Glu Ala Asp Tyr Tyr Cys Gln Val Trp Asp Ser Ser Ser Asp His
85           90           95

Tyr Val Phe Gly Thr Gly Thr Lys Val Thr Val Leu Gly Ser Arg Gly
100          105          110

Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Leu Glu
115          120          125

Met Ala Glu Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro
130          135          140

Gly Ser Ser Val Lys Val Ser Cys Lys Ala Ser Gly Gly Thr Phe Ser
145          150          155          160

Ser Tyr Ala Ile Ser Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu
165          170          175

Trp Met Gly Gly Ile Ile Pro Ile Phe Gly Thr Ala Asn Tyr Ala Gln
180          185          190

Lys Phe Gln Gly Arg Val Thr Ile Thr Ala Asp Glu Ser Thr Ser Thr
195          200          205

Ala Tyr Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr
210          215          220

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-continued

Tyr Cys Ala Arg Asp Tyr Gly Tyr Gly Asp Tyr Gly Asp Ala Phe Asp
225 230 235 240

Ile Trp Gly Gln Gly Thr Met Val Thr Val Ser Ser
245 250

<210> SEQ ID NO 246
<211> LENGTH: 121
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 246

Glu Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Glu
1 5 10 15

Ser Leu Lys Ile Ser Cys Lys Gly Ser Gly Tyr Ser Phe Thr Ser Tyr
20 25 30

Trp Ile Gly Trp Val Arg Gln Met Pro Gly Lys Gly Leu Glu Trp Met
35 40 45

Gly Ile Ile Tyr Pro Gly Asp Ser Asp Thr Arg Tyr Ser Pro Ser Phe
50 55 60

Gln Gly Gln Val Thr Ile Ser Ala Asp Lys Ser Ile Ser Thr Ala Tyr
65 70 75 80

Leu Gln Trp Ser Ser Leu Lys Ala Ser Asp Thr Ala Met Tyr Tyr Cys
85 90 95

Ala Arg Val Val Gly Thr Ile Tyr Ser Met Gln Tyr Asp Val Trp Gly
100 105 110

Gln Gly Thr Leu Val Thr Val Ser Ser
115 120

<210> SEQ ID NO 247
<211> LENGTH: 108
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 247

Ser Tyr Val Leu Thr Gln Pro Pro Ser Val Ser Val Ala Pro Gly Lys
1 5 10 15

Thr Ala Arg Ile Thr Cys Gly Gly Asn Asn Ile Gly Ser Lys Ser Val
20 25 30

His Trp Tyr Gln Gln Arg Pro Gly Gln Ala Pro Val Leu Val Val Tyr
35 40 45

Asp Asp Ser Asp Arg Pro Ser Gly Ile Pro Glu Arg Phe Ser Gly Ser
50 55 60

Asn Ser Gly Asn Thr Ala Thr Leu Thr Ile Ser Arg Val Glu Ala Gly
65 70 75 80

Asp Glu Ala Asp Tyr Ser Cys Gln Val Trp Asp Ser Ser Ser Asp His
85 90 95

Tyr Val Phe Gly Pro Gly Thr Lys Val Thr Val Leu
100 105

<210> SEQ ID NO 248
<211> LENGTH: 251
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:

-continued

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 248

```

Ser Tyr Val Leu Thr Gln Pro Pro Ser Val Ser Val Ala Pro Gly Lys
1           5           10           15

Thr Ala Arg Ile Thr Cys Gly Gly Asn Asn Ile Gly Ser Lys Ser Val
20          25          30

His Trp Tyr Gln Gln Arg Pro Gly Gln Ala Pro Val Leu Val Val Tyr
35          40          45

Asp Asp Ser Asp Arg Pro Ser Gly Ile Pro Glu Arg Phe Ser Gly Ser
50          55          60

Asn Ser Gly Asn Thr Ala Thr Leu Thr Ile Ser Arg Val Glu Ala Gly
65          70          75          80

Asp Glu Ala Asp Tyr Ser Cys Gln Val Trp Asp Ser Ser Ser Asp His
85          90          95

Tyr Val Phe Gly Pro Gly Thr Lys Val Thr Val Leu Gly Ser Arg Gly
100         105         110

Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Leu Glu
115         120         125

Met Ala Glu Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro
130         135         140

Gly Glu Ser Leu Lys Ile Ser Cys Lys Gly Ser Gly Tyr Ser Phe Thr
145         150         155         160

Ser Tyr Trp Ile Gly Trp Val Arg Gln Met Pro Gly Lys Gly Leu Glu
165         170         175

Trp Met Gly Ile Ile Tyr Pro Gly Asp Ser Asp Thr Arg Tyr Ser Pro
180         185         190

Ser Phe Gln Gly Gln Val Thr Ile Ser Ala Asp Lys Ser Ile Ser Thr
195         200         205

Ala Tyr Leu Gln Trp Ser Ser Leu Lys Ala Ser Asp Thr Ala Met Tyr
210         215         220

Tyr Cys Ala Arg Val Val Gly Thr Ile Tyr Ser Met Gln Tyr Asp Val
225         230         235         240

Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
245         250

```

<210> SEQ ID NO 249

<211> LENGTH: 130

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 249

```

Glu Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Glu
1           5           10           15

Ser Leu Lys Ile Ser Cys Lys Gly Ser Gly Tyr Ser Phe Thr Ser Tyr
20          25          30

Trp Ile Gly Trp Val Arg Gln Met Pro Gly Lys Gly Leu Glu Trp Met
35          40          45

Gly Ile Ile Tyr Pro Gly Asp Ser Asp Thr Arg Tyr Ser Pro Ser Phe
50          55          60

Gln Gly Gln Val Thr Ile Ser Ala Asp Lys Ser Ile Ser Thr Ala Tyr
65          70          75          80

```

[illegible]

```
<210> SEQ ID NO 250
<211> LENGTH: 108
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide
```

<400> SEQUENCE: 250

Leu	Pro	Val	Leu	Thr	Gln	Pro	Pro	Ser	Val	Ser	Val	Ala	Pro	Gly	Lys
1				5					10					15	
Thr	Ala	Arg	Ile	Thr	Cys	Gly	Gly	Asn	Asn	Ile	Gly	Ser	Lys	Ser	Val
			20					25					30		
His	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Gln	Ala	Pro	Val	Leu	Val	Val	Tyr
		35					40					45			
Asp	Asp	Ser	Asp	Arg	Pro	Ser	Gly	Ile	Pro	Glu	Arg	Phe	Ser	Gly	Ser
	50					55					60				
Asn	Ser	Gly	Asn	Thr	Ala	Thr	Leu	Thr	Ile	Ser	Arg	Val	Glu	Ala	Gly
65					70					75					80
Asp	Glu	Ala	Asp	Tyr	Tyr	Cys	Gln	Val	Trp	Asp	Ser	Ser	Ser	Asp	Tyr
			85						90					95	
Val	Val	Phe	Gly	Gly	Gly	Thr	Lys	Leu	Thr	Val	Leu				
			100					105							

```
<210> SEQ ID NO 251
<211> LENGTH: 260
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide
```

<400> SEQUENCE: 251

Leu	Pro	Val	Leu	Thr	Gln	Pro	Pro	Ser	Val	Ser	Val	Ala	Pro	Gly	Lys
1				5					10					15	
Thr	Ala	Arg	Ile	Thr	Cys	Gly	Gly	Asn	Asn	Ile	Gly	Ser	Lys	Ser	Val
			20					25					30		
His	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Gln	Ala	Pro	Val	Leu	Val	Val	Tyr
		35					40					45			
Asp	Asp	Ser	Asp	Arg	Pro	Ser	Gly	Ile	Pro	Glu	Arg	Phe	Ser	Gly	Ser
	50					55					60				
Asn	Ser	Gly	Asn	Thr	Ala	Thr	Leu	Thr	Ile	Ser	Arg	Val	Glu	Ala	Gly
65					70					75					80
Asp	Glu	Ala	Asp	Tyr	Tyr	Cys	Gln	Val	Trp	Asp	Ser	Ser	Ser	Asp	Tyr
			85						90					95	
Val	Val	Phe	Gly	Gly	Gly	Thr	Lys	Leu	Thr	Val	Leu	Gly	Ser	Arg	Gly
			100					105					110		
Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Leu	Glu
		115					120					125			

-continued

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Met Ala Glu Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro
 130          135          140

Gly Glu Ser Leu Lys Ile Ser Cys Lys Gly Ser Gly Tyr Ser Phe Thr
145          150          155          160

Ser Tyr Trp Ile Gly Trp Val Arg Gln Met Pro Gly Lys Gly Leu Glu
          165          170          175

Trp Met Gly Ile Ile Tyr Pro Gly Asp Ser Asp Thr Arg Tyr Ser Pro
          180          185          190

Ser Phe Gln Gly Gln Val Thr Ile Ser Ala Asp Lys Ser Ile Ser Thr
          195          200          205

Ala Tyr Leu Gln Trp Ser Ser Leu Lys Ala Ser Asp Thr Ala Met Tyr
          210          215          220

Tyr Cys Ala Arg Gln Val Trp Gly Trp Gln Gly Gly Met Tyr Pro Arg
225          230          235          240

Ser Asn Trp Trp Tyr Asn Met Asp Ser Trp Gly Gln Gly Thr Leu Val
          245          250          255

Thr Val Ser Ser
          260

```

```

<210> SEQ ID NO 252
<211> LENGTH: 121
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

```

```

<400> SEQUENCE: 252

```

```

Glu Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Glu
 1          5          10          15

Ser Leu Lys Ile Ser Cys Lys Gly Ser Gly Tyr Ser Phe Thr Ser Tyr
          20          25          30

Trp Ile Gly Trp Val Arg Gln Met Pro Gly Lys Gly Leu Glu Trp Met
          35          40          45

Gly Ile Ile Tyr Pro Gly Asp Ser Asp Thr Arg Tyr Ser Pro Ser Phe
          50          55          60

Gln Gly Gln Val Thr Ile Ser Ala Asp Lys Ser Ile Ser Thr Ala Tyr
          65          70          75          80

Leu Gln Trp Ser Ser Leu Lys Ala Ser Asp Thr Ala Met Tyr Tyr Cys
          85          90          95

Ala Arg Trp Ser Ser Thr Trp Asp Ser Met Tyr Met Asp Tyr Trp Gly
          100          105          110

Gln Gly Thr Leu Val Thr Val Ser Ser
          115          120

```

```

<210> SEQ ID NO 253
<211> LENGTH: 110
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

```

```

<400> SEQUENCE: 253

```

```

Gln Ala Val Leu Thr Gln Pro Pro Ser Val Ser Glu Ala Pro Arg Gln
 1          5          10          15

Arg Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Val Gly Asn Asn
          20          25          30

```

-continued

Ala	Val	Asn	Trp	Tyr	Gln	Gln	Val	Pro	Gly	Lys	Ala	Pro	Lys	Leu	Leu
		35					40					45			
Ile	Tyr	Tyr	Asp	Asp	Leu	Leu	Ser	Ser	Gly	Val	Ser	Asp	Arg	Phe	Ser
	50					55					60				
Gly	Ser	Lys	Ser	Gly	Thr	Ser	Ala	Ser	Leu	Ala	Ile	Ser	Gly	Leu	Gln
65					70					75					80
Ser	Glu	Asp	Glu	Ala	Asp	Tyr	Tyr	Cys	Ala	Ala	Trp	Asp	Asp	Ser	Leu
			85					90						95	
Asn	Gly	Pro	Val	Phe	Gly	Gly	Gly	Thr	Lys	Leu	Thr	Val	Leu		
			100					105					110		

<210> SEQ ID NO 254
 <211> LENGTH: 253
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 254

Gln	Ala	Val	Leu	Thr	Gln	Pro	Pro	Ser	Val	Ser	Glu	Ala	Pro	Arg	Gln
1			5					10					15		
Arg	Val	Thr	Ile	Ser	Cys	Ser	Gly	Ser	Ser	Ser	Asn	Val	Gly	Asn	Asn
		20					25					30			
Ala	Val	Asn	Trp	Tyr	Gln	Gln	Val	Pro	Gly	Lys	Ala	Pro	Lys	Leu	Leu
		35				40					45				
Ile	Tyr	Tyr	Asp	Asp	Leu	Leu	Ser	Ser	Gly	Val	Ser	Asp	Arg	Phe	Ser
	50					55				60					
Gly	Ser	Lys	Ser	Gly	Thr	Ser	Ala	Ser	Leu	Ala	Ile	Ser	Gly	Leu	Gln
65					70					75					80
Ser	Glu	Asp	Glu	Ala	Asp	Tyr	Tyr	Cys	Ala	Ala	Trp	Asp	Asp	Ser	Leu
			85					90						95	
Asn	Gly	Pro	Val	Phe	Gly	Gly	Gly	Thr	Lys	Leu	Thr	Val	Leu	Gly	Ser
			100					105					110		
Arg	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser
	115					120					125				
Leu	Glu	Met	Ala	Glu	Val	Gln	Leu	Val	Gln	Ser	Gly	Ala	Glu	Val	Lys
	130					135					140				
Lys	Pro	Gly	Glu	Ser	Leu	Lys	Ile	Ser	Cys	Lys	Gly	Ser	Gly	Tyr	Ser
145					150					155					160
Phe	Thr	Ser	Tyr	Trp	Ile	Gly	Trp	Val	Arg	Gln	Met	Pro	Gly	Lys	Gly
			165					170						175	
Leu	Glu	Trp	Met	Gly	Ile	Ile	Tyr	Pro	Gly	Asp	Ser	Asp	Thr	Arg	Tyr
		180					185						190		
Ser	Pro	Ser	Phe	Gln	Gly	Gln	Val	Thr	Ile	Ser	Ala	Asp	Lys	Ser	Ile
		195					200					205			
Ser	Thr	Ala	Tyr	Leu	Gln	Trp	Ser	Ser	Leu	Lys	Ala	Ser	Asp	Thr	Ala
	210					215					220				
Met	Tyr	Tyr	Cys	Ala	Arg	Trp	Ser	Ser	Thr	Trp	Asp	Ser	Met	Tyr	Met
225					230					235					240
Asp	Tyr	Trp	Gly	Gln	Gly	Thr	Leu	Val	Thr	Val	Ser	Ser			
			245						250						

<210> SEQ ID NO 255
 <211> LENGTH: 121
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence

-continued

<220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 255

```

Glu Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Glu
1           5           10          15
Ser Leu Arg Ile Ser Cys Lys Gly Ser Gly Tyr Ser Phe Thr Ser Tyr
          20           25          30
Trp Ile Gly Trp Val Arg Gln Met Pro Gly Lys Gly Leu Glu Trp Met
          35           40          45
Gly Ile Ile Tyr Pro Gly Asp Ser Asp Thr Arg Tyr Ser Pro Ser Phe
          50           55          60
Gln Gly Gln Val Thr Ile Ser Ala Asp Lys Ser Ile Ser Thr Ala Tyr
          65           70          75          80
Leu Gln Trp Ser Ser Leu Lys Ala Ser Asp Thr Ala Met Tyr Tyr Cys
          85           90          95
Ala Arg Val Thr Tyr Ser Met Asp Ser Tyr Tyr Phe Asp Ser Trp Gly
          100          105          110
Gln Gly Thr Leu Val Thr Val Ser Ser
          115          120

```

<210> SEQ ID NO 256
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 256

```

Gln Pro Val Leu Thr Gln Pro Pro Ser Val Ser Val Ala Pro Gly Lys
1           5           10          15
Thr Ala Arg Ile Thr Cys Gly Gly Asn Asn Ile Gly Ser Glu Ser Val
          20           25          30
His Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Met Val Val Ile Tyr
          35           40          45
Tyr Asp Ser Asn Arg Pro Ser Gly Ile Pro Glu Arg Phe Ser Gly Ser
          50           55          60
Asn Ser Gly Asn Thr Ala Thr Leu Thr Val Ser Arg Val Glu Ala Glu
          65           70          75          80
Asp Glu Ala Asp Tyr Tyr Cys Gln Val Trp Asn Ser Ser Ser Asp His
          85           90          95
Arg Gly Val Phe Gly Gly Gly Thr Lys Leu Thr Val
          100          105

```

<210> SEQ ID NO 257
 <211> LENGTH: 252
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 257

```

Gln Pro Val Leu Thr Gln Pro Pro Ser Val Ser Val Ala Pro Gly Lys
1           5           10          15
Thr Ala Arg Ile Thr Cys Gly Gly Asn Asn Ile Gly Ser Glu Ser Val
          20           25          30

```


-continued

His	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Gln	Ala	Pro	Met	Val	Val	Ile	Tyr
	35						40					45			
Tyr	Asp	Ser	Asn	Arg	Pro	Ser	Gly	Ile	Pro	Glu	Arg	Phe	Ser	Gly	Ser
	50					55					60				
Asn	Ser	Gly	Asn	Thr	Ala	Thr	Leu	Thr	Val	Ser	Arg	Val	Glu	Ala	Glu
65					70					75					80
Asp	Glu	Ala	Asp	Tyr	Tyr	Cys	Gln	Val	Trp	Asn	Ser	Ser	Ser	Asp	His
			85						90					95	
Arg	Gly	Val	Phe	Gly	Gly	Gly	Thr	Lys	Leu	Thr	Val	Leu	Gly	Ser	Arg
			100					105					110		
Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Leu
		115					120					125			
Glu	Met	Ala	Glu	Val	Gln	Leu	Val	Gln	Ser	Gly	Ala	Glu	Val	Lys	Lys
	130					135					140				
Pro	Gly	Glu	Ser	Leu	Arg	Ile	Ser	Cys	Lys	Gly	Ser	Gly	Tyr	Ser	Phe
145					150					155					160
Thr	Ser	Tyr	Trp	Ile	Gly	Trp	Val	Arg	Gln	Met	Pro	Gly	Lys	Gly	Leu
			165						170					175	
Glu	Trp	Met	Gly	Ile	Ile	Tyr	Pro	Gly	Asp	Ser	Asp	Thr	Arg	Tyr	Ser
		180						185					190		
Pro	Ser	Phe	Gln	Gly	Gln	Val	Thr	Ile	Ser	Ala	Asp	Lys	Ser	Ile	Ser
		195					200					205			
Thr	Ala	Tyr	Leu	Gln	Trp	Ser	Ser	Leu	Lys	Ala	Ser	Asp	Thr	Ala	Met
	210					215						220			
Tyr	Tyr	Cys	Ala	Arg	Val	Thr	Tyr	Ser	Met	Asp	Ser	Tyr	Tyr	Phe	Asp
225					230					235					240
Ser	Trp	Gly	Gln	Gly	Thr	Leu	Val	Thr	Val	Ser	Ser				
			245						250						

<210> SEQ ID NO 258

<211> LENGTH: 243

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 258

Asp	Ile	Lys	Leu	Gln	Gln	Ser	Gly	Ala	Glu	Leu	Ala	Arg	Pro	Gly	Ala
1			5						10					15	
Ser	Val	Lys	Met	Ser	Cys	Lys	Thr	Ser	Gly	Tyr	Thr	Phe	Thr	Arg	Tyr
		20					25					30			
Thr	Met	His	Trp	Val	Lys	Gln	Arg	Pro	Gly	Gln	Gly	Leu	Glu	Trp	Ile
	35					40						45			
Gly	Tyr	Ile	Asn	Pro	Ser	Arg	Gly	Tyr	Thr	Asn	Tyr	Asn	Gln	Lys	Phe
	50				55					60					
Lys	Asp	Lys	Ala	Thr	Leu	Thr	Thr	Asp	Lys	Ser	Ser	Ser	Thr	Ala	Tyr
65					70				75					80	
Met	Gln	Leu	Ser	Ser	Leu	Thr	Ser	Glu	Asp	Ser	Ala	Val	Tyr	Tyr	Cys
			85					90					95		
Ala	Arg	Tyr	Tyr	Asp	Asp	His	Tyr	Cys	Leu	Asp	Tyr	Trp	Gly	Gln	Gly
		100						105					110		
Thr	Thr	Leu	Thr	Val	Ser	Ser	Val	Glu	Gly	Gly	Ser	Gly	Gly	Ser	Gly
		115					120					125			
Gly	Ser	Gly	Gly	Ser	Gly	Gly	Val	Asp	Asp	Ile	Gln	Leu	Thr	Gln	Ser
	130					135					140				

-continued

Pro Ala Ile Met Ser Ala Ser Pro Gly Glu Lys Val Thr Met Thr Cys
145 150 155 160

Arg Ala Ser Ser Ser Val Ser Tyr Met Asn Trp Tyr Gln Gln Lys Ser
165 170 175

Gly Thr Ser Pro Lys Arg Trp Ile Tyr Asp Thr Ser Lys Val Ala Ser
180 185 190

Gly Val Pro Tyr Arg Phe Ser Gly Ser Gly Ser Gly Thr Ser Tyr Ser
195 200 205

Leu Thr Ile Ser Ser Met Glu Ala Glu Asp Ala Ala Thr Tyr Tyr Cys
210 215 220

Gln Gln Trp Ser Ser Asn Pro Leu Thr Phe Gly Ala Gly Thr Lys Leu
225 230 235 240

Glu Leu Lys

<210> SEQ ID NO 259

<211> LENGTH: 118

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 259

Gln Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Arg
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Lys Phe Ser Gly Tyr
20 25 30

Gly Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45

Ala Val Ile Trp Tyr Asp Gly Ser Lys Lys Tyr Tyr Val Asp Ser Val
50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65 70 75 80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95

Ala Arg Gln Met Gly Tyr Trp His Phe Asp Leu Trp Gly Arg Gly Thr
100 105 110

Leu Val Thr Val Ser Ser
115

<210> SEQ ID NO 260

<211> LENGTH: 108

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 260

Glu Ile Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Leu Ser Pro Gly
1 5 10 15

Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Val Ser Ser Tyr
20 25 30

Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
35 40 45

Tyr Asp Ala Ser Asn Arg Ala Thr Gly Ile Pro Ala Arg Phe Ser Gly
50 55 60

-continued

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Glu Pro
65 70 75 80

Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Arg Ser Asn Trp Pro Pro
85 90 95

Leu Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
100 105

<210> SEQ ID NO 261
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 261

Gly Tyr Gly Met His
1 5

<210> SEQ ID NO 262
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 262

Val Ile Trp Tyr Asp Gly Ser Lys Lys Tyr Tyr Val Asp Ser Val Lys
1 5 10 15

Gly

<210> SEQ ID NO 263
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 263

Gln Met Gly Tyr Trp His Phe Asp Leu
1 5

<210> SEQ ID NO 264
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 264

Arg Ala Ser Gln Ser Val Ser Ser Tyr Leu Ala
1 5 10

<210> SEQ ID NO 265
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 265

Asp Ala Ser Asn Arg Ala Thr

-continued

1 5

<210> SEQ ID NO 266
 <211> LENGTH: 10
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 266

Gln Gln Arg Ser Asn Trp Pro Pro Leu Thr
 1 5 10

<210> SEQ ID NO 267
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 267

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Phe
 20 25 30

Pro Met Ala Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ser Thr Ile Ser Thr Ser Gly Gly Arg Thr Tyr Tyr Arg Asp Ser Val
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
 65 70 75 80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Ala Lys Phe Arg Gln Tyr Ser Gly Gly Phe Asp Tyr Trp Gly Gln Gly
 100 105 110

Thr Leu Val Thr Val Ser Ser
 115

<210> SEQ ID NO 268
 <211> LENGTH: 101
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 268

Asp Phe Met Leu Thr Gln Pro His Ser Val Ser Glu Ser Pro Gly Lys
 1 5 10 15

Thr Val Ile Ile Ser Cys Trp Tyr Gln Gln Arg Pro Gly Arg Ala Pro
 20 25 30

Thr Thr Val Ile Phe Gly Val Pro Asp Arg Phe Ser Gly Ser Ile Asp
 35 40 45

Arg Ser Ser Asn Ser Ala Ser Leu Thr Ile Ser Gly Leu Gln Thr Glu
 50 55 60

Asp Glu Ala Asp Tyr Tyr Cys Phe Gly Gly Gly Thr Lys Leu Thr Val
 65 70 75 80

Leu Gly Gln Pro Lys Ala Ala Pro Ser Val Thr Leu Phe Pro Pro Ser
 85 90 95

-continued

Ser Glu Glu Leu Gln
100

<210> SEQ ID NO 269
<211> LENGTH: 119
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 269

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Phe
20 25 30

Pro Met Ala Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45

Ser Thr Ile Ser Thr Ser Gly Gly Arg Thr Tyr Tyr Arg Asp Ser Val
50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65 70 75 80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95

Ala Lys Phe Arg Gln Tyr Ser Gly Gly Phe Asp Tyr Trp Gly Gln Gly
100 105 110

Thr Leu Val Thr Val Ser Ser
115

<210> SEQ ID NO 270
<211> LENGTH: 111
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 270

Asp Ile Gln Leu Thr Gln Pro Asn Ser Val Ser Thr Ser Leu Gly Ser
1 5 10 15

Thr Val Lys Leu Ser Cys Thr Leu Ser Ser Gly Asn Ile Glu Asn Asn
20 25 30

Tyr Val His Trp Tyr Gln Leu Tyr Glu Gly Arg Ser Pro Thr Thr Met
35 40 45

Ile Tyr Asp Asp Asp Lys Arg Pro Asp Gly Val Pro Asp Arg Phe Ser
50 55 60

Gly Ser Ile Asp Arg Ser Ser Asn Ser Ala Phe Leu Thr Ile His Asn
65 70 75 80

Val Ala Ile Glu Asp Glu Ala Ile Tyr Phe Cys His Ser Tyr Val Ser
85 90 95

Ser Phe Asn Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu Arg
100 105 110

<210> SEQ ID NO 271
<211> LENGTH: 119
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

-continued

<400> SEQUENCE: 271

```

Asp Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala
1           5           10           15

Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Arg Tyr
          20           25           30

Thr Met His Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Ile
          35           40           45

Gly Tyr Ile Asn Pro Ser Arg Gly Tyr Thr Asn Tyr Ala Asp Ser Val
          50           55           60

Lys Gly Arg Phe Thr Ile Thr Thr Asp Lys Ser Thr Ser Thr Ala Tyr
          65           70           75           80

Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Thr Tyr Tyr Cys
          85           90           95

Ala Arg Tyr Tyr Asp Asp His Tyr Cys Leu Asp Tyr Trp Gly Gln Gly
          100          105          110

Thr Thr Val Thr Val Ser Ser
          115

```

<210> SEQ ID NO 272

<211> LENGTH: 106

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 272

```

Asp Ile Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Leu Ser Pro Gly
1           5           10           15

Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Val Ser Tyr Met
          20           25           30

Asn Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Arg Trp Ile Tyr
          35           40           45

Asp Thr Ser Lys Val Ala Ser Gly Val Pro Ala Arg Phe Ser Gly Ser
          50           55           60

Gly Ser Gly Thr Asp Tyr Ser Leu Thr Ile Asn Ser Leu Glu Ala Glu
          65           70           75           80

Asp Ala Ala Thr Tyr Tyr Cys Gln Gln Trp Ser Ser Asn Pro Leu Thr
          85           90           95

Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
          100          105

```

<210> SEQ ID NO 273

<211> LENGTH: 243

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 273

```

Asp Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala
1           5           10           15

Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Arg Tyr
          20           25           30

Thr Met His Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Ile
          35           40           45

```

-continued

Gly Tyr Ile Asn Pro Ser Arg Gly Tyr Thr Asn Tyr Ala Asp Ser Val
 50 55 60
 Lys Gly Arg Phe Thr Ile Thr Thr Asp Lys Ser Thr Ser Thr Ala Tyr
 65 70 75 80
 Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Thr Tyr Tyr Cys
 85 90 95
 Ala Arg Tyr Tyr Asp Asp His Tyr Cys Leu Asp Tyr Trp Gly Gln Gly
 100 105 110
 Thr Thr Val Thr Val Ser Ser Gly Glu Gly Thr Ser Thr Gly Ser Gly
 115 120 125
 Gly Ser Gly Gly Ser Gly Gly Ala Asp Asp Ile Val Leu Thr Gln Ser
 130 135 140
 Pro Ala Thr Leu Ser Leu Ser Pro Gly Glu Arg Ala Thr Leu Ser Cys
 145 150 155 160
 Arg Ala Ser Gln Ser Val Ser Tyr Met Asn Trp Tyr Gln Gln Lys Pro
 165 170 175
 Gly Lys Ala Pro Lys Arg Trp Ile Tyr Asp Thr Ser Lys Val Ala Ser
 180 185 190
 Gly Val Pro Ala Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Tyr Ser
 195 200 205
 Leu Thr Ile Asn Ser Leu Glu Ala Glu Asp Ala Ala Thr Tyr Tyr Cys
 210 215 220
 Gln Gln Trp Ser Ser Asn Pro Leu Thr Phe Gly Gly Gly Thr Lys Val
 225 230 235 240
 Glu Ile Lys

<210> SEQ ID NO 274
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 274

Gln Val Gln Leu Val Gln Ser Gly Gly Gly Val Val Gln Pro Gly Arg
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Arg Tyr
 20 25 30
 Thr Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile
 35 40 45
 Gly Tyr Ile Asn Pro Ser Arg Gly Tyr Thr Asn Tyr Asn Gln Lys Val
 50 55 60
 Lys Asp Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Ala Phe
 65 70 75 80
 Leu Gln Met Asp Ser Leu Arg Pro Glu Asp Thr Gly Val Tyr Phe Cys
 85 90 95
 Ala Arg Tyr Tyr Asp Asp His Tyr Cys Leu Asp Tyr Trp Gly Gln Gly
 100 105 110
 Thr Pro Val Thr Val Ser Ser
 115

<210> SEQ ID NO 275
 <211> LENGTH: 107
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:

-continued

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 275

```

Asp Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val
1          5              10              15
Gly Asp Arg Val Thr Ile Thr Cys Ser Ala Ser Ser Ser Val Ser Tyr
20          25              30
Met Asn Trp Tyr Gln Gln Thr Pro Gly Lys Ala Pro Lys Arg Trp Ile
35          40              45
Tyr Asp Thr Ser Lys Leu Ala Ser Gly Val Pro Ser Arg Phe Ser Gly
50          55              60
Ser Gly Ser Gly Thr Asp Tyr Thr Phe Thr Ile Ser Ser Leu Gln Pro
65          70              75              80
Glu Asp Ile Ala Thr Tyr Tyr Cys Gln Gln Trp Ser Ser Asn Pro Phe
85          90              95
Thr Phe Gly Gln Gly Thr Lys Leu Gln Ile Thr
100          105

```

<210> SEQ ID NO 276

<211> LENGTH: 119

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 276

```

Gln Val Gln Leu Val Gln Ser Gly Gly Gly Val Val Gln Pro Gly Arg
1          5              10              15
Ser Leu Arg Leu Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Arg Tyr
20          25              30
Thr Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile
35          40              45
Gly Tyr Ile Asn Pro Ser Arg Gly Tyr Thr Asn Tyr Asn Gln Lys Phe
50          55              60
Lys Asp Arg Phe Thr Ile Ser Thr Asp Lys Ser Lys Ser Thr Ala Phe
65          70              75              80
Leu Gln Met Asp Ser Leu Arg Pro Glu Asp Thr Ala Val Tyr Tyr Cys
85          90              95
Ala Arg Tyr Tyr Asp Asp His Tyr Cys Leu Asp Tyr Trp Gly Gln Gly
100          105              110
Thr Pro Val Thr Val Ser Ser
115

```

<210> SEQ ID NO 277

<211> LENGTH: 107

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 277

```

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1          5              10              15
Asp Arg Val Thr Ile Thr Cys Ser Ala Ser Ser Ser Val Ser Tyr Met
20          25              30
Asn Trp Tyr Gln Gln Thr Pro Gly Lys Ala Pro Lys Arg Trp Ile Tyr

```


-continued

35	40	45
Asp Thr Ser Lys Leu Ala Ser Gly Val Pro Ser Arg Phe Ser Gly Ser		
50	55	60
Gly Ser Gly Thr Asp Tyr Thr Phe Thr Ile Ser Ser Leu Gln Pro Glu		
65	70	75
Asp Ile Ala Thr Tyr Tyr Cys Gln Gln Trp Ser Ser Asn Pro Phe Thr		
85	90	95
Phe Gly Gln Gly Thr Lys Leu Gln Ile Thr Arg		
100	105	

<210> SEQ ID NO 278
 <211> LENGTH: 125
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 278

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly		
1	5	10
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Thr Tyr		
20	25	30
Ala Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val		
35	40	45
Ser Arg Ile Arg Ser Lys Tyr Asn Asn Tyr Ala Thr Tyr Tyr Ala Asp		
50	55	60
Ser Val Lys Gly Arg Phe Thr Ile Ser Arg Asp Asp Ser Lys Asn Thr		
65	70	75
Leu Tyr Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr		
85	90	95
Tyr Cys Val Arg His Gly Asn Phe Gly Asn Ser Tyr Val Ser Trp Phe		
100	105	110
Ala Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser		
115	120	125

<210> SEQ ID NO 279
 <211> LENGTH: 109
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 279

Gln Ala Val Val Thr Gln Glu Pro Ser Leu Thr Val Ser Pro Gly Gly		
1	5	10
Thr Val Thr Leu Thr Cys Gly Ser Ser Thr Gly Ala Val Thr Thr Ser		
20	25	30
Asn Tyr Ala Asn Trp Val Gln Glu Lys Pro Gly Gln Ala Phe Arg Gly		
35	40	45
Leu Ile Gly Gly Thr Asn Lys Arg Ala Pro Gly Thr Pro Ala Arg Phe		
50	55	60
Ser Gly Ser Leu Leu Gly Gly Lys Ala Ala Leu Thr Leu Ser Gly Ala		
65	70	75
Gln Pro Glu Asp Glu Ala Glu Tyr Tyr Cys Ala Leu Trp Tyr Ser Asn		
85	90	95
Leu Trp Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu		

-continued

100	105
-----	-----

<210> SEQ ID NO 280
 <211> LENGTH: 123
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide
 <400> SEQUENCE: 280

Glu	Val	Gln	Leu	Val	Glu	Ser	Gly	Gly	Gly	Leu	Val	Gln	Pro	Gly	Arg
1				5						10				15	
Ser	Leu	Arg	Leu	Ser	Cys	Ala	Ala	Ser	Gly	Phe	Thr	Phe	Asp	Asp	Tyr
			20					25					30		
Thr	Met	His	Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Val
		35					40					45			
Ser	Gly	Ile	Ser	Trp	Asn	Ser	Gly	Ser	Ile	Gly	Tyr	Ala	Asp	Ser	Val
	50					55				60					
Lys	Gly	Arg	Phe	Thr	Ile	Ser	Arg	Asp	Asn	Ala	Lys	Lys	Ser	Leu	Tyr
65					70					75				80	
Leu	Gln	Met	Asn	Ser	Leu	Arg	Ala	Glu	Asp	Thr	Ala	Leu	Tyr	Tyr	Cys
			85						90					95	
Ala	Lys	Asp	Asn	Ser	Gly	Tyr	Gly	His	Tyr	Tyr	Tyr	Gly	Met	Asp	Val
			100					105					110		
Trp	Gly	Gln	Gly	Thr	Thr	Val	Thr	Val	Ala	Ser					
		115				120									

<210> SEQ ID NO 281
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide
 <400> SEQUENCE: 281

Ala	Glu	Ile	Val	Met	Thr	Gln	Ser	Pro	Ala	Thr	Leu	Ser	Val	Ser	Pro
1				5					10					15	
Gly	Glu	Arg	Ala	Thr	Leu	Ser	Cys	Arg	Ala	Ser	Gln	Ser	Val	Ser	Ser
			20					25					30		
Asn	Leu	Ala	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Gln	Ala	Pro	Arg	Leu	Leu
		35					40					45			
Ile	Tyr	Gly	Ala	Ser	Thr	Arg	Ala	Thr	Gly	Ile	Pro	Ala	Arg	Phe	Ser
	50					55				60					
Gly	Ser	Gly	Ser	Gly	Thr	Glu	Phe	Thr	Leu	Thr	Ile	Ser	Ser	Leu	Gln
65					70					75				80	
Ser	Glu	Asp	Phe	Ala	Val	Tyr	Tyr	Cys	Gln	His	Tyr	Ile	Asn	Trp	Pro
			85					90					95		
Leu	Thr	Phe	Gly	Gly	Gly	Thr	Lys	Val	Glu	Ile	Lys				
			100				105								

<210> SEQ ID NO 282
 <211> LENGTH: 45
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide
 <400> SEQUENCE: 282

-continued

Leu Lys Glu Ala Lys Glu Lys Ala Ile Glu Glu Leu Lys Lys Ala Gly
1 5 10 15

Ile Thr Ser Asp Tyr Tyr Phe Asp Leu Ile Asn Lys Ala Lys Thr Val
20 25 30

Glu Gly Val Asn Ala Leu Lys Asp Glu Ile Leu Lys Ala
35 40 45

<210> SEQ ID NO 283

<211> LENGTH: 123

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 283

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Lys Tyr
20 25 30

Trp Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45

Ser Ser Ile Asp Phe Met Gly Pro His Thr Tyr Tyr Ala Asp Ser Val
50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65 70 75 80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95

Ala Lys Gly Arg Thr Ser Met Leu Pro Met Lys Gly Lys Phe Asp Tyr
100 105 110

Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
115 120

<210> SEQ ID NO 284

<211> LENGTH: 118

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 284

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1 5 10 15

Ser Leu Arg Leu Ser Cys Thr Ala Ser Gly Phe Thr Phe Asp Glu Tyr
20 25 30

Asn Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45

Ser Thr Ile Leu Pro His Gly Asp Arg Thr Tyr Tyr Ala Asp Ser Val
50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65 70 75 80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95

Ala Lys Gln Asp Pro Leu Tyr Arg Phe Asp Tyr Trp Gly Gln Gly Thr
100 105 110

Leu Val Thr Val Ser Ser
115

-continued

<210> SEQ ID NO 285
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 285

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
 1 5 10 15

Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Lys Ile Ala Thr Tyr
 20 25 30

Leu Asn Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
 35 40 45

Tyr Arg Ser Ser Ser Leu Gln Ser Ala Val Pro Ser Arg Phe Ser Gly
 50 55 60

Ser Gly Ser Gly Thr Val Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
 65 70 75 80

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Thr Tyr Ala Val Pro Pro
 85 90 95

Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg
 100 105

<210> SEQ ID NO 286
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 286

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
 1 5 10 15

Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Ser Ile Ser Ser Tyr
 20 25 30

Leu Asn Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
 35 40 45

Tyr Arg Asn Ser Pro Leu Gln Ser Gly Val Pro Ser Arg Phe Ser Gly
 50 55 60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
 65 70 75 80

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Thr Tyr Arg Val Pro Pro
 85 90 95

Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg
 100 105

<210> SEQ ID NO 287
 <211> LENGTH: 115
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 287

Gln Val Gln Leu Gln Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15

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Ser Leu Arg Leu Ser Cys Glu Ala Ser Gly Phe Thr Phe Ser Arg Phe
 20 25 30

Gly Met Thr Trp Val Arg Gln Ala Pro Gly Lys Gly Val Glu Trp Val
 35 40 45

Ser Gly Ile Ser Ser Leu Gly Asp Ser Thr Leu Tyr Ala Asp Ser Val
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn Thr Leu Tyr
 65 70 75 80

Leu Gln Met Asn Ser Leu Lys Pro Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Thr Ile Gly Gly Ser Leu Asn Pro Gly Gly Gln Gly Thr Gln Val Thr
 100 105 110

Val Ser Ser
 115

<210> SEQ ID NO 288
 <211> LENGTH: 121
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 288

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Val Ser Gly Ile Asp Leu Ser Asn Tyr
 20 25 30

Ala Ile Asn Trp Val Arg Gln Ala Pro Gly Lys Cys Leu Glu Trp Ile
 35 40 45

Gly Ile Ile Trp Ala Ser Gly Thr Thr Phe Tyr Ala Thr Trp Ala Lys
 50 55 60

Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Val Tyr Leu
 65 70 75 80

Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
 85 90 95

Arg Thr Val Pro Gly Tyr Ser Thr Ala Pro Tyr Phe Asp Leu Trp Gly
 100 105 110

Gln Gly Thr Leu Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 289
 <211> LENGTH: 112
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 289

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Val Ser Ala Ser Val Gly
 1 5 10 15

Asp Arg Val Thr Ile Thr Cys Gln Ser Ser Pro Ser Val Trp Ser Asn
 20 25 30

Phe Leu Ser Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu
 35 40 45

Ile Tyr Glu Ala Ser Lys Leu Thr Ser Gly Val Pro Ser Arg Phe Ser
 50 55 60

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Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln
65 70 75 80

Pro Glu Asp Phe Ala Thr Tyr Tyr Cys Gly Gly Gly Tyr Ser Ser Ile
85 90 95

Ser Asp Thr Thr Phe Gly Cys Gly Thr Lys Val Glu Ile Lys Arg Thr
100 105 110

<210> SEQ ID NO 290
 <211> LENGTH: 121
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 290

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Val Ser Gly Ile Asp Leu Ser Asn Tyr
20 25 30

Ala Ile Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile
35 40 45

Gly Ile Ile Trp Ala Ser Gly Thr Thr Phe Tyr Ala Thr Trp Ala Lys
50 55 60

Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Val Tyr Leu
65 70 75 80

Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala
85 90 95

Arg Thr Val Pro Gly Tyr Ser Thr Ala Pro Tyr Phe Asp Leu Trp Gly
100 105 110

Gln Gly Thr Leu Val Thr Val Ser Ser
115 120

<210> SEQ ID NO 291
 <211> LENGTH: 112
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 291

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Val Ser Ala Ser Val Gly
1 5 10 15

Asp Arg Val Thr Ile Thr Cys Gln Ser Ser Pro Ser Val Trp Ser Asn
20 25 30

Phe Leu Ser Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu
35 40 45

Ile Tyr Glu Ala Ser Lys Leu Thr Ser Gly Val Pro Ser Arg Phe Ser
50 55 60

Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln
65 70 75 80

Pro Glu Asp Phe Ala Thr Tyr Tyr Cys Gly Gly Gly Tyr Ser Ser Ile
85 90 95

Ser Asp Thr Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys Arg Thr
100 105 110

<210> SEQ ID NO 292
 <211> LENGTH: 20

-continued

<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 292

Gly Ser Gly Ser Gly Ser Gly Ser Gly Ser Gly Ser Gly Ser
1 5 10 15

Gly Ser Gly Ser
 20

<210> SEQ ID NO 293
<211> LENGTH: 30
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 293

Gly Gly Ser Gly Gly Ser Gly Gly Ser Gly Gly Ser Gly Gly Ser Gly
1 5 10 15

Gly Ser Gly Gly Ser Gly Gly Ser Gly Gly Ser Gly Gly Ser
 20 25 30

<210> SEQ ID NO 294
<211> LENGTH: 40
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 294

Gly Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly Ser
1 5 10 15

Gly Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly Ser
 20 25 30

Gly Gly Gly Ser Gly Gly Gly Ser
 35 40

<210> SEQ ID NO 295
<211> LENGTH: 40
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 295

Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly Ser Gly
1 5 10 15

Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly Ser Gly
 20 25 30

Gly Gly Ser Gly Gly Gly Ser Gly
 35 40

<210> SEQ ID NO 296
<211> LENGTH: 50
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

-continued

<400> SEQUENCE: 296

Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
 1 5 10 15

Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly
 20 25 30

Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
 35 40 45

Gly Gly
 50

<210> SEQ ID NO 297

<211> LENGTH: 50

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 297

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly
 1 5 10 15

Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly
 20 25 30

Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
 35 40 45

Gly Ser
 50

<210> SEQ ID NO 298

<211> LENGTH: 20

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 298

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly
 1 5 10 15

Gly Gly Gly Ser
 20

<210> SEQ ID NO 299

<211> LENGTH: 15

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 299

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
 1 5 10 15

<210> SEQ ID NO 300

<211> LENGTH: 100

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 300

-continued

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly
 1 5 10 15
 Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly
 20 25 30
 Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
 35 40 45
 Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly
 50 55 60
 Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
 65 70 75 80
 Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly
 85 90 95
 Gly Gly Gly Ser
 100

<210> SEQ ID NO 301
 <211> LENGTH: 100
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 301

Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
 1 5 10 15
 Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly
 20 25 30
 Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser
 35 40 45
 Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser Gly
 50 55 60
 Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser Gly Gly
 65 70 75 80
 Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
 85 90 95
 Gly Ser Gly Gly
 100

<210> SEQ ID NO 302
 <211> LENGTH: 9
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 302

Gly Gly Gly Gly Ser Gly Gly Gly Ser
 1 5

<210> SEQ ID NO 303
 <211> LENGTH: 642
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 303

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Gln	Val	Gln	Leu	Val	Gln	Ser	Gly	Ala	Glu	Val	Lys	Lys	Pro	Gly	Ala	
1			5						10					15		
Ser	Val	Lys	Val	Ser	Cys	Lys	Ala	Ser	Gly	Tyr	Thr	Phe	Thr	Asp	Tyr	
		20					25						30			
Ile	Met	His	Trp	Val	Arg	Gln	Ala	Pro	Gly	Gln	Gly	Leu	Glu	Trp	Met	
		35					40					45				
Gly	Tyr	Ile	Asn	Pro	Tyr	Asn	Asp	Gly	Ser	Lys	Tyr	Thr	Glu	Lys	Phe	
	50					55					60					
Gln	Gly	Arg	Val	Thr	Met	Thr	Ser	Asp	Thr	Ser	Ile	Ser	Thr	Ala	Tyr	
65					70					75					80	
Met	Glu	Leu	Ser	Arg	Leu	Arg	Ser	Asp	Asp	Thr	Ala	Val	Tyr	Tyr	Cys	
			85					90						95		
Ala	Arg	Gly	Thr	Tyr	Tyr	Tyr	Gly	Pro	Gln	Leu	Phe	Asp	Tyr	Trp	Gly	
			100					105						110		
Gln	Gly	Thr	Thr	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	
		115					120						125			
Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Asp	Ile	Val	Met	Thr	Gln	Thr	Pro	
	130					135					140					
Leu	Ser	Leu	Ser	Val	Thr	Pro	Gly	Gln	Pro	Ala	Ser	Ile	Ser	Cys	Lys	
145					150					155					160	
Ser	Ser	Gln	Ser	Leu	Glu	Thr	Ser	Thr	Gly	Thr	Thr	Tyr	Leu	Asn	Trp	
			165						170					175		
Tyr	Leu	Gln	Lys	Pro	Gly	Gln	Ser	Pro	Gln	Leu	Leu	Ile	Tyr	Arg	Val	
			180					185						190		
Ser	Lys	Arg	Phe	Ser	Gly	Val	Pro	Asp	Arg	Phe	Ser	Gly	Ser	Gly	Ser	
		195					200					205				
Gly	Thr	Asp	Phe	Thr	Leu	Lys	Ile	Ser	Arg	Val	Glu	Ala	Glu	Asp	Val	
	210					215					220					
Gly	Val	Tyr	Tyr	Cys	Leu	Gln	Leu	Leu	Glu	Asp	Pro	Tyr	Thr	Phe	Gly	
225					230					235					240	
Gln	Gly	Thr	Lys	Leu	Glu	Ile	Lys	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	
			245						250					255		
Ser	Asp	Ile	Val	Met	Thr	Gln	Ser	Pro	Asp	Ser	Leu	Ala	Val	Ser	Leu	
			260					265						270		
Gly	Glu	Arg	Ala	Thr	Ile	Asn	Cys	Lys	Ser	Ser	Gln	Ser	Leu	Leu	Asn	
		275					280					285				
Ala	Arg	Thr	Gly	Lys	Asn	Tyr	Leu	Ala	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	
		290				295					300					
Gln	Pro	Pro	Lys	Leu	Leu	Ile	Tyr	Trp	Ala	Ser	Thr	Arg	Glu	Ser	Gly	
305					310					315					320	
Val	Pro	Asp	Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	
			325						330					335		
Thr	Ile	Ser	Ser	Leu	Gln	Ala	Glu	Asp	Val	Ala	Val	Tyr	Tyr	Cys	Lys	
			340					345						350		
Gln	Ser	Tyr	Ser	Arg	Arg	Thr	Phe	Gly	Gly	Gly	Thr	Lys	Val	Glu	Ile	
		355					360						365			
Lys	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	
	370					375					380					
Gly	Gly	Gly	Gly	Ser	Gln	Val	Gln	Leu	Val	Gln	Ser	Gly	Ala	Glu	Val	
385					390					395					400	
Lys	Lys	Pro	Gly	Ala	Ser	Val	Lys	Val	Ser	Cys	Lys	Ala	Ser	Gly	Phe	
			405						410					415		
Asn	Ile	Lys	Asp	Tyr	Tyr	Met	His	Trp	Val	Arg	Gln	Ala	Pro	Gly	Gln	

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420					425					430					
Arg	Leu	Glu	Trp	Met	Gly	Trp	Ile	Asp	Leu	Glu	Asn	Ala	Asn	Thr	Ile
	435						440					445			
Tyr	Asp	Ala	Lys	Phe	Gln	Gly	Arg	Val	Thr	Ile	Thr	Arg	Asp	Thr	Ser
	450					455					460				
Ala	Ser	Thr	Ala	Tyr	Met	Glu	Leu	Ser	Ser	Leu	Arg	Ser	Glu	Asp	Thr
	465				470					475				480	
Ala	Val	Tyr	Tyr	Cys	Ala	Arg	Asp	Ala	Tyr	Gly	Arg	Tyr	Phe	Tyr	Asp
				485					490					495	
Val	Trp	Gly	Gln	Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly
		500					505						510		
Ser	Gly	Gly	Gly	Ser	Glu	Val	Gln	Leu	Val	Glu	Ser	Gly	Gly	Gly	Leu
	515					520						525			
Val	Gln	Pro	Gly	Asn	Ser	Leu	Arg	Leu	Ser	Cys	Ala	Ala	Ser	Gly	Phe
	530					535					540				
Thr	Phe	Ser	Ser	Phe	Gly	Met	Ser	Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys
	545				550					555					560
Gly	Leu	Glu	Trp	Val	Ser	Ser	Ile	Ser	Gly	Ser	Gly	Ser	Asp	Thr	Leu
			565					570						575	
Tyr	Ala	Asp	Ser	Val	Lys	Gly	Arg	Phe	Thr	Ile	Ser	Arg	Asp	Asn	Ala
			580					585					590		
Lys	Thr	Thr	Leu	Tyr	Leu	Gln	Met	Asn	Ser	Leu	Arg	Pro	Glu	Asp	Thr
	595					600						605			
Ala	Val	Tyr	Tyr	Cys	Thr	Ile	Gly	Gly	Ser	Leu	Ser	Arg	Ser	Ser	Gln
	610					615					620				
Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	His	His	His	His	His	His	His	His
	625				630					635					640
His	His														

<210> SEQ ID NO 304

<211> LENGTH: 642

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 304

Asp	Ile	Val	Met	Thr	Gln	Ser	Pro	Asp	Ser	Leu	Ala	Val	Ser	Leu	Gly
1			5						10					15	
Glu	Arg	Ala	Thr	Ile	Asn	Cys	Lys	Ser	Ser	Gln	Ser	Leu	Leu	Asn	Ala
		20					25						30		
Arg	Thr	Gly	Lys	Asn	Tyr	Leu	Ala	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Gln
	35					40						45			
Pro	Pro	Lys	Leu	Leu	Ile	Tyr	Trp	Ala	Ser	Thr	Arg	Glu	Ser	Gly	Val
	50				55						60				
Pro	Asp	Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	Thr
	65			70						75				80	
Ile	Ser	Ser	Leu	Gln	Ala	Glu	Asp	Val	Ala	Val	Tyr	Tyr	Cys	Lys	Gln
			85					90						95	
Ser	Tyr	Ser	Arg	Arg	Thr	Phe	Gly	Gly	Gly	Thr	Lys	Val	Glu	Ile	Lys
			100					105					110		
Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly
	115						120					125			
Gly	Gly	Gly	Ser	Gln	Val	Gln	Leu	Val	Gln	Ser	Gly	Ala	Glu	Val	Lys

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130	135	140
Lys Pro Gly Ala Ser Val	Lys Val Ser Cys Lys Ala Ser Gly Phe Asn	
145	150	155 160
Ile Lys Asp Tyr Tyr Met His Trp Val Arg Gln Ala Pro Gly Gln Arg		
	165	170 175
Leu Glu Trp Met Gly Trp Ile Asp Leu Glu Asn Ala Asn Thr Ile Tyr		
	180	185 190
Asp Ala Lys Phe Gln Gly Arg Val Thr Ile Thr Arg Asp Thr Ser Ala		
	195	200 205
Ser Thr Ala Tyr Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala		
	210	215 220
Val Tyr Tyr Cys Ala Arg Asp Ala Tyr Gly Arg Tyr Phe Tyr Asp Val		
	225	230 235 240
Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser Gly Gly Gly Gly Ser		
	245	250 255
Gly Gly Gly Ser Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys		
	260	265 270
Lys Pro Gly Ala Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Thr		
	275	280 285
Phe Thr Asp Tyr Ile Met His Trp Val Arg Gln Ala Pro Gly Gln Gly		
	290	295 300
Leu Glu Trp Met Gly Tyr Ile Asn Pro Tyr Asn Asp Gly Ser Lys Tyr		
	305	310 315 320
Thr Glu Lys Phe Gln Gly Arg Val Thr Met Thr Ser Asp Thr Ser Ile		
	325	330 335
Ser Thr Ala Tyr Met Glu Leu Ser Arg Leu Arg Ser Asp Asp Thr Ala		
	340	345 350
Val Tyr Tyr Cys Ala Arg Gly Thr Tyr Tyr Tyr Gly Pro Gln Leu Phe		
	355	360 365
Asp Tyr Trp Gly Gln Gly Thr Thr Val Thr Val Ser Ser Gly Gly Gly		
	370	375 380
Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Asp Ile Val Met		
	385	390 395 400
Thr Gln Thr Pro Leu Ser Leu Ser Val Thr Pro Gly Gln Pro Ala Ser		
	405	410 415
Ile Ser Cys Lys Ser Ser Gln Ser Leu Glu Thr Ser Thr Gly Thr Thr		
	420	425 430
Tyr Leu Asn Trp Tyr Leu Gln Lys Pro Gly Gln Ser Pro Gln Leu Leu		
	435	440 445
Ile Tyr Arg Val Ser Lys Arg Phe Ser Gly Val Pro Asp Arg Phe Ser		
	450	455 460
Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Lys Ile Ser Arg Val Glu		
	465	470 475 480
Ala Glu Asp Val Gly Val Tyr Tyr Cys Leu Gln Leu Leu Glu Asp Pro		
	485	490 495
Tyr Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys Gly Gly Gly Gly		
	500	505 510
Ser Gly Gly Gly Ser Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu		
	515	520 525
Val Gln Pro Gly Asn Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe		
	530	535 540
Thr Phe Ser Ser Phe Gly Met Ser Trp Val Arg Gln Ala Pro Gly Lys		
	545	550 555 560

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Gly Leu Glu Trp Val Ser Ser Ile Ser Gly Ser Gly Ser Asp Thr Leu
565 570 575

Tyr Ala Asp Ser Val Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala
580 585 590

Lys Thr Thr Leu Tyr Leu Gln Met Asn Ser Leu Arg Pro Glu Asp Thr
595 600 605

Ala Val Tyr Tyr Cys Thr Ile Gly Gly Ser Leu Ser Arg Ser Ser Gln
610 615 620

Gly Thr Leu Val Thr Val Ser Ser His His His His His His His His
625 630 635 640

His His

<210> SEQ ID NO 305

<211> LENGTH: 642

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 305

Asp Ile Val Met Thr Gln Ser Pro Asp Ser Leu Ala Val Ser Leu Gly
1 5 10 15

Glu Arg Ala Thr Ile Asn Cys Lys Ser Ser Gln Ser Leu Leu Asn Ala
20 25 30

Arg Thr Gly Lys Asn Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln
35 40 45

Pro Pro Lys Leu Leu Ile Tyr Trp Ala Ser Thr Arg Glu Ser Gly Val
50 55 60

Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr
65 70 75 80

Ile Ser Ser Leu Gln Ala Glu Asp Val Ala Val Tyr Tyr Cys Lys Gln
85 90 95

Ser Tyr Ser Arg Arg Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
100 105 110

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly
115 120 125

Gly Gly Gly Ser Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys
130 135 140

Lys Pro Gly Ala Ser Val Lys Val Ser Cys Lys Ala Ser Gly Phe Asn
145 150 155 160

Ile Lys Asp Tyr Tyr Met His Trp Val Arg Gln Ala Pro Gly Gln Arg
165 170 175

Leu Glu Trp Met Gly Trp Ile Asp Leu Glu Asn Ala Asn Thr Ile Tyr
180 185 190

Asp Ala Lys Phe Gln Gly Arg Val Thr Ile Thr Arg Asp Thr Ser Ala
195 200 205

Ser Thr Ala Tyr Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala
210 215 220

Val Tyr Tyr Cys Ala Arg Asp Ala Tyr Gly Arg Tyr Phe Tyr Asp Val
225 230 235 240

Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser Gly Gly Gly Gly Ser
245 250 255

Gly Gly Gly Ser Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val
260 265 270

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Gln Pro Gly Asn Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr
 275 280 285
 Phe Ser Ser Phe Gly Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly
 290 295 300
 Leu Glu Trp Val Ser Ser Ile Ser Gly Ser Gly Ser Asp Thr Leu Tyr
 305 310 315 320
 Ala Asp Ser Val Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys
 325 330 335
 Thr Thr Leu Tyr Leu Gln Met Asn Ser Leu Arg Pro Glu Asp Thr Ala
 340 345 350
 Val Tyr Tyr Cys Thr Ile Gly Gly Ser Leu Ser Arg Ser Ser Gln Gly
 355 360 365
 Thr Leu Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser
 370 375 380
 Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala
 385 390 395 400
 Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Asp Tyr
 405 410 415
 Ile Met His Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Met
 420 425 430
 Gly Tyr Ile Asn Pro Tyr Asn Asp Gly Ser Lys Tyr Thr Glu Lys Phe
 435 440 445
 Gln Gly Arg Val Thr Met Thr Ser Asp Thr Ser Ile Ser Thr Ala Tyr
 450 455 460
 Met Glu Leu Ser Arg Leu Arg Ser Asp Asp Thr Ala Val Tyr Tyr Cys
 465 470 475 480
 Ala Arg Gly Thr Tyr Tyr Tyr Gly Pro Gln Leu Phe Asp Tyr Trp Gly
 485 490 495
 Gln Gly Thr Thr Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly
 500 505 510
 Gly Gly Ser Gly Gly Gly Gly Ser Asp Ile Val Met Thr Gln Thr Pro
 515 520 525
 Leu Ser Leu Ser Val Thr Pro Gly Gln Pro Ala Ser Ile Ser Cys Lys
 530 535 540
 Ser Ser Gln Ser Leu Glu Thr Ser Thr Gly Thr Thr Tyr Leu Asn Trp
 545 550 555 560
 Tyr Leu Gln Lys Pro Gly Gln Ser Pro Gln Leu Leu Ile Tyr Arg Val
 565 570 575
 Ser Lys Arg Phe Ser Gly Val Pro Asp Arg Phe Ser Gly Ser Gly Ser
 580 585 590
 Gly Thr Asp Phe Thr Leu Lys Ile Ser Arg Val Glu Ala Glu Asp Val
 595 600 605
 Gly Val Tyr Tyr Cys Leu Gln Leu Leu Glu Asp Pro Tyr Thr Phe Gly
 610 615 620
 Gln Gly Thr Lys Leu Glu Ile Lys His His His His His His His His
 625 630 635 640
 His His

<210> SEQ ID NO 306

<211> LENGTH: 642

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic

-continued

polypeptide

<400> SEQUENCE: 306

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Asn
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Phe
 20 25 30
 Gly Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45
 Ser Ser Ile Ser Gly Ser Gly Ser Asp Thr Leu Tyr Ala Asp Ser Val
 50 55 60
 Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Thr Thr Leu Tyr
 65 70 75 80
 Leu Gln Met Asn Ser Leu Arg Pro Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95
 Thr Ile Gly Gly Ser Leu Ser Arg Ser Ser Gln Gly Thr Leu Val Thr
 100 105 110
 Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser Asp Ile Val Met
 115 120 125
 Thr Gln Ser Pro Asp Ser Leu Ala Val Ser Leu Gly Glu Arg Ala Thr
 130 135 140
 Ile Asn Cys Lys Ser Ser Gln Ser Leu Leu Asn Ala Arg Thr Gly Lys
 145 150 155 160
 Asn Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu
 165 170 175
 Leu Ile Tyr Trp Ala Ser Thr Arg Glu Ser Gly Val Pro Asp Arg Phe
 180 185 190
 Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu
 195 200 205
 Gln Ala Glu Asp Val Ala Val Tyr Tyr Cys Lys Gln Ser Tyr Ser Arg
 210 215 220
 Arg Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys Gly Gly Gly Gly
 225 230 235 240
 Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
 245 250 255
 Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala
 260 265 270
 Ser Val Lys Val Ser Cys Lys Ala Ser Gly Phe Asn Ile Lys Asp Tyr
 275 280 285
 Tyr Met His Trp Val Arg Gln Ala Pro Gly Gln Arg Leu Glu Trp Met
 290 295 300
 Gly Trp Ile Asp Leu Glu Asn Ala Asn Thr Ile Tyr Asp Ala Lys Phe
 305 310 315 320
 Gln Gly Arg Val Thr Ile Thr Arg Asp Thr Ser Ala Ser Thr Ala Tyr
 325 330 335
 Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
 340 345 350
 Ala Arg Asp Ala Tyr Gly Arg Tyr Phe Tyr Asp Val Trp Gly Gln Gly
 355 360 365
 Thr Leu Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser
 370 375 380
 Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala
 385 390 395 400

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Ser	Val	Lys	Val	Ser	Cys	Lys	Ala	Ser	Gly	Tyr	Thr	Phe	Thr	Asp	Tyr
				405					410					415	
Ile	Met	His	Trp	Val	Arg	Gln	Ala	Pro	Gly	Gln	Gly	Leu	Glu	Trp	Met
			420					425					430		
Gly	Tyr	Ile	Asn	Pro	Tyr	Asn	Asp	Gly	Ser	Lys	Tyr	Thr	Glu	Lys	Phe
		435					440					445			
Gln	Gly	Arg	Val	Thr	Met	Thr	Ser	Asp	Thr	Ser	Ile	Ser	Thr	Ala	Tyr
	450					455					460				
Met	Glu	Leu	Ser	Arg	Leu	Arg	Ser	Asp	Asp	Thr	Ala	Val	Tyr	Tyr	Cys
465					470					475					480
Ala	Arg	Gly	Thr	Tyr	Tyr	Tyr	Gly	Pro	Gln	Leu	Phe	Asp	Tyr	Trp	Gly
			485					490						495	
Gln	Gly	Thr	Thr	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly
			500					505					510		
Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Asp	Ile	Val	Met	Thr	Gln	Thr	Pro
		515					520					525			
Leu	Ser	Leu	Ser	Val	Thr	Pro	Gly	Gln	Pro	Ala	Ser	Ile	Ser	Cys	Lys
	530					535					540				
Ser	Ser	Gln	Ser	Leu	Glu	Thr	Ser	Thr	Gly	Thr	Thr	Tyr	Leu	Asn	Trp
545					550					555					560
Tyr	Leu	Gln	Lys	Pro	Gly	Gln	Ser	Pro	Gln	Leu	Leu	Ile	Tyr	Arg	Val
			565					570						575	
Ser	Lys	Arg	Phe	Ser	Gly	Val	Pro	Asp	Arg	Phe	Ser	Gly	Ser	Gly	Ser
			580					585				590			
Gly	Thr	Asp	Phe	Thr	Leu	Lys	Ile	Ser	Arg	Val	Glu	Ala	Glu	Asp	Val
		595					600					605			
Gly	Val	Tyr	Tyr	Cys	Leu	Gln	Leu	Leu	Glu	Asp	Pro	Tyr	Thr	Phe	Gly
	610					615					620				
Gln	Gly	Thr	Lys	Leu	Glu	Ile	Lys	His	His	His	His	His	His	His	His
625					630					635					640
His	His														

<210> SEQ ID NO 307

<211> LENGTH: 642

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 307

Glu	Val	Gln	Leu	Val	Glu	Ser	Gly	Gly	Gly	Leu	Val	Gln	Pro	Gly	Asn
1				5				10					15		
Ser	Leu	Arg	Leu	Ser	Cys	Ala	Ala	Ser	Gly	Phe	Thr	Phe	Ser	Ser	Phe
		20						25					30		
Gly	Met	Ser	Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Val
		35					40					45			
Ser	Ser	Ile	Ser	Gly	Ser	Gly	Ser	Asp	Thr	Leu	Tyr	Ala	Asp	Ser	Val
		50				55					60				
Lys	Gly	Arg	Phe	Thr	Ile	Ser	Arg	Asp	Asn	Ala	Lys	Thr	Thr	Leu	Tyr
65					70					75					80
Leu	Gln	Met	Asn	Ser	Leu	Arg	Pro	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	Cys
			85						90					95	
Thr	Ile	Gly	Gly	Ser	Leu	Ser	Arg	Ser	Ser	Gln	Gly	Thr	Leu	Val	Thr
		100						105					110		

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Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Ser	Gln	Val	Gln	Leu
		115					120					125			
Val	Gln	Ser	Gly	Ala	Glu	Val	Lys	Lys	Pro	Gly	Ala	Ser	Val	Lys	Val
	130					135					140				
Ser	Cys	Lys	Ala	Ser	Gly	Tyr	Thr	Phe	Thr	Asp	Tyr	Ile	Met	His	Trp
145					150					155					160
Val	Arg	Gln	Ala	Pro	Gly	Gln	Gly	Leu	Glu	Trp	Met	Gly	Tyr	Ile	Asn
				165					170					175	
Pro	Tyr	Asn	Asp	Gly	Ser	Lys	Tyr	Thr	Glu	Lys	Phe	Gln	Gly	Arg	Val
			180					185					190		
Thr	Met	Thr	Ser	Asp	Thr	Ser	Ile	Ser	Thr	Ala	Tyr	Met	Glu	Leu	Ser
		195					200					205			
Arg	Leu	Arg	Ser	Asp	Asp	Thr	Ala	Val	Tyr	Tyr	Cys	Ala	Arg	Gly	Thr
	210					215					220				
Tyr	Tyr	Tyr	Gly	Pro	Gln	Leu	Phe	Asp	Tyr	Trp	Gly	Gln	Gly	Thr	Thr
225					230					235					240
Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly
				245					250					255	
Gly	Gly	Gly	Ser	Asp	Ile	Val	Met	Thr	Gln	Thr	Pro	Leu	Ser	Leu	Ser
			260					265					270		
Val	Thr	Pro	Gly	Gln	Pro	Ala	Ser	Ile	Ser	Cys	Lys	Ser	Ser	Gln	Ser
		275					280					285			
Leu	Glu	Thr	Ser	Thr	Gly	Thr	Thr	Tyr	Leu	Asn	Trp	Tyr	Leu	Gln	Lys
	290					295					300				
Pro	Gly	Gln	Ser	Pro	Gln	Leu	Leu	Ile	Tyr	Arg	Val	Ser	Lys	Arg	Phe
305					310					315					320
Ser	Gly	Val	Pro	Asp	Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe
				325					330					335	
Thr	Leu	Lys	Ile	Ser	Arg	Val	Glu	Ala	Glu	Asp	Val	Gly	Val	Tyr	Tyr
			340					345					350		
Cys	Leu	Gln	Leu	Leu	Glu	Asp	Pro	Tyr	Thr	Phe	Gly	Gln	Gly	Thr	Lys
		355					360					365			
Leu	Glu	Ile	Lys	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Ser	Asp	Ile	Val
	370					375					380				
Met	Thr	Gln	Ser	Pro	Asp	Ser	Leu	Ala	Val	Ser	Leu	Gly	Glu	Arg	Ala
385					390					395					400
Thr	Ile	Asn	Cys	Lys	Ser	Ser	Gln	Ser	Leu	Leu	Asn	Ala	Arg	Thr	Gly
				405					410					415	
Lys	Asn	Tyr	Leu	Ala	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Gln	Pro	Pro	Lys
			420					425					430		
Leu	Leu	Ile	Tyr	Trp	Ala	Ser	Thr	Arg	Glu	Ser	Gly	Val	Pro	Asp	Arg
		435					440					445			
Phe	Ser	Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	Thr	Ile	Ser	Ser
	450					455					460				
Leu	Gln	Ala	Glu	Asp	Val	Ala	Val	Tyr	Tyr	Cys	Lys	Gln	Ser	Tyr	Ser
465					470					475					480
Arg	Arg	Thr	Phe	Gly	Gly	Gly	Thr	Lys	Val	Glu	Ile	Lys	Gly	Gly	Gly
				485					490					495	
Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly
			500					505					510		
Ser	Gln	Val	Gln	Leu	Val	Gln	Ser	Gly	Ala	Glu	Val	Lys	Lys	Pro	Gly
		515					520					525			
Ala	Ser	Val	Lys	Val	Ser	Cys	Lys	Ala	Ser	Gly	Phe	Asn	Ile	Lys	Asp

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530					535					540					
Tyr	Tyr	Met	His	Trp	Val	Arg	Gln	Ala	Pro	Gly	Gln	Arg	Leu	Glu	Trp
545					550					555					560
Met	Gly	Trp	Ile	Asp	Leu	Glu	Asn	Ala	Asn	Thr	Ile	Tyr	Asp	Ala	Lys
			565						570					575	
Phe	Gln	Gly	Arg	Val	Thr	Ile	Thr	Arg	Asp	Thr	Ser	Ala	Ser	Thr	Ala
			580					585					590		
Tyr	Met	Glu	Leu	Ser	Ser	Leu	Arg	Ser	Glu	Asp	Thr	Ala	Val	Tyr	Tyr
		595					600					605			
Cys	Ala	Arg	Asp	Ala	Tyr	Gly	Arg	Tyr	Phe	Tyr	Asp	Val	Trp	Gly	Gln
610					615					620					
Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	His	His	His	His	His	His	His	His
625					630					635					640

His His

<210> SEQ ID NO 308

<211> LENGTH: 636

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 308

Gln	Val	Gln	Leu	Gln	Glu	Ser	Gly	Pro	Gly	Leu	Val	Lys	Pro	Ser	Gln
1			5						10					15	
Thr	Leu	Ser	Leu	Thr	Cys	Thr	Val	Ser	Gly	Gly	Ser	Ile	Ser	Thr	Ser
			20					25					30		
Gly	Met	Gly	Val	Gly	Trp	Ile	Arg	Gln	His	Pro	Gly	Lys	Gly	Leu	Glu
		35					40					45			
Trp	Ile	Gly	His	Ile	Trp	Trp	Asp	Asp	Asp	Lys	Arg	Tyr	Asn	Pro	Ala
		50				55					60				
Leu	Lys	Ser	Arg	Val	Thr	Ile	Ser	Val	Asp	Thr	Ser	Lys	Asn	Gln	Phe
65				70						75					80
Ser	Leu	Lys	Leu	Ser	Ser	Val	Thr	Ala	Ala	Asp	Thr	Ala	Val	Tyr	Tyr
			85					90						95	
Cys	Ala	Arg	Met	Glu	Leu	Trp	Ser	Tyr	Tyr	Phe	Asp	Tyr	Trp	Gly	Gln
			100					105					110		
Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly
		115					120					125			
Gly	Ser	Gly	Gly	Gly	Gly	Ser	Glu	Ile	Val	Leu	Thr	Gln	Ser	Pro	Ala
		130				135					140				
Thr	Leu	Ser	Leu	Ser	Pro	Gly	Glu	Arg	Ala	Thr	Leu	Ser	Cys	Ser	Ala
145				150						155					160
Ser	Ser	Ser	Val	Ser	Tyr	Met	His	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Gln
			165						170					175	
Ala	Pro	Arg	Leu	Leu	Ile	Tyr	Asp	Thr	Ser	Lys	Leu	Ala	Ser	Gly	Ile
			180					185					190		
Pro	Ala	Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	Thr
		195				200						205			
Ile	Ser	Ser	Leu	Glu	Pro	Glu	Asp	Phe	Ala	Val	Tyr	Tyr	Cys	Phe	Gln
		210				215					220				
Gly	Ser	Val	Tyr	Pro	Phe	Thr	Phe	Gly	Gln	Gly	Thr	Lys	Leu	Glu	Ile
225				230						235				240	
Lys	Arg	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Ser	Asp	Ile	Val	Met	Thr

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245					250					255					
Gln	Ser	Pro	Asp	Ser	Leu	Ala	Val	Ser	Leu	Gly	Glu	Arg	Ala	Thr	Ile
			260					265					270		
Asn	Cys	Lys	Ser	Ser	Gln	Ser	Leu	Leu	Asn	Ala	Arg	Thr	Gly	Lys	Asn
		275					280					285			
Tyr	Leu	Ala	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Gln	Pro	Pro	Lys	Leu	Leu
	290					295					300				
Ile	Tyr	Trp	Ala	Ser	Thr	Arg	Glu	Ser	Gly	Val	Pro	Asp	Arg	Phe	Ser
305						310					315				320
Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	Thr	Ile	Ser	Ser	Leu	Gln
			325						330					335	
Ala	Glu	Asp	Val	Ala	Val	Tyr	Tyr	Cys	Lys	Gln	Ser	Tyr	Ser	Arg	Arg
			340					345					350		
Thr	Phe	Gly	Gly	Gly	Thr	Lys	Val	Glu	Ile	Lys	Gly	Gly	Gly	Gly	Ser
		355					360					365			
Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gln
	370					375					380				
Val	Gln	Leu	Val	Gln	Ser	Gly	Ala	Glu	Val	Lys	Lys	Pro	Gly	Ala	Ser
385						390					395				400
Val	Lys	Val	Ser	Cys	Lys	Ala	Ser	Gly	Phe	Asn	Ile	Lys	Asp	Tyr	Tyr
			405						410					415	
Met	His	Trp	Val	Arg	Gln	Ala	Pro	Gly	Gln	Arg	Leu	Glu	Trp	Met	Gly
			420					425					430		
Trp	Ile	Asp	Leu	Glu	Asn	Ala	Asn	Thr	Ile	Tyr	Asp	Ala	Lys	Phe	Gln
		435					440					445			
Gly	Arg	Val	Thr	Ile	Thr	Arg	Asp	Thr	Ser	Ala	Ser	Thr	Ala	Tyr	Met
	450					455					460				
Glu	Leu	Ser	Ser	Leu	Arg	Ser	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	Cys	Ala
465						470					475				480
Arg	Asp	Ala	Tyr	Gly	Arg	Tyr	Phe	Tyr	Asp	Val	Trp	Gly	Gln	Gly	Thr
			485						490					495	
Leu	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Ser	Glu
			500				505						510		
Val	Gln	Leu	Val	Glu	Ser	Gly	Gly	Gly	Leu	Val	Gln	Pro	Gly	Asn	Ser
	515						520					525			
Leu	Arg	Leu	Ser	Cys	Ala	Ala	Ser	Gly	Phe	Thr	Phe	Ser	Ser	Phe	Gly
	530					535					540				
Met	Ser	Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Val	Ser
545						550					555				560
Ser	Ile	Ser	Gly	Ser	Gly	Ser	Asp	Thr	Leu	Tyr	Ala	Asp	Ser	Val	Lys
			565					570					575		
Gly	Arg	Phe	Thr	Ile	Ser	Arg	Asp	Asn	Ala	Lys	Thr	Thr	Leu	Tyr	Leu
			580				585						590		
Gln	Met	Asn	Ser	Leu	Arg	Pro	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	Cys	Thr
	595						600					605			
Ile	Gly	Gly	Ser	Leu	Ser	Arg	Ser	Ser	Gln	Gly	Thr	Leu	Val	Thr	Val
	610					615					620				
Ser	Ser	His	His	His	His	His	His	His	His	His	His	His	His	His	His
625						630					635				

<210> SEQ ID NO 309

<211> LENGTH: 636

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

-continued

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 309

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Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Gln
 1             5             10             15

Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Gly Ser Ile Ser Thr Ser
      20             25             30

Gly Met Gly Val Gly Trp Ile Arg Gln His Pro Gly Lys Gly Leu Glu
      35             40             45

Trp Ile Gly His Ile Trp Trp Asp Asp Asp Lys Arg Tyr Asn Pro Ala
      50             55             60

Leu Lys Ser Arg Val Thr Ile Ser Val Asp Thr Ser Lys Asn Gln Phe
      65             70             75             80

Ser Leu Lys Leu Ser Ser Val Thr Ala Ala Asp Thr Ala Val Tyr Tyr
      85             90             95

Cys Ala Arg Met Glu Leu Trp Ser Tyr Tyr Phe Asp Tyr Trp Gly Gln
      100            105            110

Gly Thr Leu Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly
      115            120            125

Gly Ser Gly Gly Gly Gly Ser Glu Ile Val Leu Thr Gln Ser Pro Ala
      130            135            140

Thr Leu Ser Leu Ser Pro Gly Glu Arg Ala Thr Leu Ser Cys Ser Ala
      145            150            155            160

Ser Ser Ser Val Ser Tyr Met His Trp Tyr Gln Gln Lys Pro Gly Gln
      165            170            175

Ala Pro Arg Leu Leu Ile Tyr Asp Thr Ser Lys Leu Ala Ser Gly Ile
      180            185            190

Pro Ala Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr
      195            200            205

Ile Ser Ser Leu Glu Pro Glu Asp Phe Ala Val Tyr Tyr Cys Phe Gln
      210            215            220

Gly Ser Val Tyr Pro Phe Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile
      225            230            235            240

Lys Arg Gly Gly Gly Gly Ser Gly Gly Gly Ser Glu Val Gln Leu Val
      245            250            255

Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Asn Ser Leu Arg Leu Ser
      260            265            270

Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Phe Gly Met Ser Trp Val
      275            280            285

Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val Ser Ser Ile Ser Gly
      290            295            300

Ser Gly Ser Asp Thr Leu Tyr Ala Asp Ser Val Lys Gly Arg Phe Thr
      305            310            315            320

Ile Ser Arg Asp Asn Ala Lys Thr Thr Leu Tyr Leu Gln Met Asn Ser
      325            330            335

Leu Arg Pro Glu Asp Thr Ala Val Tyr Tyr Cys Thr Ile Gly Gly Ser
      340            345            350

Leu Ser Arg Ser Ser Gln Gly Thr Leu Val Thr Val Ser Ser Gly Gly
      355            360            365

Gly Gly Ser Gly Gly Gly Ser Asp Ile Val Met Thr Gln Ser Pro Asp
      370            375            380

Ser Leu Ala Val Ser Leu Gly Glu Arg Ala Thr Ile Asn Cys Lys Ser

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385	390	395	400
Ser Gln Ser Leu Leu Asn Ala Arg Thr Gly Lys Asn Tyr Leu Ala Trp	405	410	415
Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile Tyr Trp Ala	420	425	430
Ser Thr Arg Glu Ser Gly Val Pro Asp Arg Phe Ser Gly Ser Gly Ser	435	440	445
Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Ala Glu Asp Val	450	455	460
Ala Val Tyr Tyr Cys Lys Gln Ser Tyr Ser Arg Arg Thr Phe Gly Gly	465	470	475
Gly Thr Lys Val Glu Ile Lys Gly Gly Gly Ser Gly Gly Gly Gly	485	490	495
Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gln Val Gln Leu Val	500	505	510
Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala Ser Val Lys Val Ser	515	520	525
Cys Lys Ala Ser Gly Phe Asn Ile Lys Asp Tyr Tyr Met His Trp Val	530	535	540
Arg Gln Ala Pro Gly Gln Arg Leu Glu Trp Met Gly Trp Ile Asp Leu	545	550	555
Glu Asn Ala Asn Thr Ile Tyr Asp Ala Lys Phe Gln Gly Arg Val Thr	565	570	575
Ile Thr Arg Asp Thr Ser Ala Ser Thr Ala Tyr Met Glu Leu Ser Ser	580	585	590
Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys Ala Arg Asp Ala Tyr	595	600	605
Gly Arg Tyr Phe Tyr Asp Val Trp Gly Gln Gly Thr Leu Val Thr Val	610	615	620
Ser Ser His His His His His His His His His His	625	630	635

<210> SEQ ID NO 310

<211> LENGTH: 636

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 310

Asp Ile Val Met Thr Gln Ser Pro Asp Ser Leu Ala Val Ser Leu Gly	1	5	10	15
Glu Arg Ala Thr Ile Asn Cys Lys Ser Ser Gln Ser Leu Leu Asn Ala	20	25	30	
Arg Thr Gly Lys Asn Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln	35	40	45	
Pro Pro Lys Leu Leu Ile Tyr Trp Ala Ser Thr Arg Glu Ser Gly Val	50	55	60	
Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr	65	70	75	80
Ile Ser Ser Leu Gln Ala Glu Asp Val Ala Val Tyr Tyr Cys Lys Gln	85	90	95	
Ser Tyr Ser Arg Arg Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys	100	105	110	

Gly 115	Gly 130	Lys 145	Ile 160	Leu 185	Asp 205	Ser 210	Val 225	Trp 255	Gly 260	Lys 275	Ile 290	Lys 305	Tyr 335	Lys 355	Tyr 370	Ser 385	Gln 400	Ser 415	Ser 420	Lys 435	Ala 450	Phe 465	Tyr 480	Lys 495	Lys 500	Val 515	Leu 520	Leu 525
Gly 120	Gly 135	Lys 150	Met 165	Trp 180	Arg 200	Leu 215	Arg 230	Val 245	Gln 265	Leu 280	Val 295	Gly 310	His 330	Met 360	Val 375	Gly 390	Leu 405	Ser 425	Val 440	Arg 455	Phe 470	Gly 485	Gly 490	Gly 505	Gly 510	Gly 520	Gly 525	Gly 530
Ser 115	Ser 130	Pro 145	Asp 160	Glu 185	Thr 205	Thr 210	Cys 225	Ser 255	Ser 260	Thr 275	Thr 290	Leu 305	Trp 335	Ala 355	Trp 370	Gly 385	Pro 400	Met 415	His 420	Trp 435	Ile 450	Thr 465	Phe 480	Gln 495	Gly 500	Val 515	Leu 520	Phe 525
Gly 125	Gly 140	Ala 155	Gly 175	Asn 190	Asp 205	Glu 220	Phe 235	Gly 255	Pro 270	Ser 285	Gln 300	Asp 315	Asp 335	Ala 350	Gly 370	Ile 385	Val 400	Ala 415	Trp 430	Ser 445	Gly 460	Phe 475	Gly 490	Gly 510	Gly 525	Pro 540	Ser 545	Phe 550
Gly 130	Gly 145	Ala 160	Glu 175	Thr 190	Thr 205	Asp 220	Tyr 235	Gly 255	Gly 270	Gly 285	His 300	Lys 315	Asp 335	Phe 350	Gly 370	Val 385	Leu 400	Thr 415	Gln 430	Lys 445	Thr 460	Val 475	Gly 490	Ser 510	Asn 525	Ser 540	Gly 545	Gly 550
Ser 120	Ser 135	Cys 155	Gln 170	Asn 185	Thr 200	Leu 215	Arg 235	Ser 250	Ser 265	Thr 280	Arg 300	Asp 315	Asp 335	Val 350	Ser 370	Ile 385	Val 400	Ala 415	Trp 430	Ser 445	Gly 460	Phe 475	Gly 490	Gly 510	Gly 525	Pro 540	Ser 545	Phe 550
Gly 125	Gly 140	Ala 155	Gly 175	Asn 190	Asp 205	Glu 220	Phe 235	Gly 255	Pro 270	Ser 285	Gln 300	Asp 315	Asp 335	Ala 350	Gly 370	Ile 385	Val 400	Ala 415	Trp 430	Ser 445	Gly 460	Phe 475	Gly 490	Gly 510	Gly 525	Pro 540	Ser 545	Phe 550
Gly 130	Gly 145	Ala 160	Glu 175	Thr 190	Thr 205	Asp 220	Tyr 235	Gly 255	Gly 270	Gly 285	His 300	Lys 315	Asp 335	Phe 350	Gly 370	Val 385	Leu 400	Thr 415	Gln 430	Lys 445	Thr 460	Val 475	Gly 490	Ser 510	Asn 525	Ser 540	Gly 545	Gly 550
Ser 120	Ser 135	Cys 155	Gln 170	Asn 185	Thr 200	Leu 215	Arg 235	Ser 250	Ser 265	Thr 280	Arg 300	Asp 315	Asp 335	Val 350	Ser 370	Ile 385	Val 400	Ala 415	Trp 430	Ser 445	Gly 460	Phe 475	Gly 490	Gly 510	Gly 525	Pro 540	Ser 545	Phe 550
Gly 125	Gly 140	Ala 155	Gly 175	Asn 190	Asp 205	Glu 220	Phe 235	Gly 255	Pro 270	Ser 285	Gln 300	Asp 315	Asp 335	Val 350	Ser 370	Ile 385	Val 400	Ala 415	Trp 430	Ser 445	Gly 460	Phe 475	Gly 490	Gly 510	Gly 525	Pro 540	Ser 545	Phe 550
Gly 130	Gly 145	Ala 160	Glu 175	Thr 190	Thr 205	Asp 220	Tyr 235	Gly 255	Gly 270	Gly 285	His 300	Lys 315	Asp 335	Phe 350	Gly 370	Val 385	Leu 400	Thr 415	Gln 430	Lys 445	Thr 460	Val 475	Gly 490	Ser 510	Asn 525	Ser 540	Gly 545	Gly 550
Ser 120	Ser 135	Cys 155	Gln 170	Asn 185	Thr 200	Leu 215	Arg 235	Ser 250	Ser 265	Thr 280	Arg 300	Asp 315	Asp 335	Val 350	Ser 370	Ile 385	Val 400	Ala 415	Trp 430	Ser 445	Gly 460	Phe 475	Gly 490	Gly 510	Gly 525	Pro 540	Ser 545	Phe 550
Gly 125	Gly 140	Ala 155	Gly 175	Asn 190	Asp 205	Glu 220	Phe 235	Gly 255	Pro 270	Ser 285	Gln 300	Asp 315	Asp 335	Val 350	Ser 370	Ile 385	Val 400	Ala 415	Trp 430	Ser 445	Gly 460	Phe 475	Gly 490	Gly 510	Gly 525	Pro 540	Ser 545	Phe 550
Gly 130	Gly 145	Ala 160	Glu 175	Thr 190	Thr 205	Asp 220	Tyr 235	Gly 255	Gly 270	Gly 285	His 300	Lys 315	Asp 335	Phe 350	Gly 370	Val 385	Leu 400	Thr 41										

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530	535	540
Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val Ser		
545	550	555 560
Ser Ile Ser Gly Ser Gly Ser Asp Thr Leu Tyr Ala Asp Ser Val Lys		
	565	570 575
Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Thr Thr Leu Tyr Leu		
	580	585 590
Gln Met Asn Ser Leu Arg Pro Glu Asp Thr Ala Val Tyr Tyr Cys Thr		
	595	600 605
Ile Gly Gly Ser Leu Ser Arg Ser Ser Gln Gly Thr Leu Val Thr Val		
	610	615 620
Ser Ser His His His His His His His His His His		
625	630	635
<210> SEQ ID NO 311		
<211> LENGTH: 636		
<212> TYPE: PRT		
<213> ORGANISM: Artificial Sequence		
<220> FEATURE:		
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide		
<400> SEQUENCE: 311		
Asp Ile Val Met Thr Gln Ser Pro Asp Ser Leu Ala Val Ser Leu Gly		
1	5	10 15
Glu Arg Ala Thr Ile Asn Cys Lys Ser Ser Gln Ser Leu Leu Asn Ala		
	20	25 30
Arg Thr Gly Lys Asn Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln		
	35	40 45
Pro Pro Lys Leu Leu Ile Tyr Trp Ala Ser Thr Arg Glu Ser Gly Val		
	50	55 60
Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr		
65	70	75 80
Ile Ser Ser Leu Gln Ala Glu Asp Val Ala Val Tyr Tyr Cys Lys Gln		
	85	90 95
Ser Tyr Ser Arg Arg Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys		
	100	105 110
Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly		
	115	120 125
Gly Gly Gly Ser Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys		
	130	135 140
Lys Pro Gly Ala Ser Val Lys Val Ser Cys Lys Ala Ser Gly Phe Asn		
145	150	155 160
Ile Lys Asp Tyr Tyr Met His Trp Val Arg Gln Ala Pro Gly Gln Arg		
	165	170 175
Leu Glu Trp Met Gly Trp Ile Asp Leu Glu Asn Ala Asn Thr Ile Tyr		
	180	185 190
Asp Ala Lys Phe Gln Gly Arg Val Thr Ile Thr Arg Asp Thr Ser Ala		
	195	200 205
Ser Thr Ala Tyr Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala		
	210	215 220
Val Tyr Tyr Cys Ala Arg Asp Ala Tyr Gly Arg Tyr Phe Tyr Asp Val		
225	230	235 240
Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser Gly Gly Gly Gly Ser		
	245	250 255

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Gly	Gly	Gly	Ser	Glu	Val	Gln	Leu	Val	Glu	Ser	Gly	Gly	Gly	Leu	Val
			260					265						270	
Gln	Pro	Gly	Asn	Ser	Leu	Arg	Leu	Ser	Cys	Ala	Ala	Ser	Gly	Phe	Thr
		275					280					285			
Phe	Ser	Ser	Phe	Gly	Met	Ser	Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys	Gly
	290					295					300				
Leu	Glu	Trp	Val	Ser	Ser	Ile	Ser	Gly	Ser	Gly	Ser	Asp	Thr	Leu	Tyr
305					310					315				320	
Ala	Asp	Ser	Val	Lys	Gly	Arg	Phe	Thr	Ile	Ser	Arg	Asp	Asn	Ala	Lys
				325					330					335	
Thr	Thr	Leu	Tyr	Leu	Gln	Met	Asn	Ser	Leu	Arg	Pro	Glu	Asp	Thr	Ala
		340						345					350		
Val	Tyr	Tyr	Cys	Thr	Ile	Gly	Gly	Ser	Leu	Ser	Arg	Ser	Ser	Gln	Gly
		355					360					365			
Thr	Leu	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Ser
	370					375					380				
Gln	Val	Gln	Leu	Gln	Glu	Ser	Gly	Pro	Gly	Leu	Val	Lys	Pro	Ser	Gln
385					390					395					400
Thr	Leu	Ser	Leu	Thr	Cys	Thr	Val	Ser	Gly	Gly	Ser	Ile	Ser	Thr	Ser
			405						410					415	
Gly	Met	Gly	Val	Gly	Trp	Ile	Arg	Gln	His	Pro	Gly	Lys	Gly	Leu	Glu
			420					425					430		
Trp	Ile	Gly	His	Ile	Trp	Trp	Asp	Asp	Asp	Lys	Arg	Tyr	Asn	Pro	Ala
		435					440					445			
Leu	Lys	Ser	Arg	Val	Thr	Ile	Ser	Val	Asp	Thr	Ser	Lys	Asn	Gln	Phe
	450					455					460				
Ser	Leu	Lys	Leu	Ser	Ser	Val	Thr	Ala	Ala	Asp	Thr	Ala	Val	Tyr	Tyr
465					470					475				480	
Cys	Ala	Arg	Met	Glu	Leu	Trp	Ser	Tyr	Tyr	Phe	Asp	Tyr	Trp	Gly	Gln
			485						490					495	
Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly
			500					505					510		
Gly	Ser	Gly	Gly	Gly	Gly	Ser	Glu	Ile	Val	Leu	Thr	Gln	Ser	Pro	Ala
		515					520					525			
Thr	Leu	Ser	Leu	Ser	Pro	Gly	Glu	Arg	Ala	Thr	Leu	Ser	Cys	Ser	Ala
	530					535					540				
Ser	Ser	Ser	Val	Ser	Tyr	Met	His	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Gln
545					550					555					560
Ala	Pro	Arg	Leu	Leu	Ile	Tyr	Asp	Thr	Ser	Lys	Leu	Ala	Ser	Gly	Ile
			565						570					575	
Pro	Ala	Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	Thr
			580					585					590		
Ile	Ser	Ser	Leu	Glu	Pro	Glu	Asp	Phe	Ala	Val	Tyr	Tyr	Cys	Phe	Gln
		595					600					605			
Gly	Ser	Val	Tyr	Pro	Phe	Thr	Phe	Gly	Gln	Gly	Thr	Lys	Leu	Glu	Ile
	610					615					620				
Lys	Arg	His	His	His	His	His	His	His	His	His	His	His	His	His	His
625					630						635				

<210> SEQ ID NO 312

<211> LENGTH: 636

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic

-continued

polypeptide

<400> SEQUENCE: 312

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Asn
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Phe
 20 25 30
 Gly Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45
 Ser Ser Ile Ser Gly Ser Gly Ser Asp Thr Leu Tyr Ala Asp Ser Val
 50 55 60
 Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Thr Thr Leu Tyr
 65 70 75 80
 Leu Gln Met Asn Ser Leu Arg Pro Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95
 Thr Ile Gly Gly Ser Leu Ser Arg Ser Ser Gln Gly Thr Leu Val Thr
 100 105 110
 Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser Asp Ile Val Met
 115 120 125
 Thr Gln Ser Pro Asp Ser Leu Ala Val Ser Leu Gly Glu Arg Ala Thr
 130 135 140
 Ile Asn Cys Lys Ser Ser Gln Ser Leu Leu Asn Ala Arg Thr Gly Lys
 145 150 155 160
 Asn Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu
 165 170 175
 Leu Ile Tyr Trp Ala Ser Thr Arg Glu Ser Gly Val Pro Asp Arg Phe
 180 185 190
 Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu
 195 200 205
 Gln Ala Glu Asp Val Ala Val Tyr Tyr Cys Lys Gln Ser Tyr Ser Arg
 210 215 220
 Arg Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys Gly Gly Gly Gly
 225 230 235 240
 Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
 245 250 255
 Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala
 260 265 270
 Ser Val Lys Val Ser Cys Lys Ala Ser Gly Phe Asn Ile Lys Asp Tyr
 275 280 285
 Tyr Met His Trp Val Arg Gln Ala Pro Gly Gln Arg Leu Glu Trp Met
 290 295 300
 Gly Trp Ile Asp Leu Glu Asn Ala Asn Thr Ile Tyr Asp Ala Lys Phe
 305 310 315 320
 Gln Gly Arg Val Thr Ile Thr Arg Asp Thr Ser Ala Ser Thr Ala Tyr
 325 330 335
 Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
 340 345 350
 Ala Arg Asp Ala Tyr Gly Arg Tyr Phe Tyr Asp Val Trp Gly Gln Gly
 355 360 365
 Thr Leu Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser
 370 375 380
 Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Gln
 385 390 395 400

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Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Gly Ser Ile Ser Thr Ser
 405 410 415
 Gly Met Gly Val Gly Trp Ile Arg Gln His Pro Gly Lys Gly Leu Glu
 420 425 430
 Trp Ile Gly His Ile Trp Trp Asp Asp Asp Lys Arg Tyr Asn Pro Ala
 435 440 445
 Leu Lys Ser Arg Val Thr Ile Ser Val Asp Thr Ser Lys Asn Gln Phe
 450 455 460
 Ser Leu Lys Leu Ser Ser Val Thr Ala Ala Asp Thr Ala Val Tyr Tyr
 465 470 475 480
 Cys Ala Arg Met Glu Leu Trp Ser Tyr Tyr Phe Asp Tyr Trp Gly Gln
 485 490 495
 Gly Thr Leu Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly
 500 505 510
 Gly Ser Gly Gly Gly Gly Ser Glu Ile Val Leu Thr Gln Ser Pro Ala
 515 520 525
 Thr Leu Ser Leu Ser Pro Gly Glu Arg Ala Thr Leu Ser Cys Ser Ala
 530 535 540
 Ser Ser Ser Val Ser Tyr Met His Trp Tyr Gln Gln Lys Pro Gly Gln
 545 550 555 560
 Ala Pro Arg Leu Leu Ile Tyr Asp Thr Ser Lys Leu Ala Ser Gly Ile
 565 570 575
 Pro Ala Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr
 580 585 590
 Ile Ser Ser Leu Glu Pro Glu Asp Phe Ala Val Tyr Tyr Cys Phe Gln
 595 600 605
 Gly Ser Val Tyr Pro Phe Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile
 610 615 620
 Lys Arg His His His His His His His His His His
 625 630 635

<210> SEQ ID NO 313

<211> LENGTH: 636

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 313

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Asn
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Phe
 20 25 30
 Gly Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45
 Ser Ser Ile Ser Gly Ser Gly Ser Asp Thr Leu Tyr Ala Asp Ser Val
 50 55 60
 Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Thr Thr Leu Tyr
 65 70 75 80
 Leu Gln Met Asn Ser Leu Arg Pro Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95
 Thr Ile Gly Gly Ser Leu Ser Arg Ser Ser Gln Gly Thr Leu Val Thr
 100 105 110
 Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser Gln Val Gln Leu
 115 120 125

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Gln	Glu	Ser	Gly	Pro	Gly	Leu	Val	Lys	Pro	Ser	Gln	Thr	Leu	Ser	Leu
130						135					140				
Thr	Cys	Thr	Val	Ser	Gly	Gly	Ser	Ile	Ser	Thr	Ser	Gly	Met	Gly	Val
145					150					155					160
Gly	Trp	Ile	Arg	Gln	His	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Ile	Gly	His
				165					170					175	
Ile	Trp	Trp	Asp	Asp	Asp	Lys	Arg	Tyr	Asn	Pro	Ala	Leu	Lys	Ser	Arg
			180					185					190		
Val	Thr	Ile	Ser	Val	Asp	Thr	Ser	Lys	Asn	Gln	Phe	Ser	Leu	Lys	Leu
		195					200					205			
Ser	Ser	Val	Thr	Ala	Ala	Asp	Thr	Ala	Val	Tyr	Tyr	Cys	Ala	Arg	Met
		210				215					220				
Glu	Leu	Trp	Ser	Tyr	Tyr	Phe	Asp	Tyr	Trp	Gly	Gln	Gly	Thr	Leu	Val
225					230					235					240
Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly
				245					250					255	
Gly	Gly	Ser	Glu	Ile	Val	Leu	Thr	Gln	Ser	Pro	Ala	Thr	Leu	Ser	Leu
			260					265					270		
Ser	Pro	Gly	Glu	Arg	Ala	Thr	Leu	Ser	Cys	Ser	Ala	Ser	Ser	Ser	Val
		275					280					285			
Ser	Tyr	Met	His	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Gln	Ala	Pro	Arg	Leu
		290				295					300				
Leu	Ile	Tyr	Asp	Thr	Ser	Lys	Leu	Ala	Ser	Gly	Ile	Pro	Ala	Arg	Phe
305					310					315					320
Ser	Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	Thr	Ile	Ser	Ser	Leu
			325					330					335		
Glu	Pro	Glu	Asp	Phe	Ala	Val	Tyr	Tyr	Cys	Phe	Gln	Gly	Ser	Val	Tyr
			340					345					350		
Pro	Phe	Thr	Phe	Gly	Gln	Gly	Thr	Lys	Leu	Glu	Ile	Lys	Arg	Gly	Gly
		355				360						365			
Gly	Gly	Ser	Gly	Gly	Gly	Ser	Asp	Ile	Val	Met	Thr	Gln	Ser	Pro	Asp
		370				375					380				
Ser	Leu	Ala	Val	Ser	Leu	Gly	Glu	Arg	Ala	Thr	Ile	Asn	Cys	Lys	Ser
385					390					395					400
Ser	Gln	Ser	Leu	Leu	Asn	Ala	Arg	Thr	Gly	Lys	Asn	Tyr	Leu	Ala	Trp
			405					410					415		
Tyr	Gln	Gln	Lys	Pro	Gly	Gln	Pro	Pro	Lys	Leu	Leu	Ile	Tyr	Trp	Ala
			420				425						430		
Ser	Thr	Arg	Glu	Ser	Gly	Val	Pro	Asp	Arg	Phe	Ser	Gly	Ser	Gly	Ser
		435				440					445				
Gly	Thr	Asp	Phe	Thr	Leu	Thr	Ile	Ser	Ser	Leu	Gln	Ala	Glu	Asp	Val
		450				455					460				
Ala	Val	Tyr	Tyr	Cys	Lys	Gln	Ser	Tyr	Ser	Arg	Arg	Thr	Phe	Gly	Gly
465					470					475					480
Gly	Thr	Lys	Val	Glu	Ile	Lys	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly
			485					490					495		
Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gln	Val	Gln	Leu	Val
			500				505						510		
Gln	Ser	Gly	Ala	Glu	Val	Lys	Lys	Pro	Gly	Ala	Ser	Val	Lys	Val	Ser
		515					520					525			
Cys	Lys	Ala	Ser	Gly	Phe	Asn	Ile	Lys	Asp	Tyr	Tyr	Met	His	Trp	Val
530						535					540				

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Arg Gln Ala Pro Gly Gln Arg Leu Glu Trp Met Gly Trp Ile Asp Leu
545          550          555          560

Glu Asn Ala Asn Thr Ile Tyr Asp Ala Lys Phe Gln Gly Arg Val Thr
          565          570          575

Ile Thr Arg Asp Thr Ser Ala Ser Thr Ala Tyr Met Glu Leu Ser Ser
          580          585          590

Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys Ala Arg Asp Ala Tyr
          595          600          605

Gly Arg Tyr Phe Tyr Asp Val Trp Gly Gln Gly Thr Leu Val Thr Val
610          615          620

Ser Ser His His His His His His His His His His
625          630          635

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<210> SEQ ID NO 314
<211> LENGTH: 124
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

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<400> SEQUENCE: 314

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Glu Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Glu
1          5          10          15

Ser Leu Gln Ile Ser Cys Lys Gly Ser Gly Tyr Thr Phe Thr Asn Tyr
20          25          30

Trp Ile Ala Trp Val Arg Gln Met Pro Gly Lys Gly Leu Glu Trp Met
35          40          45

Gly Ile Ile Tyr Pro Gly Asp Ser Asp Thr Arg Tyr Ser Pro Ser Phe
50          55          60

Gln Gly Gln Val Thr Ile Ser Ala Asp Lys Ser Ile Ser Thr Ala Tyr
65          70          75          80

Leu Gln Trp Ser Gly Leu Lys Ala Ser Asp Thr Ala Met Tyr Tyr Cys
85          90          95

Ala Arg Gln Gly Tyr Ser Ser Gly Trp Arg Ser Tyr Tyr Gly Met Gly
100         105         110

Val Trp Gly Gln Gly Thr Thr Val Thr Val Ser Ser
115         120

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<210> SEQ ID NO 315
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

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<400> SEQUENCE: 315

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Ala Ile Gln Leu Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1          5          10          15

Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Gly Ile Ser Ser Ala
20          25          30

Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
35          40          45

Tyr Asp Ala Ser Ser Leu Glu Ser Gly Val Pro Ser Arg Phe Ser Gly
50          55          60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65          70          75          80

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Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Phe Asn Ser Tyr Pro His
85 90 95

Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys
100 105

<210> SEQ ID NO 316
<211> LENGTH: 121
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 316

Glu Val Gln Leu Gln Gln Ser Gly Pro Glu Leu Val Lys Pro Gly Ala
1 5 10 15

Ser Val Lys Met Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Asn Tyr
20 25 30

Val Ile His Trp Val Lys Gln Lys Pro Gly Gln Gly Leu Glu Trp Ile
35 40 45

Gly Tyr Phe Asn Pro Tyr Asn Asp Gly Thr Glu Tyr Asn Glu Lys Phe
50 55 60

Lys Ala Lys Ala Thr Leu Thr Ser Asp Lys Ser Ser Ser Thr Ala Tyr
65 70 75 80

Met Glu Leu Ser Ser Leu Thr Ser Glu Asp Ser Ala Val Tyr Tyr Cys
85 90 95

Ala Lys Gly Pro Tyr Tyr Tyr Gly Ser Ser Pro Phe Asp Tyr Trp Gly
100 105 110

Gln Gly Thr Thr Leu Thr Val Ser Ser
115 120

<210> SEQ ID NO 317
<211> LENGTH: 112
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 317

Asp Ala Val Met Thr Gln Thr Pro Leu Ser Leu Pro Val Ser Leu Gly
1 5 10 15

Asp Gln Ala Ser Ile Ser Cys Arg Ser Ser Gln Ser Leu Glu Asn Ser
20 25 30

Asn Gly Asn Thr Tyr Leu Asn Trp Tyr Leu Gln Lys Pro Gly Gln Ser
35 40 45

Pro Gln Leu Leu Ile Tyr Arg Val Ser Asn Arg Phe Ser Gly Val Leu
50 55 60

Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Lys Ile
65 70 75 80

Ser Arg Val Glu Ala Glu Asp Leu Gly Val Tyr Phe Cys Leu Gln Val
85 90 95

Thr His Val Pro Tyr Thr Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys
100 105 110

<210> SEQ ID NO 318
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 318

Gly Tyr Thr Phe Thr Asn Tyr Val Ile His
1 5 10

<210> SEQ ID NO 319

<211> LENGTH: 17

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 319

Tyr Phe Asn Pro Tyr Asn Asp Gly Thr Glu Tyr Asn Glu Lys Phe Lys
1 5 10 15

Ala

<210> SEQ ID NO 320

<211> LENGTH: 12

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 320

Gly Pro Tyr Tyr Tyr Gly Ser Ser Pro Phe Asp Tyr
1 5 10

<210> SEQ ID NO 321

<211> LENGTH: 16

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 321

Arg Ser Ser Gln Ser Leu Glu Asn Ser Asn Gly Asn Thr Tyr Leu Asn
1 5 10 15

<210> SEQ ID NO 322

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 322

Arg Val Ser Asn Arg Phe Ser
1 5

<210> SEQ ID NO 323

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 323

Arg Val Ser Asn Arg Phe Ser
1 5

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<210> SEQ ID NO 324
<211> LENGTH: 121
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 324

Gln Val Gln Leu Gln Gln Ser Gly Ala Glu Leu Val Arg Pro Gly Ser
1 5 10 15
Ser Val Lys Ile Ser Cys Lys Ala Ser Gly Tyr Ala Phe Ser Thr Tyr
20 25 30
Trp Met Asn Trp Val Lys Gln Arg Pro Gly Gln Gly Leu Glu Trp Ile
35 40 45
Gly Gln Ile Tyr Pro Gly Asp Asp Asp Thr Lys Tyr Asn Gly Lys Phe
50 55 60
Lys Gly Lys Ala Ser Leu Thr Ala Asp Lys Ser Ser Ser Thr Ala Tyr
65 70 75 80
Met Gln Leu Ile Ser Leu Thr Ser Glu Asp Ser Ala Val Tyr Phe Cys
85 90 95
Ala Arg Arg Tyr Phe Arg Tyr Asp Tyr Trp Tyr Ser Asp Val Trp Gly
100 105 110
Ala Gly Thr Thr Val Thr Val Thr Ser
115 120

<210> SEQ ID NO 325
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 325

Asp Ile Gln Met Thr Gln Thr Thr Ser Ser Leu Ser Ala Ser Leu Gly
1 5 10 15
Asp Arg Val Thr Ile Ser Cys Arg Ala Ser Gln Asp Ile Ser Asn Tyr
20 25 30
Leu Asn Trp Tyr Gln Gln Lys Pro Asp Gly Thr Val Lys Leu Leu Ile
35 40 45
Tyr Tyr Thr Ser Arg Leu His Ser Gly Val Pro Ala Arg Phe Ser Gly
50 55 60
Ser Gly Ser Gly Thr Asp Tyr Ser Leu Thr Ile Ser Asn Leu Glu Gln
65 70 75 80
Glu Asp Ile Ala Thr Tyr Phe Cys His Gln Gly Asn Thr Leu Pro Leu
85 90 95
Thr Phe Gly Ala Gly Thr Lys Leu Glu Leu Lys
100 105

<210> SEQ ID NO 326
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 326

-continued

Gly Tyr Ala Phe Ser Thr Tyr Trp Met Asn
1 5 10

<210> SEQ ID NO 327
 <211> LENGTH: 17
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 327

Gln Ile Tyr Pro Gly Asp Asp Asp Thr Lys Tyr Asn Gly Lys Phe Lys
1 5 10 15

Gly

<210> SEQ ID NO 328
 <211> LENGTH: 12
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 328

Arg Tyr Phe Arg Tyr Asp Tyr Trp Tyr Ser Asp Val
1 5 10

<210> SEQ ID NO 329
 <211> LENGTH: 11
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 329

Arg Ala Ser Gln Asp Ile Ser Asn Tyr Leu Asn
1 5 10

<210> SEQ ID NO 330
 <211> LENGTH: 7
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 330

Tyr Thr Ser Arg Leu His Ser
1 5

<210> SEQ ID NO 331
 <211> LENGTH: 9
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 331

His Gln Gly Asn Thr Leu Pro Leu Thr
1 5

<210> SEQ ID NO 332
 <211> LENGTH: 116
 <212> TYPE: PRT

-continued

<213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 332

```

Glu Ile Gln Leu Gln Gln Ser Gly Pro Glu Leu Val Lys Pro Gly Ala
1           5           10          15
Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Ala Phe Thr Ser Tyr
          20          25          30
Asn Met Tyr Trp Val Lys Gln Ser His Gly Lys Ser Leu Glu Trp Ile
          35          40          45
Gly Tyr Ile Asp Pro Tyr Asn Gly Asp Thr Thr Tyr Asn Gln Lys Phe
          50          55          60
Lys Gly Lys Ala Thr Leu Thr Val Asp Lys Ser Ser Ser Thr Ala Tyr
          65          70          75          80
Met His Leu Asn Ser Leu Thr Ser Glu Asp Ser Ala Val Tyr Tyr Cys
          85          90          95
Leu Thr Thr Ala Tyr Ala Met Asp Tyr Trp Gly Gln Gly Thr Ser Val
          100         105         110
Thr Val Ser Ser
          115

```

<210> SEQ ID NO 333
 <211> LENGTH: 106
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 333

```

Gln Ile Val Leu Thr Gln Ser Pro Ala Ile Met Ser Ala Ser Leu Gly
1           5           10          15
Glu Glu Ile Thr Leu Thr Cys Ser Ala Ser Ser Thr Val Asn Tyr Met
          20          25          30
His Trp Tyr Gln Gln Lys Ser Gly Thr Ser Pro Lys Leu Leu Ile Tyr
          35          40          45
Ser Thr Ser Asn Leu Ala Ser Gly Val Pro Ser Arg Phe Ser Gly Ser
          50          55          60
Gly Ser Gly Thr Phe Tyr Ser Leu Thr Ile Arg Ser Val Glu Ala Glu
          65          70          75          80
Asp Ala Ala Asp Tyr Tyr Cys His Gln Trp Ser Ser Tyr Pro Tyr Thr
          85          90          95
Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys
          100         105

```

<210> SEQ ID NO 334
 <211> LENGTH: 10
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 334

```

Gly Tyr Ala Phe Thr Ser Tyr Asn Met Tyr
1           5           10

```

<210> SEQ ID NO 335

-continued

<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 335

Tyr Ile Asp Pro Tyr Asn Gly Asp Thr Thr Tyr Asn Gln Lys Phe Lys
1 5 10 15

Gly

<210> SEQ ID NO 336
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 336

Thr Ala Tyr Ala Met Asp Tyr
1 5

<210> SEQ ID NO 337
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 337

Ser Ala Ser Ser Thr Val Asn Tyr Met His
1 5 10

<210> SEQ ID NO 338
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 338

Ser Thr Ser Asn Leu Ala Ser
1 5

<210> SEQ ID NO 339
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 339

His Gln Trp Ser Ser Tyr Pro Tyr Thr
1 5

<210> SEQ ID NO 340
<211> LENGTH: 121
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

-continued

<400> SEQUENCE: 340

```

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ser
1           5           10           15
Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Asp Tyr
          20           25           30
Val Ile His Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Met
          35           40           45
Gly Tyr Phe Asn Pro Tyr Asn Asp Gly Thr Glu Tyr Asn Glu Lys Phe
          50           55           60
Lys Ala Arg Val Thr Ile Thr Ala Asp Lys Ser Thr Ser Thr Ala Tyr
          65           70           75           80
Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
          85           90           95
Ala Arg Gly Pro Tyr Tyr Tyr Gly Ser Ser Pro Phe Asp Tyr Trp Gly
          100          105          110
Gln Gly Thr Thr Val Thr Val Ser Ser
          115          120

```

<210> SEQ ID NO 341

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 341

```

Gly Tyr Thr Phe Thr Asn Tyr Val Ile His
1           5           10

```

<210> SEQ ID NO 342

<211> LENGTH: 17

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 342

```

Tyr Phe Asn Pro Tyr Asn Asp Gly Thr Glu Tyr Asn Glu Lys Phe Lys
1           5           10           15

```

Ala

<210> SEQ ID NO 343

<211> LENGTH: 12

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 343

```

Gly Pro Tyr Tyr Tyr Gly Ser Ser Pro Phe Asp Tyr
1           5           10

```

<210> SEQ ID NO 344

<211> LENGTH: 112

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

-continued

<400> SEQUENCE: 344

Asp Ile Val Met Thr Gln Thr Pro Leu Ser Leu Pro Val Thr Pro Gly
1 5 10 15
Glu Pro Ala Ser Ile Ser Cys Arg Ser Ser Gln Ser Leu Glu Asn Ser
20 25 30
Asn His Asn Thr Tyr Ile Asn Trp Tyr Leu Gln Lys Pro Gly Gln Ser
35 40 45
Pro Gln Leu Leu Ile Tyr Arg Val Ser Lys Arg Phe Ser Gly Val Pro
50 55 60
Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Lys Ile
65 70 75 80
Ser Arg Val Glu Ala Glu Asp Val Gly Val Tyr Tyr Cys His Gln Val
85 90 95
Thr His Val Pro Tyr Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys
100 105 110

<210> SEQ ID NO 345

<211> LENGTH: 16

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 345

Arg Ser Ser Gln Ser Leu Glu Asn Ser Asn His Asn Thr Tyr Ile Asn
1 5 10 15

<210> SEQ ID NO 346

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 346

Arg Val Ser Lys Arg Phe Ser
1 5

<210> SEQ ID NO 347

<211> LENGTH: 9

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 347

His Gln Val Thr His Val Pro Tyr Thr
1 5

<210> SEQ ID NO 348

<211> LENGTH: 121

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 348

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala
1 5 10 15

-continued

Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Ala Phe Ser Thr Tyr
 20 25 30

Trp Met Asn Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Met
 35 40 45

Gly Gln Ile Tyr Pro Gly Asp Asp Asp Thr Lys Tyr Ser Gly Lys Phe
 50 55 60

Lys Gly Arg Val Thr Ile Thr Ala Asp Lys Ser Thr Ser Thr Ala Tyr
 65 70 75 80

Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Ala Arg Arg Tyr Phe Arg Tyr Asp Tyr Trp Tyr Ser Asp Val Trp Gly
 100 105 110

Gln Gly Thr Thr Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 349
 <211> LENGTH: 107
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 349

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
 1 5 10 15

Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Asp Ile Ser Asn Tyr
 20 25 30

Leu Asn Trp Tyr Gln Gln Lys Pro Gly Lys Val Pro Lys Leu Leu Ile
 35 40 45

Tyr Tyr Thr Ser Arg Leu His Ser Gly Val Pro Ser Arg Phe Ser Gly
 50 55 60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
 65 70 75 80

Glu Asp Val Ala Thr Tyr Tyr Cys His Gln Gly Asn Thr Leu Pro Leu
 85 90 95

Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys
 100 105

<210> SEQ ID NO 350
 <211> LENGTH: 10
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 350

Gly Tyr Ala Phe Ser Thr Tyr Trp Met Asn
 1 5 10

<210> SEQ ID NO 351
 <211> LENGTH: 17
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 351

Gln Ile Tyr Pro Gly Asp Asp Thr Lys Tyr Ser Gly Lys Phe Lys

-continued

1	5	10	15
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Gly

<210> SEQ ID NO 352
 <211> LENGTH: 12
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 352

Arg	Tyr	Phe	Arg	Tyr	Asp	Tyr	Trp	Tyr	Ser	Asp	Val
1			5					10			

<210> SEQ ID NO 353
 <211> LENGTH: 11
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 353

Arg	Ala	Ser	Gln	Asp	Ile	Ser	Asn	Tyr	Leu	Asn
1			5					10		

<210> SEQ ID NO 354
 <211> LENGTH: 7
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 354

Tyr	Thr	Ser	Arg	Leu	His	Ser
1			5			

<210> SEQ ID NO 355
 <211> LENGTH: 9
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 355

His	Gln	Gly	Asn	Thr	Leu	Pro	Leu	Thr
1			5					

<210> SEQ ID NO 356
 <211> LENGTH: 116
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 356

Gln	Met	Gln	Leu	Val	Gln	Ser	Gly	Pro	Glu	Val	Lys	Lys	Pro	Gly	Thr
1			5					10						15	
Ser	Val	Lys	Val	Ser	Cys	Lys	Ala	Ser	Gly	Tyr	Ala	Phe	Thr	Ser	Tyr
		20						25					30		
Asn	Met	Tyr	Trp	Val	Arg	Gln	Ala	Arg	Gly	Gln	Arg	Leu	Glu	Trp	Ile
		35						40					45		

-continued

Gly Tyr Ile Asp Pro Tyr Asn Ala Asp Thr Thr Tyr Asn Gln Lys Phe
 50 55 60

Lys Gly Arg Val Thr Ile Thr Arg Asp Met Ser Thr Ser Thr Ala Tyr
 65 70 75 80

Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Leu Thr Thr Ala Tyr Ala Met Asp Tyr Trp Gly Gln Gly Thr Leu Val
 100 105 110

Thr Val Ser Ser
 115

<210> SEQ ID NO 357
 <211> LENGTH: 106
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 357

Asp Ile Gln Leu Thr Gln Ser Pro Ser Phe Leu Ser Ala Ser Val Gly
 1 5 10 15

Asp Arg Val Thr Ile Thr Cys Ser Ala Ser Ser Thr Val Asn Tyr Met
 20 25 30

His Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile Tyr
 35 40 45

Ser Thr Ser Asn Leu Ala Ser Gly Val Pro Ser Arg Phe Ser Gly Ser
 50 55 60

Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro Glu
 65 70 75 80

Asp Phe Ala Thr Tyr Tyr Cys His Gln Trp Ser Ser Tyr Pro Tyr Thr
 85 90 95

Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys
 100 105

<210> SEQ ID NO 358
 <211> LENGTH: 10
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 358

Gly Tyr Ala Phe Thr Ser Tyr Asn Met Tyr
 1 5 10

<210> SEQ ID NO 359
 <211> LENGTH: 17
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 359

Tyr Ile Asp Pro Tyr Asn Ala Asp Thr Thr Tyr Asn Gln Lys Phe Lys
 1 5 10 15

Gly

-continued

```

<210> SEQ ID NO 360
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      peptide

```

```

<400> SEQUENCE: 360

```

```

Thr Ala Tyr Ala Met Asp Tyr
1           5

```

```

<210> SEQ ID NO 361
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      peptide

```

```

<400> SEQUENCE: 361

```

```

Ser Ala Ser Ser Thr Val Asn Tyr Met His
1           5           10

```

```

<210> SEQ ID NO 362
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      peptide

```

```

<400> SEQUENCE: 362

```

```

Ser Thr Ser Asn Leu Ala Ser
1           5

```

```

<210> SEQ ID NO 363
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      peptide

```

```

<400> SEQUENCE: 363

```

```

His Gln Trp Ser Ser Tyr Pro Tyr Thr
1           5

```

```

<210> SEQ ID NO 364
<211> LENGTH: 117
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

```

```

<400> SEQUENCE: 364

```

```

Gln Val Lys Leu Glu Glu Ser Gly Gly Glu Leu Val Gln Pro Gly Gly
1           5           10           15

```

```

Pro Leu Arg Leu Ser Cys Ala Ala Ser Gly Asn Ile Phe Ser Ile Asn
20           25           30

```

```

Arg Met Gly Trp Tyr Arg Gln Ala Pro Gly Lys Gln Arg Ala Phe Val
35           40           45

```

```

Ala Ser Ile Thr Val Arg Gly Ile Thr Asn Tyr Ala Asp Ser Val Lys
50           55           60

```

```

Gly Arg Phe Thr Ile Ser Val Asp Lys Ser Lys Asn Thr Ile Tyr Leu

```


-continued

65	70	75	80
Gln Met Asn Ala Leu Lys Pro Glu Asp Thr Ala Val Tyr Tyr Cys Asn			
	85	90	95
Ala Val Ser Ser Asn Arg Asp Pro Asp Tyr Trp Gly Gln Gly Thr Gln			
	100	105	110
Val Thr Val Ser Ser			
	115		

<210> SEQ ID NO 365
 <211> LENGTH: 5
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 365

Ile Asn Arg Met Gly
1 5

<210> SEQ ID NO 366
 <211> LENGTH: 16
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 366

Ser Ile Thr Val Arg Gly Ile Thr Asn Tyr Ala Asp Ser Val Lys Gly
1 5 10 15

<210> SEQ ID NO 367
 <211> LENGTH: 9
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 367

Val Ser Ser Asn Arg Asp Pro Asp Tyr
1 5

<210> SEQ ID NO 368
 <211> LENGTH: 12
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 368

Ala Arg Asp Ser Tyr Gly Arg Tyr Phe Tyr Asp Val
1 5 10

<210> SEQ ID NO 369
 <211> LENGTH: 12
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 369

Ala Arg Asp Val Tyr Gly Arg Tyr Phe Tyr Asp Leu

-continued

1 5 10

<210> SEQ ID NO 370
 <211> LENGTH: 12
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 peptide

<400> SEQUENCE: 370

Ala Arg Asp Ala Tyr Gly Gly Tyr Phe Tyr Asp Val
 1 5 10

<210> SEQ ID NO 371
 <211> LENGTH: 125
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 371

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Asp Thr Tyr
 20 25 30

Ala Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ala Arg Ile Arg Ser Lys Tyr Asn Asn Tyr Ala Thr Tyr Tyr Ala Asp
 50 55 60

Ser Val Lys Asp Arg Phe Thr Ile Ser Arg Asp Asp Ser Lys Ser Thr
 65 70 75 80

Leu Tyr Leu Gln Met Glu Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr
 85 90 95

Tyr Cys Val Arg His Gly Asn Phe Gly Asn Tyr Ala Val Ser Trp Phe
 100 105 110

Ala His Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
 115 120 125

<210> SEQ ID NO 372
 <211> LENGTH: 109
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 372

Gln Thr Val Val Thr Gln Glu Pro Ser Leu Ser Val Ser Pro Gly Gly
 1 5 10 15

Thr Val Thr Leu Thr Cys Gly Ser Ser Thr Gly Ala Val Thr Thr Ser
 20 25 30

Asn Tyr Ala Asn Trp Val Gln Gln Thr Pro Gly Gln Ala Pro Arg Gly
 35 40 45

Leu Ile Gly Gly Thr Asp Lys Arg Ala Pro Gly Val Pro Asp Arg Phe
 50 55 60

Ser Gly Ser Leu Leu Gly Asp Lys Ala Ala Leu Thr Ile Thr Gly Ala
 65 70 75 80

Gln Ala Glu Asp Glu Ala Asp Tyr Tyr Cys Ala Leu Trp Tyr Ser Asn
 85 90 95

-continued

His Trp Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
100 105

<210> SEQ ID NO 373
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 373

Phe Thr Phe Asp Thr Tyr Ala Met Asn
1 5

<210> SEQ ID NO 374
<211> LENGTH: 19
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 374

Arg Ile Arg Ser Lys Tyr Asn Asn Tyr Ala Thr Tyr Tyr Ala Asp Ser
1 5 10 15

Val Lys Asp

<210> SEQ ID NO 375
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 375

Val Arg His Gly Asn Phe Gly Asn Tyr Ala Val Ser Trp Phe Ala His
1 5 10 15

<210> SEQ ID NO 376
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 376

Gly Ser Ser Thr Gly Ala Val Thr Thr Ser Asn Tyr Ala Asn
1 5 10

<210> SEQ ID NO 377
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 377

Gly Thr Asp Lys Arg Ala Pro
1 5

<210> SEQ ID NO 378
<211> LENGTH: 9

-continued

<212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 378

Ala Leu Trp Tyr Ser Asn His Trp Val
 1 5

<210> SEQ ID NO 379
 <211> LENGTH: 125
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 379

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Asp Thr Tyr
 20 25 30

Ala Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ala Arg Ile Arg Ser Lys Tyr Asn Asn Tyr Ala Thr Tyr Tyr Ala Asp
 50 55 60

Ser Val Lys Asp Arg Phe Thr Ile Ser Arg Asp Asp Ser Lys Ser Thr
 65 70 75 80

Leu Tyr Leu Gln Met Glu Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr
 85 90 95

Tyr Cys Val Arg His Gly Ser Phe Gly Asn His Ile Val Ser Trp Phe
 100 105 110

Ala His Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
 115 120 125

<210> SEQ ID NO 380
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 380

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ser
 1 5 10 15

Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Ser Phe Thr Thr Tyr
 20 25 30

Tyr Ile His Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Met
 35 40 45

Gly Trp Ile Phe Pro Gly Asn Asp Asn Ile Lys Tyr Ser Glu Lys Phe
 50 55 60

Lys Gly Arg Val Thr Ile Thr Ala Asp Lys Ser Thr Ser Thr Ala Tyr
 65 70 75 80

Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Ala Ile Asp Ser Val Ser Ile Tyr Tyr Phe Asp Tyr Trp Gly Gln Gly
 100 105 110

Thr Leu Val Thr Val Ser Ser

-continued

115

<210> SEQ ID NO 381
<211> LENGTH: 112
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 381

Asp Ile Val Met Thr Gln Ser Pro Asp Ser Leu Ala Val Ser Leu Gly
1 5 10 15
Glu Arg Ala Thr Ile Asn Cys Lys Ser Ser Gln Ser Leu Leu Asn Ser
20 25 30
Arg Thr Arg Lys Asn Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln
35 40 45
Pro Pro Lys Leu Leu Ile Tyr Trp Ala Ser Thr Arg Lys Ser Gly Val
50 55 60
Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr
65 70 75 80
Ile Ser Ser Leu Gln Ala Glu Asp Val Ala Val Tyr Tyr Cys Thr Gln
85 90 95
Ser Phe Ile Leu Arg Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
100 105 110

<210> SEQ ID NO 382
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 382

Gly Tyr Ser Phe Thr Thr Tyr Tyr Ile His
1 5 10

<210> SEQ ID NO 383
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 383

Trp Ile Phe Pro Gly Asn Asp Asn Ile Lys Tyr Ser Glu Lys Phe Lys
1 5 10 15
Gly

<210> SEQ ID NO 384
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 384

Asp Ser Val Ser Ile Tyr Tyr Phe Asp Tyr
1 5 10

-continued

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<210> SEQ ID NO 385
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      peptide

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<400> SEQUENCE: 385

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```

Lys Ser Ser Gln Ser Leu Leu Asn Ser Arg Thr Arg Lys Asn Tyr Leu
1           5           10           15

```

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Ala

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<210> SEQ ID NO 386
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      peptide

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<400> SEQUENCE: 386

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Trp Ala Ser Thr Arg Lys Ser
1           5

```

```

<210> SEQ ID NO 387
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      peptide

```

```

<400> SEQUENCE: 387

```

```

Thr Gln Ser Phe Ile Leu Arg Thr
1           5

```

```

<210> SEQ ID NO 388
<211> LENGTH: 119
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

```

```

<400> SEQUENCE: 388

```

```

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ser
1           5           10           15

```

```

Ser Val Lys Val Ser Cys Lys Ala Ser Gly Phe Ala Phe Thr Asp Tyr
20           25           30

```

```

Tyr Ile His Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Met
35           40           45

```

```

Gly Trp Ile Ser Pro Gly Asn Val Asn Thr Lys Tyr Asn Glu Asn Phe
50           55           60

```

```

Lys Gly Arg Val Thr Ile Thr Ala Asp Lys Ser Thr Ser Thr Ala Tyr
65           70           75           80

```

```

Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
85           90           95

```

```

Ala Arg Asp Gly Tyr Ser Leu Tyr Tyr Phe Asp Tyr Trp Gly Gln Gly
100          105          110

```

```

Thr Leu Val Thr Val Ser Ser
115

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-continued

<210> SEQ ID NO 389
 <211> LENGTH: 112
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 389

```

Asp Ile Val Met Thr Gln Ser Pro Asp Ser Leu Ala Val Ser Leu Gly
1             5             10             15

Glu Arg Ala Thr Ile Asn Cys Lys Ser Ser Gln Ser Leu Leu Asn Ser
                20             25             30

Arg Thr Arg Lys Asn Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln
            35             40             45

Pro Pro Lys Leu Leu Ile Tyr Trp Ala Ser Thr Arg Gln Ser Gly Val
            50             55             60

Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr
65             70             75             80

Ile Ser Ser Leu Gln Ala Glu Asp Val Ala Val Tyr Tyr Cys Thr Gln
            85             90             95

Ser His Thr Leu Arg Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
            100            105            110

```

<210> SEQ ID NO 390
 <211> LENGTH: 10
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 390

```

Gly Phe Ala Phe Thr Asp Tyr Tyr Ile His
1             5             10

```

<210> SEQ ID NO 391
 <211> LENGTH: 17
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 391

```

Trp Ile Ser Pro Gly Asn Val Asn Thr Lys Tyr Asn Glu Asn Phe Lys
1             5             10             15

Gly

```

<210> SEQ ID NO 392
 <211> LENGTH: 10
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 392

```

Asp Gly Tyr Ser Leu Tyr Tyr Phe Asp Tyr
1             5             10

```

<210> SEQ ID NO 393
 <211> LENGTH: 7
 <212> TYPE: PRT

-continued

<213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 393

Trp Ala Ser Thr Arg Gln Ser
 1 5

<210> SEQ ID NO 394
 <211> LENGTH: 8
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 394

Thr Gln Ser His Thr Leu Arg Thr
 1 5

<210> SEQ ID NO 395
 <211> LENGTH: 120
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 395

Met Ala Glu Ser Gly Gly Gly Ser Val Gln Thr Gly Gly Ser Leu Arg
 1 5 10 15
 Leu Ser Cys Ala Tyr Thr Ala Ser Ser Val Cys Met Ala Trp Phe Arg
 20 25 30
 Gln Ala Pro Gly Lys Glu Arg Glu Gly Val Ala Val Thr Arg Glu Gly
 35 40 45
 Leu Thr Lys Thr Gly Tyr Ala Asp Ser Val Lys Gly Arg Phe Ala Ile
 50 55 60
 Ser Gln Asp Tyr Ala Lys Lys Thr Leu Tyr Leu Gln Met Ser Ser Leu
 65 70 75 80
 Lys Pro Glu Asp Thr Ala Arg Tyr Tyr Cys Ala Ala Arg Pro Thr Ser
 85 90 95
 Pro Cys Thr Val Asp Gly Glu Leu Leu Ala Ser Thr Tyr Asn Tyr Trp
 100 105 110
 Gly Gln Gly Thr Gln Val Thr Val
 115 120

<210> SEQ ID NO 396
 <211> LENGTH: 120
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 396

Met Ala Glu Ser Gly Gly Gly Ser Val Gln Thr Gly Gly Ser Leu Arg
 1 5 10 15
 Leu Ser Cys Ala Tyr Thr Ala Ser Ser Val Cys Met Ala Trp Phe Arg
 20 25 30
 Gln Ala Pro Gly Lys Glu Arg Glu Gly Val Ala Val Thr Arg Glu Gly
 35 40 45

-continued

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Leu Thr Lys Thr Gly Tyr Ala Asp Ser Val Lys Gly Arg Phe Ala Ile
 50                      55                      60

Ser Gln Asp Tyr Ala Lys Lys Thr Leu Tyr Leu Gln Met Ser Ser Leu
65                      70                      75                      80

Lys Pro Glu Asp Thr Ala Arg Tyr Tyr Cys Ala Ala Arg Pro Thr Ser
                        85                      90                      95

Pro Cys Thr Val Asp Gly Glu Leu Leu Ala Ser Thr Tyr Asp Tyr Trp
          100                      105                      110

Gly Gln Gly Thr Gln Val Thr Val
 115                      120

```

```

<210> SEQ ID NO 397
<211> LENGTH: 120
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
                        polypeptide

```

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<400> SEQUENCE: 397

```

```

Met Ala Glu Ser Gly Gly Gly Ser Val Gln Thr Gly Gly Ser Leu Arg
 1                      5                      10                      15

Leu Ser Cys Ala Tyr Thr Ala Ser Ser Val Cys Met Ala Trp Phe Arg
          20                      25                      30

Gln Ala Pro Gly Lys Glu Arg Glu Gly Val Ala Val Thr Arg Glu Gly
          35                      40                      45

Leu Thr Gln Thr Gly Tyr Ala Asp Ser Val Lys Gly Arg Phe Ala Ile
 50                      55                      60

Ser Gln Asp Tyr Ala Lys Lys Thr Leu Tyr Leu Gln Met Ser Ser Leu
65                      70                      75                      80

Lys Pro Glu Asp Thr Ala Arg Tyr Tyr Cys Ala Ala Arg Pro Thr Ser
          85                      90                      95

Pro Cys Thr Val Asp Gly Glu Leu Leu Ala Ser Thr Tyr Asn Tyr Trp
          100                      105                      110

Gly Gln Gly Thr Gln Val Thr Val
 115                      120

```

```

<210> SEQ ID NO 398
<211> LENGTH: 120
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
                        polypeptide

```

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<400> SEQUENCE: 398

```

```

Met Ala Glu Ser Gly Gly Gly Ser Val Gln Thr Gly Gly Ser Leu Arg
 1                      5                      10                      15

Leu Ser Cys Ala Tyr Thr Ala Ser Ser Val Cys Met Ala Trp Phe Arg
          20                      25                      30

Gln Ala Pro Gly Lys Glu Arg Glu Gly Val Ala Val Thr Arg Glu Gly
          35                      40                      45

Leu Thr Gln Thr Gly Tyr Ala Asp Ser Val Lys Gly Arg Phe Ala Ile
 50                      55                      60

Ser Gln Asp Tyr Ala Lys Lys Thr Leu Tyr Leu Gln Met Ser Ser Leu
65                      70                      75                      80

Lys Pro Glu Asp Thr Ala Arg Tyr Tyr Cys Ala Ala Arg Pro Thr Ser
          85                      90                      95

```

-continued

Pro	Cys	Thr	Val	Asp	Gly	Glu	Leu	Leu	Ala	Ser	Thr	Tyr	Asp	Tyr	Trp
			100					105					110		

Gly	Gln	Gly	Thr	Gln	Val	Thr	Val
	115					120	

<210> SEQ ID NO 399
 <211> LENGTH: 16
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 399

Val	Arg	His	Gly	Ser	Phe	Gly	Asn	His	Ile	Val	Ser	Trp	Phe	Ala	His
1			5					10					15		

<210> SEQ ID NO 400
 <211> LENGTH: 40
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (1) .. (40)
 <223> OTHER INFORMATION: This sequence may encompass 1-20 "Gly Ser" repeating units

<400> SEQUENCE: 400

Gly	Ser	Gly	Ser	Gly	Ser	Gly	Ser	Gly	Ser	Gly	Ser	Gly	Ser	Gly	Ser
1			5					10					15		

Gly	Ser	Gly	Ser	Gly	Ser	Gly	Ser	Gly	Ser	Gly	Ser	Gly	Ser	Gly	Ser
		20				25				30					

Gly	Ser	Gly	Ser	Gly	Ser	Gly	Ser
	35					40	

<210> SEQ ID NO 401
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (1) .. (60)
 <223> OTHER INFORMATION: This sequence may encompass 1-20 "Gly Gly Ser" repeating units

<400> SEQUENCE: 401

Gly	Gly	Ser	Gly	Gly	Ser	Gly	Gly	Ser	Gly	Gly	Ser	Gly	Gly	Ser	Gly
1			5					10					15		

Gly	Ser	Gly	Gly	Ser	Gly	Gly	Ser	Gly	Gly	Ser	Gly	Gly	Ser	Gly	Gly
		20				25				30					

Ser	Gly	Gly	Ser	Gly	Gly	Ser	Gly	Gly	Ser	Gly	Gly	Ser	Gly	Gly	Ser
		35				40				45					

Gly	Gly	Ser	Gly	Gly	Ser	Gly	Gly	Ser	Gly	Gly	Ser
	50				55				60		

<210> SEQ ID NO 402
 <211> LENGTH: 80
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:

-continued

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (1)..(80)
 <223> OTHER INFORMATION: This sequence may encompass 1-20 "Gly Gly Gly Ser" repeating units

<400> SEQUENCE: 402

Gly Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly Ser
 1 5 10 15
 Gly Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly Ser
 20 25 30
 Gly Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly Ser
 35 40 45
 Gly Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly Ser
 50 55 60
 Gly Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly Ser
 65 70 75 80

<210> SEQ ID NO 403
 <211> LENGTH: 80
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (1)..(80)
 <223> OTHER INFORMATION: This sequence may encompass 1-20 "Gly Gly Ser Gly" repeating units

<400> SEQUENCE: 403

Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly Ser Gly
 1 5 10 15
 Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly Ser Gly
 20 25 30
 Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly Ser Gly
 35 40 45
 Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly Ser Gly
 50 55 60
 Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly Ser Gly
 65 70 75 80

<210> SEQ ID NO 404
 <211> LENGTH: 100
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (1)..(100)
 <223> OTHER INFORMATION: This sequence may encompass 1-20 "Gly Gly Ser Gly Gly" repeating units

<400> SEQUENCE: 404

Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly
 1 5 10 15
 Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly
 20 25 30
 Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly Ser

-continued

35	40	45
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Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly
 50 55 60

Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly
 65 70 75 80

Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly
 85 90 95

Gly Ser Gly Gly
 100

<210> SEQ ID NO 405
 <211> LENGTH: 100
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (1) .. (100)
 <223> OTHER INFORMATION: This sequence may encompass 1-20 "Gly Gly Gly
 Gly Ser" repeating units

<400> SEQUENCE: 405

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly
 1 5 10 15

Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly
 20 25 30

Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
 35 40 45

Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly
 50 55 60

Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
 65 70 75 80

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly
 85 90 95

Gly Gly Gly Ser
 100

<210> SEQ ID NO 406
 <211> LENGTH: 634
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 406

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15

Ser Leu Lys Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Asn Lys Tyr
 20 25 30

Ala Ile Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ala Arg Ile Arg Ser Lys Tyr Asn Asn Tyr Ala Thr Tyr Tyr Ala Asp
 50 55 60

Gln Val Lys Asp Arg Phe Thr Ile Ser Arg Asp Asp Ser Lys Asn Thr
 65 70 75 80

Ala Tyr Leu Gln Met Asn Asn Leu Lys Thr Glu Asp Thr Ala Val Tyr
 85 90 95

-continued

Tyr Cys Val Arg His Ala Asn Phe Gly Asn Ser Tyr Ile Ser Tyr Trp
 100 105 110
 Ala Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser Gly Gly Gly
 115 120 125
 Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gln Thr Val Val
 130 135 140
 Thr Gln Glu Pro Ser Leu Thr Val Ser Pro Gly Gly Thr Val Thr Leu
 145 150 155 160
 Thr Cys Ala Ser Ser Thr Gly Ala Val Thr Ser Gly Asn Tyr Pro Asn
 165 170 175
 Trp Val Gln Gln Lys Pro Gly Gln Ala Pro Arg Gly Leu Ile Gly Gly
 180 185 190
 Thr Lys Phe Leu Val Pro Gly Thr Pro Ala Arg Phe Ser Gly Ser Leu
 195 200 205
 Leu Gly Gly Lys Ala Ala Leu Thr Leu Ser Gly Val Gln Pro Glu Asp
 210 215 220
 Glu Ala Glu Tyr Tyr Cys Thr Leu Trp Tyr Ser Asn Arg Trp Val Phe
 225 230 235 240
 Gly Gly Gly Thr Lys Leu Thr Val Leu Gly Gly Gly Gly Ser Gly Gly
 245 250 255
 Gly Ser Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro
 260 265 270
 Gly Asn Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser
 275 280 285
 Ser Phe Gly Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu
 290 295 300
 Trp Val Ser Ser Ile Ser Gly Ser Gly Ser Asp Thr Leu Tyr Ala Asp
 305 310 315 320
 Ser Val Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Thr Thr
 325 330 335
 Leu Tyr Leu Gln Met Asn Ser Leu Arg Pro Glu Asp Thr Ala Val Tyr
 340 345 350
 Tyr Cys Thr Ile Gly Gly Ser Leu Ser Arg Ser Ser Gln Gly Thr Leu
 355 360 365
 Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser Gln Val
 370 375 380
 Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Gln Thr Leu
 385 390 395 400
 Ser Leu Thr Cys Thr Val Ser Gly Gly Ser Ile Ser Thr Ser Gly Met
 405 410 415
 Gly Val Gly Trp Ile Arg Gln His Pro Gly Lys Gly Leu Glu Trp Ile
 420 425 430
 Gly His Ile Trp Trp Asp Asp Asp Lys Arg Tyr Asn Pro Ala Leu Lys
 435 440 445
 Ser Arg Val Thr Ile Ser Val Asp Thr Ser Lys Asn Gln Phe Ser Leu
 450 455 460
 Lys Leu Ser Ser Val Thr Ala Ala Asp Thr Ala Val Tyr Tyr Cys Ala
 465 470 475 480
 Arg Met Glu Leu Trp Ser Tyr Tyr Phe Asp Tyr Trp Gly Gln Gly Thr
 485 490 495
 Leu Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
 500 505 510

-continued

Gly Gly Gly Gly Ser Glu Ile Val Leu Thr Gln Ser Pro Ala Thr Leu
 515 520 525
 Ser Leu Ser Pro Gly Glu Arg Ala Thr Leu Ser Cys Ser Ala Ser Ser
 530 535 540
 Ser Val Ser Tyr Met His Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro
 545 550 555 560
 Arg Leu Leu Ile Tyr Asp Thr Ser Lys Leu Ala Ser Gly Ile Pro Ala
 565 570 575
 Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser
 580 585 590
 Ser Leu Glu Pro Glu Asp Phe Ala Val Tyr Tyr Cys Phe Gln Gly Ser
 595 600 605
 Val Tyr Pro Phe Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys Arg
 610 615 620
 His His His His His His His His His His
 625 630

<210> SEQ ID NO 407
 <211> LENGTH: 10
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 10xHis tag

<400> SEQUENCE: 407

His His His His His His His His His His
 1 5 10

What is claimed is:

1. A multi-specific binding protein comprising:

(a) a first antigen-binding site that binds to human CD19 comprising heavy chain complementarity determining regions (HCDR) and light chain complementarity determining regions (LCDR), wherein:

HCDR1 comprises the amino acid sequence of SEQ ID NO: 3, HCDR2 comprises the amino acid sequence of SEQ ID NO: 4, HCDR3 comprises the amino acid sequence of SEQ ID NO: 5, LCDR1 comprises the amino acid sequence of SEQ ID NO: 6, LCDR2 comprises the amino acid sequence of SEQ ID NO: 7, and LCDR3 comprises the amino acid sequence of SEQ ID NO: 8;

(b) a second antigen-binding site that binds to human CD3 comprising heavy chain complementarity determining regions (HCDR) and light chain complementarity determining regions (LCDR), wherein HCDR1 comprises the amino acid sequence of SEQ ID NO: 99, HCDR2 comprises the amino acid sequence of SEQ ID NO: 100, HCDR3 comprises the amino acid sequence of SEQ ID NO: 101, LCDR1 comprises the amino acid sequence of SEQ ID NO: 102, LCDR2 comprises the amino acid sequence of SEQ ID NO: 103, and LCDR3 comprises the amino acid sequence of SEQ ID NO: 104;

(c) a third antigen-binding site that binds to human serum albumin (HSA) comprising a heavy chain complementarity determining regions (HCDR) wherein HCDR1 comprises the amino acid sequence of SEQ ID NO: 122 or 123, HCDR2 comprises the amino acid sequence of SEQ ID NO: 124 or 125, HCDR3 comprises the amino acid sequence of SEQ ID NO: 126,

35 and wherein the first antigen-binding site and second antigen-binding site comprise a single-chain variable fragment (scFv) and the third antigen-binding site comprises a single-domain antibody (sdAb).

2. The multi-specific binding protein of claim 1, wherein the third antigen-binding site is not positioned between the first antigen-binding site and the second antigen-binding site in the polypeptide chain.

3. The multi-specific binding protein of claim 2, wherein the third antigen-binding site is positioned N-terminal to both the first antigen-binding site and the second antigen-binding site in the polypeptide chain.

4. The multi-specific binding protein of claim 3, wherein the third antigen-binding site is positioned N-terminal to the first antigen-binding site, and the first antigen-binding site is positioned N-terminal to the second antigen-binding site in the polypeptide chain.

5. The multi-specific binding protein of claim 3, wherein the third antigen-binding site is positioned N-terminal to the second antigen-binding site, and the second antigen-binding site is positioned N-terminal to the first antigen-binding site in the polypeptide chain.

6. The multi-specific binding protein of claim 2, wherein the third antigen-binding site is positioned C-terminal to both the first antigen-binding site and the second antigen-binding site in the polypeptide chain.

7. The multi-specific binding protein of claim 6, wherein:
 (i) the first antigen-binding site is positioned N-terminal to the second antigen-binding site, and the second antigen-binding site is positioned N-terminal to the third antigen-binding site in the polypeptide chain; or
 (ii) the second antigen-binding site is positioned N-terminal to the first antigen-binding site, and the first

391

antigen-binding site is positioned N-terminal of the third antigen-binding site in the polypeptide chain.

8. The multi-specific binding protein of claim 1, wherein:

- (i) the first antigen-binding site is positioned N-terminal to the third antigen-binding site, and the third antigen-binding site is positioned N-terminal to the second antigen-binding site in the polypeptide chain; or
- (ii) the second antigen-binding site is positioned N-terminal to the third antigen-binding site, and the third antigen-binding site is positioned N-terminal to the first antigen-binding site in the polypeptide chain.

9. The multi-specific binding protein of claim 8, wherein at least two adjacent antigen-binding sites are connected by a peptide linker comprising the amino acid sequence of SEQ ID NO: 298, 299, or 302.

10. The multi-specific binding protein of claim 1, wherein the first antigen-binding site binds to human CD19 with a dissociation constant (K_D) equal to or lower than 20 nM.

11. The multi-specific binding protein of claim 1, wherein the first antigen-binding site has a melting temperature of at least 60° C.

12. The multi-specific binding protein of claim 1, wherein the second antigen-binding site binds to human CD3 with a K_D equal to or lower than 10 nM.

13. The multi-specific binding protein of claim 1, wherein the second antigen-binding site has a melting temperature of at least 60° C.

14. The multi-specific binding protein of claim 1, wherein the third antigen-binding site binds to HSA with a K_D equal to or lower than 20 nM.

15. The multi-specific binding protein of claim 1, wherein the third antigen-binding site has a melting temperature of at least 60° C.

16. The multi-specific binding protein of claim 1, wherein the multi-specific binding protein does not comprise an antibody Fc region.

17. The multi-specific binding protein of claim 1, wherein the molecular weight of the multi-specific binding protein is at least 65 kD.

18. The multi-specific binding protein of claim 1, wherein the serum half-life of the multi-specific binding protein is at least 24 hours.

392

19. A pharmaceutical composition comprising the multi-specific binding protein of claim 1 and a pharmaceutically acceptable carrier.

20. A complex comprising a T cell expressing CD3, a B cell expressing CD19, and the multi-specific binding protein of claim 1, wherein the multi-specific binding protein is capable of simultaneously binding to both the T cell and the B cell.

21. An isolated polynucleotide encoding the multi-specific binding protein of claim 1.

22. A vector comprising the polynucleotide of claim 21.

23. An isolated recombinant host cell comprising the polynucleotide of claim 21.

24. A method of producing a multi-specific binding protein, the method comprising culturing the host cell of claim 23 under suitable conditions that allow expression of the multi-specific binding protein.

25. A method of stimulating an immune response against a cell expressing CD19, the method comprising exposing the cell and a T lymphocyte to the multi-specific binding protein of claim 1.

26. A multi-specific binding protein comprising:

(a) a first antigen-binding site that binds to CD19 comprising

a heavy chain variable domain (VH) and a light chain variable domain (VL), wherein

(i) the VH comprises the amino acid sequence of SEQ ID NO: 1, and the VL comprises the amino acid sequence of SEQ ID NO: 2; or

(ii) the VH comprises the amino acid sequence of SEQ ID NO: 10 and the VL comprises the amino acid sequence of SEQ ID NO: 11;

(b) a second antigen-binding site that binds to human CD3 comprising a VH, wherein the VH comprises the amino acid sequence of SEQ ID NO: 97 and a VL wherein the VL comprises the amino acid sequence of SEQ ID NO: 98; and

a third antigen-binding site that binds to human serum albumin (HSA) comprising a VH, wherein the VH comprises the amino acid sequence of SEQ ID NO: 121.

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